MIDDLE YEARS PROGRAMME
YEAR 3 · MATHEMATICS

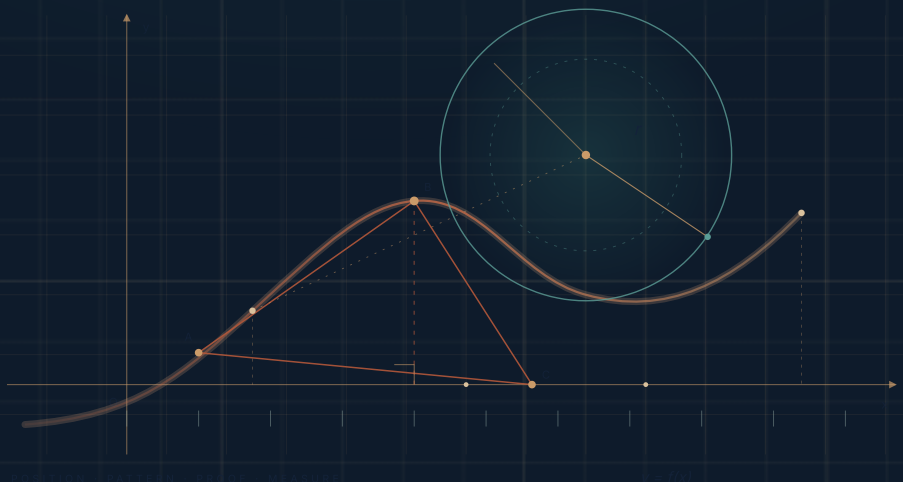
FIRST EDITION
2026

MYP

MATHEMATICS 3

3 *Student Coursebook*

An inquiry-led course in number, algebra, proof, space and evidence — built around concepts, contexts and criteria.



Real numbers · Powers and scientific notation · Functions and algebra
Linear relationships · Equations and systems · Transformations and congruence
Similarity and trigonometry · Solids and measurement · Bivariate data · Probability and evidence

SADIKOU Moukoddim
HEAD OF MATHEMATICS DEPARTMENT | MATHEMATICS TEACHER | CATEGORY 1

STUDENT EDITION
MYP YEAR 3

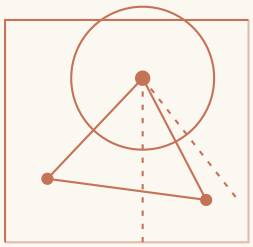
STUDENT COURSEBOOK
FIRST EDITION · 2026

MYP

Mathematics 3

Student Coursebook

An inquiry-led course in number systems, algebraic reasoning, linear relationships, proof, space, measurement and evidence — designed around the concepts, contexts and assessment criteria of the Middle Years Programme.



SADIKOU Moukoddim

HEAD OF MATHEMATICS DEPARTMENT |
MATHEMATICS TEACHER | CATEGORY 1

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MYP Mathematics 3 — Student Coursebook

FIRST EDITION · 2026

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Author and publisher: SADIKOU Moukoddim | Head of Mathematics Department | Mathematics Teacher | Category 1.

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Scope and sequence. The order of chapters is one defensible school sequence, not an official required sequence. MYP schools design their own units around conceptual understandings and local contexts. Teachers should check the sequence against their programme documentation and their students’ prior learning.

Data and contexts. Real-world scenarios use rounded, simplified or classroom-generated data. Figures are chosen to support the mathematics and may not reflect the most recent published statistics unless a source is named. Students should verify current data from a reliable source when a task requires it.

Assessment. Criterion tasks, rubrics and level descriptors in this book are teaching tools written for formative classroom use. They paraphrase and interpret the MYP assessment criteria; they are not official IB descriptors and do not replace the current subject guide.

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AUTHOR CREDIT: SADIKOU MOUKODDIM | HEAD OF MATHEMATICS DEPARTMENT | MATHEMATICS TEACHER | CATEGORY 1.

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How to use this coursebook

Year 3 is where several familiar ideas become tools for general reasoning. A letter starts to stand for a family of numbers, a table of values becomes a relationship, a drawing becomes evidence for a claim, and a set of measurements becomes an argument about uncertainty.

You already know how to calculate with whole numbers, fractions and decimals. You have met negative numbers, simple equations and the basic properties of shapes. This course asks you to work with those ideas at a different level: to explain why a rule always works, to choose an efficient representation, to connect an equation with its graph, and to decide when a numerical answer is precise enough to be useful.

The book is arranged in four parts. Part I builds the number system and algebraic structure that the rest of the course depends on: irrational numbers, exponent laws, scientific notation, functions and algebraic manipulation. Part II turns that structure into relationships: gradient and intercept, linear models, equations, inequalities and systems. Part III is geometry as a deductive discipline: transformations, congruence, similarity, right-triangle trigonometry, and the measurement of solids. Part IV examines data and chance, with particular attention to the difference between a pattern in data and a causal claim.

Every chapter follows the same inquiry rhythm. A situation raises a question; definitions and representations give you the tools to investigate it; worked examples make the reasoning visible; graded practice makes the method dependable; an investigation, an application and a criterion task ask you to use the mathematics beyond the original example. The answer section confirms your reasoning rather than replacing it.

ANATOMY OF A CHAPTER

SIX MOVEMENTS, REPEATED IN EVERY CHAPTER

01

Chapter opener

A visual context and one central question that makes the chapter's mathematics worth developing.

02

MYP unit planner

Key and related concepts, global context, statement of inquiry, inquiry questions and criterion mapping.

03

Intentions and prerequisites

Learning intentions, success criteria, a prior-knowledge check and the boundary of the Year 3 treatment.

04

Concept sequence

Definitions, representations, worked examples, common errors and graduated practice for each concept.

05

Investigate and apply

A conjecture-building investigation, a practical application and a mini-task mapped to MYP criteria.

06

Review and mixed practice

Summary, reflection, revision checklist and cumulative questions combining the chapter's ideas.

MATHEMATICAL HABITS THAT THIS COURSE DEVELOPS

- → **Represent.** Move between symbols, tables, graphs, diagrams, words and physical models.
- → **Generalise.** Move from examples to a rule, then test the rule against a new case.
- → **Connect.** Ask how an equation, a graph and a context describe the same relationship.
- → **Justify.** Say why a method works, not only what answer it produces.
- → **Estimate.** Predict the size and sign of an answer before calculating it.
- → **Reflect.** Name what you understand, what is still fragile, and what you will do next.

HOW THE LABELS WORK

Definition boxes state precise meanings. **Key idea** boxes connect a method to a larger principle. **Strategy** boxes give a reusable procedure. **Common error** boxes show a mistake worth anticipating. **ATL** boxes make a learning skill explicit and give you a way to practise it. **Criterion** strips name the MYP criterion and strands developed by a section.

EXERCISE TYPES

F

FLUENCY

Short, deliberate practice to make a method accurate and efficient.

R

REASONING

Explain, compare, justify, find a counterexample or diagnose an error.

P

PROBLEM SOLVING

Multi-step and unfamiliar contexts that require selection and planning.

E

EXTENSION

Deeper structure, generalisation or a more open task.

ATL · USING THE ANSWER KEY WELL

Attempt the question fully before checking. If your answer differs, locate the first line where your reasoning diverges. Mark the correction in a different colour and write one sentence explaining why the alternative was wrong. That sentence is often more valuable than the correct number.

ACADEMIC HONESTY

Copying a solution without understanding it creates the appearance of progress and hides a gap. Use sample answers to compare *methods*, not to replace your attempt. In investigations, report your own data and reasoning, and acknowledge any source you use.

The MYP framework in this course

The Middle Years Programme asks students to learn mathematics as a connected, conceptual and applicable discipline. Concepts organise the content, contexts give it purpose, and criteria make the quality of thinking visible.

Aims of MYP mathematics

The MYP mathematics framework aims to help students enjoy mathematics, develop curiosity and confidence, build conceptual understanding, apply mathematics to authentic situations, communicate reasoning, think critically and creatively, and appreciate the international and historical dimensions of the subject. In Year 3, the emphasis shifts from describing procedures to explaining structure: why exponent laws follow from repeated multiplication, why a linear function has constant gradient, why congruence can be established from limited information, and why correlation alone cannot establish causation.

KEY CONCEPT 01

Form

The shape, structure and organisation of mathematical objects: the form of a prime factorisation, the form of a linear equation, the form of a geometrical proof. Form helps us notice what is mathematically the same beneath different appearances.

KEY CONCEPT 02

Logic

The reasoning that connects claims to conclusions. Logic asks not only "Is it true?" but "How do we know, and for which cases?" It is the discipline behind classification, deduction, counterexample and generalisation.

KEY CONCEPT 03

Relationships

Connections between quantities, variables, representations and ideas. A function, a graph, a ratio, a similarity statement and a trend line are different ways of expressing relationships.

Related concepts used in this course

CHANGE
How one quantity responds when another changes.

EQUIVALENCE
Different forms with the same value or meaning.

GENERALISATION
Extending a relationship to a whole class of cases.

JUSTIFICATION
Reasons and evidence that support a claim.

MEASUREMENT
Assigning numbers and units to attributes of objects.

MODELS
Simplified representations of a real situation.

PATTERNS
Regular structure that repeats or grows predictably.

QUANTITY
Number, magnitude, measure and unit.

REPRESENTATION
Words, symbols, tables, diagrams and graphs.

SIMPLIFICATION
Rewriting an expression or result in a more efficient equivalent form.

SPACE
Position, direction, transformation and the properties of figures in two and three dimensions.

SYSTEMS
Interacting parts governed by rules — number systems, coordinate systems, systems of equations.

Global contexts

Each chapter is anchored in one of the six MYP global contexts. The context is not decoration: it shapes the questions asked, the data used and the kind of solution that counts as useful.

GLOBAL CONTEXT	HOW MATHEMATICS ENTERS THE INQUIRY
Identities and relationships	How categories, groups and belonging can be described and interrogated; how similarity and scale connect objects while preserving identity of shape.
Orientation in space and time	How number systems, calendars and measurement developed across cultures; how coordinates, transformations and scale structure space.
Personal and cultural expression	How proportion, pattern and geometry shape music, design, craft and architecture; how tessellation and symmetry create visual order.
Scientific and technical innovation	How accurate measurement, formulas, scientific notation and data displays support invention, engineering and scientific claims.
Fairness and development	How budgets, percentages, comparisons and data can support or distort fair decisions; how samples and claims can mislead.
Globalization and sustainability	How quantities are compared across communities and environments, and how claims about resources and risk are communicated.

Inquiry in a mathematics lesson

Inquiry does not mean that students are left to rediscover every definition without guidance. It means that a question genuinely drives the sequence of work, that students make and test conjectures, and that they can explain how a result was reached. The course uses a repeating pathway.

1 PROVOKE
Notice a situation, pattern or claim that is not yet explainable.

2 QUESTION
Turn the noticing into a mathematical question with quantities or objects.

3 REPRESENT
Choose diagrams, tables, symbols or materials that reveal structure.

4 GENERALISE
Move from examples to a rule, testing it against further cases.

5 APPLY
Use the rule in a context and judge whether the answer makes sense.

6 REFLECT
Evaluate the method, the evidence and the limits of the conclusion.

Approaches to learning

ATL skills are not an extra syllabus. They are the practices that make mathematical learning transferable. Each chapter names one ATL focus and gives a concrete way to practise it.

ATL CATEGORY	WHAT IT LOOKS LIKE IN THIS COURSE
Communication	Writing a chain of reasoning that another student can follow; choosing notation and units deliberately; labelling axes and diagrams.
Research	Collecting or locating data, checking its provenance, recording sources and describing limitations.
Thinking	Testing conjectures, constructing counterexamples, comparing representations and transferring a method to a new context.
Social	Explaining a method to a partner, asking a clarifying question and reaching agreement on a justified answer.
Self-management	Planning a multi-step task, checking a result by an independent method and reflecting on an error.

ATL FOCUS BY CHAPTER

CHAPTER	ATL FOCUS	HOW IT IS PRACTISED
1 · The real number system	Critical thinking	Testing universal claims, finding counterexamples and constructing a proof by contradiction.
2 · Powers, roots and scientific notation	Research	Locating, comparing and presenting very large and very small quantities from reliable sources.
3 · Functions and algebraic reasoning	Thinking	Moving between a pattern, a table, an expression and a graph; generalising a rule.
4 · Linear relationships	Communication	Explaining the meaning of gradient and intercept in a real context with correct units.
5 · Equations, inequalities and systems	Self-management	Planning a multi-step solution, checking by substitution and reviewing the reasonableness of a result.
6 · Transformations and congruence	Thinking	Justifying congruence from limited information and distinguishing a demonstration from a proof.
7 · Similarity, proof and trigonometry	Research	Using indirect measurement and evaluating the precision of a calculated length.
8 · Three-dimensional solids	Self-management	Organising a multi-stage volume or surface-area calculation and checking units.
9 · Bivariate data and causation	Research	Evaluating the provenance of data and separating a statistical association from a causal claim.
10 · Probability and evidence	Communication	Reporting a sample, its limitations and a probability estimate in language a non-specialist can interpret.

Assessment criteria in practice

MYP mathematics uses four criteria. They describe different qualities of mathematical work, not four separate syllabuses. A single rich task can provide evidence for more than one criterion.

A

KNOWING AND UNDERSTANDING

- Select appropriate mathematics for familiar and unfamiliar problems.
- Apply the selected mathematics successfully.
- Solve problems correctly across a range of contexts.
- In this course: number systems, exponent laws, function notation, linear equations, angle and circle facts, trigonometry, surface area and volume.

B

INVESTIGATING PATTERNS

- Apply problem-solving techniques to recognise patterns.
- Describe patterns as relationships or general rules consistent with findings.
- Verify whether a pattern works for other examples.
- In this course: growth patterns, gradient and rate of change, transformation mappings, similarity ratios, scaling of volume, trends in bivariate data.

C

COMMUNICATING

- Use appropriate mathematical language, notation and symbols.
- Use different forms of representation and move between them.
- Present complete, coherent and concise lines of reasoning.
- Organise information using a logical structure.
- In this course: function notation, set and interval notation, congruence statements, two-column and paragraph proofs, labelled graphs and units.

D

APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

- Identify relevant elements of authentic situations.
- Select and apply mathematical strategies.
- Justify the degree of accuracy of a solution.
- Justify whether the solution makes sense in context.
- In this course: data plans, ramp design, packaging, urban planning, survey sampling, risk communication and evidence evaluation.

How the criteria work together

A criterion task in this book names the criteria it is designed to evidence. A task may, for example, ask you to model a situation (A), identify and generalise a pattern (B), present the reasoning (C) and evaluate the accuracy and meaning of the result (D). Not every exercise needs all four criteria. Short fluency exercises usually provide evidence for A only; an investigation may provide evidence for B and C; a modelling task may provide evidence for A, C and D.

WHAT TO SHOW	WHAT STRONG WORK LOOKS LIKE
Selection	The chosen method fits the structure of the problem, and an alternative was considered where relevant.
Application	Each step is mathematically valid, notation is accurate, and the result is checked against the context.
Justification	Claims are supported by reasons, definitions or evidence; a generalisation is distinguished from a set of examples.
Communication	A reader can follow the reasoning without reconstructing missing steps; diagrams and units are purposeful.
Evaluation	Limitations, assumptions and the appropriateness of the accuracy are stated, not merely implied.

Formative practice is not a formal assessment

Most questions in this book are practice, not summative assessment. Their purpose is to produce feedback: an error is information about what to practise next. Criterion tasks at the end of each chapter are longer and are accompanied by a planning frame and a course rubric. Teachers should apply their school's official assessment descriptors and procedures; the bands in this book are teaching summaries, not official IB level descriptors.

> **A NOTE ON COMMAND TERMS**
>
> **State** requires a short answer without justification. **Describe** requires relevant features. **Explain** requires reasons or mechanisms. **Justify** requires evidence or a valid argument. **Evaluate** requires a judgment supported by criteria. **Investigate** requires a planned process, data or examples, a pattern, and a conclusion. The command term tells you how much reasoning to show, not how long the answer must be.

Course map and progression

The sequence below is one coherent route through Year 3. Each chapter names the conceptual focus that organises it, the global context that gives it purpose, and the criteria most likely to be evidenced by its main task.

PART	CHAPTER	CONCEPTUAL FOCUS	GLOBAL CONTEXT	MAIN CRITERION TASK
I Number and algebraic structure	**1 · The real number system**	Form, equivalence, proof	Orientation in space and time	C — a mathematical exposition of irrationality
I Number and algebraic structure	**2 · Powers, roots and scientific notation**	Patterns, representation, quantity	Scientific and technical innovation	D — communicating very large and very small quantities
I Number and algebraic structure	**3 · Functions, expressions and algebraic reasoning**	Relationships, generalisation, equivalence	Identities and relationships	B — generalising a growing pattern
II Relationships and equations	**4 · Linear relationships**	Change, relationships, models	Fairness and development	D — comparing two pricing models
II Relationships and equations	**5 · Equations, inequalities and systems**	Equivalence, balance, systems	Globalization and sustainability	A and D — planning a resource allocation
III Geometry, proof and space	**6 · Transformations, congruence and symmetry**	Space, form, justification	Personal and cultural expression	C — justifying congruence and symmetry
III Geometry, proof and space	**7 · Similarity, proof and trigonometry**	Relationships, justification, measurement	Orientation in space and time	B and D — indirect measurement
III Geometry, proof and space	**8 · Three-dimensional solids and measurement**	Measurement, models, change	Globalization and sustainability	D — packaging and material use
IV Data, chance and evidence	**9 · Bivariate data, correlation and causation**	Relationships, models, justification	Fairness and development	D — evaluating a public claim
IV Data, chance and evidence	**10 · Probability, sampling and evidence**	Quantity, models, justification	Identities and relationships	B and C — designing and reporting a sample

What Year 3 assumes

BEFORE YEAR 3

Fluency with integers, fractions and decimals; percentage of a quantity; simple algebraic substitution; solving one- and two-step linear equations; angle facts for lines and triangles; area and perimeter of rectangles and

triangles; mean, median, mode and range.	**YEAR 3 · THIS COURSE** Rational and irrational numbers; exponent laws; scientific notation; function ideas; linear functions and models; equation solving, inequalities and systems; transformations and congruence; similarity and trigonometry; volume and surface area; bivariate data; probability and sampling.	**YEARS 4–5** Quadratic and exponential functions; algebraic fractions; simultaneous equations in more general forms; similarity proofs and the unit circle; vectors; complex numbers; statistical inference, probability distributions and modelling with functions.

How the parts build on one another

Part I

Structure before method

Irrational numbers extend the number line; exponent laws compress repeated multiplication; algebraic notation names a function. Without this structure, later procedures are memorised rather than understood.

Part II

Relationships as objects

Linear functions connect tables, graphs, equations and contexts. Solving equations and systems is then a question about intersection and balance, not a sequence of arbitrary moves.

Part III

Reasoning about space

Transformations make congruence precise. Similarity makes trigonometric ratios possible. Solids turn those ratios into volume and surface area, including the surprising cube law for scaling.

Part IV

Evidence and uncertainty

Bivariate data connects algebra and statistics. Probability quantifies uncertainty. Sampling explains why a claim from data needs a method, a margin of doubt and a causal argument.

ATL · PLANNING YOUR YEAR

At the start of each chapter, look at the learning intentions and the criterion task. Write one question you expect the chapter to answer and one skill you want to become more reliable at. At the end, return to both. This small routine turns a list of topics into a plan for your own learning.

CONVENTIONS

Notation, reasoning and habits

Mathematical notation is a form of compression: it lets a complex idea be held in a single line. Learn the conventions now, because a small notation error can change the meaning of an entire argument.

Core notation

x^2	the square of x; x multiplied by itself
$\sqrt{x}$	the non-negative number whose square is x; the square root of x
$\sqrt[3]{x}$	the cube root of x
a^{-n}	the reciprocal of a^n, so $a^{-n} = \frac{1}{a^n}$ for $a \neq 0$
3.2×10^5	scientific notation: 3.2 multiplied by 10 five times
$f(x) = 2x + 1$	the function f assigns the output $2x + 1$ to the input x
$\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$	natural, integer, rational and real number sets
$\in$, $\notin$	is an element of; is not an element of
$3 \leq x < 8$	the set of real numbers greater than or equal to 3 and less than 8
Δx	a change in x; the difference between two x-values
$m = \frac{\Delta y}{\Delta x}$	gradient: vertical change divided by horizontal change
$y \propto x$	y is proportional to x; there is a nonzero constant k with $y = kx$
$\triangle ABC \cong \triangle DEF$	triangles are congruent: corresponding vertices, sides and angles match
$\triangle ABC \sim \triangle DEF$	triangles are similar: corresponding angles are equal and sides are in a constant ratio
$P(A)$	the probability of event A
$P(A \mid B)$	the conditional probability of A given that B has occurred
$\bar{x}$, μ	sample mean; population mean
σ	standard deviation, a measure of spread

Three habits of precise reasoning

HABIT 01

State the rule

When you use a result, name it. “The sum of angles on a straight line is 180°” is stronger than “the angle is 40” because it shows the reason, not only the answer.

HABIT 02

Distinguish example from proof

Three examples support a conjecture; they do not prove it. A single counterexample refutes a universal claim. Write “for example” when you mean an example, and “because” when you mean a reason.

HABIT 03

Keep the units

A number without a unit may be meaningless. Carry units through calculations and check that the unit of the answer matches the quantity asked for.

Writing a solution

A complete solution does not need to be long. It needs to show the selection of a method, the execution of that method, and a check or conclusion. In this course, the pattern is:

1
Given and required. What information is supplied, and what must be found or shown?
2
Method. Which definition, rule or representation applies, and why?
3
Working. Carry out the steps without skipping a transformation that changes meaning.
4
Answer and check. State the answer with units, then test it against the context or by an independent method.

A NOTE ON CALCULATORS

Unless a question says otherwise, a calculator is allowed. Calculator use does not replace estimation. Before entering a calculation, estimate the answer; after entering it, compare the display with your estimate. For irrational results, keep enough digits during working and round only at the end unless the question specifies otherwise.

ATL · BUILDING A PERSONAL GLOSSARY

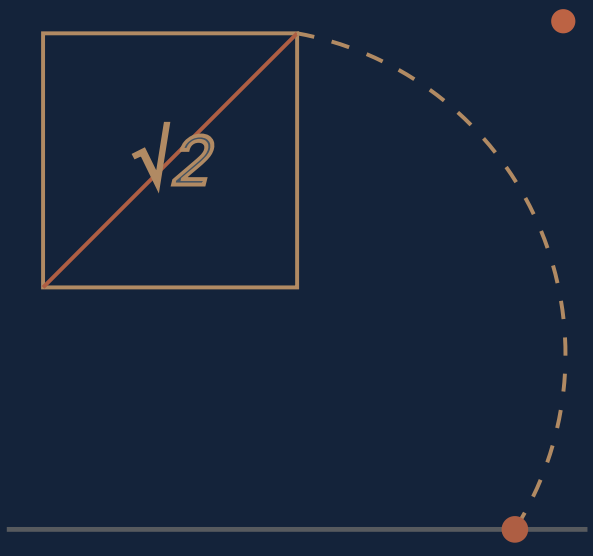
Start a glossary in the back of your notebook. For each new term, write a definition, a diagram or example, and one contrast: “different from …”. A term you can define and contrast is a term you can use in an explanation.

PART I · THE STRUCTURE BENEATH THE SYMBOLS

I

Number and algebraic structure

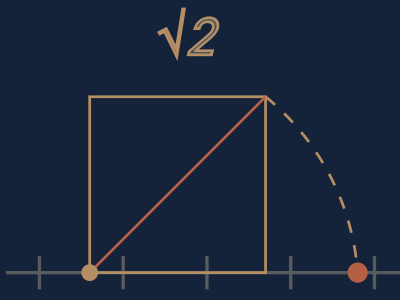
The number line is not a list of labels; it is a system with gaps, extensions and rules. This part extends the numbers you know, compresses repeated multiplication into powers, and shows how a function organises a relationship between two quantities. The aim is not to memorise more rules, but to see why the rules are forced by the structure.



01 The real number system

02 Powers, roots and scientific notation

03 Functions, expressions and algebraic reasoning



1

NUMBER, STRUCTURE AND PROOF

The real number system

A number line drawn with a ruler looks continuous, yet the numbers that fill it are of strikingly different kinds. Some can be written as exact fractions; some cannot be written as fractions at all. Understanding that difference changes what "exact" means.

KEY CONCEPT · FORM | RELATED · QUANTITY | RELATED · EQUIVALENCE | CONTEXT · ORIENTATION IN SPACE AND TIME

INQUIRY

If every point on a line corresponds to a number, how can we describe the numbers that lie between the fractions — and how can we prove that some of them can never be written as fractions?

ATL · CRITICAL THINKING IN ACTION

Consider the claim "every number can be written as a fraction". Test it on integers, terminating decimals and repeating decimals. Then try $\sqrt{2}$. When a single counterexample appears, decide whether the claim is false, or whether the definition of "number" needs to be widened. In this chapter you will do both: reject the universal claim and extend the number system.

MYP UNIT PLANNER

Framing the inquiry

UNIT 1 · APPROX. 14 HOURS
ATL FOCUS · CRITICAL THINKING

KEY AND RELATED CONCEPTS

Form. A number has more than one possible written form: fraction, terminating decimal, repeating decimal, surd or approximation. The form reveals which operations are exact and which are limited.

Quantity. The real number line assigns a position to every decimal, including those with infinitely many digits. Order and distance can be described without listing every digit.

Equivalence. Different decimals, fractions and radical expressions can name the same real number. Recognising equivalence is central to simplifying and to proving.

GLOBAL CONTEXT AND EXPLORATION

Orientation in space and time. Exploration: how different cultures represented and approximated quantities, and why the discovery of incommensurable lengths challenged the idea that all lengths could be expressed as whole-number ratios.

Greek mathematicians initially assumed that any two lengths could be compared using a common unit. The diagonal of a square exposed a length with no such common unit. That discovery did not break mathematics; it forced a broader conception of number. Today, irrational numbers appear in geometry, music theory, engineering tolerances and computer graphics.

STATEMENT OF INQUIRY

The real number system contains both rational and irrational numbers; precise definitions and logical proof let us distinguish exact forms from approximations and reason about quantities that cannot be written as fractions.

FACTUAL

What is a rational number? How can a repeating decimal be written as a fraction? What does the square-root symbol mean?

CONCEPTUAL

Why is the square root of 2 not rational, and how does a proof by contradiction establish that claim?

DEBATABLE

When engineers use 1.414 for $\sqrt{2}$, are they working with the number itself or with a different, rational number?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Classify numbers; convert between forms; simplify surds; order values; solve interval and tolerance problems.
B · Investigating patterns	i, ii, iii	Investigate decimal expansions and successive rational approximations to $\sqrt{2}$; describe and verify the pattern.
C · Communicating	i, ii, iii, iv	Write a structured proof; use set, interval and radical notation; explain the meaning of an approximation.
D · Applying mathematics	i, ii, iii, iv, v	Interpret a manufacturing tolerance as an interval and justify accept/reject decisions.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** the relationships among natural, integer, rational and real numbers.
- **Know** how to recognise terminating and repeating decimal expansions.
- **Be able to** convert between fractions, decimals and simple surd forms.
- **Be able to** order real numbers, including irrational values, using exact reasoning or careful approximation.
- **Be able to** interpret and use intervals, absolute value and tolerance.
- **Be able to** construct a short proof, including a proof by contradiction.

SUCCESS CRITERIA

- ☐ I can classify a number as natural, integer, rational or irrational and justify the classification.
- ☐ I can decide whether a fraction terminates or repeats without performing a long division.
- ☐ I can convert a simple repeating decimal to a fraction.
- ☐ I can simplify a surd and rationalise a one-term denominator.
- ☐ I can place values such as $\sqrt{5}$, π and $-\frac{7}{3}$ in order on a number line.
- ☐ I can write a tolerance statement as an inequality and as an absolute-value statement.
- ☐ I can explain why $\sqrt{2}$ is irrational using a valid chain of reasoning.

PREREQUISITE CHECK · 12 MINUTES, NO CALCULATOR

1. Simplify $\frac{18}{24}$.
2. Write 0.35 as a fraction in lowest terms.
3. Calculate 7^2 and $(-7)^2$.
4. Which is greater, $\frac{3}{4}$ or $\frac{5}{7}$? Show a reason.
5. Calculate 2.5×4.
6. Write the first five square numbers.

Answers are in the Chapter 1 answer section. If more than two are incorrect, review fraction equivalence and square numbers before continuing.

BEFORE YEAR 3

Integers, fractions, decimals, percentages; square numbers; simple roots; ordering positive numbers.

YEAR 3 · THIS CHAPTER

Rational and irrational numbers; decimal expansions; surds; intervals; absolute value; proof by contradiction.

LATER

Real exponents, logarithms, limits of sequences, rigorous ε-δ arguments and the completeness of the real numbers.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Ancient measurement and the diagonal of a square expose a length with no common unit.	Which numbers can be written as fractions, and which cannot?	Rational form, decimal expansion, irrationality, order and interval.	Approximate $\sqrt{2}$ by rationals and examine the error.	Write a proof by contradiction and interpret a tolerance interval.	What does “exact” mean when the decimal expansion never ends?

1.1 Rational numbers and decimal form

CRITERION A · I, II ◆ FRACTIONS ◆ TERMINATING AND REPEATING DECIMALS

A rational number is a quotient of two integers. That simple definition unifies whole numbers, fractions, terminating decimals and repeating decimals: they are different surface forms of the same underlying structure.

DEFINITION

A **rational number** is a number that can be written in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$. The set of rational numbers is denoted $\mathbb{Q}$.

Every integer is rational: $5 = \frac{5}{1}$ and $-12 = \frac{-12}{1}$. Every terminating decimal is rational because it can be written over a power of 10: $0.375 = \frac{375}{1000} = \frac{3}{8}$. A repeating decimal is also rational, although that is less obvious. The repetition is the clue: it allows a finite algebraic calculation to recover an exact fraction.

Not all rational numbers have terminating decimals. In lowest terms, the denominator decides. If the denominator has only prime factors 2 and 5, the decimal terminates, because multiplying numerator and denominator by a suitable power of 10 produces an equivalent fraction with denominator 10^n. If the denominator has any other prime factor, the decimal repeats forever.

TERMINATING

$\frac{3}{8} = 0.375$

$\frac{7}{40} = 0.175$

$\frac{9}{25} = 0.36$

Denominators $8 = 2^3$, $40 = 2^3 \times 5$, $25 = 5^2$ have only 2 and 5 as prime factors.

REPEATING

$\frac{5}{6} = 0.8\overline{3}$

$\frac{2}{11} = 0.\overline{18}$

$\frac{7}{9} = 0.\overline{7}$

Denominators $6 = 2 \times 3$, 11, $9 = 3^2$ contain a prime factor other than 2 or 5.

WORKED EXAMPLE 1.1.1 — FROM REPEATING DECIMAL TO FRACTION

PROBLEM

Write $0.\overline{36}$ as a fraction in lowest terms.

1. Let $x = 0.\overline{36} = 0.363636\ldots$.
2. Two digits repeat, so multiply by 100: $100x = 36.363636\ldots$.
3. Subtract the first equation from the second: $100x - x = 36.\overline{36} - 0.\overline{36} = 36$.
4. So $99x = 36$, giving $x = \frac{36}{99}$.
5. Divide numerator and denominator by 9: $x = \frac{4}{11}$.

ANSWER $\frac{4}{11}$

Check. $4 \div 11 = 0.363636\ldots$, so the result is consistent with the original decimal.

WORKED EXAMPLE 1.1.2 — A DECIMAL WITH A NON-REPEATING PREFIX

PROBLEM

Write $0.2\overline{7}$ as a fraction in lowest terms.

1. Let $x = 0.27777\ldots$. The repeating block has one digit, but it starts after the first decimal place.
2. Multiply by 10 to move the non-repeating part: $10x = 2.7777\ldots$.
3. Multiply the original by 100: $100x = 27.7777\ldots$.
4. Subtract: $100x - 10x = 27.7777\ldots - 2.7777\ldots = 25$.
5. So $90x = 25$ and $x = \frac{25}{90} = \frac{5}{18}$.

ANSWER $\frac{5}{18}$

Alternative. $0.2\overline{7} = 0.2 + 0.0\overline{7} = \frac{1}{5} + \frac{7}{90} = \frac{18}{90} + \frac{7}{90} = \frac{25}{90} = \frac{5}{18}$.

COMMON ERROR

Writing $0.\overline{36}$ as $\frac{36}{100}$ treats a repeating decimal as if it terminated. The value $\frac{36}{100} = 0.36$ is slightly smaller than $0.\overline{36}$. When the digits repeat forever, the denominator must be 9, 99, 999, … or a related value, not a power of 10.

KEY IDEA · DENSITY

Between any two distinct rational numbers there is another rational number. If $a < b$, the midpoint $\frac{a+b}{2}$ is rational and lies strictly between them. Repeating the process shows there are infinitely many rationals in any interval, however small. Yet the number line is not exhausted by rationals; irrational numbers fill the remaining positions.

PRACTICE 1.1 · RATIONAL FORM AND DECIMAL EXPANSION

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Where a decimal repeats, use bar notation, for example $0.1\overline{6}$.

1. Classify each number as natural, integer or rational. More than one label may apply: (a) -8 (b) $\frac{7}{3}$ (c) 0.25 (d) 15. F
2. Write each fraction as a terminating decimal: (a) $\frac{7}{20}$ (b) $\frac{13}{8}$ (c) $\frac{9}{50}$. F
3. Write each fraction as a repeating decimal using bar notation: (a) $\frac{2}{3}$ (b) $\frac{5}{11}$ (c) $\frac{7}{12}$. F
4. Without dividing, decide whether each fraction terminates or repeats: (a) $\frac{3}{14}$ (b) $\frac{11}{40}$ (c) $\frac{17}{30}$ (d) $\frac{9}{64}$. Explain your method. R
5. Convert each repeating decimal to a fraction in lowest terms: (a) $0.\overline{5}$ (b) $0.\overline{27}$ (c) $0.4\overline{1}$. F
6. Find a rational number strictly between $\frac{1}{3}$ and $\frac{1}{2}$. Then find a second rational between your answer and $\frac{1}{3}$. R
7. Explain why $0.999\ldots$ is equal to 1, not "just less than 1". Use the same algebraic method as Worked example 1.1.1. R
8. A student claims that every fraction with a denominator less than 10 has a terminating decimal. Find a counterexample and explain why the claim fails. E

1.2 Irrational numbers and surds

CRITERION A · I ◆ IRRATIONALITY ◆ RADICALS

A rational number has a decimal expansion that eventually terminates or repeats. An irrational number does neither: its decimal continues forever without a repeating block. This is not a limitation of our notation but a property of the number itself.

> DEFINITION
>
> An **irrational number** is a real number that cannot be written as a fraction of two integers. The square root of a positive integer is irrational unless that integer is a perfect square. The set of real numbers $\mathbb{R}$ consists of all rational and irrational numbers.

Familiar examples include $\sqrt{2}$, $\sqrt{3}$, π and e. Their decimal expansions never terminate and never settle into a repeating block. A calculator display such as 1.414213562 is an approximation, not the number itself. No finite string of digits can equal $\sqrt{2}$, because a finite decimal is rational.

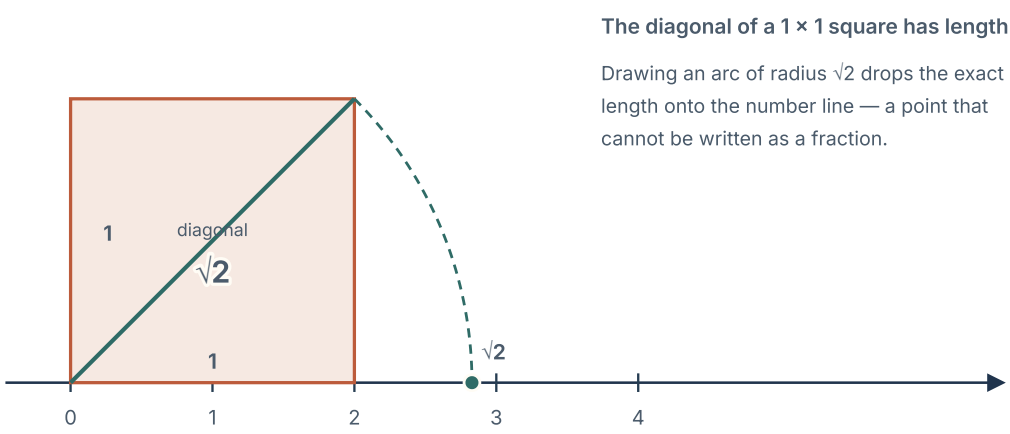


Figure 1.1 A square of side 1 has diagonal length $\sqrt{2}$. Copying that diagonal onto the number line locates an irrational point using only a straight edge and compass.

The construction in Figure 1.1 makes the existence of $\sqrt{2}$ visible. A unit square has diagonal d satisfying $d^2 = 1^2 + 1^2 = 2$, so $d = \sqrt{2}$. The length is not a measurement error or an artefact of drawing: it is forced by the geometry. The next question is whether that length can be expressed as a ratio of integers.

THEOREM · IRRATIONALITY OF $\sqrt{2}$

There are no integers a and b, with $b \neq 0$, such that $\sqrt{2} = \frac{a}{b}$.

PROOF BY CONTRADICTION

Suppose, for contradiction, that $\sqrt{2}$ is rational. Then we can write $\sqrt{2} = \frac{a}{b}$, where a and b are integers, $b \neq 0$, and the fraction is in lowest terms: a and b are not both even.
Squaring both sides gives $2 = \frac{a^2}{b^2}$, so $a^2 = 2b^2$. This means a^2 is even. The square of an odd integer is odd, so a must be even. Write $a = 2k$ for some integer k.
Substituting: $(2k)^2 = 2b^2$, so $4k^2 = 2b^2$, and therefore $b^2 = 2k^2$. This means b^2 is even, so b is even. But a and b cannot both be even if the fraction is in lowest terms. Contradiction.
Therefore the assumption that $\sqrt{2}$ is rational must be false. Hence $\sqrt{2}$ is irrational. □

The proof uses three ideas worth naming: a clear assumption, valid algebraic consequences, and a contradiction with an established fact. It also uses a parity argument — an integer is either even or odd, and the square of an odd number is odd. That small fact does real work.

WORKED EXAMPLE 1.2.1 — ESTIMATING AND COMPARING SURDS

PROBLEM

Without a calculator, estimate $\sqrt{50}$ to one decimal place and place 7.1, $\sqrt{50}$ and 7.05 in order.

1. The perfect squares either side of 50 are $49 = 7^2$ and $64 = 8^2$, so $7 < \sqrt{50} < 8$.
2. Because 50 is much closer to 49 than to 64, $\sqrt{50}$ is close to 7. A first estimate is 7.07.
3. Check: $7.07^2 = 49.9849$, which is slightly below 50. Try $7.08^2 = 50.1264$, which is above 50. So $7.07 < \sqrt{50} < 7.08$.
4. Therefore $7.05 < \sqrt{50} < 7.1$.

ANSWER $\sqrt{50} \approx 7.07$**; the order is** $7.05 < \sqrt{50} < 7.1$**.**

WORKED EXAMPLE 1.2.2 — SIMPLIFYING A SURD

PROBLEM

Simplify $\sqrt{72}$ and $\sqrt{50}$, then write $3\sqrt{2} + \sqrt{50}$ in the form $k\sqrt{2}$.

1. Look for the largest perfect-square factor: $72 = 36 \times 2$, so $\sqrt{72} = \sqrt{36}\sqrt{2} = 6\sqrt{2}$.
2. Similarly $50 = 25 \times 2$, so $\sqrt{50} = 5\sqrt{2}$.
3. Add like surds: $3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$.

ANSWER $\sqrt{72} = 6\sqrt{2}$**,** $\sqrt{50} = 5\sqrt{2}$**, and** $3\sqrt{2} + \sqrt{50} = 8\sqrt{2}$**.**

MISCONCEPTION

$\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ for all non-negative a and b.

CORRECT

Test $a = 9$, $b = 16$: $\sqrt{9+16} = 5$, but $\sqrt{9} + \sqrt{16} = 7$. The sum rule fails. Use $\sqrt{ab} = \sqrt{a}\sqrt{b}$ for products and quotients only.

PRACTICE 1.2 · IRRATIONAL NUMBERS AND SURDS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

Give exact answers unless an approximation is requested.

1. State whether each number is rational or irrational: (a) $\sqrt{9}$ (b) $\sqrt{10}$ (c) $0.\overline{4}$ (d) π. F
2. Between which two consecutive integers does each surd lie? (a) $\sqrt{20}$ (b) $\sqrt{75}$ (c) $\sqrt{103}$. F
3. Simplify: (a) $\sqrt{18}$ (b) $\sqrt{48}$ (c) $\sqrt{200}$ (d) $\sqrt{45}$. F
4. Order from least to greatest: $\sqrt{8}$, 2.9, $\sqrt{7}$, 2.65. Justify each comparison briefly. R
5. Explain why $\sqrt{25}$ is rational while $\sqrt{26}$ is irrational. R
6. Use the proof of $\sqrt{2}$ as a model to prove that $\sqrt{3}$ is irrational. State clearly where you use the fact that if a^2 is divisible by 3 then a is divisible by 3. P
7. Find two irrational numbers whose product is rational. Explain why your example works. E
8. The number $0.101001000100001\ldots$ contains a growing block of zeros between ones. Explain why it is irrational, even though its pattern is describable. E

1.3 Order, absolute value and intervals

CRITERION A · II, III ◆ NUMBER LINE ◆ INEQUALITIES

Real numbers are ordered: for any two distinct values, exactly one is smaller. A number line turns that order into geometry, so comparisons become statements about position and distance.

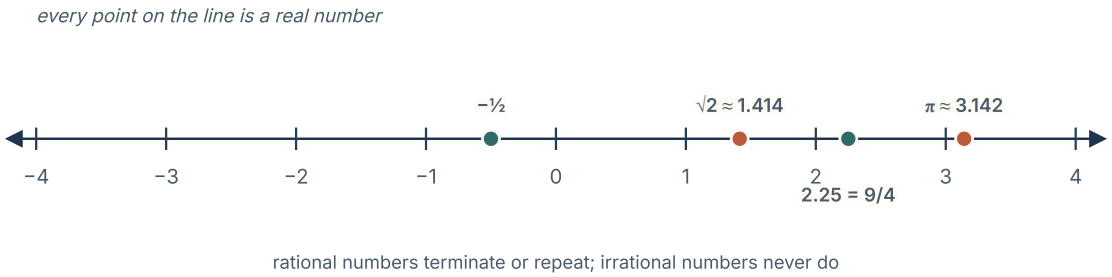


Figure 1.2 Every real number corresponds to exactly one point on the number line. The rational numbers are dense, but they do not fill the line; irrational points such as $\sqrt{2}$ occupy positions between them.

To compare numbers, locate them. The number further to the right is greater. This rule covers negative numbers as well: -7 lies to the left of -2, so $-7 < -2$, even though 7 is greater than 2. The symbol always points to the smaller value, like an arrowhead pointing downhill.

DEFINITIONS

The **absolute value** of a real number x, written $|x|$, is its distance from zero on the number line. Thus $|5| = 5$, $|-5| = 5$ and $|0| = 0$. The distance between real numbers a and b is $|a - b|$.

An **interval** is a connected portion of the number line. The interval of values between a and b is written with brackets: (a, b) excludes both endpoints; $[a, b]$ includes both; $(a, b]$ includes b but not a. Unbounded intervals use $-\infty$ or ∞, always with a parenthesis because infinity is not a number.

Intervals are a compact language for constraints. “The temperature stayed above 18°C but never exceeded 25°C” is $(18, 25]$. “The load must be at least 400 kg” is $[400, \infty)$. Absolute value expresses a symmetric constraint: $|x - 30| \leq 2$ means that x is at most 2 units from 30, so $28 \leq x \leq 32$.

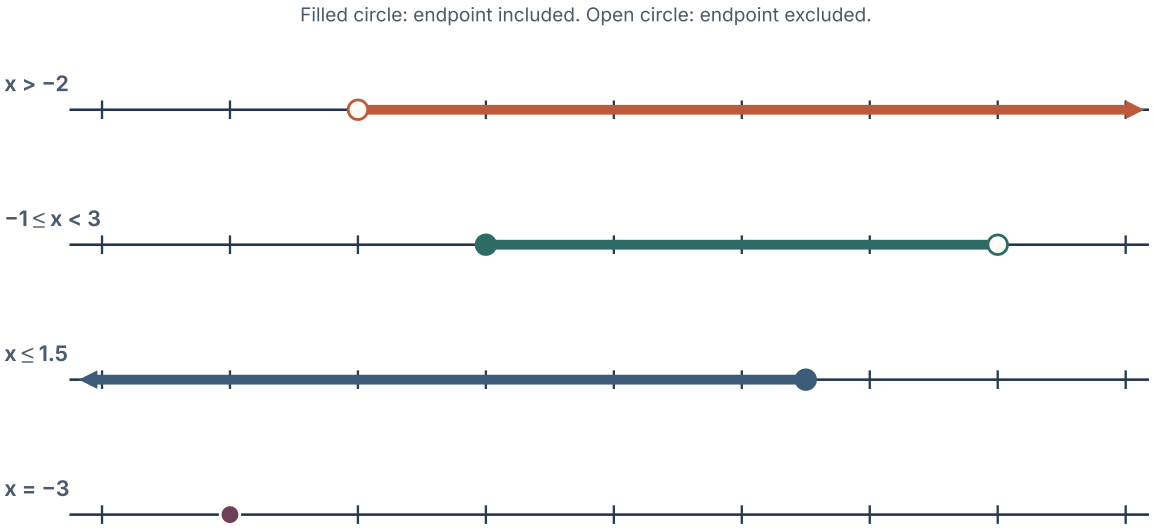


Figure 1.3 Open circles mark excluded endpoints; filled circles mark included endpoints. An absolute-value inequality describes a symmetric interval around a centre.

WORKED EXAMPLE 1.3.1 DISTANCE AND ABSOLUTE VALUE

PROBLEM

Find all real numbers x such that $|x - 12| \leq 3$, and write the solution as an interval.

1. The expression $|x - 12|$ is the distance from x to 12.
2. The inequality says that this distance is at most 3, so x lies from $12 - 3$ to $12 + 3$.
3. Thus $9 \leq x \leq 15$, or $x \in [9, 15]$.

ANSWER $[9, 15]$

WORKED EXAMPLE 1.3.2 ORDERING A MIXED SET

PROBLEM

Order $-\frac{7}{3}$, -2.2, $\sqrt{5}$, 2.3 and $\pi - 1$ from least to greatest.

1. Approximate the non-decimal values: $-\frac{7}{3} = -2.\overline{3}$, $\sqrt{5} \approx 2.236$ and $\pi - 1 \approx 2.142$.
2. Compare the negatives first: $-2.\overline{3} < -2.2$, because $-2.333\ldots$ lies further left.
3. Compare the positives: $2.142 < 2.236 < 2.3$, so $\pi - 1 < \sqrt{5} < 2.3$.
4. Combine the two groups.

ANSWER $-\frac{7}{3},\ -2.2,\ \pi - 1,\ \sqrt{5},\ 2.3$

COMMON ERROR

Assuming that a longer decimal is larger. Digits beyond the first differing place cannot reverse an inequality. 2.0999 is less than 2.1 even though it has more digits. Compare place by place, or place both values approximately on a number line before deciding.

PRACTICE 1.3 · ORDER, DISTANCE AND INTERVALS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Replace each box with $<$, $>$ or $=$: (a) $-6 \square -1$ (b) $|-9| \square 9$ (c) $\sqrt{17} \square 4.1$. F
2. Find each distance on the number line: (a) between -4 and 7 (b) between -2.5 and -8 (c) $|3-11|$. F
3. Write each statement as an inequality and as an interval: (a) x is at least -3 and less than 5 (b) x is greater than 4 (c) x is within 1.5 of 10. F
4. Order from least to greatest: -3.5, $-\sqrt{10}$, -3, $-\frac{22}{7}$. R
5. A machine part must have length L satisfying $|L-45| \le 0.2$ mm. Write the acceptable interval and state whether lengths 44.75 mm and 45.15 mm are acceptable. P
6. Explain why $|x-5| \le 2$ and $3 \le x \le 7$ describe the same set of numbers. R
7. Find two irrational numbers between 3 and 4. Justify that each lies in the interval and that each is irrational. E
8. Is it true that if $a < b$ then $a^2 < b^2$? Give a counterexample or prove the statement. E

1.4 Working with surds

CRITERION A · II ◆ EXACT ARITHMETIC ◆ RATIONALISING

A surd is an irrational root left in exact form. Writing $\sqrt{2}$ instead of 1.414 preserves information that rounding destroys. Surd arithmetic follows the ordinary rules of multiplication and division, with one extra skill: removing roots from denominators.

RULES FOR RADICALS

For non-negative real numbers a and b:

$$\sqrt{a}\sqrt{b} = \sqrt{ab}, \qquad \frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}\ (b \neq 0), \qquad (\sqrt{a})^2 = a.$$

Like surds have the same radicand and can be added or subtracted: $3\sqrt{5} + 2\sqrt{5} = 5\sqrt{5}$. Surds with different radicands cannot be combined by addition, although they may simplify to like surds first.

Multiplying surds is straightforward because the product rule works in both directions. Adding them requires the radicands to match. This is exactly analogous to algebra: $3x + 2x = 5x$, but $3x + 2y$ cannot be simplified. The radicand plays the role of the variable.

WORKED EXAMPLE 1.4.1 PRODUCTS AND QUOTIENTS

PROBLEM

Simplify: (a) $3\sqrt{2} \times 2\sqrt{8}$ (b) $\frac{\sqrt{75}}{\sqrt{3}}$.

1. (a) Multiply the coefficients and the radicands separately: $3 \times 2 \times \sqrt{2 \times 8} = 6\sqrt{16}$.
2. $\sqrt{16} = 4$, so the product is $6 \times 4 = 24$.
3. (b) Combine the roots: $\frac{\sqrt{75}}{\sqrt{3}} = \sqrt{\frac{75}{3}} = \sqrt{25} = 5$.

ANSWER **(a) 24 (b) 5**

WORKED EXAMPLE 1.4.2 RATIONALISING A DENOMINATOR

PROBLEM

Write $\frac{6}{\sqrt{3}}$ and $\frac{5}{2\sqrt{7}}$ with rational denominators.

1. For the first, multiply numerator and denominator by $\sqrt{3}$: $\frac{6\sqrt{3}}{\sqrt{3}\sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3}$.
2. For the second, multiply by $\sqrt{7}$: $\frac{5\sqrt{7}}{2\times 7} = \frac{5\sqrt{7}}{14}$.
3. Multiplying by $\frac{\sqrt{7}}{\sqrt{7}}$ is multiplying by 1, so the value is unchanged.

ANSWER $2\sqrt{3}$ **and** $\frac{5\sqrt{7}}{14}$

STRATEGY · SIMPLIFY BEFORE YOU COMBINE

When a surd expression looks complicated, simplify each radical first. For example, $\sqrt{12} + \sqrt{27} = 2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$. Without simplification, the terms look unlike and the sum appears impossible. The same principle applies to algebraic expressions: factor first, then combine.

PRACTICE 1.4 · SURD ARITHMETIC

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Simplify: (a) $\sqrt{12} + \sqrt{27}$ (b) $5\sqrt{2} - \sqrt{8}$ (c) $\sqrt{50} + \sqrt{18}$. F
2. Simplify: (a) $\sqrt{3} \times \sqrt{12}$ (b) $2\sqrt{5} \times 3\sqrt{10}$ (c) $\frac{\sqrt{96}}{\sqrt{6}}$. F
3. Rationalise each denominator: (a) $\frac{4}{\sqrt{2}}$ (b) $\frac{9}{\sqrt{6}}$ (c) $\frac{7}{3\sqrt{5}}$. F
4. Expand and simplify: (a) $(\sqrt{3}+2)(\sqrt{3}-2)$ (b) $(2+\sqrt{5})^2$. F
5. A student writes $\sqrt{8} + \sqrt{2} = 4$. Is the student correct? Explain by simplifying $\sqrt{8}$. R
6. Show that $\frac{\sqrt{20}}{\sqrt{5}}$ is an integer, and explain which radical rule you used. R
7. Rationalise $\frac{1}{\sqrt{5}-1}$ by multiplying by $\frac{\sqrt{5}+1}{\sqrt{5}+1}$. Show that the result is $\frac{\sqrt{5}+1}{4}$. E
8. Find the exact area of a square whose side length is $3 + \sqrt{2}$ cm. Then find the side length of a square with area $14 + 6\sqrt{5}$ cm^2 and explain your reasoning. E

1.5 Sets, logic and proof

CRITERION A · I ◆ CRITERION C · I, III ◆ CLASSIFICATION

Mathematical statements are not judged by how plausible they sound. They are judged by whether they follow from definitions and previously established results. This section makes the rules of that judgment explicit.

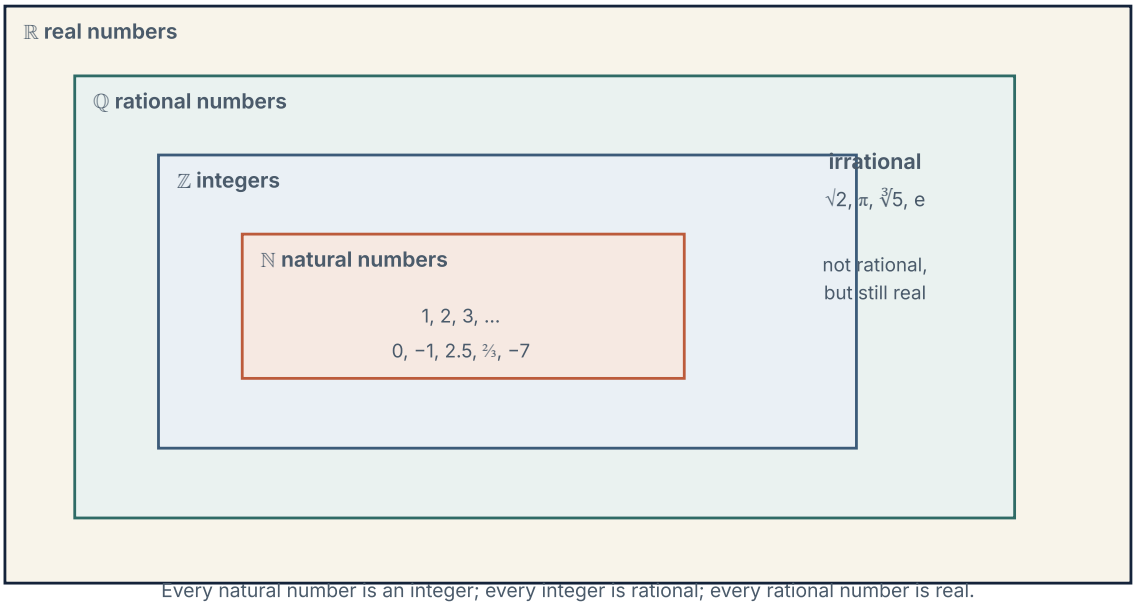


Figure 1.4 The number sets are nested: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$. Irrational numbers lie in $\mathbb{R}$ but outside $\mathbb{Q}$, so the real numbers are the union of the rational and irrational numbers.

DEFINITIONS

The **natural numbers** are $\mathbb{N} = \{1, 2, 3, \ldots\}$. The **integers** are $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$. The **rational numbers** are $\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z},\ q \neq 0 \right\}$. The **irrational numbers** are the real numbers not in $\mathbb{Q}$. The **real numbers** $\mathbb{R}$ are all rational and irrational numbers together.

A counterexample is a single case that makes a universal statement false. The statement “every prime number is odd” is refuted by 2. One counterexample is enough because “every” claims all cases without exception. A proof, by contrast, must cover every case allowed by the definitions. Checking several examples can suggest a proof; it cannot replace one.

WORKED EXAMPLE 1.5.1 — A DIRECT PROOF

PROBLEM

Prove that the sum of two rational numbers is rational.

1. Let x and y be rational. By definition, $x = \frac{a}{b}$ and $y = \frac{c}{d}$, where a, b, c, d are integers and $b \neq 0$, $d \neq 0$.
2. Then $x + y = \frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$.
3. The numerator $ad + bc$ is an integer because integers are closed under addition and multiplication. The denominator bd is a nonzero integer because neither b nor d is zero.
4. Therefore $x + y$ is a quotient of integers, so it is rational. □

ANSWER **The proof is complete: the sum has the required form.**

WORKED EXAMPLE 1.5.2 — DISPROVING A UNIVERSAL CLAIM

PROBLEM

Is the statement "the square of every irrational number is irrational" true? Justify your answer.

1. Try a simple irrational number: $\sqrt{2}$. Its square is $(\sqrt{2})^2 = 2$, which is rational.
2. So an irrational number can have a rational square.
3. The claim is universal ("every"), so one counterexample is enough to show it is false.

ANSWER **The statement is false. Counterexample: $\sqrt{2}$ is irrational but $(\sqrt{2})^2 = 2$ is rational.**

COMMON ERROR · PROVING BY EXAMPLE

Showing that a statement works for 2, 5 and 17 does not prove it works for every number. It may reveal a pattern, but a proof must use the definitions in a way that covers all permitted values. Conversely, a single counterexample is sufficient to disprove a universal statement. Know which direction your evidence supports.

PRACTICE 1.5 · CLASSIFICATION, COUNTEREXAMPLE AND PROOF

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Place each number in the smallest set that contains it: (a) -4 (b) 0.75 (c) $\sqrt{16}$ (d) $\sqrt{7}$. F
2. Decide whether each statement is always true, sometimes true or never true. Give a reason or counterexample: (a) The product of two integers is an integer. (b) The quotient of two integers is an integer. (c) The sum of two irrational numbers is irrational. R
3. Write the negation of "all rational numbers have terminating decimals". Is the negation true? R
4. Prove that the product of two rational numbers is rational. P
5. Prove that the sum of a rational number and an irrational number is irrational. Use proof by contradiction. P
6. Explain why $\sqrt{2} + \sqrt{3}$ cannot be simplified into a single surd of the form $\sqrt{n}$ for an integer n. Test by squaring. P
7. Find a counterexample to "if x^2 is rational then x is rational". Explain what your example shows about the converse of a true statement. E
8. Investigate the claim "the sum of two irrational numbers is always irrational". Find two different counterexamples and state the strongest revised claim you can support. E

Investigation · Chasing $\sqrt{2}$ with fractions

CRITERION B

Long before calculators, mathematicians approximated square roots using only fractions and arithmetic. A Babylonian method starts with a guess x_1, then repeatedly computes

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right).$$

The method is a feedback process: if x_n is too large, then $2/x_n$ is too small, and the average moves closer to $\sqrt{2}$.

n	x_n (8 decimal places)	x_n^2	$\lvert\sqrt{2} - x_n\rvert$
1	1.50000000	2.25000000	0.08578644
2	1.41666667	2.00694445	0.00245311
3	1.41421569	2.00000601	0.00000212
4	1.41421356	2.00000000	0.00000000

1. **Compute.** Verify the value of x_2 from $x_1 = 1.5$. Then calculate x_3 yourself and compare with the table.
2. **Describe.** How does the error change from one row to the next? Is the change approximately the same each time, or does the error shrink much faster?
3. **Explain.** Why does the method require x_n and $2/x_n$ to lie on opposite sides of $\sqrt{2}$? Use the fact that their product is 2.
4. **Generalise.** Adapt the method to approximate $\sqrt{7}$. Start with $x_1 = 3$ and compute three iterations.
5. **Verify.** Use your generalised formula to approximate $\sqrt{10}$. Compare with a calculator and describe the size of the error after three iterations.
6. **Conclude.** The sequence produces rational numbers forever. Explain why no term can ever equal $\sqrt{2}$, even though the terms get arbitrarily close. What does this show about the completeness of the rational numbers on the number line?

Report your table, your formula for the general case, and a conclusion of three or four sentences. A strong conclusion distinguishes evidence from proof.

Application · Tolerance is an interval

CRITERION D · I, II, III ◆ MANUFACTURING ◆ ACCURACY

A bicycle workshop produces seat posts with a nominal diameter of 27.2 mm. A post that is too wide will not fit; one that is too narrow will slip. The specification allows a deviation of at most 0.05 mm in either direction.

Measured diameter (mm)	Device resolution (mm)	Recorded value	Decision
27.24	0.01	27.24	accept
27.16	0.01	27.16	accept
27.252	0.01	27.25	borderline
27.148	0.01	27.15	borderline

1. Write the acceptable interval using inequalities and using absolute value.

2. Which recorded values in the table are definitely acceptable? Which are definitely rejects? Explain using the interval.
3. Why does the device resolution matter for the borderline cases? If the true value is 27.252 and the display rounds to 27.25, can the workshop be certain the part is acceptable? What measurement or rule would remove the ambiguity?
4. A second workshop specifies the same nominal diameter with a tolerance of ± 0.04 mm. Compare the two intervals. Is the second specification stricter? How many parts from the table would be acceptable under each specification?
5. Suggest two reasons why a manufacturer might choose a tolerance that is wider than the precision of its measuring device. Relate your answer to cost, function and the difference between exact values and measured values.

CRITERION D FOCUS

Interpret the situation, select a mathematical representation, justify the accuracy of the decision and evaluate whether the decision makes sense in the workshop context.

Criterion task · Why $\sqrt{2}$ cannot be written as a fraction

CRITERION C COMMUNICATING 24 MARKS

Write a short mathematical exposition for a student who has met fractions and square roots but has never seen a proof by contradiction. Your exposition should explain, with a diagram and a proof, why $\sqrt{2}$ is not rational.

1. Introduce the question with a diagram of a unit square and its diagonal. State the length of the diagonal and explain why it is $\sqrt{2}$. *(4 marks)*
2. State the definition of a rational number and explain what it would mean for $\sqrt{2}$ to be rational. *(4 marks)*
3. Present the proof by contradiction. Define every symbol, justify each algebraic step, and make the contradiction explicit. *(8 marks)*
4. Explain why the proof also shows that no finite decimal can equal $\sqrt{2}$. *(4 marks)*
5. Write a concluding paragraph that distinguishes between "$\sqrt{2} \approx 1.414$" and "$\sqrt{2} = 1.414$". *(4 marks)*

PLANNING FRAME

Audience: a Year 2 student. **Purpose:** to convince, not merely to state. **Structure:** question → diagram → definition → assumption → contradiction → conclusion. **Notation:** define a, b, k. **Check:** every sentence either defines, justifies or concludes.

BAND	DESCRIPTOR	MARKS
4	Complete, coherent exposition with correct definitions and a fully justified proof. Diagram and conclusion are integrated and notation is precise.	19–24
3	Sound proof with minor gaps in justification or notation. Explanation is clear and mostly complete.	13–18
2	Correct outline of the proof with significant gaps in logic or explanation. Some correct notation and structure.	7–12
1	States the result or reproduces fragments without a coherent argument.	1–6

CRITERIA: C **ATL:** CRITICAL THINKING **TIME:** 45 MINUTES **CONDITIONS:** NOTES ALLOWED

MAPPED TO: Criterion C strands i–iv. This task also provides evidence for A i, because it requires accurate classification and use of definitions. The rubric is a course summary, not an official IB descriptor.

Chapter summary and review

KEY IDEAS

- A rational number can be written as $\frac{p}{q}$ with integers p, q and $q \neq 0$.
- Fractions terminate in decimal form exactly when the denominator in lowest terms has no prime factor other than 2 or 5.
- A repeating decimal can be converted to a fraction by subtracting two multiples of the decimal.
- Irrational numbers are real numbers that are not rational; their decimals neither terminate nor repeat.
- $\sqrt{2}$ is irrational. A proof by contradiction assumes the opposite and derives an impossibility.
- Surds can be multiplied and divided using radical rules; only like surds can be added or subtracted.
- Absolute value measures distance; $|x - a| \leq r$ describes an interval centred at a with radius r.

REVISION CHECKLIST

- ☐ I can classify numbers using $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$.
- ☐ I can convert a repeating decimal to a fraction and check it.
- ☐ I can simplify surds and rationalise a one-term denominator.
- ☐ I can order a mixed set of rational and irrational values.
- ☐ I can express a tolerance as an interval and make an accept/reject decision.
- ☐ I can distinguish a counterexample from a proof.
- ☐ I can reproduce the proof that $\sqrt{2}$ is irrational with justification at each step.

VOCABULARY

rational number
A number of the form $\frac{p}{q}$ with integers p, q and $q \neq 0$.

irrational number
A real number that is not rational.

surd
An irrational root, such as $\sqrt{2}$, left in exact form.

radicand
The number under a radical sign.

terminating decimal
A decimal that ends after finitely many digits.

repeating decimal
A decimal with a block of digits that repeats forever.

absolute value
The distance of a number from zero.

interval
A connected set of real numbers described by endpoints.

counterexample
A case that makes a universal statement false.

proof by contradiction
A proof that assumes the negation and derives a contradiction.

dense
Having another element between any two distinct elements.

rationalise
Rewrite a quotient so that no surd remains in the denominator.

MIXED PRACTICE · CHAPTER 1

Connecting the ideas

FLUENCY Core procedures

1. Write $\frac{11}{8}$ as a decimal and $\frac{7}{12}$ as a repeating decimal. F
2. Convert $0.\overline{45}$ to a fraction in lowest terms. F
3. Simplify: (a) $\sqrt{98}$ (b) $2\sqrt{18} - \sqrt{8}$ (c) $\frac{10}{\sqrt{5}}$. F
4. Write $x \in [-4, 9)$ as an inequality. F

REASONING Justify and connect

1. Without calculating, decide whether $\sqrt{40}$ is closer to 6.3 or 6.4. Explain. R
2. Order $-\sqrt{3}$, -1.7, $-\frac{5}{3}$ and -1.65. R
3. Explain why the product of a nonzero rational number and an irrational number is irrational. R
4. A student says "$\frac{1}{3} = 0.333$, so $\frac{1}{3}$ is terminating." Identify the error and give the correct decimal expansion. R

APPLICATION Use and evaluate

1. A cylindrical pin must have diameter d satisfying $|d - 8| \leq 0.03$ mm. Write the acceptable interval and state whether a pin of diameter 7.96 mm should be accepted. P
2. A rectangular tile has sides $\sqrt{5}$ cm and $\sqrt{20}$ cm. Find its exact area and its area to one decimal place. P
3. An online shop rounds parcel weights to the nearest 0.1 kg. A parcel is displayed as 2.4 kg. Write an interval of possible true weights. If postage is charged for weights above 2.35 kg, can you be certain which band applies? Explain. E
4. Find a rational number between $\sqrt{2}$ and $\sqrt{3}$, and an irrational number between the same two values. Justify both. E

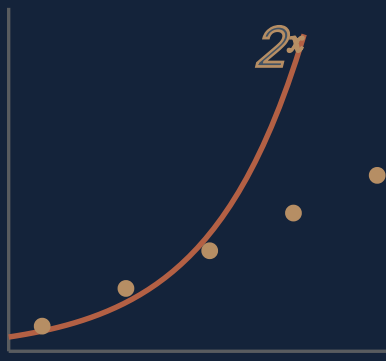
Check your progress

Use the table to record your own judgment. A skill marked "not yet" should be paired with a specific action: revisit a worked example, redo three practice questions without notes, or ask for a different explanation.

I CAN …	CONFIDENCE	EVIDENCE	NEXT ACTION
Classify numbers and justify the classification.	1 / 2 / 3		
Convert between fractions and decimal expansions.	1 / 2 / 3		
Simplify and combine surds.	1 / 2 / 3		
Order and interpret intervals.	1 / 2 / 3		
Write a proof by contradiction.	1 / 2 / 3		

Reflection

The decimal expansion of $\sqrt{2}$ never ends. What does this imply about the possibility of writing $\sqrt{2}$ exactly in decimal notation, and why is the symbol $\sqrt{2}$ still useful?



2

NUMBER, STRUCTURE AND PROOF

Powers, roots and scientific notation

Multiplication repeated many times becomes a power; a power undone becomes a root. The same notation that compresses $10 \times 10 \times 10$ *also lets astronomers write a distance of 4.5 billion kilometres without counting zeros.*

KEY CONCEPT · PATTERNS | RELATED · QUANTITY | RELATED · REPRESENTATION | CONTEXT · SCIENTIFIC AND TECHNICAL INNOVATION

INQUIRY

How can a compact notation for repeated multiplication make extremely large and extremely small quantities easier to compare, calculate and communicate accurately?

ATL · RESEARCH IN ACTION

Find three published quantities from different scales: the mass of a grain of sand, the population of a city and the distance to a nearby star. Record the value, the unit and the source. Rewrite each in scientific notation and identify how many significant figures the source actually supports. A source that reports “about 8 billion” does not support nine significant figures.

MYP UNIT PLANNER

Framing the inquiry

UNIT 2 · APPROX. 13 HOURS
ATL FOCUS · RESEARCH

KEY AND RELATED CONCEPTS

Patterns. The exponent laws are not arbitrary rules; they record what happens when we count repeated factors. The pattern continues consistently from positive to zero to negative exponents.

Quantity. Scientific notation separates the size of a quantity (the power of 10) from its significant digits (the mantissa), making comparisons of scale immediate.

Representation. A number can be written in standard form, factor form, index form or scientific notation. Each representation highlights a different property.

GLOBAL CONTEXT AND EXPLORATION

Scientific and technical innovation. Exploration: how scientists and engineers communicate measurements across radically different scales, from the diameter of an atom to the distance between galaxies, and why a shared notation is essential for collaboration.

Scientific notation is a technology of communication as much as a mathematical convenience. It makes the order of magnitude visible, supports reliable calculation with calculators and computers, and forces the writer to state how many digits are actually meaningful.

STATEMENT OF INQUIRY

Patterns in repeated multiplication generate the exponent laws; these laws, together with scientific notation, provide an efficient and precise language for representing quantities across an enormous range of scales.

FACTUAL

What does a^n mean when n is positive, zero or negative? How do we convert a number to scientific notation?

CONCEPTUAL

Why must $a^0 = 1$ for $a \neq 0$, and why does $a^{1/n}$ represent an nth root?

DEBATABLE

Is a result reported to six significant figures always more trustworthy than one reported to two?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Apply exponent laws; evaluate roots and rational powers; calculate with scientific notation; round to significant figures.
B · Investigating patterns	i, ii, iii	Investigate doubling and halving patterns; describe growth using a general rule and test its predictions.
C · Communicating	i, ii, iii	Report quantities with correct notation, units and significant figures; explain the meaning of an order of magnitude.
D · Applying mathematics	i, ii, iii, iv, v	Produce a scale model and a public briefing using extremely large or small quantities, justifying the precision chosen.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** why the exponent laws follow from the meaning of repeated multiplication.
- **Know** how to use positive, zero and negative integer exponents.
- **Be able to** evaluate roots and expressions with rational exponents.
- **Be able to** convert between standard form and scientific notation, and calculate with numbers in scientific notation.
- **Be able to** compare quantities by order of magnitude and estimate with powers of 10.
- **Be able to** round to a stated number of significant figures and judge the appropriate precision of a result.

SUCCESS CRITERIA

- ☐ I can simplify products, quotients and powers of powers using the exponent laws.
- ☐ I can explain why $a^0 = 1$ and $a^{-n} = 1/a^n$ are consistent with the pattern of exponents.
- ☐ I can evaluate expressions such as $27^{2/3}$ without a calculator.
- ☐ I can multiply, divide, add and subtract numbers in scientific notation and adjust the result to standard scientific form.
- ☐ I can estimate the ratio of two quantities using powers of 10.
- ☐ I can state the number of significant figures in a measurement and round appropriately.

PREREQUISITE CHECK · 10 MINUTES, NO CALCULATOR

1. Calculate 3^4.
2. Calculate $\sqrt{81}$.
3. Write 10^5 as a whole number.
4. Simplify $\frac{8}{12}$.
5. Calculate 2.5×10^2.
6. Round 7.684 to one decimal place.

Answers are in the Chapter 2 answer section. If more than two are incorrect, review powers and decimal place value before continuing.

BEFORE YEAR 3
Square and cube numbers; simple powers; square roots; multiplying by powers of 10; rounding to decimal places.

YEAR 3 · THIS CHAPTER
Integer and rational exponents; exponent laws; scientific notation; orders of magnitude; significant figures and relative error.

LATER
Exponential functions, logarithms, compound interest, exponential growth and decay models, curve fitting on logarithmic scales.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Scientists compare quantities from atomic to astronomical scale.	How can notation make extreme quantities manageable?	Repeated multiplication, inverse operations, powers of 10.	Model repeated doubling and predict when a thin sheet reaches the Moon.	Write magnitudes, compare scales and choose significant figures.	When does a precise-looking number overstate what is known?

2.1 Integer exponents and the exponent laws

CRITERION A · I, II ◆ PATTERNS ◆ STRUCTURE

A power records repeated multiplication. The notation a^n is efficient, but its real power comes from the rules that follow when powers are multiplied, divided or raised to another power.

DEFINITION

For a real number a and a positive integer n, $a^n = \underbrace{a \times a \times \cdots \times a}_{n \text{ factors}}$. The number a is the **base**; n is the **exponent** or index. For $a \neq 0$, $a^0 = 1$ and $a^{-n} = \frac{1}{a^n}$.

Consider $a^3 \times a^4 = (a \times a \times a)(a \times a \times a \times a)$. There are seven factors of a in total, so the product is a^7. The exponent law $a^m a^n = a^{m+n}$ is simply a count. Similarly, dividing a^7 by a^3 cancels three factors and leaves four, so $a^7 / a^3 = a^4$. These laws are not new definitions; they are consequences of what exponents mean.

KEY IDEA · THE EXPONENT LAWS

For $a, b \neq 0$ and integers m, n:

$$a^m a^n = a^{m+n}, \qquad \frac{a^m}{a^n} = a^{m-n}, \qquad (a^m)^n = a^{mn},$$

$$(ab)^n = a^n b^n, \qquad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}, \qquad a^{-n} = \frac{1}{a^n}.$$

The zero and negative exponents are forced by the pattern. If $a^3 / a^3 = a^{3-3} = a^0$, and the same quotient is 1 whenever $a \neq 0$, then a^0 must equal 1. If $a^2 / a^5 = a^{2-5} = a^{-3}$, and direct cancellation gives $1/a^3$, then a^{-3} must mean $1/a^3$. Definitions are chosen so that the laws remain consistent.

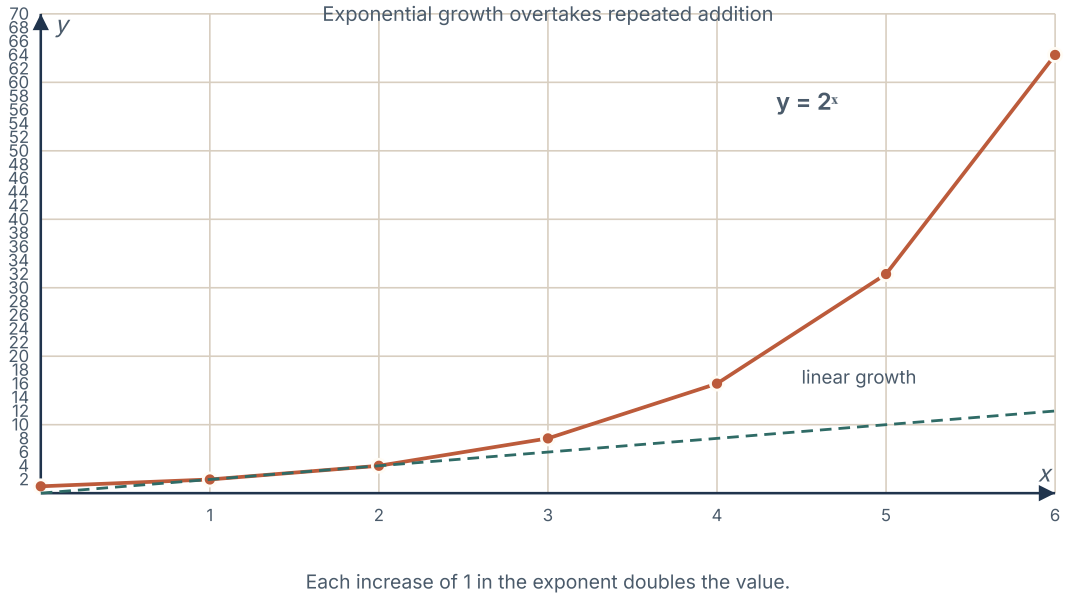


Figure 2.1 Powers grow at very different rates. A linear expression grows by a constant amount; a square grows by increasing amounts; a cube grows faster still. The exponent controls the rate of growth.

WORKED EXAMPLE 2.1.1 APPLYING THE LAWS

PROBLEM

Simplify: (a) $x^5 \times x^3$ (b) $\frac{y^9}{y^4}$ (c) $(2z^3)^4$ (d) $\frac{6a^5}{3a^2}$.

1. (a) Add the exponents: $x^{5+3} = x^8$.
2. (b) Subtract the exponents: $y^{9-4} = y^5$.
3. (c) The exponent applies to both factors: $2^4(z^3)^4 = 16z^{12}$.
4. (d) Divide the coefficients and subtract the exponents: $\frac{6}{3}a^{5-2} = 2a^3$.

ANSWER **(a)** x^8 **(b)** y^5 **(c)** $16z^{12}$ **(d)** $2a^3$

WORKED EXAMPLE 2.1.2 ZERO AND NEGATIVE EXPONENTS

PROBLEM

Evaluate: (a) 7^0 (b) 2^{-4} (c) $\left(\frac{3}{4}\right)^{-2}$ (d) $5x^0$ when $x = 3$.

1. (a) Any nonzero base to the power zero is 1: $7^0 = 1$.
2. (b) $2^{-4} = \frac{1}{2^4} = \frac{1}{16}$.
3. (c) A negative exponent on a fraction inverts the fraction: $\left(\frac{3}{4}\right)^{-2} = \left(\frac{4}{3}\right)^2 = \frac{16}{9}$.
4. (d) The exponent applies only to x, not to 5: $5x^0 = 5 \times 1 = 5$.

ANSWER **(a)** **1** **(b)** $\frac{1}{16}$ **(c)** $\frac{16}{9}$ **(d)** **5**

MISCONCEPTION

$2^3 \times 2^4 = 2^{12}$ and $(x^2)^3 = x^5$.

CORRECT

Multiply powers with the same base by adding exponents: $2^3 \times 2^4 = 2^7 = 128$. A power of a power multiplies exponents: $(x^2)^3 = x^6$.

PRACTICE 2.1 · EXPONENT LAWS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Simplify: (a) a^4a^6 (b) $\frac{b^{11}}{b^3}$ (c) $(c^2)^5$ (d) $(3d)^2$. F
2. Evaluate: (a) 5^3 (b) 4^{-2} (c) $(-3)^4$ (d) $(-2)^5$. F
3. Simplify: (a) $\frac{12m^7}{4m^2}$ (b) $\frac{10p^3q^4}{5pq}$ (c) $(2x^2y)^3$. F
4. Explain why $2^3 \times 2^4$ is not 4^7. Use the meaning of each expression. R
5. Show that $\frac{5^8}{5^8} = 5^0$ and also equals 1. What does this demonstrate about the definition of a^0? R
6. Find the value of n: (a) $3^n = 81$ (b) $2^{n+1} = 32$ (c) $5^{2n} = 625$. P
7. Simplify $\left(\frac{2a^3}{b^2}\right)^2 \times \frac{b^6}{a^2}$, giving your answer with positive exponents. E
8. Is $(-2)^4$ equal to -2^4? Explain the difference between the two expressions, and state when the distinction matters. E

2.2 Roots and rational exponents

CRITERION A · I, II ◆ INVERSE OPERATIONS ◆ REPRESENTATION

Roots undo powers. The square root asks which number squared gives the target; the cube root asks which number cubed gives it. Rational exponents make the same idea available inside the exponent laws.

DEFINITIONS

For $a \geq 0$ and a positive integer n, the **principal nth root** $\sqrt[n]{a}$ is the non-negative number whose nth power is a. For odd n, $\sqrt[n]{a}$ is defined for negative a as well. A **rational exponent** is defined by

$$a^{1/n} = \sqrt[n]{a}, \qquad a^{m/n} = \left(\sqrt[n]{a}\right)^m = \sqrt[n]{a^m},$$

provided the root exists and the base is non-negative when the denominator is even.

The notation is consistent with the exponent law $(a^{1/n})^n = a^1 = a$. Taking the nth power of $a^{1/n}$ returns a, which is exactly what an nth root must do. For $16^{3/4}$, the denominator 4 says "take the fourth root" and the numerator 3 says "cube". The order can be chosen for convenience: $\left(\sqrt[4]{16}\right)^3 = 2^3 = 8$, whereas $\sqrt[4]{16^3} = \sqrt[4]{4096} = 8$.

WORKED EXAMPLE 2.2.1 — EVALUATING RATIONAL EXPONENTS

PROBLEM

Evaluate without a calculator: (a) $27^{1/3}$ (b) $16^{3/4}$ (c) $8^{-2/3}$ (d) $\left(\frac{25}{36}\right)^{1/2}$.

1. (a) The cube root of 27 is 3, so $27^{1/3} = 3$.
2. (b) The fourth root of 16 is 2, and $2^3 = 8$, so $16^{3/4} = 8$.
3. (c) The negative sign inverts, and $8^{2/3} = \left(\sqrt[3]{8}\right)^2 = 2^2 = 4$, so $8^{-2/3} = \frac{1}{4}$.
4. (d) The square root of the fraction is the quotient of the roots: $\frac{\sqrt{25}}{\sqrt{36}} = \frac{5}{6}$.

ANSWER **(a) 3 (b) 8 (c) $\frac{1}{4}$ (d) $\frac{5}{6}$**

WORKED EXAMPLE 2.2.2 SOLVING A SIMPLE POWER EQUATION

PROBLEM

Solve $x^3 = 64$ and $x^2 = 49$ for real x.

1. For $x^3 = 64$, take the cube root of both sides: $x = \sqrt[3]{64} = 4$. There is only one real cube root of a positive number.
2. For $x^2 = 49$, taking the square root gives two possibilities because both 7^2 and $(-7)^2$ equal 49.
3. So $x = 7$ or $x = -7$, often written $x = \pm 7$.

ANSWER $x = 4$ **for the cube equation;** $x = \pm 7$ **for the square equation.**

COMMON ERROR · EVEN ROOTS OF NEGATIVE NUMBERS

$\sqrt{-9}$ is not -3. Squaring either 3 or -3 gives 9, never -9. An even root of a negative real number is not a real number. Odd roots do accept negative inputs: $\sqrt[3]{-8} = -2$, because $(-2)^3 = -8$.

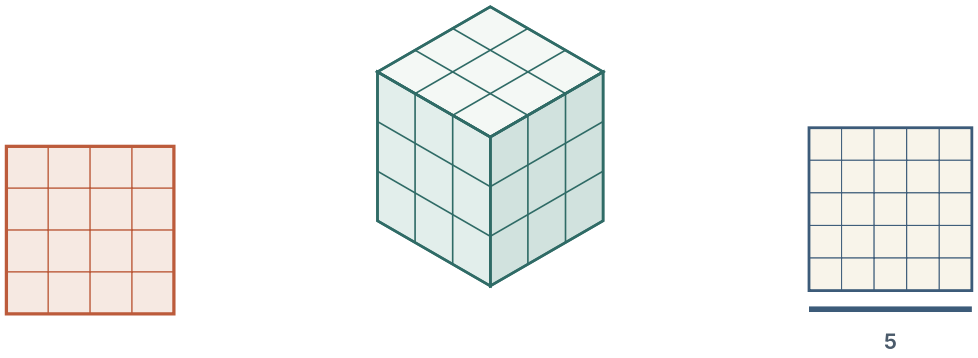


Figure 2.2 A square with side x has area x^2; a cube with edge x has volume x^3. Geometry gives the exponents 2 and 3 a concrete meaning.

PRACTICE 2.2 · ROOTS AND RATIONAL EXPONENTS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Evaluate: (a) $\sqrt{121}$ (b) $\sqrt[3]{27}$ (c) $\sqrt[3]{-64}$ (d) $\sqrt[4]{81}$. F
2. Write each as a power: (a) $\sqrt{7}$ (b) $\sqrt[3]{5}$ (c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt[5]{x^2}$. F
3. Evaluate without a calculator: (a) $64^{1/2}$ (b) $125^{1/3}$ (c) $16^{1/4}$ (d) $9^{3/2}$. F
4. Evaluate: (a) $32^{2/5}$ (b) $64^{-1/3}$ (c) $\left(\frac{8}{27}\right)^{2/3}$. F
5. Solve for x: (a) $x^2 = 144$ (b) $x^3 = 125$ (c) $x^{1/2} = 9$. P
6. Explain why $(-8)^{1/3}$ is a real number but $(-8)^{1/2}$ is not. Refer to the meaning of the exponent. R
7. Show that $a^{m/n} = \sqrt[n]{a^m}$ by using the exponent law $(a^m)^{1/n} = a^{m/n}$. E
8. A cube has volume $343\ \text{cm}^3$. Find its edge length, then find the side length of a square with the same area as one face of the cube. E

2.3 Scientific notation

CRITERION A · II, III ◆ CRITERION C · I ◆ SCALE

Scientific notation writes a number as a mantissa between 1 and 10 multiplied by a power of 10. The power of 10 carries the order of magnitude; the mantissa carries the meaningful digits.

DEFINITION

A number is in **scientific notation** when it is written as $a \times 10^n$, where $1 \leq |a| < 10$ and n is an integer. The number a is the **mantissa** or coefficient, and n is the **exponent**.

To convert $45{,}300$, move the decimal point left until the mantissa lies between 1 and 10: 4.53. The point moved four places, so $45{,}300 = 4.53 \times 10^4$. To convert 0.00072, move the point right four places to get 7.2; moving right corresponds to a negative exponent, so $0.00072 = 7.2 \times 10^{-4}$. The number of places moved is always the absolute value of the exponent.

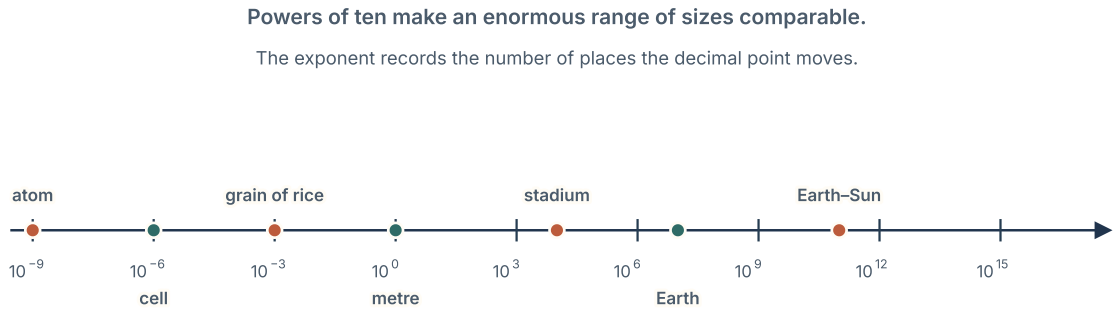


Figure 2.3 Each step of one on the scale represents a factor of 10. Scientific notation compresses this vast range into a mantissa and an exponent.

Multiplication and division use the exponent laws directly. Multiply the mantissas, then add the exponents; divide the mantissas, then subtract the exponents. If the resulting mantissa is not between 1 and 10, adjust it by moving the decimal point and changing the exponent. Addition and subtraction require a common exponent before the mantissas can be combined.

WORKED EXAMPLE 2.3.1 MULTIPLICATION AND DIVISION

PROBLEM

Calculate and write in scientific notation: (a) $(3.2 \times 10^5)(4.0 \times 10^{-2})$ (b) $\frac{8.4 \times 10^7}{2.1 \times 10^{-3}}$.

1. (a) Multiply mantissas: $3.2 \times 4.0 = 12.8$. Add exponents: $10^{5+(-2)} = 10^3$.
2. The mantissa 12.8 is too large, so write $12.8 = 1.28 \times 10^1$. Then $1.28 \times 10^1 \times 10^3 = 1.28 \times 10^4$.
3. (b) Divide mantissas: $8.4 \div 2.1 = 4$. Subtract exponents: $10^{7-(-3)} = 10^{10}$.
4. The mantissa is between 1 and 10, so no adjustment is needed.

ANSWER **(a)** 1.28×10^4 **(b)** 4.0×10^{10}

WORKED EXAMPLE 2.3.2 ADDITION WITH DIFFERENT EXPONENTS

PROBLEM

Calculate $3.4 \times 10^5 + 2.1 \times 10^4$, giving the answer in scientific notation.

1. The exponents differ. Rewrite the smaller power to match the larger: $2.1 \times 10^4 = 0.21 \times 10^5$.
2. Now add the mantissas: $3.4 + 0.21 = 3.61$.
3. Keep the common power: 3.61×10^5.

ANSWER 3.61×10^5

Check. $340{,}000 + 21{,}000 = 361{,}000 = 3.61 \times 10^5$.

MISCONCEPTION

$(2 \times 10^3) + (3 \times 10^4) = 5 \times 10^7$.

CORRECT

Addition does not add exponents. Align powers first: $0.2 \times 10^4 + 3 \times 10^4 = 3.2 \times 10^4$.

PRACTICE 2.3 · SCIENTIFIC NOTATION

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Write in scientific notation: (a) 8,250 (b) 0.00046 (c) 91,000,000 (d) 0.000000305. F
2. Write in standard form: (a) 5.7×10^3 (b) 9.02×10^{-5} (c) 1.6×10^8 (d) 4.5×10^{-1}. F
3. Calculate and write in scientific notation: (a) $(2.5 \times 10^6)(4 \times 10^3)$ (b) $(6.4 \times 10^{-4})(2.5 \times 10^7)$. F
4. Calculate: (a) $\frac{9.6 \times 10^8}{3.2 \times 10^2}$ (b) $\frac{4.5 \times 10^{-5}}{9 \times 10^{-9}}$. F
5. Calculate $4.7 \times 10^6 + 3.8 \times 10^5$, giving the answer in scientific notation. F
6. Order from smallest to largest: 3.4×10^{-3}, 4.1×10^{-4}, 2.9×10^{-2}, 3.3×10^{-4}. R
7. The mass of a dust particle is 7.5×10^{-9} kg. A grain of sand is 40 times heavier. Find the mass of the grain in scientific notation. P
8. A computer stores 2.0×10^{12} bytes. A file has size 4.0×10^8 bytes. How many files of that size can be stored? Express the answer in scientific notation and as a whole number. E

2.4 Orders of magnitude and estimation

CRITERION A · III ◆ CRITERION D · II ◆ SCALE

An order of magnitude is a factor of 10. Comparing orders of magnitude reveals which quantities dominate a problem and which are negligible, often before any detailed calculation.

DEFINITION

Two quantities differ by one **order of magnitude** when one is about 10 times the other. A number $a \times 10^n$ has order of magnitude 10^n if $1 \le a < 3.16$, and 10^{n+1} if $3.16 \le a < 10$. The threshold is $10^{0.5} \approx 3.16$, the midpoint on a logarithmic scale.

A useful estimation skill is to round each quantity to one significant figure and then to the nearest power of 10. The thickness of a sheet of paper is about 1×10^{-4} m; the height of a room is about 3×10^{0} m. The ratio is about 3×10^{4}, so roughly thirty thousand sheets are needed to reach the ceiling. The estimate is not exact, but it is stable: it identifies the scale of the answer.

WORKED EXAMPLE 2.4.1 — COMPARING SCALES

PROBLEM

The diameter of a human hair is about 7×10^{-5} m. The diameter of a red blood cell is about 7×10^{-6} m. How many times larger is the hair?

1. Form the ratio: $\frac{7 \times 10^{-5}}{7 \times 10^{-6}}$.
2. Divide the mantissas: $7 \div 7 = 1$. Subtract the exponents: $-5 - (-6) = 1$.
3. The ratio is $1 \times 10^{1} = 10$.

ANSWER **The hair is about 10 times wider than the red blood cell.**

WORKED EXAMPLE 2.4.2 — A FERMI ESTIMATE

PROBLEM

Estimate the number of breaths a person takes in one year. State the assumptions.

1. Estimate a breathing rate: about 15 breaths per minute.
2. Estimate the number of minutes in a year: $60 \times 24 \times 365 \approx 5.26 \times 10^{5}$.
3. Multiply: $15 \times 5.26 \times 10^{5} \approx 7.9 \times 10^{6}$.
4. Round to one significant figure: about 8×10^{6} breaths per year.

ANSWER **Approximately 8×10^{6} breaths per year, assuming 15 breaths per minute and no pauses.**

Evaluate. The rate and the number of minutes are both estimates. Reporting 7,884,000 would imply a precision the assumptions do not support.

PRACTICE 2.4 · ORDERS OF MAGNITUDE AND ESTIMATION

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. State the order of magnitude of each quantity: (a) 4.8×10^{3} (b) 8.1×10^{6} (c) 2.2×10^{-4} (d) 5.0×10^{-2}. F
2. How many times larger is 6×10^{8} than 3×10^{5}? F
3. Estimate the number of seconds in a day, giving your answer in scientific notation. F
4. Estimate the number of grains of rice in a 1 kg bag if one grain has mass about 0.02 g. State your assumptions. P
5. A molecule is about 10^{-9} m across. Estimate how many molecules could fit across a 1 cm gap. P
6. Compare the mass of an electron (9.1×10^{-31} kg) with the mass of a proton (1.67×10^{-27} kg) as a ratio. Which is heavier, and by roughly what factor? R
7. A data centre stores 3.2×10^{15} bytes. A smartphone stores 1.28×10^{11} bytes. How many smartphones would hold the same data? Comment on the reasonableness of the answer. E
8. Estimate the volume of water in a swimming pool 25 m by 10 m by 2 m, in litres. Then estimate the number of 500 mL bottles that could be filled. Compare the order of magnitude with the number of people in your school. E

2.5 Significant figures and precision

CRITERION A · III ◆ CRITERION D · III ◆ ACCURACY

A measurement is not a pure number. It carries a claim about how much is known. Significant figures are a compact way of stating that claim.

> **RULES FOR SIGNIFICANT FIGURES**
>
> All nonzero digits are significant. Zeros between nonzero digits are significant. Leading zeros are never significant. Trailing zeros after a decimal point are significant. Trailing zeros in a whole number are ambiguous unless the number is written in scientific notation.

Thus 0.004507 has four significant figures (4, 5, 0 and 7); the leading zeros only position the decimal point. 12.30 has four significant figures because the final zero is a deliberate claim of precision. 1,500 is ambiguous: it could have two, three or four significant figures. Writing 1.5×10^{3} makes the claim explicit.

WORKED EXAMPLE 2.5.1 — ROUNDING TO SIGNIFICANT FIGURES

PROBLEM

Round (a) 0.004567 to 2 significant figures, (b) 12.05 to 3 significant figures and (c) 8.995 to 3 significant figures.

1. (a) The first two significant figures are 4 and 5. The next digit is 6, so round up: 0.0046.
2. (b) The first three significant figures are 1, 2 and 0. The next digit is 5, so round up: 12.1.
3. (c) The first three significant figures are 8, 9 and 9. The next digit is 5, so round up; the 9s carry: 9.00. The zeros are significant because they record the requested precision.

ANSWER **(a)** 0.0046 **(b)** 12.1 **(c)** 9.00

WORKED EXAMPLE 2.5.2 ABSOLUTE AND RELATIVE ERROR

PROBLEM

A length is measured as 4.8 cm, but the true length is 4.75 cm. Find the absolute error and the relative error as a percentage.

1. Absolute error $= |4.8 - 4.75| = 0.05$ cm.
2. Relative error $= \frac{0.05}{4.75} \approx 0.01053$.
3. As a percentage: $0.01053 \times 100\% \approx 1.1\%$.

ANSWER **Absolute error 0.05 cm; relative error approximately 1.1%.**

Interpret. An absolute error of 0.05 cm is small on a 4.75 cm object but large on a 0.1 cm object. Relative error expresses the error as a proportion, so it supports comparison across scales.

STRATEGY · DO NOT MANUFACTURE PRECISION

When multiplying or dividing measurements, the result should be reported to the same number of significant figures as the least precise input. When adding or subtracting, the result is limited by the least precise decimal place. In a multi-step calculation, keep extra digits during working and round only the final answer.

PRACTICE 2.5 · SIGNIFICANT FIGURES AND PRECISION

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. State the number of significant figures: (a) 0.0034 (b) 2.070 (c) 1500 (d) 6.02×10^{23}. F
2. Round to 3 significant figures: (a) 4.5678 (b) 0.009876 (c) 12,345 (d) 8.9949. F
3. Round 7.85 to 1 decimal place, then explain why the result differs from rounding 7.849 to 1 decimal place. R
4. A measurement is recorded as 12.4 cm. State a reasonable interval of possible true lengths if the measuring device rounds to the nearest 0.1 cm. P
5. A mass is measured as 250 g, but the balance has an uncertainty of ± 2 g. Find the relative error as a percentage using the measured value. P
6. A rectangle measures 4.2 cm by 7.5 cm. Calculate its area and state the appropriate number of significant figures in the answer. P
7. A student calculates $2.5 \times 4.0 = 10.0$ and reports the answer as 10.000. Explain the error and give the correct significant-figure claim. R
8. A distance is reported as 3.84×10^8 m. A student says "it is exactly 384,000,000 m". Explain why this interpretation is not justified. E

Investigation · Forty-two folds to the Moon

CRITERION B

A sheet of paper is about 0.1 mm thick. Each fold doubles the thickness, so after n folds the thickness is 0.1×2^n mm, assuming a perfect fold each time.

Number of folds n	Thickness (mm)	Thickness (m)	Comparison
1	0.2	2.0×10^{-4}	two sheets
5	3.2	3.2×10^{-3}	about 3 mm
10	102.4	1.024×10^{-1}	about 10 cm
20	104,857.6	1.048576×10^{2}	about 105 m
30	107,374,182.4	$1.073741824 \times 10^{5}$	about 107 km
42	4.398×10^{11}	4.398×10^{8}	about the Earth–Moon distance

1. **Verify.** Show that the formula 0.1×2^n gives the value for $n = 10$ and $n = 20$. Convert both to metres.
2. **Predict.** The average Earth–Moon distance is 3.84×10^8 m. Use the formula to find the smallest whole number of folds for which the thickness exceeds this distance. Show the calculation.
3. **Compare.** How many folds are needed to exceed the height of a 300 m tower? How many for Mount Everest (8,849 m)? Present your answers in a table.
4. **Generalise.** Write a formula for the number of folds needed to reach a height H metres. You may use the fact that $2^{10} \approx 10^3$ to build an estimate rather than an exact logarithmic solution.
5. **Evaluate.** Explain why the model breaks down before 42 folds in practice. Consider the area of the sheet, the size of a fold and the strength of the paper. What does the investigation reveal about the difference between a mathematical model and a physical procedure?

A strong report states the formula, shows the substitution, reports units and separates the mathematical result from the feasibility of the physical action.

Application · A scale model of the solar system

CRITERION D · I, II, III, IV ◆ SCALE ◆ COMMUNICATION

A science museum wants a scale model along a 500 m outdoor path. The distances from the Sun are given in kilometres and in astronomical units (AU), where $1\ \text{AU} = 1.496 \times 10^8$ km.

Object	Distance from Sun (AU)	Distance from Sun (km)	Diameter (km)
Mercury	0.39	5.8×10^{7}	4.88×10^{3}
Earth	1.00	1.50×10^{8}	1.27×10^{4}
Mars	1.52	2.27×10^{8}	6.79×10^{3}
Jupiter	5.20	7.78×10^{8}	1.40×10^{5}
Neptune	30.1	4.50×10^{9}	4.92×10^{4}

1. Choose a scale that places Neptune at approximately 450 m from the Sun. Calculate the scale factor in the form $1 : n$ using the distance in metres.
2. Use your scale to calculate the model distance of each planet from the Sun and the model diameter of each planet. Present the results in a table with sensible units (metres or centimetres).

3. Is the model diameter of Earth visible to a visitor? Is Jupiter's? Comment on the practical problem of showing both distance and size at the same scale.
4. If the museum instead chooses a scale that makes Jupiter 30 cm across, where would Neptune be placed? Is the path long enough? Show your reasoning.
5. Write a 120-word label for the model that explains the scale, the two different visual compromises and why scientific notation is useful for the design. Justify the number of significant figures you use.

CRITERION D FOCUS

Identify the relevant quantities, select and apply a scale, justify the degree of accuracy, and evaluate whether the model communicates the intended relationships.

Criterion task · A public briefing on very small particles

CRITERION D
APPLYING
MATHEMATICS
24 MARKS

A city health office is preparing a briefing on airborne particles. The briefing must be accurate, readable and honest about uncertainty. The following data are supplied.

Quantity	Value
Diameter of a PM2.5 particle	up to 2.5×10^{-6} m
Diameter of a human hair	about 7×10^{-5} m
Concentration limit (annual mean)	5.0×10^{-9} kg/m^3
Measured annual mean at one site	8.3×10^{-9} kg/m^3
Uncertainty of the measurement	$\pm 6\%$

1. How many PM2.5 particles would fit across the diameter of a human hair? Use the maximum particle diameter and give your answer to an appropriate number of significant figures. *(4 marks)*
2. Express the measured concentration and the limit in micrograms per cubic metre, using $1\ \mu\text{g} = 10^{-9}$ kg. *(3 marks)*
3. Calculate the percentage by which the measured value exceeds the limit. *(4 marks)*
4. Using the stated uncertainty, write an interval for the true annual mean and decide whether the site definitely exceeds the limit. *(5 marks)*
5. Write a 150-word briefing for a non-specialist audience. It must state the comparison of particle size to hair, the measured value, the uncertainty and the conclusion. Use scientific notation at least once and justify the precision of every number you report. *(8 marks)*

PLANNING FRAME

Identify: particle diameter, hair diameter, limit, measurement, uncertainty. **Select:** ratio for size, unit conversion for concentration, percentage difference, interval for uncertainty. **Justify:** why the conclusion changes or does not change once uncertainty is included. **Evaluate:** is the briefing useful without overstating certainty?

BAND	DESCRIPTOR	MARKS
4	Correct and well-justified use of ratio, conversion and uncertainty; communication is precise, readable and appropriately cautious.	19–24
3	Most calculations correct; uncertainty handled with minor errors; communication is clear but precision is uneven.	13–18
2	Some relevant calculations; limited justification of accuracy or context; communication may overstate certainty.	7–12
1	Little correct mathematics or an answer without context.	1–6

CRITERIA: D (A, C) **ATL:** RESEARCH **TIME:** 50 MINUTES **CONDITIONS:** CALCULATOR ALLOWED

MAPPED TO: D strands i–v, with supporting evidence for A iii and C i. The rubric is a course summary, not an official IB descriptor.

Chapter summary and review

KEY IDEAS

- a^n means n factors of a; the exponent laws follow from counting factors.
- $a^0 = 1$ and $a^{-n} = 1/a^n$ are forced by the requirement that the exponent laws remain consistent.
- $\sqrt[n]{a} = a^{1/n}$ and $a^{m/n} = (\sqrt[n]{a})^m$.
- Scientific notation has the form $a \times 10^n$ with $1 \leq |a| < 10$.
- Multiplication adds exponents; division subtracts them; addition requires a common exponent.
- An order of magnitude is a factor of 10; it supports rapid comparison and estimation.
- Significant figures communicate the precision of a measurement; calculations should not manufacture extra precision.

REVISION CHECKLIST

- ☐ I can simplify expressions using all the exponent laws, including negative exponents.
- ☐ I can evaluate roots and rational powers without a calculator.
- ☐ I can convert fluently between standard form and scientific notation.
- ☐ I can multiply, divide, add and subtract in scientific notation.
- ☐ I can estimate a ratio of quantities by rounding to powers of 10.
- ☐ I can round to a stated number of significant figures and identify ambiguous trailing zeros.
- ☐ I can calculate absolute and relative error and interpret both.

VOCABULARY

base
The number that is repeatedly multiplied in a power.

exponent
The number of factors; the power to which the base is raised.

index form
A number written using a base and an exponent.

root
A value that produces a given number when raised to a power.

principal root
The non-negative root when an even root is taken.

rational exponent
An exponent of the form m/n, equivalent to a root and a power.

scientific notation
The form $a \times 10^n$ with $1 \leq |a| < 10$.

mantissa
The coefficient a in scientific notation.

order of magnitude
A factor of 10 used to compare scales.

significant figures
The digits in a measurement that carry meaningful information.

absolute error
The size of the difference between a measurement and the true value.

relative error
The absolute error divided by the true value, often expressed as a percentage.

MIXED PRACTICE · CHAPTER 2

Connecting the ideas

FLUENCY Core procedures

1. Simplify: (a) $x^{7}x^{-3}$ (b) $(3a^{2})^{3}$ (c) $\frac{8m^{6}}{2m^{-2}}$. F
2. Evaluate: (a) $81^{3/4}$ (b) $27^{-2/3}$ (c) $\sqrt[3]{-125}$. F
3. Write 0.000903 and $28{,}400{,}000$ in scientific notation. F
4. Calculate $(2.4 \times 10^{-4})(5.0 \times 10^{7})$ and write the result in scientific notation. F

REASONING Justify and connect

1. Explain why $2^{-3} = \frac{1}{8}$ using the exponent law for division. R
2. Is 5.2×10^{4} larger than 4.9×10^{5}? Explain without converting both to standard form. R
3. A student writes $3.6 \times 10^{5} + 4.0 \times 10^{4} = 7.6 \times 10^{9}$. Identify the error and correct it. R
4. How many significant figures does 0.07080 have? Explain each digit. R

APPLICATION Use and evaluate

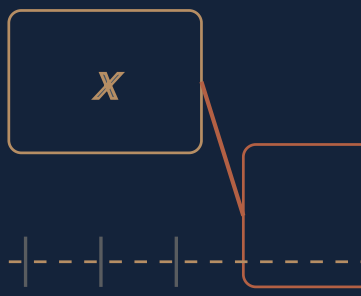
1. A single human cell has diameter about 2×10^{-5} m. Estimate how many cells would span a 10 cm line. P
2. A hard drive holds 4.0×10^{12} bytes. A photo has average size 3.2×10^{6} bytes. Estimate the number of photos the drive can hold. P
3. A bacterial culture doubles every hour. Starting with 5.0×10^{3} bacteria, find the population after 8 hours in scientific notation. P
4. A measurement of 8.4 g has an uncertainty of ± 0.1 g. Calculate the relative error and explain whether this is a precise measurement for a 100 g sample. E

Check your progress

I CAN …	CONFIDENCE	EVIDENCE	NEXT ACTION
Apply the exponent laws with integer exponents.	1 / 2 / 3		
Evaluate roots and rational exponents.	1 / 2 / 3		
Calculate fluently in scientific notation.	1 / 2 / 3		
Estimate and compare orders of magnitude.	1 / 2 / 3		
Report a result with appropriate significant figures.	1 / 2 / 3		

Reflection

Scientific notation separates the size of a quantity from its significant digits. How does this separation help you avoid both underestimating and overstating what a measurement tells you?



3

NUMBER, STRUCTURE AND PROOF

Functions, expressions and algebraic reasoning

Algebra begins when a letter stands for a number we have not chosen yet. It becomes powerful when the letter describes a relationship: one quantity determines another, and the rule can be written, tabulated, graphed and transformed.

KEY CONCEPT · RELATIONSHIPS | RELATED · GENERALISATION | RELATED · EQUIVALENCE | CONTEXT · IDENTITIES AND RELATIONSHIPS

INQUIRY

How can a single expression capture a whole family of cases — and how do we know that two different-looking expressions describe exactly the same relationship?

ATL · THINKING IN ACTION

When you meet a new expression, ask three questions: What are the inputs? What operations are performed, and in what order? What changes if one operation changes? Treat an expression as a description of a process, not a string of symbols to memorise. This habit makes expanding, factorising and rearranging easier to reconstruct when memory fails.

MYP UNIT PLANNER

Framing the inquiry

UNIT 3 · APPROX. 15 HOURS
ATL FOCUS · THINKING

KEY AND RELATED CONCEPTS

Relationships. A function assigns each input exactly one output. The relationship can be represented by a rule, a table, a graph, a machine diagram or a set of ordered pairs.

Generalisation. An expression such as $2n + 1$ stands for infinitely many numerical cases at once. Generalising is the move from "this pattern works here" to "this rule works for every position".

Equivalence. Expressions that look different can have the same value for every input. Expanding and factorising are tools for revealing that equivalence.

GLOBAL CONTEXT AND EXPLORATION

Identities and relationships. Exploration: how a function describes a dependable relationship between two quantities, and how the same relationship can appear in a table, a graph, a formula or a machine.

Functions are the language of dependence: the cost depends on the quantity, the height depends on the time, the area depends on the side length. Learning to name the input, the output and the rule is the first step in modelling any relationship.

STATEMENT OF INQUIRY

A function is a relationship in which each input determines exactly one output; algebraic expressions and equivalent forms allow that relationship to be represented, generalised and used to make predictions.

FACTUAL

What is a function? What do the symbols $f(x)$, domain and range mean? How do you expand and factorise an expression?

CONCEPTUAL

Why does a graph represent a function only if every vertical line meets it at most once?

DEBATABLE

Is a mathematical model still useful when it describes only part of a real relationship?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii, iii	Substitute and simplify; evaluate functions; expand, factorise and rearrange formulas.
B · Investigating patterns	i, ii, iii	Generalise a growing pattern, express it symbolically and verify the rule for further cases.
C · Communicating	i, ii, iii, iv	Use function notation, domain and range language, and coherent algebraic working.
D · Applying mathematics	i, ii, iii, iv, v	Model a painting job with formulas, interpret inputs and outputs, and judge the domain.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** the difference between an expression, an equation, a formula and a function.
- **Know** how to evaluate and simplify algebraic expressions.
- **Be able to** use function notation, identify domain and range, and interpret graphs.
- **Be able to** expand and factorise linear and quadratic expressions.
- **Be able to** rearrange a formula to make a chosen variable the subject.
- **Be able to** generalise a sequence or pattern using a position-to-term rule.

SUCCESS CRITERIA

- ☐ I can distinguish like and unlike terms and simplify correctly.
- ☐ I can evaluate a function at a given input and solve $f(x) = k$.
- ☐ I can determine whether a graph represents a function using the vertical line test.
- ☐ I can expand $(x + a)(x + b)$ and factorise $x^2 + bx + c$ when the factors are integers.
- ☐ I can rearrange a formula and justify each step as an equivalent transformation.
- ☐ I can find the nth term of a linear sequence and explain the meaning of the formula.

PREREQUISITE CHECK · 12 MINUTES, NO CALCULATOR

1. Evaluate $4x + 3$ when $x = 5$.
2. Simplify $7a - 2a + 3a$.
3. Expand $3(2x + 4)$.
4. Calculate $(-3)^2$.
5. Solve $2x + 5 = 17$.
6. Write the next two terms: 4, 7, 10, 13, …

Answers are in the Chapter 3 answer section. If more than two are incorrect, review substitution and the distributive law before continuing.

BEFORE YEAR 3

Substitution; simple algebraic notation; one- and two-step equations; plotting coordinates; recognising simple number patterns.

YEAR 3 · THIS CHAPTER

Function notation; domain and range; expansion and factorisation; formula rearrangement; nth-term rules.

LATER

Quadratic, exponential and trigonometric functions; composite and inverse functions; algebraic fractions; polynomial factorisation.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Costs, heights and areas depend on quantities that change.	How can a rule describe a relationship for every input?	Variable, function, domain, equivalence and generalisation.	Build a growing matchstick pattern and find its nth-term rule.	Move between rule, table, graph and machine diagram.	What makes an algebraic model trustworthy?

3.1 Variables, expressions and substitution

CRITERION A · I ◆ NOTATION ◆ LIKE TERMS

Algebra uses letters to stand for numbers. The letters are not mysterious: they are placeholders that let a statement describe a whole family of numerical cases at once.

DEFINITIONS

A **variable** is a symbol that represents a number that may change. A **constant** is a fixed number. A **term** is a product of numbers and variables, such as $3x^2$ or $-5y$. The **coefficient** is the numerical factor of a term. An **expression** combines terms with addition and subtraction; it does not contain an equals sign. An **equation** states that two expressions are equal.

In the expression $4x^2 - 3x + 7$, the terms are $4x^2$, $-3x$ and 7. The coefficient of x^2 is 4; the coefficient of x is -3; 7 is a constant term. The exponent belongs only to x, so $4x^2$ means $4 \times x \times x$, not $(4x)^2$.

Like terms have exactly the same variable part. $5x$ and $-2x$ are like terms; $5x$ and $5x^2$ are not. Combining like terms uses the distributive law in reverse: $5x - 2x = (5 - 2)x = 3x$. This is why $3x + 4$ cannot be simplified further: the terms are unlike, and their sum is not $7x$.

WORKED EXAMPLE 3.1.1 — SIMPLIFYING EXPRESSIONS

PROBLEM

Simplify: (a) $7a + 3b - 2a + 5b$ (b) $3(2x - 5) + 4x$ (c) $5m^2 - 3m + 2m^2 + m$.

1. (a) Group like terms: $(7a - 2a) + (3b + 5b) = 5a + 8b$.
2. (b) Expand first: $6x - 15 + 4x$. Then combine: $10x - 15$.
3. (c) Group the squared terms and the linear terms: $(5m^2 + 2m^2) + (-3m + m) = 7m^2 - 2m$.

ANSWER **(a)** $5a + 8b$ **(b)** $10x - 15$ **(c)** $7m^2 - 2m$

WORKED EXAMPLE 3.1.2 — SUBSTITUTION WITH NEGATIVE VALUES

PROBLEM

Evaluate $3x^2 - 2x + 1$ when $x = -2$.

1. Replace each x with -2, using brackets: $3(-2)^2 - 2(-2) + 1$.
2. Evaluate the power: $(-2)^2 = 4$, so the first term is $3 \times 4 = 12$.
3. The second term is $-2 \times (-2) = +4$.
4. Add: $12 + 4 + 1 = 17$.

ANSWER **17**

Common slip. Writing -2^2 without brackets gives -4, because the exponent applies only to 2. The brackets are essential when substituting a negative number.

COMMON ERROR

Combining unlike terms: $3x + 4$ does not equal $7x$, and $x^2 + x$ does not equal x^3 or $2x^2$. Only terms with identical variable parts can be added or subtracted by combining coefficients.

PRACTICE 3.1 · EXPRESSIONS AND SUBSTITUTION

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. State the coefficient of x and the constant term: (a) $7x - 4$ (b) $-2x + 9$ (c) $x^2 - 5x + 1$. F
2. Simplify: (a) $8y - 3y + 2y$ (b) $4p + 7q - p - 3q$ (c) $6x^2 - x^2 + 2x$. F
3. Expand and simplify: (a) $2(3a + 4) + 5a$ (b) $5(2x - 1) - 3x$ (c) $4(3m + 2) - 2(m - 5)$. F
4. Evaluate $2x^2 + 3x - 4$ for (a) $x = 2$ (b) $x = -1$ (c) $x = 0.5$. F
5. Explain why $5x + 2x = 7x$ but $5x + 2$ cannot be simplified. R
6. A student evaluates x^2 at $x = -3$ and writes -9. Identify the error and give the correct value. R
7. The perimeter of a rectangle is $2(l + w)$. Write and simplify an expression for the perimeter when the length is $3x + 1$ and the width is $x - 2$. P
8. Find a pair of expressions in x that are not identical but produce the same value for $x = 2$. Explain why one pair of matching values is not enough to prove equivalence. E

3.2 Functions, domain and range

CRITERION A · II ◆ CRITERION C · I, II ◆ REPRESENTATION

A function is a rule with a guarantee: every permitted input produces exactly one output. The guarantee is what makes prediction possible.

DEFINITION

A **function** from a set X to a set Y assigns to each element of X exactly one element of Y. The set X is the **domain**; the set of outputs actually produced is the **range**. The notation $f(x)$ means the output of function f at input x.

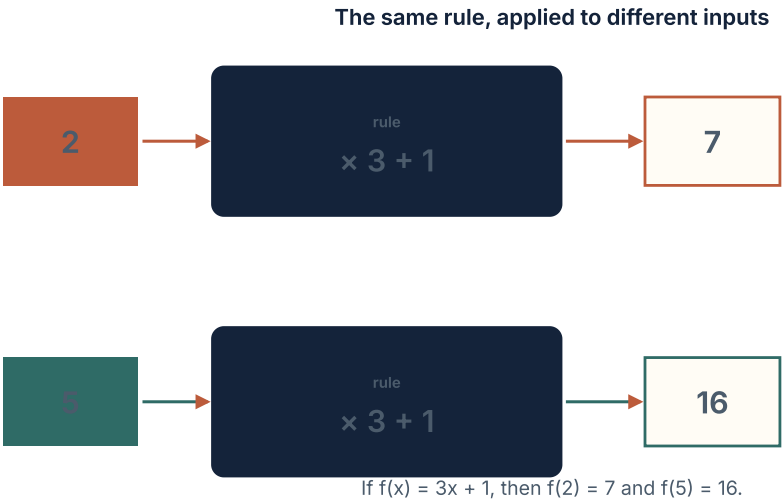


Figure 3.1 A function machine makes the input–process–output structure visible. The rule $f(x) = 2x + 1$ doubles the input and adds 1.

If $f(x) = 2x + 1$, then $f(4) = 2(4) + 1 = 9$. The symbol $f(4)$ is not f multiplied by 4; it means "apply the function f to the input 4". To solve $f(x) = 13$, set the rule equal to 13: $2x + 1 = 13$, so $x = 6$. The input 6 produces the output 13.

The domain is not always all real numbers. For $g(x) = \frac{1}{x-2}$, $x = 2$ must be excluded because division by zero is undefined, so the domain is all real numbers except 2. For $h(x) = \sqrt{x}$, the domain is $x \geq 0$ if we require real outputs. In a context, the domain may be further restricted: a function giving the cost of n tickets only makes sense for whole-number values of n.

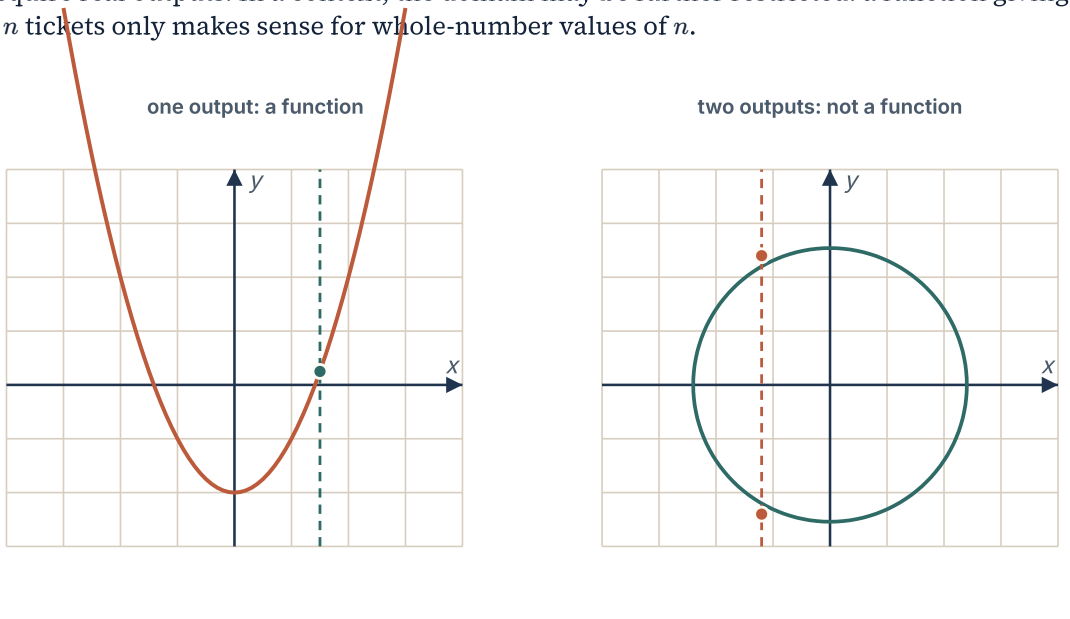


Figure 3.2 The vertical line test: a graph represents a function of x if and only if no vertical line meets the graph more than once.

WORKED EXAMPLE 3.2.1 EVALUATING AND SOLVING

PROBLEM

For $f(x) = x^2 - 3x$, find $f(5)$, $f(-2)$, and all x such that $f(x) = 0$.

1. $f(5) = 5^2 - 3(5) = 25 - 15 = 10$.
2. $f(-2) = (-2)^2 - 3(-2) = 4 + 6 = 10$.
3. Solve $x^2 - 3x = 0$. Factorise: $x(x - 3) = 0$.
4. So $x = 0$ or $x = 3$.

ANSWER $f(5) = 10$, $f(-2) = 10$, **and** $f(x) = 0$ **when** $x = 0$ **or** $x = 3$.

WORKED EXAMPLE 3.2.2 DOMAIN FROM CONTEXT

PROBLEM

A taxi charges a fixed 4 units plus 2 units per kilometre. Write the cost as a function of distance d, state a sensible domain, and find the cost of a 9 km journey.

1. The rule is $C(d) = 4 + 2d$.
2. Distances cannot be negative, and the context is a single journey, so a sensible domain is $d \geq 0$.
3. $C(9) = 4 + 2(9) = 22$.

ANSWER $C(d) = 4 + 2d$, **with** $d \geq 0$; **the cost is 22 units.**

PRACTICE 3.2 · FUNCTIONS, DOMAIN AND RANGE SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. For $f(x) = 3x - 5$, find (a) $f(2)$ (b) $f(0)$ (c) $f(-4)$. F
2. For $g(x) = x^2 + 1$, find (a) $g(3)$ (b) $g(-3)$ (c) $g(0.5)$. F
3. For $h(x) = 10 - 2x$, solve $h(x) = 4$. F
4. State the domain of each function: (a) $f(x) = \frac{1}{x}$ (b) $g(x) = \sqrt{x - 3}$ (c) $h(x) = 5x + 1$. F
5. Explain why the graph of $x^2 + y^2 = 25$ does not represent y as a function of x. R
6. A function machine takes an input, doubles it, then subtracts 3. Write the rule and find the input that produces output 11. P
7. The cost of printing n posters is $C(n) = 25 + 1.5n$, in the currency used by the print shop. State a sensible domain and find the cost of 40 posters. P
8. Find a function whose domain is all real numbers except $x = 4$ and whose output at $x = 0$ is 1. Explain how you constructed it. E

3.3 Expanding and factorising

CRITERION A · I, II ◆ EQUIVALENCE ◆ STRUCTURE

Expanding turns a product into a sum; factorising turns a sum into a product. The two processes are inverse, and both reveal structure that the other form hides.

KEY IDEA · THE DISTRIBUTIVE LAW

For all real numbers a, b, c: $a(b + c) = ab + ac$. For two binomials, $(x + a)(x + b) = x^2 + (a + b)x + ab$. The coefficient of x is the sum of the constants; the constant term is their product.

Expanding $3(2x - 5)$ gives $6x - 15$. Expanding $(x + 3)(x + 4)$ requires four products: $x^2 + 4x + 3x + 12 = x^2 + 7x + 12$. The two middle terms combine because they are like terms. A useful check is to substitute a simple value, such as $x = 1$, into both the original and expanded forms: $(1 + 3)(1 + 4) = 20$ and $1 + 7 + 12 = 20$.

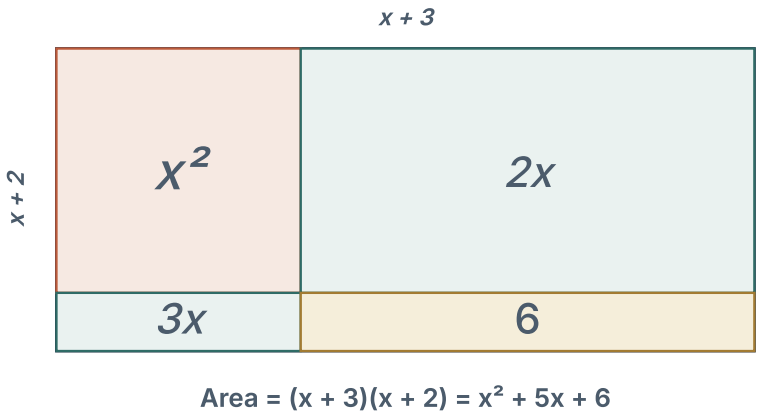


Figure 3.3 An area model for $(x + a)(x + b)$. The four regions correspond to the four terms produced by expansion. The diagram makes the product structure visible.

Factorising reverses the process. Always remove a common factor first. For $6x^2 + 9x$, both terms contain $3x$, so $6x^2 + 9x = 3x(2x + 3)$. For a quadratic $x^2 + bx + c$, look for two integers whose product is c and whose sum is b. For $x^2 + 8x + 15$, the pair 3 and 5 has sum 8 and product 15, so the factorisation is $(x + 3)(x + 5)$.

WORKED EXAMPLE 3.3.1 EXPANDING TWO BINOMIALS

PROBLEM

Expand and simplify: (a) $(x + 6)(x - 2)$ (b) $(2x - 3)(x + 5)$ (c) $(x + 7)(x - 7)$.

1. (a) $x^2 - 2x + 6x - 12 = x^2 + 4x - 12$.
2. (b) $2x^2 + 10x - 3x - 15 = 2x^2 + 7x - 15$.
3. (c) $x^2 - 7x + 7x - 49 = x^2 - 49$. The middle terms cancel; this is the difference of two squares.

ANSWER **(a)** $x^2 + 4x - 12$ **(b)** $2x^2 + 7x - 15$ **(c)** $x^2 - 49$

WORKED EXAMPLE 3.3.2 FACTORISING

PROBLEM

Factorise fully: (a) $4x^2 - 12x$ (b) $x^2 - 36$ (c) $x^2 - 2x - 15$.

1. (a) The common factor is $4x$: $4x^2 - 12x = 4x(x - 3)$.
2. (b) This is a difference of squares: $x^2 - 36 = (x - 6)(x + 6)$.
3. (c) Find two integers with product -15 and sum -2: they are -5 and 3. So $x^2 - 2x - 15 = (x - 5)(x + 3)$.
4. Check by expanding: $x^2 + 3x - 5x - 15 = x^2 - 2x - 15$.

ANSWER **(a)** $4x(x - 3)$ **(b)** $(x - 6)(x + 6)$ **(c)** $(x - 5)(x + 3)$

COMMON ERROR

$(x + 5)^2$ is not $x^2 + 25$. It means $(x + 5)(x + 5) = x^2 + 10x + 25$. The middle term is essential. Likewise, $(x - 3)(x + 3) = x^2 - 9$, not $x^2 + 9$. Test with a value such as $x = 2$ before trusting an algebraic habit.

PRACTICE 3.3 · EXPANDING AND FACTORISING SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Expand: (a) $4(2x + 3)$ (b) $-3(x - 5)$ (c) $2x(3x + 4)$. F
2. Expand and simplify: (a) $(x + 2)(x + 7)$ (b) $(x - 4)(x + 5)$ (c) $(x - 6)(x - 2)$. F
3. Expand: (a) $(3x + 1)(x - 2)$ (b) $(2x - 5)(2x + 5)$ (c) $(x + 8)^2$. F
4. Factorise fully: (a) $5x^2 + 15x$ (b) $x^2 - 49$ (c) $x^2 + 9x + 20$ (d) $x^2 - x - 12$. F
5. Explain why $x^2 + 6x + 9$ can be written as $(x + 3)^2$, and why $x^2 + 9$ cannot. R
6. A student factors $x^2 + 5x + 6$ as $(x + 2)(x + 3)$. Check the factorisation by expanding and by substituting $x = 1$. R
7. Find a quadratic expression in the form $x^2 + bx + c$ that factorises as $(x - 7)(x + 2)$, and state b and c. P
8. Explain why $x^2 + 4x + 7$ cannot be factorised into $(x + a)(x + b)$ with integer a and b. What does this imply about the graph of $y = x^2 + 4x + 7$? E

3.4 Rearranging formulas

CRITERION A · II, III ◆ EQUIVALENCE ◆ REASONING

A formula is an equation that describes a relationship between quantities. Rearranging it changes which quantity is expressed in terms of the others without changing the relationship itself.

DEFINITION

The **subject** of a formula is the variable expressed in terms of the others, usually isolated on one side. Rearranging uses inverse operations: to undo addition, subtract; to undo multiplication, divide; to undo a square, take a root (with attention to sign and domain).

From $y = 3x - 7$, make x the subject. Add 7 to both sides: $y + 7 = 3x$. Divide by 3: $x = \frac{y+7}{3}$. Each step produces an equivalent equation because the same operation is applied to both sides. The final formula gives x for any chosen y.

When the variable appears twice, first collect it on one side. From $y = ax + bx$, factor out x: $y = x(a + b)$, so $x = \frac{y}{a+b}$ provided $a + b \neq 0$. When the variable is squared, taking a square root introduces a sign choice; in a geometric context, only the positive root may be meaningful.

WORKED EXAMPLE 3.4.1 CHANGING THE SUBJECT

PROBLEM

Make w the subject of $P = 2(l + w)$. Then find w when $P = 34$ and $l = 10$.

1. Divide both sides by 2: $\frac{P}{2} = l + w$.
2. Subtract l: $w = \frac{P}{2} - l$.
3. Substitute: $w = \frac{34}{2} - 10 = 17 - 10 = 7$.

ANSWER $w = \frac{P}{2} - l$; **when** $P = 34$ **and** $l = 10$, $w = 7$.

WORKED EXAMPLE 3.4.2 A SQUARE ROOT IN THE FORMULA

PROBLEM

Make r the subject of $A = \pi r^2$, where r represents a radius.

1. Divide by π: $\frac{A}{\pi} = r^2$.
2. Take the square root: $r = \pm\sqrt{\frac{A}{\pi}}$.
3. Since a radius is non-negative, choose the positive root: $r = \sqrt{\frac{A}{\pi}}$.

ANSWER $r = \sqrt{\frac{A}{\pi}}$ **for a radius.**

STRATEGY · REARRANGING SAFELY

Work from the outside in. Undo operations in reverse order: addition and subtraction first, then multiplication and division, then powers and roots. Keep the target variable visible, and check the final formula by substituting a simple numerical case.

PRACTICE 3.4 · REARRANGING FORMULAS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Make x the subject: (a) $y = x + 5$ (b) $y = 4x$ (c) $y = 2x - 3$. F
2. Make t the subject of $v = u + at$. F
3. Make h the subject of $A = \frac{1}{2}bh$. F
4. Make r the subject of $C = 2\pi r$. F
5. Make x the subject of $y = 5(x - 2)$. P
6. Make b the subject of $A = \frac{a+b}{2}h$. P
7. Make x the subject of $y = ax + bx$, stating any condition on the constants. E
8. The surface area of a cylinder is $S = 2\pi r^2 + 2\pi rh$. Make h the subject and find h when $S = 300$ cm^2 and $r = 4$ cm, giving your answer to 2 significant figures. E

3.5 Sequences, patterns and generalisation

CRITERION B · I, II, III ◆ CRITERION C · II ◆ GENERALISATION

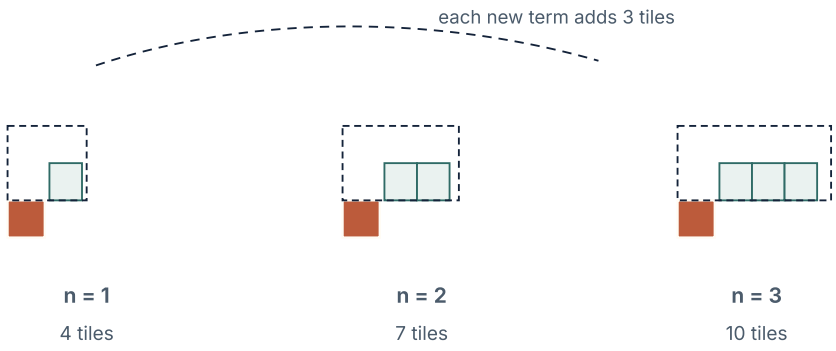
A sequence is an ordered list of numbers. A pattern is a relationship between the position of a term and its value. Finding that relationship turns a finite list into a rule that can predict terms never written down.

DEFINITIONS

A **sequence** is an ordered list of numbers called terms. A **term-to-term rule** describes how to get from one term to the next. A **position-to-term rule** or **nth term** gives the value directly from the position n. A sequence with a constant difference between consecutive terms is **linear** or arithmetic.

The sequence $5, 8, 11, 14, \ldots$ has a constant difference of 3. If the rule is $3n + c$, the first term ($n = 1$) gives $3 + c = 5$, so $c = 2$. The nth term is $3n + 2$. Check the fourth term: $3(4) + 2 = 14$. The coefficient 3 is the common difference; the constant 2 adjusts the starting value.

Not every sequence is linear. The square numbers $1, 4, 9, 16, \ldots$ have differences $3, 5, 7, \ldots$, which are not constant; their nth term is n^2. The pattern of differences can itself reveal structure: constant first differences indicate a linear rule, while constant second differences indicate a quadratic rule. Recognising the type of growth is part of generalising responsibly.



A linear sequence has a constant first difference; its rule is of the form 3n + 1.

Figure 3.4 A growing tile pattern. The visual structure explains why the number of tiles increases by a constant amount and supports the nth-term formula.

WORKED EXAMPLE 3.5.1 — FROM SEQUENCE TO NTH TERM

PROBLEM

Find the nth term of $20, 17, 14, 11, \ldots$ and use it to find the 10th term.

1. The common difference is -3. A linear rule has the form $-3n + c$.
2. For $n = 1$: $-3 + c = 20$, so $c = 23$.
3. The nth term is $23 - 3n$.
4. For $n = 10$: $23 - 30 = -7$.

ANSWER $23 - 3n$**; the 10th term is** -7**.**

WORKED EXAMPLE 3.5.2 — JUSTIFYING A PATTERN RULE

PROBLEM

A pattern uses 4 tiles in stage 1, 7 in stage 2 and 10 in stage 3. Find a rule for stage n and explain why it must hold for every stage.

1. The numbers 4, 7, 10 increase by 3 each time, so a linear rule $3n + c$ is plausible.
2. Stage 1: $3 + c = 4$, so $c = 1$. The rule is $3n + 1$.
3. Structural explanation: stage 1 has 1 central tile and 3 arms of 1 tile; each new stage adds one tile to each of the 3 arms. So stage n has $1 + 3n$ tiles.
4. Check stage 4: $3(4) + 1 = 13$, which matches adding 3 arms of 4 tiles plus the centre.

ANSWER $3n + 1$ **tiles. The rule follows from the structure, not only from the first three numbers.**

COMMON ERROR · THREE TERMS ARE NOT A PROOF

Infinitely many sequences begin 4, 7, 10. A polynomial or a changing rule can be invented to fit any finite list. A generalisation needs a structural reason, or a proof that the rule continues to hold. "It looks like it goes up by 3" is a conjecture; the explanation of why each stage adds 3 is the justification.

PRACTICE 3.5 · SEQUENCES AND GENERALISATION

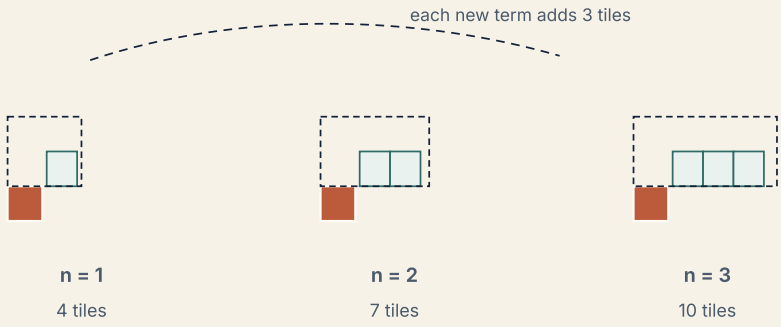
SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Find the next two terms: (a) 6, 11, 16, … (b) 30, 26, 22, … (c) 2, 6, 18, … F
2. Find the nth term: (a) 4, 9, 14, 19, … (b) 12, 9, 6, 3, … (c) 1, 3, 5, 7, … F
3. The nth term of a sequence is $5n - 2$. Find the 1st, 5th and 20th terms. F
4. Is 71 a term of the sequence $4n + 3$? Justify your answer. R
5. Explain why the sequence $1, 4, 9, 16, \ldots$ is not linear. Use first differences. R
6. A pattern has 6 tiles in stage 1, 10 in stage 2 and 14 in stage 3. Find a rule in terms of n and explain the structure. P
7. A sequence begins 2, 4, 8, 16. Write a rule that fits these four terms, then write a different rule that also fits them. What does this show about generalising from limited data? E
8. Find the nth term of the sequence $3, 8, 15, 24, 35, \ldots$. Use second differences to identify the type of rule. E

Investigation · The matchstick triangles

CRITERION B

Triangles are made from identical matchsticks and placed in a row, sharing sides where they meet. Stage 1 uses 3 matchsticks; stage 2 uses 5; stage 3 uses 7.



A linear sequence has a constant first difference; its rule is of the form 3n + 1.

Stage 1 has one triangle, stage 2 has two triangles sharing a side, and stage 3 has three. The shared sides reduce the number of new matchsticks added each time.

Stage n	Triangles	Matchsticks	First difference
1	1	3	—
2	2	5	2
3	3	7	2
4	4	9	2

1. **Continue.** Draw stages 5 and 6 and record the number of matchsticks. Confirm that the first differences remain constant.
2. **Find a rule.** Write the number of matchsticks M as a function of the number of triangles n. Explain the rule using the structure: how many matchsticks does the first triangle use, and how many does each additional triangle add?
3. **Predict.** Use your rule to find the number of matchsticks for 20 triangles and for 100 triangles.
4. **Reverse.** A student has 61 matchsticks. How many complete triangles can be made in the row? Will any matchsticks remain? Explain.
5. **Change the pattern.** Suppose the triangles are arranged in a ring so that every triangle shares two sides. Find the rule for n triangles, state the domain, and compare it with the row pattern.
6. **Generalise.** What happens if each new triangle shares exactly one side with the previous triangle but is placed on the other side, forming a zigzag? Does the number of matchsticks change? Justify your answer with a diagram.

Your report should include a table, two diagrams, a formula and a structural explanation. A correct formula without a reason is a conjecture, not yet a generalisation.

Application · Painting a community hall

CRITERION D · I, II, III, IV ◆ FORMULAS ◆ MODELLING

A community hall is a rectangular room of length L metres, width W metres and height H metres. It has one door of area $2.0\ \text{m}^2$ and four windows, each of area $1.5\ \text{m}^2$. The walls are to receive two coats of paint. One litre covers $10\ \text{m}^2$ per coat.

1. Write an expression for the total wall area, excluding the door and windows. Simplify it.
2. Write a formula for the number of litres N needed for two coats. Explain why the answer must be rounded up to buy whole litres.
3. The hall has $L = 18$ m, $W = 12$ m and $H = 4.0$ m. Calculate the wall area and the litres required. State the number of 5-litre tins to buy.
4. Paint costs 42 units per 5-litre tin. Calculate the paint cost. Add 8% for brushes, tape and covers, and find the total budget.
5. Rearrange your formula to make H the subject. Use it to find the height of a hall with $L = 15$ m, $W = 9$ m that requires 44 litres for two coats, assuming the same door and window areas.
6. Evaluate the model. What does it ignore? How could the estimate be improved without measuring every wall separately? Comment on whether rounding up litres is the right kind of accuracy for a purchasing decision.

CRITERION D FOCUS

Translate a context into a formula, apply it with units, rearrange it to answer a reverse question, and evaluate the assumptions behind the number.

Criterion task · The growing square frame

CRITERION B
INVESTIGATING PATTERNS
24 MARKS

A square frame is made from unit tiles. Frame 1 is a 2×2 square with no centre; Frame 2 is a 3×3 square with a 1×1 hole; Frame 3 is a 4×4 square with a 2×2 hole.

1. Draw Frames 1, 2 and 3. Record the number of tiles in a table for $n = 1, 2, 3, 4, 5$. *(4 marks)*
2. Find the first differences. Describe the pattern and state whether it is linear. *(4 marks)*
3. Find a formula for the number of tiles T in Frame n. Explain the formula using the structure of the frame, not only the numbers. *(6 marks)*
4. Show that your formula is equivalent to $4n$ and to $4(n-1)+4$. Explain what each form reveals about the frame. *(4 marks)*
5. Use your formula to predict the number of tiles in Frame 25, and to find the largest frame that can be made from 100 tiles. *(4 marks)*
6. Investigate what happens if the frame is rectangular rather than square, with width $n+1$ and height $n+2$, and the centre removed. Find a formula and explain whether it is linear. *(2 marks)*

PLANNING FRAME

Represent: draw the frames, then make a table. **Investigate:** calculate first differences. **Generalise:** write a rule and connect it to the picture. **Verify:** test the rule on Frame 4 and Frame 5. **Justify:** explain why the rule must hold for all n.

BAND	DESCRIPTOR	MARKS
4	Accurate table and diagram; correct general rule with structural justification; equivalence explained; predictions verified.	19–24
3	Correct rule with some justification; table and predictions mostly accurate; equivalence partially explained.	13–18
2	Pattern recognised and rule attempted; limited structural explanation or verification.	7–12
1	Isolated correct terms or a rule that fits only the given cases.	1–6

CRITERIA: B (C) **ATL:** THINKING **TIME:** 45 MINUTES **CONDITIONS:** NOTES ALLOWED

MAPPED TO: B strands i–iii and C strands ii–iii. The rubric is a course summary, not an official IB descriptor.

Chapter summary and review

KEY IDEAS

- Like terms have identical variable parts and can be combined by adding coefficients.
- A function assigns exactly one output to each input; $f(x)$ denotes the output at x.
- The domain is the set of permitted inputs; the range is the set of actual outputs.
- The vertical line test identifies whether a graph represents a function of x.
- Expanding and factorising are inverse processes based on the distributive law.
- Rearranging a formula uses inverse operations and preserves equivalence.
- A linear sequence has a constant first difference and an nth term of the form $an + b$.

REVISION CHECKLIST

- ☐ I can simplify expressions with several variables and powers.
- ☐ I can evaluate a function and solve $f(x) = k$.
- ☐ I can state a domain from an expression or a context.
- ☐ I can expand $(x + a)(x + b)$ and factorise simple quadratics.
- ☐ I can rearrange a formula, including one containing a square.
- ☐ I can find and justify the nth term of a linear pattern.
- ☐ I can distinguish a conjecture from a justified generalisation.

VOCABULARY

variable
A symbol representing a number that may change.

expression
A combination of terms without an equals sign.

domain
The set of permitted inputs.

expand
Write a product as a sum.

subject
The variable isolated in a formula.

nth term
A rule giving the value at position n.

coefficient
The numerical factor of a term.

function
A rule assigning exactly one output to each input.

range
The set of actual outputs.

factorise
Write a sum as a product.

sequence
An ordered list of numbers.

generalisation
A rule claimed to hold for a whole class of cases.

MIXED PRACTICE · CHAPTER 3

Connecting the ideas

FLUENCY Core procedures

1. Simplify $6x^2 - 2x + 3 - 4x^2 + 7x - 8$. F
2. For $f(x) = 4 - 3x$, find $f(0)$, $f(2)$ and $f(-1)$. F
3. Expand $(2x - 1)(x + 6)$. F
4. Factorise $x^2 + 11x + 24$ and $x^2 - 64$. F
5. Make x the subject of $3x + 2y = 12$. F
6. Find the nth term of 9, 13, 17, 21, ... F

REASONING Justify and connect

1. Show that $(x + 5)^2$ and $x^2 + 25$ are not equivalent by substituting $x = 1$. R
2. Explain how the graph of a function can be used to find $f(3)$ and to solve $f(x) = 2$. R
3. Why is the domain of $g(x) = \frac{5}{x+1}$ not all real numbers? R
4. A sequence has nth term $7 - 2n$. Without calculating every term, determine whether the sequence is increasing or decreasing and explain. R

APPLICATION Use and evaluate

1. A mobile plan costs 12 units per month plus 0.08 units per megabyte. Write a cost function for m megabytes and find the cost for 500 MB. P
2. A rectangle has length $x + 3$ and width $x - 1$. Write and expand an expression for its area. Find the area when $x = 6$. P
3. The sum of the first n odd numbers is n^2. Use this to find the sum of the first 30 odd numbers and describe the pattern in a table. P
4. A formula for the surface area of a cube is $A = 6s^2$. Make s the subject and find the side length of a cube with surface area $150\ \text{cm}^2$. E

Check your progress

I CAN ...	CONFIDENCE	EVIDENCE	NEXT ACTION
Simplify expressions and substitute values.	1 / 2 / 3		
Use function notation and identify domain and range.	1 / 2 / 3		
Expand and factorise quadratic expressions.	1 / 2 / 3		
Rearrange formulas.	1 / 2 / 3		
Generalise a pattern and justify the rule.	1 / 2 / 3		

Reflection

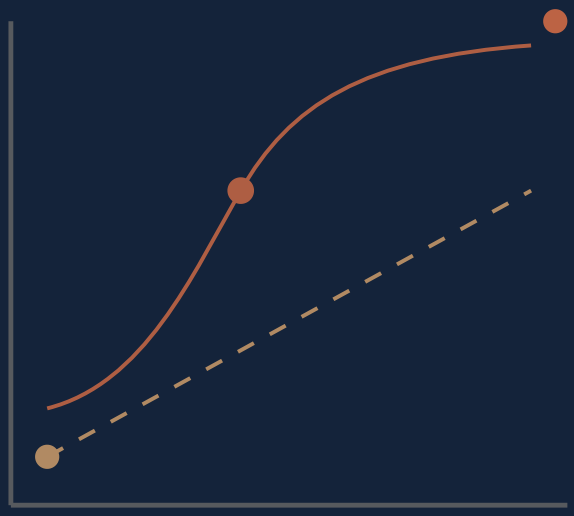
Two expressions can produce the same value for every input but look completely different. How can a diagram help you see why they are equivalent?

PART II · DESCRIBING CHANGE

II

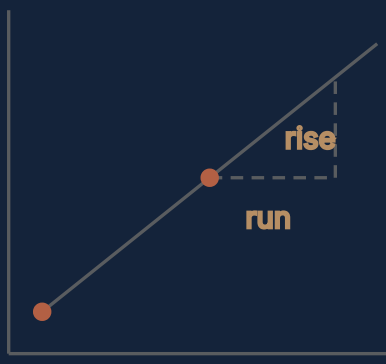
Relationships and equations

A linear relationship is one of the most powerful ideas in mathematics: a constant rate of change generates a straight line, and a straight line can answer questions about cost, distance, temperature and time. This part moves between tables, graphs and equations, then uses those representations to solve equations, inequalities and systems in context.



04 Linear relationships

05 Equations, inequalities and systems



4

CHANGE, RELATIONSHIPS AND PREDICTION

Linear relationships and graphs

Some quantities change at a steady rate. Equal steps in one produce equal steps in the other, and a graph makes that regularity visible as a straight line. The line is not just a picture: it is a rule for predicting beyond the data.

KEY CONCEPT · RELATIONSHIPS | RELATED · CHANGE | RELATED · REPRESENTATION | CONTEXT · FAIRNESS AND DEVELOPMENT

INQUIRY

How can a straight line turn a table of measurements into a rule that predicts values we have not measured?

ATL · COMMUNICATING WITH REPRESENTATIONS

A graph is an argument, not decoration. Always label both axes with quantities and units, choose a scale that makes the relationship readable, and mark key points such as intercepts. When you read a gradient aloud, name both units: “18 kilometres per hour”, not “18”. A reader should be able to reconstruct your model from your table, graph and equation without guessing.

MYP UNIT PLANNER

Framing the inquiry

UNIT 4 · APPROX. 15 HOURS
ATL FOCUS · COMMUNICATION

KEY AND RELATED CONCEPTS

Relationships. A linear relationship connects two quantities by a constant rate of change. The same relationship can be represented by a table, a graph, an equation or a description in words.

Change. Gradient measures how quickly the dependent quantity changes as the independent quantity increases by one unit. A zero gradient and an undefined gradient describe very different situations.

Representation. Each representation exposes different information. A table shows individual values, a graph shows shape and direction, and an equation gives the exact rule.

GLOBAL CONTEXT AND EXPLORATION

Fairness and development. Exploration: how linear cost models can make charges predictable, and how a fixed fee, a rate or a cap affects different users unequally.

Two businesses may describe their prices with straight lines that look equally simple. The intercept says how much someone pays before using anything; the gradient says how much each additional unit costs. Reading those two numbers together is a first step in judging whether a price is fair.

STATEMENT OF INQUIRY

A constant rate of change produces a linear relationship whose table, graph and equation are equivalent representations; gradient and intercept communicate the rate and the starting value, and together they allow predictions to be made and compared.

FACTUAL

What are gradient and y-intercept? How do you plot a line from a table or an equation? What does it mean for two quantities to be directly proportional?

CONCEPTUAL

Why does a constant rate of change necessarily produce a straight line, and why does a straight line necessarily have a constant gradient?

DEBATABLE

When a linear model fits only part of a real situation, how far should we trust its predictions about fairness?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii	Calculate gradients, identify intercepts, find equations of lines and use direct proportion.
B · Investigating patterns	i, ii, iii	Investigate a candle-burning pattern, test a linear rule and evaluate its limiting assumptions.
C · Communicating	i, ii, iii, iv	Move between tables, graphs, equations and words; state the meaning of gradient and intercept with units.
D · Applying mathematics	i, ii, iii, iv, v	Compare two pricing models, justify a recommendation and evaluate fairness and model limitations.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** how a table, graph and equation can represent the same linear relationship.
- **Know** how to calculate gradient from a graph, a table or two points.
- **Be able to** interpret gradient and intercept in context, with correct units.
- **Be able to** find the equation of a line from a graph, a table or two points.
- **Be able to** recognise direct proportion and use a constant of proportionality.
- **Be able to** build, compare and critique a linear model of a real cost or motion.

SUCCESS CRITERIA

- ☐ I can plot points accurately and read coordinates from a grid.
- ☐ I can calculate gradient using $\frac{y_2-y_1}{x_2-x_1}$ and interpret its sign.
- ☐ I can explain the meaning of m and c in $y = mx + c$, including units.
- ☐ I can find a linear equation from two points and check it by substitution.
- ☐ I can decide whether a relationship is proportional or merely linear.
- ☐ I can use a model to interpolate, extrapolate and discuss its limitations.

PREREQUISITE CHECK · 12 MINUTES, NO CALCULATOR

1. Plot the points $(0, 0)$, $(2, 3)$ and $(-1, 4)$ on a coordinate grid.
2. Evaluate $3x + 2$ when $x = 4$.
3. Solve $2x - 6 = 0$.
4. Write the next two terms: 5, 9, 13, 17, ...
5. Calculate $\frac{8-2}{5-1}$.
6. Convert 90 minutes to hours.

Answers are in the Chapter 4 answer section. If more than two are incorrect, review coordinates, substitution and simple equations before continuing.

BEFORE YEAR 3
Coordinates in four quadrants; substitution; simple equations; ratio and unit rates; recognising number patterns.

YEAR 3 · THIS CHAPTER
Gradient as rate of change; $y = mx + c$; equations from graphs and points; direct proportion; linear models and critique.

LATER
Simultaneous equations; systems of inequalities; quadratic and exponential models; calculus of change; linear regression.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Prices, distances and heights change at measurable rates.	How can a graph show a constant rate and predict unseen values?	Gradient, intercept, equivalence of representations and direct proportion.	Model a burning candle and test where the linear model stops working.	Explain the meaning of gradient and intercept in a fair-pricing context.	When is a straight-line prediction responsible, and when is it misleading?

4.1 From tables to straight lines

CRITERION A · I ◆ CRITERION C · I ◆ REPRESENTATION

A graph is a correspondence between algebra and geometry. Each ordered pair becomes a point, and a collection of points can reveal a pattern that the numbers alone might hide.

DEFINITIONS

The **Cartesian plane** is formed by a horizontal x-axis and a vertical y-axis crossing at the **origin** $(0, 0)$. A point is named by an ordered pair (x, y), where x is the horizontal displacement and y the vertical displacement. The axes divide the plane into four **quadrants**. A relationship is **linear** when its graph is a straight line; equivalently, equal changes in x produce equal changes in y.

In a linear relationship, the quantity we choose is usually called the independent variable and is plotted on the horizontal axis. The quantity that responds is the dependent variable and is plotted on the vertical axis. Time is often independent because we choose when to measure; distance, cost or height is often dependent because it responds to time or to the amount used.

A table of values is a list of ordered pairs. To plot $y = 2x - 1$, substitute a few values of x, calculate the corresponding values of y, and plot the resulting pairs. Because the rule is linear, two correctly plotted points would determine the line, but a third point is a useful check.

WORKED EXAMPLE 4.1.1 FROM EQUATION TO TABLE TO GRAPH

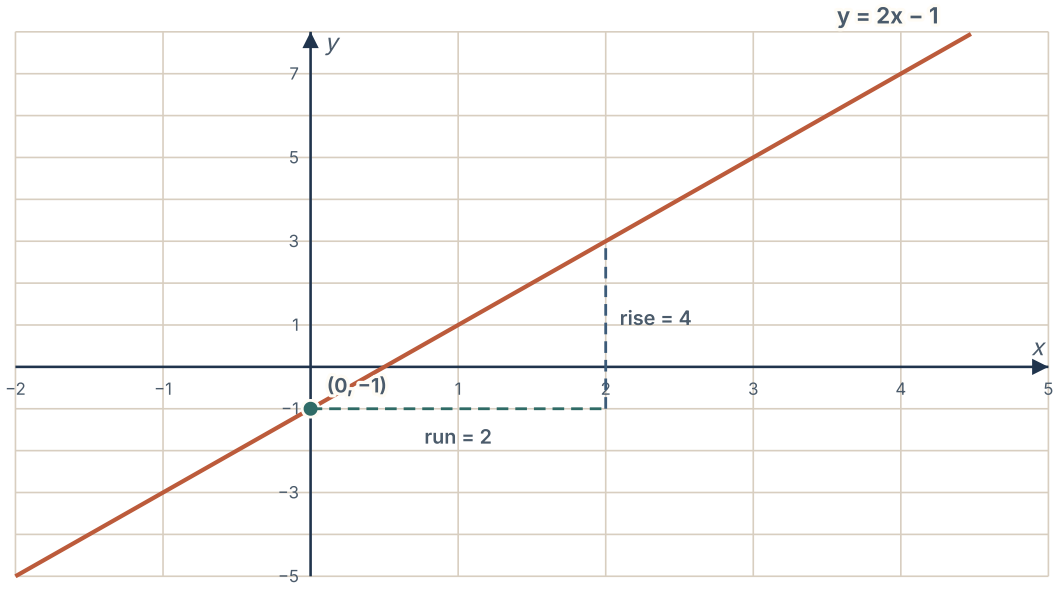
PROBLEM

Complete the table for $y = 2x - 1$ and describe the graph.

x	−2	−1	0	1	2	3
$y = 2x - 1$	—	—	—	—	—	—

1. Substitute each x-value. For example, when $x = -2$, $y = 2(-2) - 1 = -5$; when $x = 3$, $y = 2(3) - 1 = 5$.
2. The completed values are $-5, -3, -1, 1, 3, 5$. The change in y is always 2 when x increases by 1.
3. Plot the six ordered pairs and join them with a straight line. Extend the line beyond the plotted points to show that the relationship continues.

ANSWER **The points lie on a straight line; as x increases by 1, y increases by 2. The line crosses the y-axis at $(0, -1)$.**



The gradient triangle shows that a run of 2 produces a rise of 4.

Figure 4.1 The graph of $y = 2x - 1$. Every step of 1 to the right raises the line by 2; the intercept $(0, -1)$ is where the line meets the y-axis.

A table can also tell us whether a relationship is linear. If the values of x increase by equal steps, compare the corresponding changes in y. A constant first difference in y indicates a linear pattern. If the first differences change, the relationship is not linear.

WORKED EXAMPLE 4.1.2 TESTING A TABLE FOR LINEARITY

PROBLEM

Decide which table represents a linear relationship.

Table A: x	0	1	2	3	4
y	7	10	13	16	19
Table B: x	0	1	2	3	4
y	1	4	9	16	25

1. Table A: the changes in y are $3, 3, 3, 3$. The first differences are constant, so the table represents a linear relationship. When $x = 0$, $y = 7$, so the rule is $y = 3x + 7$.
2. Table B: the changes in y are $3, 5, 7, 9$. They are not constant, so the relationship is not linear. The values are in fact $y = (x + 1)^2$.
3. A constant first difference tells us that the rate of change is constant. For Table A that rate is 3 units of y for each unit of x.

ANSWER **Table A is linear with rule $y = 3x + 7$; Table B is not linear.**

COMMON ERROR · READING COORDINATES IN THE WRONG ORDER

The point $(3, 5)$ means $x = 3$, $y = 5$, not the other way round. Also check the scale: one square might represent 2 units, not 1. A line can look straight but be plotted incorrectly if the axes are not labelled or the scale changes partway along an axis.

PRACTICE 4.1 · COORDINATES, TABLES AND GRAPHS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Plot $A(3,2)$, $B(-2,4)$, $C(-3,-1)$, $D(4,-3)$ and $E(0,-2)$. State the quadrant or axis on which each point lies. F
2. Complete the table for $y = 3x - 2$ for $x = -1, 0, 1, 2, 3$, then plot the graph. F

x	−1	0	1	2	3
y	—	—	—	—	—

3. A table has $x = 0, 1, 2, 3, 4$ and $y = 5, 8, 11, 14, 17$. Find the first differences, state whether the relationship is linear, and write its equation. F
4. Which tables represent linear relationships? Explain using first differences. R
   a. $x = 0, 1, 2, 3$; $y = 4, 7, 10, 13$
   b. $x = 0, 1, 2, 3$; $y = 2, 4, 8, 16$
   c. $x = 1, 2, 3, 4$; $y = 9, 7, 5, 3$
   d. $x = -1, 0, 1, 2$; $y = 1, 3, 6, 10$
5. A plant is 12 cm tall at week 0 and grows 2.5 cm each week. Make a table for weeks 0 to 4, write a formula for its height h after w weeks, and predict its height at week 10. P
6. Explain why the graph of $y = 3$ is horizontal, while the graph of $x = 3$ is vertical and does not represent y as a function of x. R
7. A taxi company charges a fixed amount plus a rate per kilometre. The table shows the cost for several distances. Find the fixed charge and the rate per kilometre, write an equation, and predict the cost of an 11 km journey. P

Distance (km)	0	2	4	6
Cost (units)	4.50	9.00	13.50	18.00

8. Invent a real context that could be modelled by $C = 15 + 0.4x$. State what x and C represent, include their units, and give one reason the linear model might eventually fail. E

4.2 Gradient as a rate of change

CRITERION A · II ◆ CRITERION C · II ◆ CHANGE

Gradient is the central measurement of a linear relationship. It answers a practical question: how much does the dependent quantity change for each additional unit of the independent quantity?

> **DEFINITION**
>
> For two points (x_1, y_1) and (x_2, y_2) on a line, the **gradient** is $m = \frac{y_2 - y_1}{x_2 - x_1}$, provided $x_1 \neq x_2$. The numerator is the vertical change or **rise**; the denominator is the horizontal change or **run**. In context, gradient is a **rate of change** and carries units: units of y per unit of x.

A positive gradient means the line rises from left to right. A negative gradient means it falls. A horizontal line has gradient 0 because the vertical change is 0. A vertical line has an undefined gradient because the horizontal change is 0 and division by zero is undefined. “No slope” is not the same as “slope zero”.

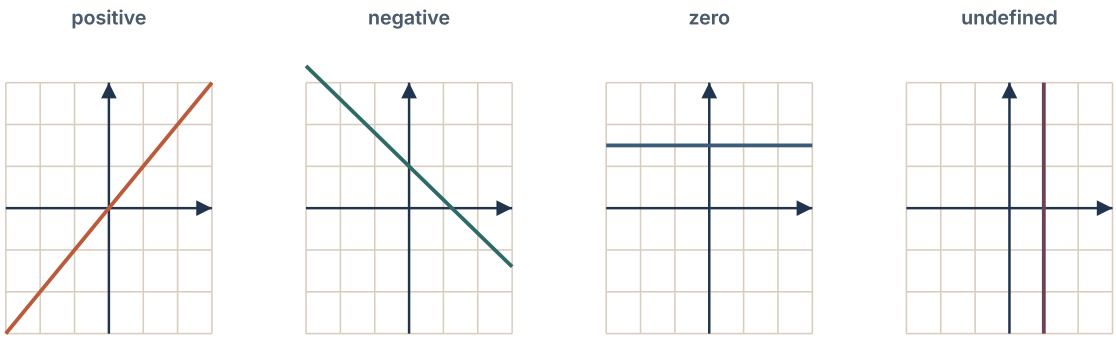


Figure 4.2 The four possible types of gradient for a straight line. Only the horizontal and vertical cases require special care: zero is a number, while undefined is not.

WORKED EXAMPLE 4.2.1 — GRADIENT FROM TWO POINTS

PROBLEM

Find the gradient of the line through $(-2, 3)$ and $(4, -1)$.

1. Label the points: $(x_1, y_1) = (-2, 3)$ and $(x_2, y_2) = (4, -1)$.
2. Substitute: $m = \frac{-1-3}{4-(-2)} = \frac{-4}{6}$.
3. Simplify: $m = -\frac{2}{3}$.
4. The negative sign means the line falls from left to right. For every 3 units moved right, the line falls 2 units.

ANSWER $m = -\frac{2}{3}$

WORKED EXAMPLE 4.2.2 — GRADIENT FROM A TABLE

PROBLEM

Find the gradient of the linear relationship shown in the table, and interpret it.

x	0	2	4	6
y	7	4	1	−2

1. Choose two points, for example $(0, 7)$ and $(6, -2)$.
2. $m = \frac{-2-7}{6-0} = \frac{-9}{6} = -1.5$.
3. Check with another pair: $\frac{1-4}{4-2} = \frac{-3}{2} = -1.5$. The gradient is constant, as expected for a linear relationship.
4. Each increase of 1 in x produces a decrease of 1.5 in y.

ANSWER $m = -1.5$; **y decreases by 1.5 for every unit increase in x.**

WORKED EXAMPLE 4.2.3 GRADIENT WITH UNITS

PROBLEM

A cyclist is 6 km from the start at 09:00 and 12 km from the start at 09:20. Find the gradient of the distance–time graph and interpret it.

1. Let t be the time in minutes after 09:00 and d the distance in kilometres. The points are $(0, 6)$ and $(20, 12)$.
2. $m = \frac{12-6}{20-0} = \frac{6}{20} = 0.3$.
3. The units are kilometres per minute: 0.3 km/min. This is $0.3 \times 60 = 18$ km/h.
4. The gradient is the cyclist's average speed over that interval. The starting value 6 km is the intercept, not part of the speed.

ANSWER 0.3 km/min $= 18$ km/h.

COMMON ERROR · INCONSISTENT SUBTRACTION

Subtract the coordinates in the same order on top and bottom. Using $\frac{y_2-y_1}{x_1-x_2}$ reverses only one sign and gives the wrong answer. If you swap the points, both numerator and denominator change sign, so the gradient is unchanged: $\frac{y_1-y_2}{x_1-x_2} = \frac{y_2-y_1}{x_2-x_1}$.

PRACTICE 4.2 · GRADIENT AND RATE OF CHANGE

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Find the gradient of the line through each pair of points, or state that it is undefined: (a) $(2, 1)$, $(6, 5)$ (b) $(-1, 4)$, $(3, -4)$ (c) $(-2, -3)$, $(2, 5)$ (d) $(4, 0)$, $(7, 0)$ (e) $(5, -2)$, $(5, 3)$. F
2. A linear table has $x = 1, 3, 5, 7$ and $y = 2, 11, 20, 29$. Find the gradient and use it to predict y when $x = 9$. F
3. A tank contains 500 litres at $t = 0$ and 340 litres after 8 minutes. Find the rate of change in litres per minute, including its sign, and determine when the tank will be empty if the rate stays constant. F
4. Sketch four lines with gradients that are positive, negative, zero and undefined. Label each line and explain in one sentence what its gradient means. F
5. Explain why a gradient of 0 is different from an undefined gradient. Give an equation of a line of each type. R
6. A student calculates the gradient of the line through $(2, 3)$ and $(5, -1)$ as $\frac{-1-3}{2-5} = \frac{-4}{-3} = \frac{4}{3}$. Identify the error and give the correct gradient. R
7. Cyclist A travels 12 km in 40 minutes. Cyclist B travels 9 km in 30 minutes. Compare their average speeds. Cyclist B started 2 km ahead of A; explain how this affects the graph but not the gradient. P
8. A ramp has a rise of 0.6 m over a horizontal run of 7.2 m. Calculate its gradient. An accessibility guideline recommends a gradient no steeper than $\frac{1}{12}$. Is the ramp within the guideline? What run would be needed for a rise of 0.6 m at exactly $\frac{1}{12}$? E

4.3 The equation of a straight line

CRITERION A · I, II ◆ CRITERION C · III ◆ EQUIVALENCE

Every non-vertical straight line can be described by an equation of the form $y = mx + c$. The two numbers m and c determine the line completely: one controls its direction, the other its starting height.

KEY IDEA · GRADIENT–INTERCEPT FORM

In $y = mx + c$, the coefficient m is the gradient and c is the y-intercept, the value of y when $x = 0$. The line crosses the y-axis at $(0, c)$. A horizontal line has equation $y = k$; a vertical line has equation $x = k$ and is not the graph of a function of x.

The equation can be read in either direction. Given $y = 4x - 3$, the gradient is 4 and the intercept is -3. Given a line that passes through $(0, 2)$ and rises 5 units for every 1 unit right, its equation is $y = 5x + 2$. The intercept is the value at the start; the gradient is the repeated change.

An equation may be written in another form. To rewrite $3x + 2y = 12$ in gradient–intercept form, isolate y: $2y = 12 - 3x$, so $y = 6 - \frac{3}{2}x$. The gradient is $-\frac{3}{2}$ and the intercept is 6. The same line can also be drawn from its intercepts: when $x = 0$, $y = 6$; when $y = 0$, $x = 4$.

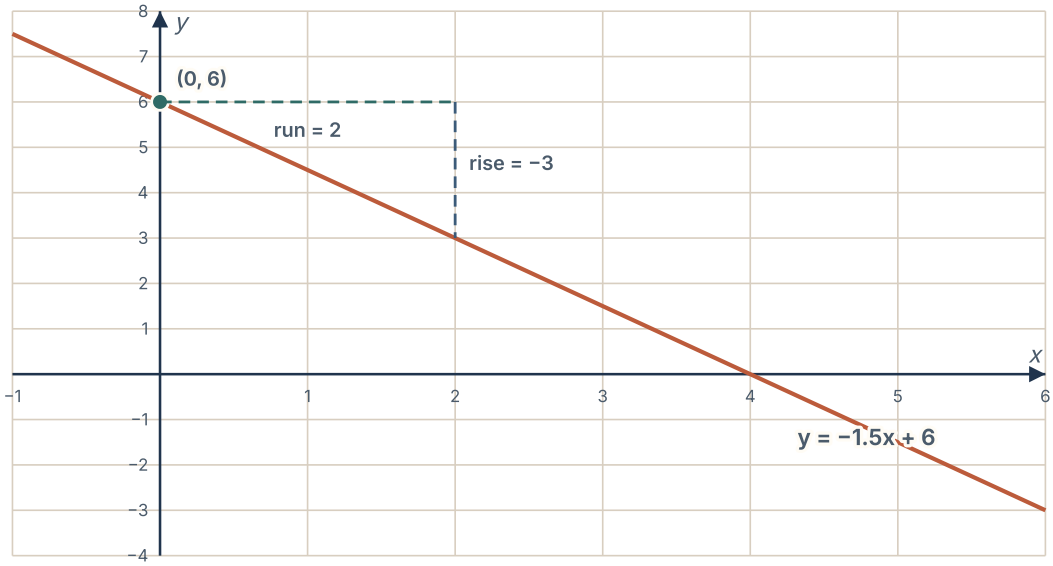
WORKED EXAMPLE 4.3.1 READING AND REWRITING AN EQUATION

PROBLEM

For the line $3x + 2y = 12$, state the gradient and the intercepts, then sketch it.

1. Rearrange: $2y = 12 - 3x$, so $y = 6 - \frac{3}{2}x$. The gradient is $-\frac{3}{2}$ and the y-intercept is 6.
2. Find the x-intercept by setting $y = 0$: $3x = 12$, so $x = 4$. The intercepts are $(0, 6)$ and $(4, 0)$.
3. Plot the two intercepts and join them. Because the gradient is negative, the line falls from left to right.

ANSWER $m = -\frac{3}{2}$; **y-intercept** $(0, 6)$; **x-intercept** $(4, 0)$.



A run of 2 produces a rise of −3, so the gradient is −1.5.

Figure 4.3 The line $3x + 2y = 12$, rewritten as $y = 6 - \frac{3}{2}x$. The intercepts $(0, 6)$ and $(4, 0)$ are enough to draw the line.

WORKED EXAMPLE 4.3.2 EQUATION FROM TWO POINTS

PROBLEM

Find the equation of the line through $(2, -1)$ and $(6, 7)$.

1. Calculate the gradient: $m = \frac{7-(-1)}{6-2} = \frac{8}{4} = 2$.
2. Write the partial equation $y = 2x + c$. Substitute one point to find c. Using $(2, -1)$: $-1 = 2(2) + c$, so $-1 = 4 + c$ and $c = -5$.
3. The equation is $y = 2x - 5$.
4. Check with the other point: $2(6) - 5 = 7$, as required.

ANSWER $y = 2x - 5$

WORKED EXAMPLE 4.3.3 A PARALLEL LINE

PROBLEM

Find the equation of the line parallel to $y = 3x + 1$ that passes through $(4, -2)$.

1. Parallel lines have the same gradient, so the new line has $m = 3$.
2. Write $y = 3x + c$ and substitute $(4, -2)$: $-2 = 3(4) + c$, so $c = -14$.
3. The equation is $y = 3x - 14$.

ANSWER $y = 3x - 14$

Extension. Perpendicular lines have gradients whose product is -1: if $m_1 m_2 = -1$, the lines meet at a right angle. For example, a line perpendicular to $y = 3x + 1$ has gradient $-\frac{1}{3}$. This is developed further in Chapter 7.

COMMON ERROR · DIVIDING ONLY ONE TERM

From $2y = 6x - 8$, dividing by 2 gives $y = 3x - 4$, not $y = 3x - 8$. Every term on the right must be divided. Similarly, the intercept of $y = 5 - 2x$ is 5, not -2; rewrite it as $y = -2x + 5$ before reading the intercept.

PRACTICE 4.3 · EQUATIONS OF LINES

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. State the gradient and y-intercept: (a) $y = 4x - 7$ (b) $y = -0.5x + 2$ (c) $y = 5 - 2x$ (d) $3x + y = 5$ (e) $2y = 6x - 8$. F
2. Write the equation of the line with: (a) gradient 3 and y-intercept -2 (b) gradient $-\frac{2}{3}$ and y-intercept 4 (c) gradient 0 through $(0, -3)$ (d) undefined gradient through $(5, 0)$. F
3. Find the equation of the line through each pair of points: (a) $(0, 2)$, $(4, 10)$ (b) $(1, 5)$, $(3, 1)$ (c) $(-2, 4)$, $(2, 0)$ (d) $(3, -1)$, $(6, -1)$. F
4. A table has $x = 0, 1, 2, 3$ and $y = 6, 4, 2, 0$. Find the equation of the line. F
5. Which of these points lie on the line $y = 2x - 3$: $(1, -1)$, $(2, 1)$, $(3, 4)$, $(0, -3)$? Show a substitution for each. R
6. Find the equation of the line parallel to $y = -x + 4$ through $(2, 5)$. Then find the equation of the line perpendicular to $y = -x + 4$ through the same point. P
7. A tank drains so that its height after t minutes is $h = 2.4 - 0.15t$ metres. Interpret the gradient and intercept, find the height after 8 minutes, and determine when the tank is empty. State the domain. P

8. How many different lines pass through $(2, 5)$? Write equations for three of them with different gradients. Is there a line through $(2, 5)$ with gradient 0? With undefined gradient? Explain. E

4.4 Direct proportion

CRITERION A · II ◆ CHANGE ◆ PROPORTIONALITY

Direct proportion is the simplest linear relationship: doubling one quantity doubles the other, and zero input produces zero output. Its graph is a straight line through the origin.

DEFINITION

Two quantities x and y are in **direct proportion** if $y = kx$ for some non-zero constant k. The number k is the **constant of proportionality**. Equivalently, the ratio $\frac{y}{x}$ is constant for all $x \neq 0$, and the graph of y against x is a straight line through the origin.

Every direct proportion is linear with y-intercept 0. The converse is false: $y = 2x + 3$ is linear but not proportional. Its graph does not pass through the origin, and the ratio $\frac{y}{x}$ changes with x. At $x = 1$, $\frac{y}{x} = 5$; at $x = 3$, $\frac{y}{x} = 3$. A constant gradient alone does not guarantee direct proportion; the intercept must also be zero.

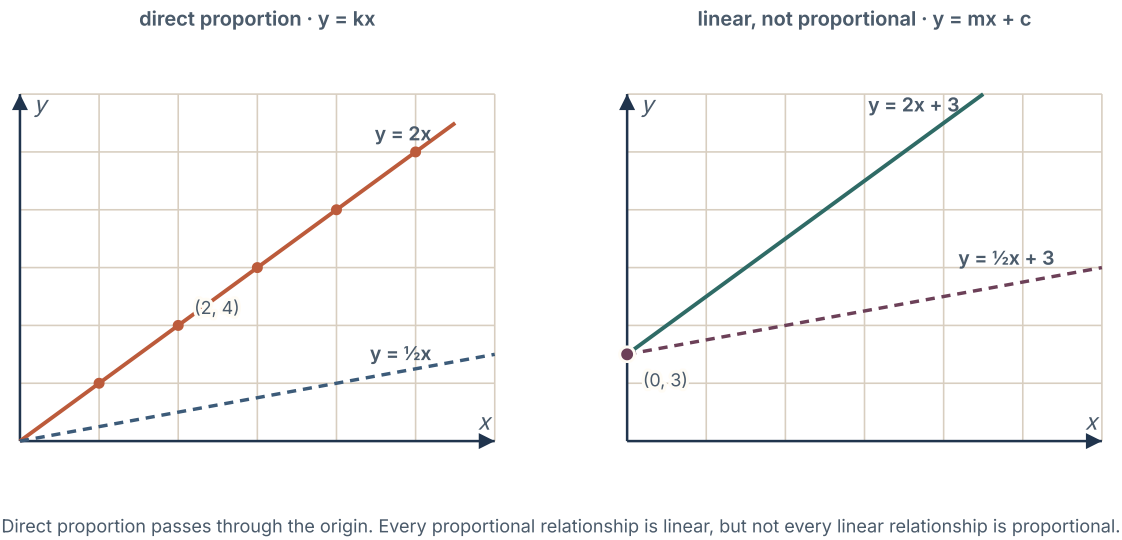


Figure 4.4 Proportional relationships pass through the origin and have a constant quotient. Linear relationships with a non-zero intercept have a constant gradient but not a constant quotient.

WORKED EXAMPLE 4.4.1 FINDING THE CONSTANT OF PROPORTIONALITY

PROBLEM

y is directly proportional to x, and $y = 18$ when $x = 6$. Find the constant of proportionality and the value of x when $y = 45$.

1. Write $y = kx$. Substitute the known pair: $18 = k(6)$.
2. Divide by 6: $k = 3$. The relationship is $y = 3x$.
3. When $y = 45$: $45 = 3x$, so $x = 15$.

ANSWER $k = 3$; **the relationship is** $y = 3x$; **when** $y = 45$, $x = 15$.

WORKED EXAMPLE 4.4.2 A CURRENCY CONVERSION

PROBLEM

One US dollar is exchanged for 0.92 euros. Write the euro value E as a function of the dollar value D, convert 250 dollars, and find how many dollars 500 euros will buy.

1. The relationship is $E = 0.92D$. This is direct proportion with $k = 0.92$.
2. For $D = 250$: $E = 0.92(250) = 230$ euros.
3. For $E = 500$: $500 = 0.92D$, so $D = \frac{500}{0.92} \approx 543.48$ dollars.
4. The domain is $D \geq 0$. In practice, a currency kiosk may charge a commission, so the linear model is only an approximation unless the rate is guaranteed.

ANSWER $E = 0.92D$; **250 dollars buys 230 euros; 500 euros buys about 543.48 dollars.**

STRATEGY · TESTING FOR DIRECT PROPORTION

Check three things: the graph passes through the origin; the ratio $\frac{y}{x}$ is constant for every non-zero pair; and the relationship can be written as $y = kx$. A single pair is never enough to prove proportionality — it only determines k if proportionality is already known or justified by the context.

PRACTICE 4.4 · DIRECT PROPORTION

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Which relationships are direct proportions? Explain each answer. (a) $C = 4n$ (b) $C = 4n + 10$ (c) $y = 0.5x$ (d) $y = \frac{2}{x}$ (e) $y = 3 - x$ F
2. For $y = 7x$, find y when $x = 8$, and find x when $y = 63$. F
3. y is directly proportional to x, and $y = 45$ when $x = 9$. Find k, write the relationship, and find y when $x = 12$ and x when $y = 100$. F
4. A recipe uses 3 cups of flour for 12 muffins. Write a proportional relationship between cups c and muffins M. How many cups are needed for 20 muffins? How many muffins can be made from 7.5 cups? P
5. The exchange rate is 1 US dollar $= 0.93$ euros. Write an equation, convert 250 dollars, and convert 500 euros. State the domain and one reason the model may not be exact. P
6. Explain why $y = 2x + 3$ is linear but not directly proportional. Use a table of three values and the ratios $\frac{y}{x}$ in your explanation. R
7. A car travels at a constant 80 km/h, so $d = 80t$. Explain why this is direct proportion, find the time for 300 km, and state a sensible domain. How does the equation change if the car starts 20 km from home? Is the new relationship still proportional? P

8. Prove that if $y = kx$, doubling x doubles y. Does $y = 2x + 3$ have this property? Give a counterexample. What does your answer show about the difference between linear and proportional reasoning? E

4.5 Linear models, prediction and fairness

CRITERION D · I, II, III ◆ CRITERION C · IV ◆ MODELS

A linear model is a deliberately simple description of a relationship. Its value lies in prediction and comparison, but its simplicity also limits what it can claim.

DEFINITIONS

A **mathematical model** uses mathematics to describe a real situation. **Interpolation** predicts within the range of the data or the stated domain; **extrapolation** predicts outside it. The **break-even point** is the value at which two cost models give the same total. A model's **domain** is the set of inputs for which it is intended and meaningful.

To build a linear model, name the variables and units, identify the fixed amount as the intercept and the repeated amount as the gradient, then write $y = mx + c$. In a cost model, the intercept is the amount paid before any use and the gradient is the marginal cost of one more unit. Comparing two models means comparing both numbers: a lower intercept favours light users, while a lower gradient favours heavy users.

WORKED EXAMPLE 4.5.1 COMPARING TWO MEMBERSHIPS

PROBLEM

Gym A charges a 20-unit joining fee plus 15 units per month. Gym B charges 50 units to join plus 10 units per month. Write a cost model for each, compare the first year, and state when each is cheaper.

1. For m months, Gym A costs $A = 20 + 15m$ and Gym B costs $B = 50 + 10m$.
2. At $m = 0$, A costs 20 and B costs 50. At $m = 6$, both cost 110. At $m = 12$, A costs 200 and B costs 170.
3. Gym A is cheaper for fewer than 6 months; Gym B is cheaper for more than 6 months; they are equal at 6 months.
4. The gradients show why: A charges 5 more per month, so its lower joining fee is gradually overtaken by its higher monthly rate.

ANSWER $A = 20 + 15m$, $B = 50 + 10m$; **both cost 110 at 6 months. A is cheaper for** $m < 6$**, B for** $m > 6$**.**

CHECKPOINT

Using the models above, calculate the cost of 3 months and of 12 months under each gym. Which number in each equation causes the ranking to change as increases? Explain in one sentence.

WORKED EXAMPLE 4.5.2 INTERPOLATION, EXTRAPOLATION AND CRITIQUE

PROBLEM

A phone plan costs $C = 12 + 0.06m$ units for m megabytes, where the model is stated to apply for $0 \le m \le 500$. Find the cost at 400 MB, predict the cost at 900 MB using the equation, and evaluate that prediction.

1. At 400 MB: $C = 12 + 0.06(400) = 12 + 24 = 36$ units. This is interpolation because 400 is inside the stated domain.
2. At 900 MB the equation gives $12 + 0.06(900) = 66$ units. This is extrapolation because 900 lies outside the stated domain.
3. The extrapolation may be unreliable: the provider may cap the charge, change the rate, slow the connection or apply a different tariff above 500 MB.
4. A responsible answer reports the calculation but labels it as a conditional prediction, not a fact about the bill.

ANSWER **36 units at 400 MB; the equation gives 66 units at 900 MB, but this extrapolation is outside the model's stated domain.**

STRATEGY · EVALUATING A LINEAR MODEL

Ask four questions. What do the gradient and intercept mean, with units? What domain was measured? Is the prediction interpolation or extrapolation? What real effects — caps, discounts, fixed capacity, measurement error — does the model ignore? A model can be mathematically correct and still be the wrong model for a decision.

PRACTICE 4.5 · BUILDING AND EVALUATING MODELS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Write a linear model: (a) a plumber charges 45 units call-out plus 30 units per hour; (b) a taxi charges 3.50 units plus 2.20 units per kilometre; (c) an internet service costs 18 units per month plus 0.40 units per gigabyte. F
2. For each model in Question 1, state the gradient and the intercept, including their units, and interpret both in one sentence. F
3. Plan A costs $10 + 4m$ units for m months. Plan B costs $25 + 2m$ units. Complete a table for $m = 0, 3, 6, 9, 12$, identify the cheaper plan for each value, and estimate the break-even point. P
4. Sketch the graphs of $C = 15 + 3n$ and $C = 30 + 1.5n$ on the same axes. Estimate the value of n at which they meet and explain what that value means. P
5. A membership costs 120 units per year and includes unlimited use; a pay-as-you-go option costs 3 units per visit. Write both models, give the break-even number of visits, and explain why the membership model has gradient 0. R
6. A study-time model predicts a score $S = 40 + 5h$ for $0 \le h \le 20$, where h is hours of study. Interpret the gradient and intercept, predict the score for 12 hours, and discuss a prediction for 40 hours. R
7. A community internet scheme offers (A) 20 units per month plus 1 unit per GB, or (B) 45 units per month for unlimited data. Which option is more affordable for a low-income household that uses 5 GB per month? Which is better for a household that uses 40 GB? Discuss fairness and state two assumptions behind your advice. P
8. Design a linear pricing model that is cheaper than $C = 30 + 2x$ when $x < 10$ and more expensive when $x > 10$. Explain how the intercept and gradient achieve this, and state one group of users who would benefit. E

Investigation · The candle clock

CRITERION B

Before mechanical clocks, people used candles to measure time. A candle of initial height 18.0 cm is lit and its height is measured every hour. The results are shown in the table.

Time t (hours)	0	1	2	3	4	5
Height h (cm)	18.0	16.4	14.8	13.2	11.6	10.0

1. **Represent.** Plot the points on a graph with time on the horizontal axis. Describe the pattern and calculate the first differences in height.
2. **Measure the rate.** Calculate the gradient using two different pairs of points. Explain why the units are centimetres per hour and why the gradient is negative.
3. **Model.** Find an equation of the form $h = mt + c$. Interpret the intercept in terms of the candle and verify the equation using $t = 3$.
4. **Predict.** Use the model to find the height after 2.5 hours, and the time at which the candle would burn out. State the domain of the model.
5. **Reverse.** Find the time at which the height is 7.2 cm. Show your method and check that the answer lies within the domain.
6. **Critique.** The candle does not burn perfectly evenly. List three physical effects that could make the true graph depart from a straight line, and suggest how the experiment could be improved.
7. **Generalise.** A candle has initial height H cm and burns at a constant r cm per hour. Write a formula for its height after t hours and a formula for its burning time. Explain the meaning of each symbol.
8. **Extend.** A second candle is 24.0 cm tall and burns at 2.4 cm per hour. Use tables or graphs to find when the two candles are the same height. Explain whether your answer is exact or approximate.

Your report should include a labelled graph, a table of first differences, an equation, a domain, and an evaluation of the model's limits. A straight line through the points is evidence for a linear pattern, not proof that the candle will behave linearly forever.

Application · A library of things

CRITERION D · I, II, III, IV ◆ COST MODELS ◆ FAIRNESS

A community tool library lends drills, sewing machines and gardening equipment. It is choosing between two pricing models. **Option A:** a 36-unit annual membership plus 3 units per hire. **Option B:** no membership, but 9 units per hire. Let h be the number of hires in a year and C the annual cost in units.

1. Write a linear equation for each option. Interpret the intercept and gradient of each model, including their units.
2. Complete a table of costs for $h = 0, 2, 4, 6, 8, 10, 12$ for both options.
3. Plot both lines on the same axes. Estimate the number of hires at which the costs are equal, and interpret this point in context.
4. Calculate the cost of 3 hires and of 15 hires under each option. Which option would you recommend in each case? Justify with the numbers.

5. Option A requires money before any tool is borrowed. Discuss how this affects a household on a very low income. Is the option with the lower annual cost for light users necessarily the fairer option?
6. The library proposes a third option: 12 units membership plus 6 units per hire. Write its equation, compare it with both existing options using a table, and explain who benefits.
7. State the assumptions in your models. What costs or behaviours do they ignore? How could the library collect data to test whether the models are realistic?
8. Recommend a pricing structure, with one modification if necessary, that balances access for occasional users with enough income to maintain the tools. Justify your recommendation using your graphs and equations.

> **CRITERION D FOCUS**
>
> Build models from a context, use them to compare options, interpret the gradient and intercept as access and usage costs, and evaluate the assumptions behind a fairness judgement.

THE THREE OPTIONS AT A GLANCE

Option	Equation	Intercept (units)	Gradient (units per hire)	Cheapest for
A · membership	$C_A = 36 + 3h$	36	3	more than 8 hires
B · pay as you borrow	$C_B = 9h$	0	9	fewer than 4 hires
C · proposed	$C_C = 12 + 6h$	12	6	4 to 8 hires

C meets B at $12 + 6h = 9h$, so $h = 4$. C meets A at $12 + 6h = 36 + 3h$, so $h = 8$. Between those values C is the cheapest of the three, although A becomes cheaper again for very frequent users.

Criterion task · Two ways to pay for the school bus

CRITERION D
APPLYING MATHEMATICS
24 MARKS

A school is reviewing its bus payment system. **Option A** is an annual pass costing 240 units and allowing unlimited trips. **Option B** is a registration fee of 40 units plus 2 units for every trip. Let t be the number of trips taken in a year.

1. Write an equation for the annual cost of each option in terms of t. State the units of the gradient and intercept of each. *(3 marks)*
2. Complete a table of costs for $t = 0, 20, 40, 60, 80, 100, 120$. *(4 marks)*
3. Draw both graphs on the same axes, label the intercepts, and describe how the two lines behave as t increases. *(5 marks)*
4. Determine the number of trips for which the two options cost the same. Show how the table or graph supports your answer, and interpret the point in context. *(4 marks)*
5. A student makes 30 trips in a year; another makes 110. Calculate the cost under each option for both students and recommend an option for each. Justify using the equations. *(4 marks)*
6. Evaluate the fairness of the two options. Consider students who travel rarely, families with several children, and students who cannot pay a large sum at the start of the year. State the assumptions in the models and propose one modification to the pricing system. *(4 marks)*

PLANNING FRAME

Represent: define the variable and write both equations. **Organise:** make a table and choose sensible axes. **Compare:** look for the intersection and interpret it. **Apply:** calculate for the two students. **Evaluate:** state assumptions and connect the model to fairness.

BAND	DESCRIPTOR	MARKS
4	Accurate equations, table and graphs; intersection correctly identified and interpreted; recommendations well justified; assumptions and fairness evaluated with insight.	19–24
3	Correct models and mostly accurate graph; intersection found; recommendations supported; some evaluation of assumptions.	13–18
2	Models or table partly correct; graph readable but errors in interpretation; limited justification or evaluation.	7–12
1	Isolated correct calculations; little connection between the models, graph and context.	1–6

CRITERIA: D (C) **ATL:** COMMUNICATION **TIME:** 45 MINUTES **CONDITIONS:** CALCULATOR ALLOWED

MAPPED TO: D strands i–v and C strands i–iv. The rubric is a course summary, not an official IB descriptor.

Chapter summary and review

KEY IDEAS

- An ordered pair (x, y) locates a point; a linear relationship has a constant rate of change.
- A constant first difference in a table is evidence of a linear pattern.
- Gradient $m = \frac{y_2 - y_1}{x_2 - x_1}$ measures change in y per unit change in x.
- A horizontal line has gradient 0; a vertical line has undefined gradient.
- In $y = mx + c$, m is the gradient and c is the y-intercept.
- Parallel lines have equal gradients; perpendicular gradients multiply to -1.
- Direct proportion means $y = kx$: a straight line through the origin with a constant quotient.
- Linear models can compare costs, but extrapolation and unstated assumptions limit their authority.

REVISION CHECKLIST

- ☐ I can construct a table and plot a linear graph accurately.
- ☐ I can calculate a gradient from two points, a table or a graph.
- ☐ I can interpret gradient and intercept with units.
- ☐ I can find the equation of a line from two points or a graph.
- ☐ I can distinguish direct proportion from a general linear relationship.
- ☐ I can use a model to interpolate and extrapolate, and label which is which.
- ☐ I can compare two pricing models and evaluate their fairness.

VOCABULARY

Cartesian plane
A grid formed by two perpendicular number axes.

origin
The point $(0, 0)$.

independent variable
The input, plotted on the horizontal axis.

dependent variable
The output, plotted on the vertical axis.

linear relationship
A relationship with a constant rate of change.

gradient
Change in y per unit change in x.

rate of change
A gradient interpreted with units in context.

y-intercept
The value of y when $x = 0$.

direct proportion
A relationship of the form $y = kx$.

constant of proportionality
The constant k in $y = kx$.

interpolation
Prediction inside the known range.

extrapolation
Prediction outside the known range.

break-even point
The value at which two cost models are equal.

model
A simplified mathematical description of a real situation.

MIXED PRACTICE · CHAPTER 4

Connecting the ideas

FLUENCY Core procedures

1. State the gradient and y-intercept of $y = 5 - 2x$. F
2. Find the gradient of the line through $(-3, 2)$ and $(5, 10)$. F
3. Find the equation of the line with gradient 3 that passes through $(0, -4)$. F
4. Find the equation of the line through $(2, 1)$ and $(6, 9)$. F
5. The table shows a direct proportion. Find k and the missing value. F

x	2	5	8
y	7	17.5	—

6. A hire company charges $C = 25 + 0.4n$. State the gradient and intercept with units, and find the cost for 60 units of use. F

REASONING Justify and connect

1. Explain why $y = 4x + 1$ is not a direct proportion. Use a table and the ratio $\frac{y}{x}$. R
2. The lines $y = 3x - 2$ and $y = 3x + 5$ are parallel. Will they ever meet? Explain using gradients and intercepts. R
3. A student claims the gradient of the line through $(1, 2)$ and $(4, 8)$ is 6 because $8 - 2 = 6$. Identify the missing step and give the correct gradient. R
4. A distance–time graph is horizontal for 10 minutes. Describe the motion and state the gradient during that interval. Does a zero gradient mean the object has disappeared? Explain. R

APPLICATION Use and evaluate

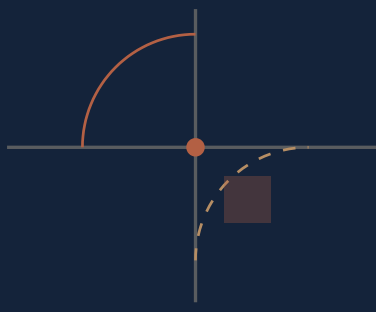
1. A phone plan costs $C = 15 + 0.08m$ units for m megabytes. Find the cost for 250 MB and the usage when the cost is 31 units. Interpret the gradient. P
2. Plan X costs $12 + 3n$; Plan Y costs $24 + 1.5n$. Compare the plans at $n = 4$, $n = 8$ and $n = 12$, and state when each is cheaper. P
3. A candle has height $h = 20 - 2.5t$ cm after t hours. State its initial height and burning rate, find when it is 5 cm tall, and state the domain. P
4. A shop's weekly sales are modelled by $S = 800 + 40w$ for the first 6 weeks. Predict sales at week 30 and evaluate the reliability of that prediction. What extra information would you request? E

Check your progress

I CAN …	CONFIDENCE	EVIDENCE	NEXT ACTION
Plot points and recognise a linear pattern in a table.	1 / 2 / 3		
Calculate and interpret gradient, including units.	1 / 2 / 3		
Find the equation of a line from a graph, table or two points.	1 / 2 / 3		
Recognise and use direct proportion.	1 / 2 / 3		
Build, compare and critique a linear model.	1 / 2 / 3		

Reflection

Two pricing plans can have the same break-even cost but very different effects on poor and wealthy users. Which representation — table, graph or equation — best communicates that difference to a decision-maker, and why?



5

BALANCE, CONSTRAINT AND DECISION

Equations, inequalities and systems

An equation asks when two expressions balance. An inequality asks which values keep one side larger. A system asks when several conditions must hold at once — the mathematics behind sharing limited land, water and time.

KEY CONCEPT · RELATIONSHIPS

RELATED · EQUIVALENCE

RELATED · SYSTEMS

CONTEXT · GLOBALIZATION AND SUSTAINABILITY

INQUIRY

How can equations and inequalities turn competing demands into a plan that is both feasible and fair?

ATL · SELF-MANAGEMENT IN MULTI-STEP WORK

Long solutions fail for small reasons: a lost negative sign, a fraction divided on one side only, or a check that was never performed. Use a written plan: simplify, isolate, solve, substitute, interpret. Finish every problem by checking the original condition, not your own rearranged version. A solution that has not been checked is only a candidate.

MYP UNIT PLANNER

Framing the inquiry

UNIT 5 · APPROX. 16 HOURS
ATL FOCUS · SELF-MANAGEMENT

KEY AND RELATED CONCEPTS

Relationships. An equation or inequality is a condition on a variable. Solving it means finding every value that makes the condition true, not merely producing a number that looks plausible.

Equivalence. Each step in a solution must preserve the solution set. Adding, subtracting, multiplying or dividing both sides by the same non-zero number produces an equivalent equation; multiplying or dividing an inequality by a negative number calls for special care.

Systems. Two conditions at once restrict the possible values more than either condition alone. Geometrically, a system of two linear equations asks where two lines meet; a system of inequalities describes a region of possible allocations.

GLOBAL CONTEXT AND EXPLORATION

Globalization and sustainability. Exploration: how communities allocate limited land, water, energy and labour among competing uses, and how mathematical constraints make the trade-offs visible.

Sustainability is not only an environmental idea; it is a problem of allocation. A garden, a clinic or a workshop has finite resources. Writing the limits as inequalities turns a debate about priorities into a set of feasible plans that can be compared, tested and questioned.

STATEMENT OF INQUIRY

Equations, inequalities and systems express conditions that must hold together; solving them reveals which allocations are possible, while comparing the feasible options makes trade-offs — and questions of fairness — visible.

FACTUAL

How do you solve a linear equation, a linear inequality and a pair of simultaneous equations? What is a feasible region?

CONCEPTUAL

Why does multiplying an inequality by a negative number reverse its direction, and why does the intersection of two graphs solve a system?

DEBATABLE

Should a mathematical optimum decide how a community shares limited resources?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii	Solve linear equations, inequalities and systems; evaluate an objective at the vertices of a feasible region.
B · Investigating patterns	i, ii, iii	Investigate when a system has one, none or infinitely many solutions using balance puzzles and graphs.
C · Communicating	i, ii, iii, iv	Present solution sets on number lines, explain each equivalent step and interpret a feasible region.
D · Applying mathematics	i, ii, iii, iv, v	Model resource constraints, optimise an allocation and evaluate the fairness of the recommendation.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** what it means for two equations to be equivalent and why each solving step must preserve the solution set.
- **Know** how to solve linear equations with brackets, fractions and variables on both sides.
- **Be able to** solve linear inequalities and represent their solution sets on a number line.
- **Be able to** solve simultaneous linear equations by graphing, substitution and elimination.
- **Be able to** model constraints with a system of inequalities and evaluate an objective over a feasible region.
- **Be able to** interpret and critique a resource allocation for feasibility and fairness.

SUCCESS CRITERIA

- [ ] I can solve a multi-step equation and verify the solution in the original equation.
- [ ] I can explain why an equation has one, no or infinitely many solutions.
- [ ] I can solve an inequality, including one that requires reversing the sign.
- [ ] I can solve a system by substitution or elimination and check both equations.
- [ ] I can graph a system of inequalities and identify its feasible region.
- [ ] I can evaluate an objective at the vertices and justify an allocation with reference to the constraints.

PREREQUISITE CHECK · 12 MINUTES, NO CALCULATOR

1. Solve $2x + 5 = 17$.
2. Expand $3(x - 2)$.
3. Solve $5 - 2x = 1$.
4. Plot $y = 2x - 1$ for $x = 0, 1, 2$.
5. Evaluate $3x + 2y$ when $x = 4$ and $y = 1$.
6. Write the next two terms: 7, 11, 15, ...

Answers are in the Chapter 5 answer section. If more than two are incorrect, review inverse operations, substitution and plotting before continuing.

BEFORE YEAR 3

One- and two-step equations; simple inequalities; plotting linear graphs; substitution; coordinates.

YEAR 3 · THIS CHAPTER

Multi-step equations; inequalities and solution sets; simultaneous equations; systems of inequalities and feasible regions.

LATER

Quadratic systems; linear programming in several variables; matrices; algebraic proof of solution-set equivalence.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**APPLICATION**	**REFLECTION**
A community must divide limited land, water and labour.	Which allocations satisfy every constraint at once?	Equivalence, balance, systems and feasible regions.	Use balance puzzles to discover when a system has one, none or many solutions.	Solve a real two-variable problem and examine the fairness of the result.	Who decides what counts as the best allocation?

5.1 Solving linear equations

CRITERION A · I ◆ EQUIVALENCE ◆ REASONING

An equation is a balance. Whatever is done to one side must be done to the other, and the goal is to isolate the variable without changing the solution set.

DEFINITIONS

A **linear equation** in one variable can be written in the form $ax + b = c$ with $a \neq 0$. A **solution** is a value of the variable that makes the equation true. Two equations are **equivalent** if they have exactly the same solutions. An **identity** is true for every permitted value; a **contradiction** is true for none.

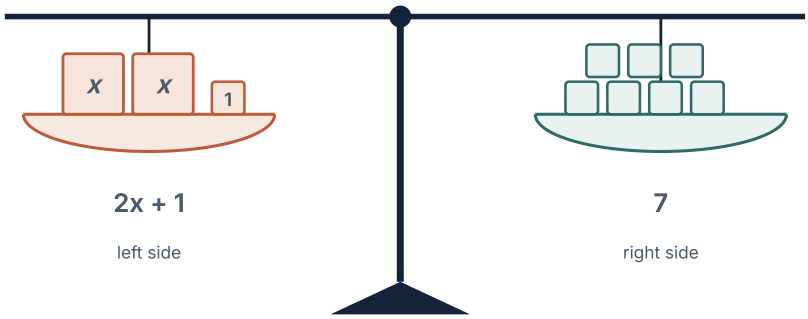


Figure 5.1 The balance model of $2x + 1 = 7$. Removing the same amount from both pans, or dividing both sides into the same number of equal groups, keeps the balance.

To solve $2x + 1 = 7$, subtract 1 from both sides: $2x = 6$. Divide both sides by 2: $x = 3$. Substitution checks the result: $2(3) + 1 = 7$. Each step is reversible, so no solution is gained or lost. For longer equations, first remove brackets, then collect variable terms on one side and constants on the other.

When fractions appear, multiply every term by a common denominator. This clears the fractions without changing the solution set, provided the denominator is not zero. A useful habit is to write the common denominator explicitly before multiplying.

WORKED EXAMPLE 5.1.1 — BRACKETS AND VARIABLES ON BOTH SIDES

PROBLEM

Solve $3(x - 4) = 2x + 5$.

1. Expand the left side: $3x - 12 = 2x + 5$.
2. Subtract $2x$ from both sides: $x - 12 = 5$.
3. Add 12 to both sides: $x = 17$.
4. Check in the original equation: $3(17 - 4) = 3(13) = 39$ and $2(17) + 5 = 39$.

ANSWER $x = 17$

WORKED EXAMPLE 5.1.2 CLEARING FRACTIONS

PROBLEM

Solve $\frac{2x-1}{3} = \frac{x+4}{2}$.

1. Multiply both sides by 6, the lowest common multiple of 3 and 2: $2(2x-1) = 3(x+4)$.
2. Expand: $4x - 2 = 3x + 12$.
3. Subtract $3x$: $x - 2 = 12$. Add 2: $x = 14$.
4. Check: the left side is $\frac{28-1}{3} = 9$; the right side is $\frac{14+4}{2} = 9$.

ANSWER $x = 14$

WORKED EXAMPLE 5.1.3 NO SOLUTION OR EVERY SOLUTION

PROBLEM

Solve (a) $2(x+3) = 2x + 6$ and (b) $2(x+3) = 2x + 1$.

1. (a) Expanding gives $2x + 6 = 2x + 6$, which is true for every value of x. The equation is an identity; the solution set is all real numbers.
2. (b) Expanding gives $2x + 6 = 2x + 1$. Subtracting $2x$ leaves $6 = 1$, which is false. There is no solution.
3. The two cases look similar before simplification. The difference is whether the variable terms cancel to a true statement or a false one.

ANSWER **(a) infinitely many solutions; (b) no solution.**

COMMON ERROR · DISTRIBUTING A NEGATIVE SIGN

$5 - 2(x-1)$ is $5 - 2x + 2$, not $5 - 2x - 2$. The negative sign multiplies both terms inside the bracket. Write the expansion in a separate line before collecting terms. Most "mysterious" wrong answers in equation solving come from a lost sign, not from the main method.

PRACTICE 5.1 · LINEAR EQUATIONS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Solve: (a) $4x - 7 = 25$ (b) $5 - 2x = 17$ (c) $3(x+2) = 21$. F
2. Solve: (a) $2x + 5 = 4x - 9$ (b) $5(x-1) = 3x + 7$ (c) $4 - 3(x+1) = 10$. F
3. Solve $\frac{2x+1}{3} = \frac{x-2}{4}$. F
4. Solve $0.4x + 1.2 = 2.8$. F
5. Explain why $3(x+2) = 3x + 6$ has infinitely many solutions, while $3(x+2) = 3x + 1$ has none. R
6. A student solves $5 - 2(x-1) = 9$ and writes $5 - 2x - 2 = 9$. Identify the error, complete the solution correctly, and check it. R
7. The perimeter of a rectangle is 38 cm. Its length is 3 cm more than its width. Write and solve an equation to find the dimensions. P
8. Write an equation with solution $x = 4$ that has variables on both sides, at least one bracket and a negative coefficient. Exchange with a partner and solve each other's equation. E

5.2 Inequalities and solution sets

CRITERION A · I ◆ CRITERION C · II ◆ REPRESENTATION

An inequality describes a range of possibilities rather than a single value. Its solution is a set, and a number line is often the clearest way to see that set.

DEFINITIONS

An **inequality** compares two expressions using $<$, $>$, $\leq$ or $\geq$. Its **solution set** is the set of all values that make it true. A filled circle on a number line includes the endpoint ($\leq$ or $\geq$); an open circle excludes it ($<$ or $>$). The solution can also be written in **interval notation**: for example, $x \leq 3$ is $(-\infty, 3]$.

Solving an inequality uses the same inverse operations as solving an equation, with one extra rule. Adding or subtracting the same number preserves the direction of the inequality. Multiplying or dividing by a positive number also preserves it. But multiplying or dividing by a negative number **reverses** the direction: if $2 < 5$, then $-2 > -5$.

WORKED EXAMPLE 5.2.1 — A BASIC INEQUALITY

PROBLEM

Solve $2x + 1 \leq 7$ and represent the solution on a number line.

1. Subtract 1: $2x \leq 6$.
2. Divide by 2, a positive number, so the direction stays the same: $x \leq 3$.
3. Shade the number line to the left of 3 and use a filled circle at 3 because equality is allowed.

ANSWER $x \leq 3$, **or** $(-\infty, 3]$.

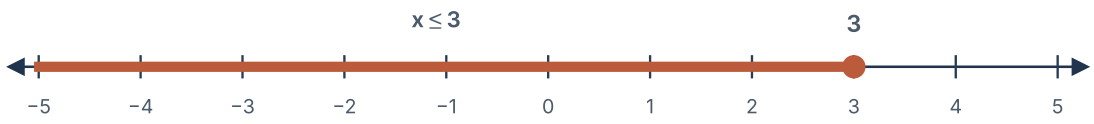


Figure 5.2 The solution set of $2x + 1 \leq 7$. The endpoint 3 is included, so the circle is filled; every number to its left satisfies the inequality.

WORKED EXAMPLE 5.2.2 REVERSING THE SIGN

PROBLEM

Solve $-2x + 7 \geq 1$.

1. Subtract 7 from both sides: $-2x \geq -6$.
2. Divide both sides by -2. Because -2 is negative, reverse the inequality: $x \leq 3$.
3. Check with a value: $x = 0$ gives $7 \geq 1$, true. The excluded value $x = 4$ gives $-1 \geq 1$, false, as expected.

ANSWER $x \leq 3$

WORKED EXAMPLE 5.2.3 A COMPOUND INEQUALITY

PROBLEM

Solve $-3 < 2x + 1 \leq 7$.

1. Subtract 1 from all three parts: $-4 < 2x \leq 6$.
2. Divide all three parts by 2: $-2 < x \leq 3$.
3. The solution is the interval $(-2, 3]$: open at -2, closed at 3.

ANSWER $-2 < x \leq 3$, **or** $(-2, 3]$.

COMMON ERROR · FORGETTING TO REVERSE

From $-3x < 12$, dividing by -3 gives $x > -4$, not $x < -4$. Test a number on each side if you are unsure: $x = 0$ makes $-3(0) = 0 < 12$ true, so 0 must be in the solution set; the inequality $x > -4$ includes 0, while $x < -4$ does not.

PRACTICE 5.2 · INEQUALITIES

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Solve and graph: (a) $x + 4 < 9$ (b) $3x \geq 12$ (c) $5 - x > 2$ (d) $-2x \leq 8$. F
2. Solve: (a) $4x - 3 > 2x + 7$ (b) $2(x - 1) \leq 5x + 4$ (c) $7 - 3x < 1$. F
3. Solve the compound inequalities: (a) $-1 < 3x + 2 < 11$ (b) $4 \leq 2x - 2 \leq 10$. F
4. Explain why $x < 5$ and $5 > x$ describe the same solution set. Why does multiplying by a negative number reverse an inequality? R
5. A student solves $-3x < 12$ and writes $x < -4$. Identify the error, give the correct solution, and test one value to justify your correction. R
6. A club has a budget of 40 units. It spends 12 units on a room and $2n$ units on materials for n people. Write and solve an inequality for the maximum number of people who can attend. P
7. A drone is at height $h = 1.8 + 0.3t$ metres after t minutes. Write and solve an inequality for the times when the drone is above 6 m. Interpret the result, including whether t can be negative. P
8. Write an inequality whose solution is $-2 \leq x < 5$. Then write an inequality with no solution, and explain why it has none. E

5.3 Simultaneous equations: graphs and substitution

CRITERION A · II ◆ CRITERION C · III ◆ SYSTEMS

A single equation with two variables has infinitely many solutions. Two equations together can pin down a single pair — if the conditions are independent and consistent.

DEFINITION

A **system of simultaneous equations** is a set of equations that must all be true at the same time. A **solution** of the system is an ordered pair (x, y) that satisfies every equation. On a graph, each linear equation is a line; a solution is a point where the lines intersect.

The graphical method makes the meaning visible: plot both lines and read the intersection. It is approximate unless the intersection lies exactly on grid lines. The substitution method is exact: solve one equation for one variable, substitute that expression into the other equation, and solve the resulting one-variable equation.

WORKED EXAMPLE 5.3.1 SOLVING GRAPHICALLY

PROBLEM

Solve the system $y = 2x + 1$ and $y = -x + 4$ by graphing.

1. Plot $y = 2x + 1$: intercept 1, gradient 2.
2. Plot $y = -x + 4$: intercept 4, gradient -1.
3. The lines intersect at $(1, 3)$.
4. Check both equations: $2(1) + 1 = 3$ and $-1 + 4 = 3$.

ANSWER $(1, 3)$

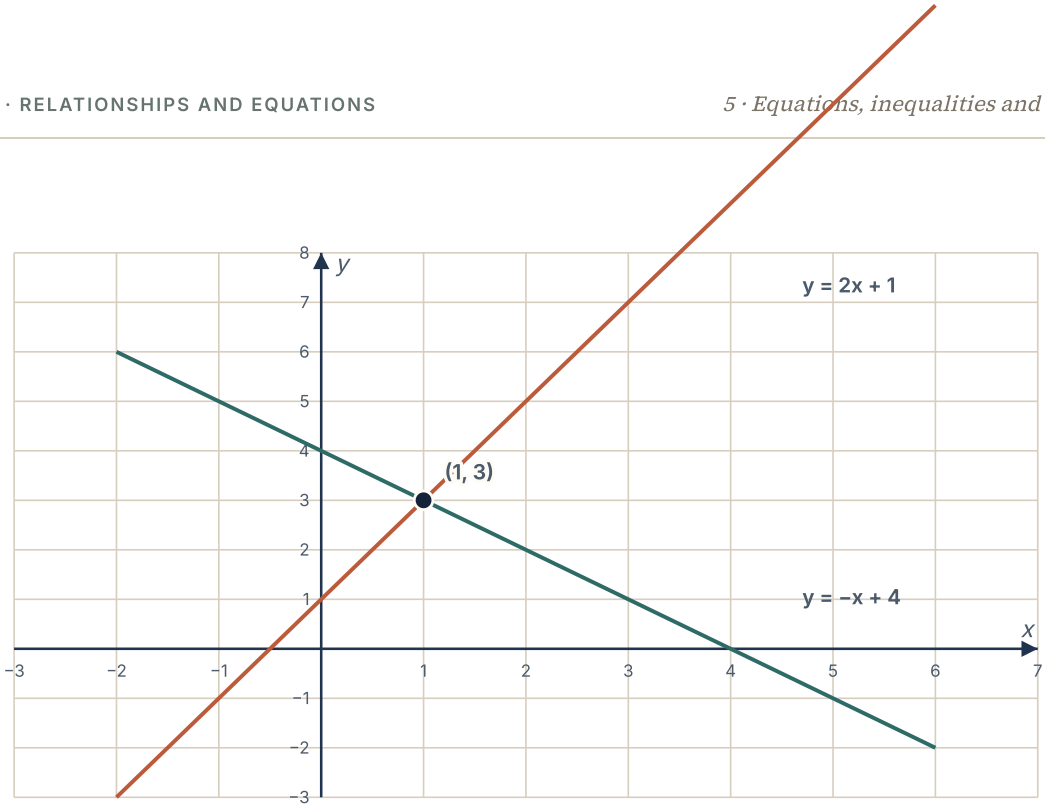


Figure 5.3 The graphs of $y = 2x + 1$ and $y = -x + 4$. Their intersection $(1, 3)$ is the only pair that satisfies both equations.

WORKED EXAMPLE 5.3.2 — SUBSTITUTION WITH A READY EXPRESSION

PROBLEM

Solve $y = 3x - 2$ and $2x + y = 8$.

1. The first equation already gives y in terms of x. Substitute $3x - 2$ for y in the second: $2x + (3x - 2) = 8$.
2. Simplify: $5x - 2 = 8$, so $5x = 10$ and $x = 2$.
3. Substitute $x = 2$ into $y = 3x - 2$: $y = 3(2) - 2 = 4$.
4. Check in the second equation: $2(2) + 4 = 8$.

ANSWER $(2, 4)$

WORKED EXAMPLE 5.3.3 — REARRANGING BEFORE SUBSTITUTION

PROBLEM

Solve $x + 2y = 10$ and $3x - y = 9$.

1. Rearrange the first equation for x: $x = 10 - 2y$. (Rearranging the second for y would also work.)
2. Substitute into the second: $3(10 - 2y) - y = 9$.
3. Expand and simplify: $30 - 6y - y = 9$, so $30 - 7y = 9$ and $-7y = -21$. Thus $y = 3$.
4. Then $x = 10 - 2(3) = 4$. Check: $3(4) - 3 = 9$.

ANSWER $(4, 3)$

COMMON ERROR · STOPPING AT ONE VARIABLE

A solution of a two-variable system is an ordered pair, not a single number. After finding x, substitute it to find y, then check both original equations. Substituting back into the equation you just rearranged is a weak check: it can conceal an error made during the rearrangement. Use the other original equation.

PRACTICE 5.3 · GRAPHICAL AND SUBSTITUTION METHODS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Solve graphically: (a) $y = x + 1$ and $y = -x + 5$ (b) $y = 2x - 3$ and $y = -x + 3$. Use a coordinate grid and check your answers. F
2. Solve by substitution: (a) $y = x + 4$, $3x + y = 16$ (b) $y = 2x - 1$, $x + 3y = 11$ (c) $x + y = 7$, $y = 2x - 5$. F
3. Solve: (a) $2x + y = 9$, $x - y = 3$ (b) $3x + 2y = 16$, $y = x + 3$. F
4. Explain what the intersection of two lines represents in a system of equations? What does it mean if the lines are parallel? R
5. A student solves $y = 2x + 1$ and $3x + y = 11$ by writing $3x + 2x + 1 = 11$, then stops with $x = 2$. What is the solution of the system? What should the student do next? R
6. Two adult tickets and three child tickets cost 46 units. One adult ticket and two child tickets cost 27 units. Write a system, solve it, and state the price of each ticket. P
7. For each system, choose substitution or graphing and explain your choice: (a) $y = 4x - 1$, $2x + y = 11$ (b) $x + y = 5$, $x - y = 1$. Then solve. P
8. Write a system of two linear equations whose solution is $(3, -2)$. Explain how you constructed it. Can you write a second system with the same solution? How many systems have that solution? E

5.4 Elimination and special systems

CRITERION A · II ◆ CRITERION C · II ◆ EQUIVALENCE

Elimination removes a variable by adding or subtracting equivalent multiples of the equations. It is often faster than substitution when both equations are in the form $ax + by = c$.

KEY IDEA · ELIMINATION

If two equations are true, their sum and difference are also true. Multiply one or both equations by a constant so that the coefficients of one variable are opposites, then add to eliminate that variable. Alternatively, make the coefficients equal and subtract.

WORKED EXAMPLE 5.4.1 ADDING TO ELIMINATE

PROBLEM

Solve $3x + 2y = 16$ and $5x - 2y = 16$.

1. The coefficients of y are opposites, so add the equations: $(3x + 5x) + (2y - 2y) = 16 + 16$.
2. This gives $8x = 32$, so $x = 4$.
3. Substitute into the first equation: $3(4) + 2y = 16$, so $12 + 2y = 16$ and $y = 2$.
4. Check in the second equation: $5(4) - 2(2) = 20 - 4 = 16$.

ANSWER $(4, 2)$

WORKED EXAMPLE 5.4.2 SCALING ONE EQUATION

PROBLEM

Solve $2x + 3y = 13$ and $x + 2y = 8$.

1. Multiply the second equation by 2: $2x + 4y = 16$.
2. Subtract the first equation from this result: $(2x + 4y) - (2x + 3y) = 16 - 13$, so $y = 3$.
3. Substitute into $x + 2y = 8$: $x + 6 = 8$, so $x = 2$.
4. Check in the first equation: $2(2) + 3(3) = 4 + 9 = 13$.

ANSWER $(2, 3)$

WORKED EXAMPLE 5.4.3 SCALING BOTH EQUATIONS

PROBLEM

Solve $2x + 3y = 18$ and $4x - y = 8$.

1. Multiply the second equation by 3: $12x - 3y = 24$.
2. Add the first equation to eliminate y: $14x = 42$, so $x = 3$.
3. Substitute into $4x - y = 8$: $12 - y = 8$, so $y = 4$.
4. Check in the first equation: $2(3) + 3(4) = 6 + 12 = 18$.

ANSWER $(3, 4)$

STRATEGY · HOW MANY SOLUTIONS?

Write both equations in the form $y = mx + c$. If the gradients differ, the lines intersect once and there is exactly one solution. If the gradients are equal but the intercepts differ, the lines are parallel and there is no solution. If both gradient and intercept agree, the equations describe the same line and there are infinitely many solutions. The same conclusions follow algebraically when a variable cancels to a false statement (no solution) or a true identity (infinitely many).

PRACTICE 5.4 · ELIMINATION AND SPECIAL SYSTEMS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Solve by elimination: (a) $x + y = 10$, $x - y = 4$ (b) $2x + y = 11$, $3x - y = 4$ (c) $4x + 3y = 25$, $2x - 3y = -1$. F
2. Solve: (a) $3x + 2y = 16$, $x + 2y = 8$ (b) $5x - 2y = 19$, $3x + 4y = 1$. F
3. Solve: $3x + 2y = 16$ and $x + 2y = 8$ by elimination, then check both equations. F

4. Explain how the graphs of $2x + y = 6$ and $4x + 2y = 12$ are related. How many solutions does the system have? R
5. Explain why the system $2x + y = 6$, $4x + 2y = 10$ has no solution. What do the graphs look like? R
6. Three adult tickets and two child tickets cost 74 units. Two adult and three child tickets cost 66 units. Write and solve a system to find each price. P
7. Solve by a method of your choice and justify the choice: (a) $y = 2x - 3$, $3x + 2y = 8$ (b) $4x + 3y = 18$, $2x - 3y = 6$. P
8. Create a system with no solution and a system with infinitely many solutions. Explain, using gradients and intercepts, why each has the number of solutions it does. E

5.5 Systems of inequalities and resource allocation

CRITERION D · I, II, III ◆ CRITERION C · IV ◆ MODELS

Real decisions rarely involve a single equation. Several limits — money, space, time, labour — must be satisfied at once. Their solution is not one point but a region of feasible possibilities.

DEFINITIONS

A **system of inequalities** is a set of inequalities in the same variables. Its **feasible region** is the set of points satisfying all of them. A **constraint** is a restriction expressed as an inequality or equation. An **objective function** is the quantity to be maximised or minimised, such as profit, participants or food produced.

To graph a system, draw the boundary line of each inequality. Use a solid line for $\leq$ or $\geq$ and a dashed line for $<$ or $>$. Test a point not on the line to decide which side to shade, then keep the overlap of all the shaded regions. For integer problems, only lattice points inside the region are feasible.

WORKED EXAMPLE 5.5.1 A COMMUNITY GARDEN

PROBLEM

A community garden can plant x vegetable plots and y herb plots. It has at most 7 plots, 12 units of water and 12 hours of volunteer labour. Vegetables need 2 water units and 1 labour hour per plot; herbs need 1 water unit and 2 labour hours. Write the constraints, graph the feasible region, and maximise the food score $P = 5x + 4y$.

1. Constraints: $x + y \leq 7$, $2x + y \leq 12$, $x + 2y \leq 12$, with $x \geq 0$ and $y \geq 0$. For whole plots, x and y are integers.
2. The boundary lines meet at $(6, 0)$, $(5, 2)$, $(2, 5)$ and $(0, 6)$, with the origin as a corner. The feasible region is the shaded polygon in Figure 5.4.
3. Evaluate $P = 5x + 4y$ at each vertex: $(0, 0) \to 0$, $(6, 0) \to 30$, $(5, 2) \to 33$, $(2, 5) \to 30$, $(0, 6) \to 24$.
4. The maximum is 33 at $(5, 2)$: 5 vegetable plots and 2 herb plots.

ANSWER **Maximum $P = 33$ at 5 vegetable plots and 2 herb plots.**

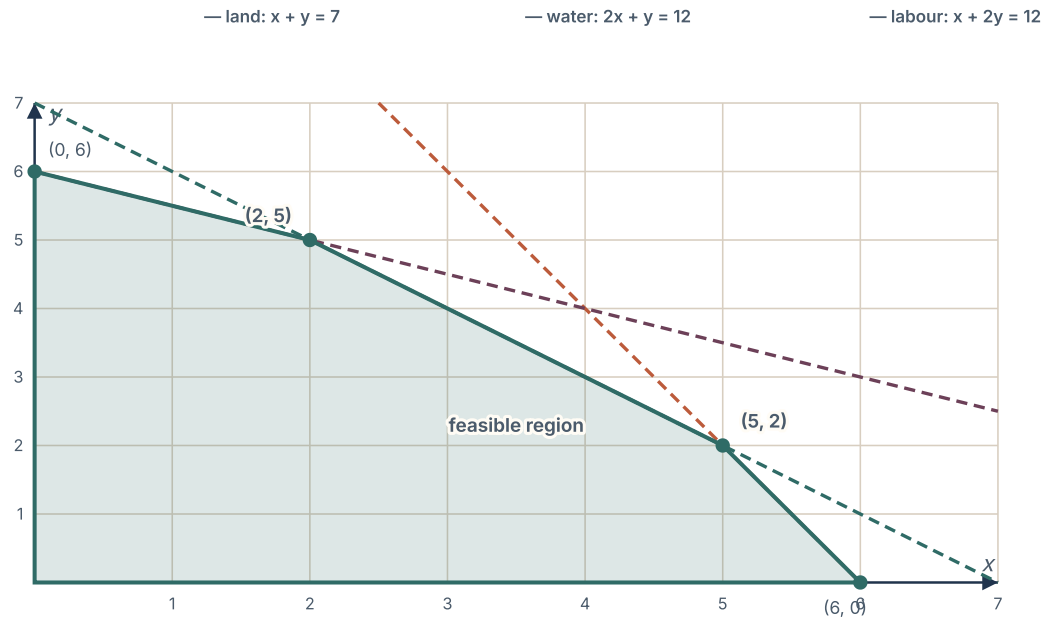


Figure 5.4 The feasible region for the garden. Every lattice point inside the polygon is a possible allocation; the vertices are the candidates for an optimum of a linear objective.

WORKED EXAMPLE 5.5.2 A DIFFERENT OBJECTIVE, A DIFFERENT CHOICE

PROBLEM

Using the same feasible region, maximise $Q = 4x + 5y$. Compare the result with $P = 5x + 4y$ and interpret the difference.

1. Evaluate Q at the vertices: $(0,0) \to 0$, $(6,0) \to 24$, $(5,2) \to 30$, $(2,5) \to 33$, $(0,6) \to 30$.
2. The maximum is 33 at $(2,5)$: 2 vegetable plots and 5 herb plots.
3. The feasible region is unchanged; only the objective has changed. P values vegetables more, so it chooses more vegetable plots; Q values herbs more, so it chooses more herb plots.
4. Both allocations use all 7 plots but distribute water and labour differently. A community must decide which objective reflects its priorities, not merely which number is larger.

ANSWER **Maximum $Q = 33$ at $(2,5)$. The optimum depends on the objective, not only on the constraints.**

STRATEGY · SOLVING A LINEAR ALLOCATION PROBLEM

1. Name the variables and units. 2. Write every constraint, including non-negativity. 3. Graph the feasible region and find its vertices. 4. Evaluate the objective at each vertex. 5. Check integer conditions and interpret the answer in context. 6. Ask who benefits from the chosen objective and what the model ignores.

PRACTICE 5.5 · FEASIBLE REGIONS AND OBJECTIVES

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Graph $x + y \le 6$, $x \ge 0$ and $y \ge 0$. Identify the vertices of the feasible region. F
2. Graph $y \ge 2x - 1$ and $y < 4$. Test the points $(2,3)$, $(0,0)$ and $(3,8)$ and state whether each is feasible. F
3. A feasible region has vertices $(0,0)$, $(4,0)$, $(3,2)$ and $(0,3)$. Evaluate $P = 3x + 2y$ at each vertex and state the maximum and minimum. F

4. A club sells x adult tickets and y child tickets. It can sell at most 120 tickets, must sell at least 30 child tickets, and cannot sell a negative number of either. Write the constraints and describe the feasible region. P
5. A garden has at most 8 plots and 18 units of water. Vegetables use 3 units per plot and herbs use 1 unit per plot. Write the inequalities, graph the region, and list two feasible integer allocations. P
6. Using the region from Question 3, evaluate $Q = 6x + 5y$ at the vertices. Which vertex is optimal, and does the answer change from Question 3? Explain. P
7. A student shades the side containing $(0, 0)$ for $x + y \geq 5$. Test $(0, 0)$ and explain the error. What is the correct region? R
8. A community requires $x \geq 10$ and $y \geq 10$ but has only 15 plots in total. Show that there is no feasible allocation. Propose one change to the constraints that would create a feasible region, and describe the trade-off. E

Investigation · The three weighings

CRITERION B

In a balance puzzle, identical circles have mass c grams and identical squares have mass s grams. The first two weighings give:

2 circles + 1 square = 14 g 1 circle + 2 squares = 13 g

1. **Explore.** Make a table of possible whole-number values of c and s that satisfy the first weighing. Which pair also satisfies the second? Verify both.
2. **Model.** Write the two weighings as equations and solve the system by substitution or elimination. Compare the algebraic solution with your table.
3. **Interpret.** What does each equation represent geometrically? Sketch both lines and mark the solution.
4. **Add information.** A third weighing gives $3c + 3s = 27$. Does this add new information? Explain using the first two equations.
5. **Break it.** Suppose the third weighing had been $3c + 3s = 28$. What would that tell you about the measurements? Does a solution exist?
6. **Generalise.** When does a system of two linear equations have one solution, no solution or infinitely many solutions? Give a condition in terms of gradients and intercepts.
7. **Create.** Design a system of two equations whose solution is $c = 3$, $s = 2$. Then design a system with no solution. Exchange with a partner and explain how you know each answer.

A balance puzzle is evidence, not proof, about the real masses: the scales have limited precision and the objects are assumed identical. State those assumptions in your report.

Application · The water refill station

CRITERION D · I, II, III, IV ◆ SYSTEMS ◆ SUSTAINABILITY

A community water refill station sells two sizes of container: 5 litres and 12 litres. On Saturday it sold 80 refills and dispensed 680 litres in total. A 5-litre refill costs 2 units; a 12-litre refill costs 5 units.

1. Define variables for the number of each size sold. Write a system of equations for the number of refills and the total volume.
2. Solve the system by elimination or substitution. Show your working.
3. Verify your solution in both original equations, then calculate the total revenue collected.

4. Find the average revenue per litre, to two decimal places. Does the larger container offer better value per litre? Explain.
5. The station raises the price of a 5-litre refill to 3 units but keeps the 12-litre price at 5 units. Recalculate the revenue for the same sales. Who is affected most by the change?
6. The station can dispense at most 700 litres in a day. If demand rises by 10%, can it meet demand from the same 80 refills? What if the mix of container sizes changes?
7. Each 5-litre refill is estimated to replace about three 1.5-litre plastic bottles. Estimate the number of bottles avoided on Saturday. State the assumption behind your estimate.
8. Evaluate the model. What does it ignore about leakage, free refills, container reuse or changes in price? How could the station collect better data?

CRITERION D FOCUS

Translate a two-variable situation into a system, solve it exactly, interpret the answer with units, and evaluate the assumptions behind a sustainability claim.

Criterion task · Planning a youth centre programme

CRITERIA A AND D
KNOWING AND APPLYING
24 MARKS

A youth centre plans x art workshops and y music sessions each month. Each art workshop uses 1 venue slot, 2 tutor hours and 1 equipment unit. Each music session uses 1 venue slot, 1 tutor hour and 2 equipment units. The centre has at most 8 venue slots, 16 tutor hours and 14 equipment units. The programme aims to maximise a participation score $P = 8x + 9y$.

1. Write a system of inequalities for the three constraints, including $x \geq 0$ and $y \geq 0$. State what each variable and each inequality represents. *(5 marks)*
2. Graph the feasible region. Find and label its vertices, and explain why the region is bounded. *(5 marks)*
3. Test the points $(4, 4)$ and $(6, 3)$ against every constraint. Which is feasible? Explain how the test supports your graph. *(4 marks)*
4. Evaluate the participation score at each vertex. State the optimal integer allocation and justify it. *(4 marks)*
5. The centre notices that the optimum provides only 2 art workshops, while demand for art is high. Discuss whether the mathematical optimum is the fairest programme. Propose one change to a constraint or to the objective that would reflect this concern, and explain its effect. *(4 marks)*
6. If the venue limit increases from 8 to 10 slots, describe how the feasible region changes. Does the original optimum remain feasible? Explain. *(2 marks)*

PLANNING FRAME

Represent: define variables, write constraints with units. **Graph:** draw boundaries, test a point, shade the overlap. **Verify:** test given points. **Optimise:** evaluate the objective at every vertex. **Evaluate:** compare the optimum with the community's priorities and state assumptions.

BAND	DESCRIPTOR	MARKS
4	All constraints correct; accurate feasible region and vertices; tests and objective evaluation correct; recommendation justified and fairness evaluated thoughtfully.	19–24
3	Most constraints and graph correct; vertices mostly accurate; optimum found; some evaluation of fairness or assumptions.	13–18
2	Some correct constraints or a readable graph; limited vertex analysis; recommendation weakly justified.	7–12
1	Isolated correct inequalities or calculations; little connection between the model and the context.	1–6

CRITERIA: A, D (C) **ATL:** SELF-MANAGEMENT **TIME:** 50 MINUTES **CONDITIONS:** CALCULATOR ALLOWED

MAPPED TO: A strands i–ii, D strands i–v and C strands i–iv. The rubric is a course summary, not an official IB descriptor.

Chapter summary and review

KEY IDEAS

- Equivalent equations have the same solution set; each operation must be applied to both sides.
- Clearing brackets and fractions before isolating the variable reduces errors.
- An identity has infinitely many solutions; a contradiction has none.
- Multiplying or dividing an inequality by a negative number reverses its direction.
- A solution set can be shown on a number line or written in interval notation.
- Simultaneous equations can be solved graphically, by substitution or by elimination.
- Parallel lines give no solution; coincident lines give infinitely many.
- A system of inequalities has a feasible region; a linear objective is optimised at a vertex, subject to integer constraints.

REVISION CHECKLIST

- ☐ I can solve equations with brackets, fractions and variables on both sides.
- ☐ I can recognise identities and contradictions.
- ☐ I can solve and graph linear inequalities, including compound inequalities.
- ☐ I can solve a system by substitution and by elimination, and check the solution.
- ☐ I can determine whether a system has one, no or infinitely many solutions.
- ☐ I can graph a feasible region and evaluate an objective at its vertices.
- ☐ I can interpret an allocation and discuss its fairness and limitations.

VOCABULARY

linear equation
An equation of the form $ax + b = c$.

equivalent equations
Equations with exactly the same solutions.

contradiction
An equation with no solution.

solution set
All values satisfying an inequality.

system
A set of equations or inequalities that must hold together.

elimination
Removing a variable by adding or subtracting equations.

constraint
A restriction on the variables.

solution
A value that makes an equation true.

identity
An equation true for every permitted value.

inequality
A statement comparing expressions with $<$, $>$, $\leq$ or $\geq$.

interval notation
A compact way to write a set of real numbers.

substitution
Replacing a variable with an equivalent expression.

feasible region
The set of points satisfying all constraints.

objective function
The quantity to be maximised or minimised.

MIXED PRACTICE · CHAPTER 5

Connecting the ideas

FLUENCY Core procedures

1. Solve $5x - 3 = 2x + 12$. F
2. Solve $4(x - 1) = 3x + 5$. F
3. Solve $7 - 2x \geq 1$ and graph the solution. F
4. Solve $-2 \leq x + 1 < 4$. F
5. Solve the system $x + y = 12$, $x - y = 4$. F
6. Solve $3x + 2y = 19$, $x - 2y = 1$. F
7. Solve $2x + 3y = 16$, $x + y = 7$. F

REASONING Justify and connect

1. Does $(2, 3)$ solve the system $y = 2x - 1$ and $x + y = 6$? Show both substitutions and explain your conclusion. R
2. A student solves $4 - 3x < 13$ and writes $x < -3$. Identify the error and give the correct solution. Test $x = 0$. R
3. A system has equations $y = 3x + 1$ and $y = 3x - 4$. Explain why it has no solution, using both graphs and algebra. R
4. Which method would you choose for $x + y = 9$ and $2x - y = 3$? Justify your choice, then solve the system. R

APPLICATION Use and evaluate

1. A feasible region has vertices $(0, 0)$, $(5, 0)$, $(3, 3)$ and $(0, 4)$. Evaluate $P = 4x + 3y$ at each vertex and state the maximum. P
2. A community centre has 120 chairs and 30 tables. Each event needs at least 2 tables and at most 120 guests. Write two constraints involving the number of guests g and tables t. P
3. A shop sells notebooks at 4 units and pens at 2 units. On Monday it sold 35 items and took 100 units. Write and solve a system to find the number of each item sold. P
4. A resource plan maximises total output but leaves one group with fewer sessions than it requested. Explain why the mathematical optimum may not be the fairest plan, and propose a constraint that would protect the group. E

Check your progress

I CAN …	CONFIDENCE	EVIDENCE	NEXT ACTION
Solve multi-step equations and check my solutions.	1 / 2 / 3		
Solve and represent linear inequalities.	1 / 2 / 3		
Solve simultaneous equations by substitution and elimination.	1 / 2 / 3		
Graph a system of inequalities and identify its feasible region.	1 / 2 / 3		
Evaluate an objective and justify an allocation.	1 / 2 / 3		

Reflection

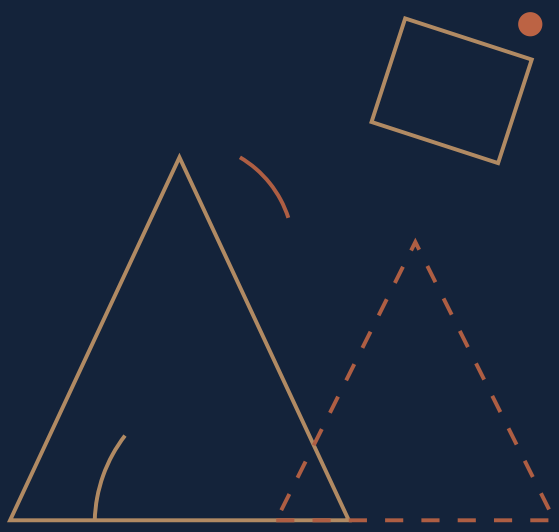
Two feasible allocations can both be mathematically optimal for different objectives. Who should choose the objective when a community allocates scarce resources — and what evidence would make that choice defensible?

PART III · REASONING ABOUT SPACE

III

Geometry, proof and space

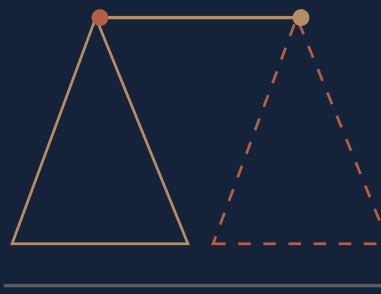
Geometry is where mathematics learns to argue. A transformation gives a precise meaning to “same shape and size”; similarity gives a precise meaning to “same shape, different size”; trigonometry turns those ratios into measurement. The part closes by extending measurement into three dimensions, where area and volume scale at different rates.



06 Transformations, congruence and symmetry

07 Similarity, proof and trigonometry

08 Three-dimensional solids and measurement



6

GEOMETRY, PROOF AND SPACE

Transformations, congruence and symmetry

A shape can slide, flip or turn and stay the same shape and size. These movements generate the symmetry of tile patterns and make congruence a question that reasoning can settle, not appearance.

KEY CONCEPT · FORM | RELATED · SPACE | RELATED · JUSTIFICATION | CONTEXT · PERSONAL AND CULTURAL EXPRESSION

INQUIRY

How much information do you need before you can be certain that two shapes are identical?

ATL · THINKING THROUGH JUSTIFICATION

A drawing can suggest that two shapes are congruent; it cannot prove it. In this chapter, separate demonstration from deduction. Describe each transformation precisely — centre, angle, direction, mirror line — so that a reader could reproduce it without seeing your diagram. When you claim congruence, name the equal parts that are given and the criterion they satisfy. When a claim fails, look for the smallest counterexample: one is enough to refute a rule.

MYP UNIT PLANNER

Framing the inquiry

UNIT 6 · APPROX. 15 HOURS
ATL FOCUS · THINKING

KEY AND RELATED CONCEPTS

Form. A shape's form is fixed by its sides, angles and proportions. Transformations show what is preserved when a figure moves; congruence criteria state which measurements fix a triangle completely.

Space. Position and orientation in the plane. Coordinates make transformations calculable, and symmetry describes an object that maps onto itself.

Justification. Congruence criteria turn measured facts into deductions; a proof records each step so that another person can check it.

GLOBAL CONTEXT AND EXPLORATION

Personal and cultural expression. Exploration: how designers and artisans use transformations, symmetry and tessellation to build pattern, and how mathematical structure limits what repeated tiles can make.

A woven band, a tiled courtyard and a printed logo can all be described by the same three moves: slide, flip, turn. Mathematics does not explain why a pattern moves us, but it explains which patterns continue without gaps and how a repetition can be specified precisely enough to reproduce.

STATEMENT OF INQUIRY

Transformations that preserve distance and angle produce congruent figures; congruence criteria justify identity from limited measurements, while symmetry and tessellation generate designed pattern.

FACTUAL

What are translations, reflections and rotations? What does congruence mean? What are the SSS, SAS, ASA, AAS and RHS criteria? How do you find lines and orders of symmetry?

CONCEPTUAL

Why do rigid transformations preserve congruence while enlargement does not? How can three measurements determine a triangle with six? Why must a regular polygon's interior angle divide $360°$ for it to tessellate alone?

DEBATABLE

Is a careful fold sufficient evidence of congruence, or is a deductive proof always required? How should a designer credit a pattern tradition that inspires a new work?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii	Identify and describe transformations; apply congruence criteria; calculate interior angles of regular polygons.
B · Investigating patterns	i, ii, iii	Investigate which regular polygons tessellate, tabulate interior angles and generalise the condition for a tiling.
C · Communicating	i, ii, iii, iv	Describe transformations precisely with coordinates and vectors; justify congruence in a logical chain; use correct notation and correspondence.
D · Applying mathematics	i, ii, iii, iv, v	Apply transformations and symmetry to a design brief; cost materials; evaluate cultural credit and model limitations.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** what an isometry preserves and why its image is congruent to the original.
- **Know** how to describe and perform translations, reflections and rotations.
- **Be able to** find image coordinates using vectors, mirror lines and the rules for rotations about the origin.
- **Be able to** justify congruence using SSS, SAS, ASA, AAS and RHS, and recognise when information is insufficient.
- **Be able to** analyse the symmetry of a shape and explain why only certain regular polygons tessellate.
- **Be able to** construct, cost and critique a symmetry-based design.

SUCCESS CRITERIA

- ☐ I can describe a translation with a column vector and calculate image coordinates.
- ☐ I can reflect a shape in a given line and state which points are invariant.
- ☐ I can rotate a shape about a point by $90°$, $180°$ or $270°$ and give the image coordinates.
- ☐ I can decide whether two triangles are congruent and justify the decision with a named criterion.
- ☐ I can find the order and angle of rotational symmetry and the number of lines of symmetry of a shape.
- ☐ I can explain why a regular pentagon does not tessellate while a regular hexagon does.

PREREQUISITE CHECK · 12 MINUTES, NO CALCULATOR

1. Plot $(2, 1)$, $(-3, 4)$ and $(0, -2)$. State the quadrant or axis on which each point lies.
2. Which point is 3 units right and 2 units down from $(1, 4)$?
3. Solve $2x + 3 = 11$.
4. State the interior angle sum of a triangle and of a quadrilateral.
5. Calculate $360 \div 6$ and $360 \div 5$.
6. A square has side length 4 cm. Find its perimeter and its area.

Answers are in the Chapter 6 answer section. If more than two are incorrect, review coordinates, angle sums and the properties of triangles and quadrilaterals before continuing.

BEFORE YEAR 3
Coordinates in four quadrants; plotting and reading points; angle facts for straight lines, triangles and quadrilaterals; properties of triangles and quadrilaterals; naming polygons.

YEAR 3 · THIS CHAPTER
Isometries; image coordinates; congruence criteria and proof; line and rotational symmetry; regular tessellations.

LATER
Enlargement and similarity; trigonometric ratios; vectors and matrices as transformations; formal congruence proofs; symmetry groups and frieze patterns.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Tilework, textiles and logos repeat motifs by sliding, flipping and turning.	What is the smallest amount of information that fixes a shape completely?	Isometry, congruence criteria, symmetry and tessellation.	Test which regular polygons close exactly around a point, and prove the pattern.	Write congruence proofs in corresponding order so that a reader can follow each step.	When is “it looks identical” acceptable, and when does certainty require a proof?

6.1 Moving shapes without changing them

CRITERION A · I ◆ CRITERION C · I ◆ SPACE

A transformation is a rule that moves every point of a shape. Some transformations stretch a figure; the three in this chapter do not. They change position or orientation while leaving every distance and every angle exactly as it was.

DEFINITIONS

A **transformation** maps each point of a shape, the **object** (or pre-image), to a new point called its **image**. A transformation that preserves every distance and every angle is an **isometry**, or rigid transformation. A **translation** slides every point of a shape through the same distance in the same direction. It is specified by a **column vector** $\begin{pmatrix} p \\ q \end{pmatrix}$: p units right (negative means left) and q units up (negative means down). The image of a point A is written A'.

Under a translation by $\begin{pmatrix} p \\ q \end{pmatrix}$, a point with coordinates (x, y) has image $(x + p,\ y + q)$. The whole shape moves as one rigid piece: every side keeps its length and direction, every angle keeps its size, and the order of the vertices around the shape is unchanged. A translation therefore produces a congruent image, but in a new position.

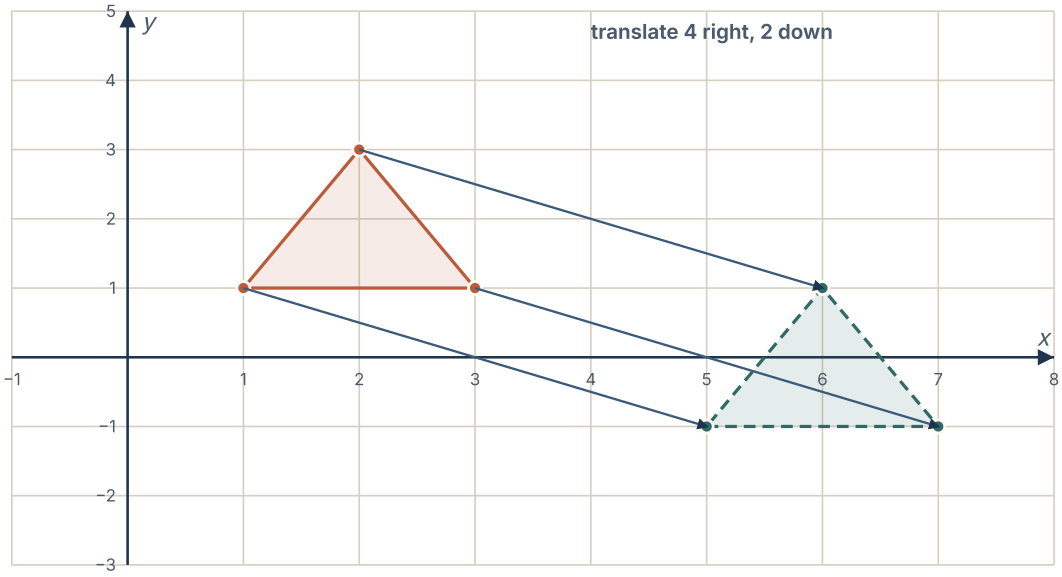


Figure 6.1 The translation $\begin{pmatrix} 4 \\ -2 \end{pmatrix}$. Each arrow has the same length and direction; the image triangle is dashed and congruent to the original.

WORKED EXAMPLE 6.1.1 TRANSLATING A TRIANGLE

PROBLEM

Triangle ABC has vertices $A(1, 1)$, $B(3, 1)$ and $C(2, 3)$. Find the image of the triangle under the translation $\begin{pmatrix} 4 \\ -2 \end{pmatrix}$ and describe the result.

1. Add 4 to each x-coordinate and subtract 2 from each y-coordinate: $A'(1 + 4,\ 1 - 2) = (5, -1)$.
2. Similarly $B'(3 + 4,\ 1 - 2) = (7, -1)$ and $C'(2 + 4,\ 3 - 2) = (6, 1)$.
3. Check that the shape has not changed: $AB = 2$ and $A'B' = 2$; the displacement from B to C is $(-1, 2)$, and the displacement from B' to C' is also $(-1, 2)$.
4. The vector from every vertex to its image is the same, $\begin{pmatrix} 4 \\ -2 \end{pmatrix}$, so the triangle has slid 4 units right and 2 units down without turning.

ANSWER $A'(5, -1)$, $B'(7, -1)$, $C'(6, 1)$. **The image is congruent to ABC, with the same orientation.**

WORKED EXAMPLE 6.1.2 FINDING THE VECTOR FROM A POINT AND ITS IMAGE

PROBLEM

A translation maps $P(-3, 2)$ to $P'(2, 5)$. Find the column vector, describe the translation in words, and find the image of $Q(1, -4)$.

1. The vector is the image coordinates minus the object coordinates: $\begin{pmatrix} 2 - (-3) \\ 5 - 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}$.
2. In words: 5 units right and 3 units up.
3. Apply the same vector to Q: $Q'(1 + 5,\ -4 + 3) = (6, -1)$.
4. Check by subtraction: $Q' - Q = (6 - 1,\ -1 - (-4)) = (5, 3)$, the same vector.

ANSWER **Vector $\begin{pmatrix} 5 \\ 3 \end{pmatrix}$; $Q'(6, -1)$.**

WORKED EXAMPLE 6.1.3 TWO TRANSLATIONS IN SUCCESSION

PROBLEM

A shape is translated by $T_1 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ and then by $T_2 = \begin{pmatrix} -5 \\ 4 \end{pmatrix}$. Find the single translation equivalent to the two moves. What is the reverse of T_1?

1. Apply both vectors to a general point:
   $(x, y) \to (x + 3,\ y - 1) \to (x + 3 - 5,\ y - 1 + 4) = (x - 2,\ y + 3)$.
2. The single equivalent translation is $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$.
3. Reversing the order gives the same result, because adding the components in either order gives the same total: $3 + (-5) = -5 + 3 = -2$ and $-1 + 4 = 4 - 1 = 3$. Translations commute.
4. The reverse of T_1 sends every image back to its object, so it is $\begin{pmatrix} -3 \\ 1 \end{pmatrix}$.

ANSWER **Combined translation $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$; reverse of T_1 is $\begin{pmatrix} -3 \\ 1 \end{pmatrix}$.**

COMMON ERROR · SUBTRACTING IN THE WRONG ORDER

The translation vector is always *image minus object*. For $A(2,5)$ and $A'(6,1)$ the vector is $\begin{pmatrix}4\\-4\end{pmatrix}$, not $\begin{pmatrix}-4\\4\end{pmatrix}$. Also read the signs: a negative first component means left, not down. When a whole shape is translated, every vertex must move by the same vector; if two vertices have different displacements, the move is not a translation.

PRACTICE 6.1 · TRANSLATIONS AND VECTORS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Find the image of each point under the translation $\begin{pmatrix}3\\-4\end{pmatrix}$: $A(2,5)$, $B(-1,0)$, $C(0,-3)$, $D(-4,-2)$. F
2. Triangle PQR has vertices $P(1,2)$, $Q(4,2)$ and $R(1,5)$. It is translated by $\begin{pmatrix}-2\\3\end{pmatrix}$. Find P', Q' and R', and describe the translation in words. F
3. A point $A(5,-1)$ maps to $A'(-2,4)$. Find the column vector and the image of $B(0,0)$ under the same translation. F
4. A square has vertices $(0,0)$, $(2,0)$, $(2,2)$ and $(0,2)$. Under a translation, $(0,0)$ maps to $(4,3)$. Find the other image vertices and the area of the image. Explain why the area is unchanged. R
5. The segment from $(-2,1)$ to $(3,1)$ is mapped onto the segment from $(1,4)$ to $(6,4)$. Find the translation vector. Is the translation unique, or could a different vector also work? Explain. R
6. A pattern of identical arrows starts at $(0,0)$, and the translation $(2,3)$ is applied repeatedly. Find the starting points of the next four arrows. What do you notice about the points, and what is the gradient of the line through them? P
7. A delivery drone is at grid position $(14,9)$ and moves by $\begin{pmatrix}-5\\-3\end{pmatrix}$ each minute. Where is it after 4 minutes? Write a single translation equivalent to the four moves. P
8. A translation T maps $(2,3)$ to $(5,1)$. Find the point that T maps to $(0,0)$. Explain your reasoning by reversing the translation. E

6.2 Reflecting across a line

CRITERION A · I ◆ CRITERION C · I, II ◆ FORM

A reflection turns a shape over a line, as a page turns about its spine. The mirror line stays where it is; every other point swaps to the opposite side, at the same perpendicular distance.

DEFINITIONS

A **reflection** maps each point to its mirror image across a line. The line is the **mirror line**, or line of reflection. For every point P and its image P', the mirror line is the perpendicular bisector of PP': it crosses PP' at a right angle and divides it into two equal parts. Points on the mirror line do not move; they are **invariant**. A reflection reverses the orientation of a shape.

The reflection rules for the coordinate axes and the lines $y = x$ and $y = -x$ are worth committing to memory, because they allow a reflection to be performed without drawing.

Mirror line	Rule	Example
x-axis	$(x, y) \to (x, -y)$	$(3, 2) \to (3, -2)$
y-axis	$(x, y) \to (-x, y)$	$(3, 2) \to (-3, 2)$
$y = x$	$(x, y) \to (y, x)$	$(3, 2) \to (2, 3)$
$y = -x$	$(x, y) \to (-y, -x)$	$(3, 2) \to (-2, -3)$
$x = k$	$(x, y) \to (2k - x,\ y)$	Mirror $x = 1$: $(4, 2) \to (-2, 2)$
$y = k$	$(x, y) \to (x,\ 2k - y)$	Mirror $y = 1$: $(3, 5) \to (3, -3)$

For a mirror at $y = k$, the vertical distance from the point to the mirror is $|y - k|$; the image lies the same distance on the other side, which is why its y-coordinate is $2k - y$. The rule for $x = k$ works in the same way horizontally. When $k = 0$, both rules reduce to the axis cases already listed.

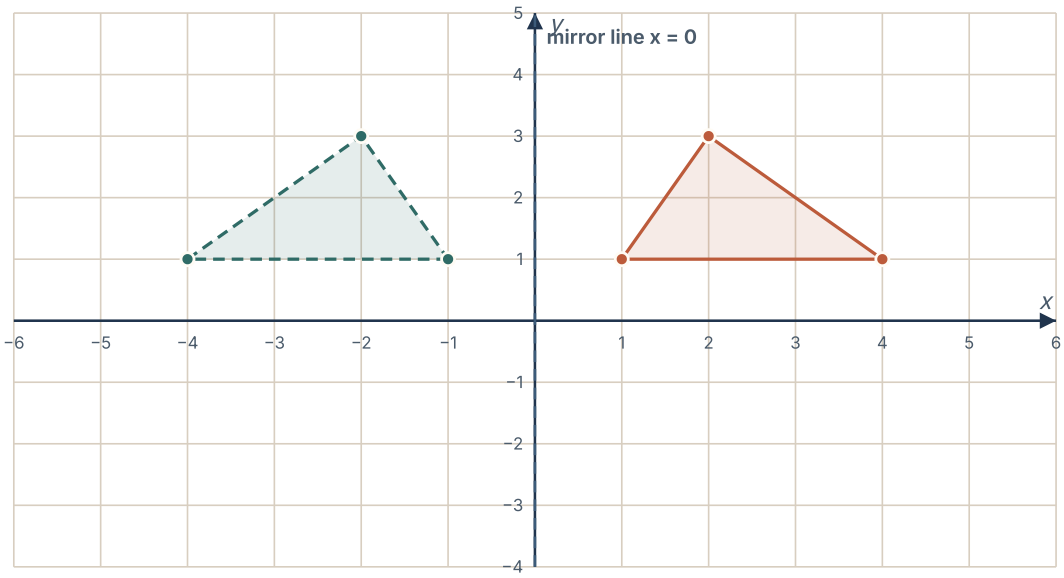


Figure 6.2 A reflection in the line $x = 0$. Corresponding points lie on opposite sides of the mirror at equal distances, and the mirror cuts each joining segment at a right angle.

WORKED EXAMPLE 6.2.1 REFLECTING A TRIANGLE IN THE Y-AXIS

PROBLEM

Triangle ABC has vertices $A(1,1)$, $B(4,1)$ and $C(2,3)$. Reflect it in the y-axis, find the image coordinates, and verify two properties of the reflection.

1. Apply $(x,y) \to (-x,y)$: $A'(-1,1)$, $B'(-4,1)$, $C'(-2,3)$.
2. Check the mirror is the perpendicular bisector of each segment joining a point to its image. For $A(1,1)$ and $A'(-1,1)$, the midpoint is $(0,1)$, which lies on the y-axis, and AA' is horizontal, hence perpendicular to the mirror.
3. The order of the vertices around the triangle is reversed: $A \to B \to C$ runs anticlockwise, while $A' \to B' \to C'$ runs clockwise. A reflection reverses orientation.
4. All side lengths and angles are unchanged, so the image is congruent to the original.

ANSWER $A'(-1,1)$, $B'(-4,1)$, $C'(-2,3)$. **The image is congruent but has reversed orientation.**

WORKED EXAMPLE 6.2.2 REFLECTIONS IN DIFFERENT LINES

PROBLEM

Find the image of $(-2,5)$ after reflection in (a) the x-axis, (b) the line $y = 1$, (c) the line $x = 3$.

1. (a) The x-axis has rule $(x,y) \to (x,-y)$, so $(-2,5) \to (-2,-5)$. The point is 5 units above the axis; its image is 5 units below.
2. (b) Use $(x, 2k - y)$ with $k = 1$: the image is $(-2,\ 2(1) - 5) = (-2,-3)$. The point is 4 units above the line $y = 1$; the image is 4 units below it.
3. (c) Use $(2k - x,\ y)$ with $k = 3$: the image is $(2(3) - (-2),\ 5) = (8,5)$. The point is 5 units left of the line $x = 3$; the image is 5 units right of it.
4. In each case, check that the midpoint of the point and its image lies on the mirror line.

ANSWER **(a)** $(-2,-5)$ **(b)** $(-2,-3)$ **(c)** $(8,5)$

WORKED EXAMPLE 6.2.3 RECOVERING THE MIRROR LINE

PROBLEM

A reflection maps $P(2,3)$ to $P'(2,-7)$. Find the mirror line, then find the image of $Q(-1,4)$ under the same reflection.

1. The midpoint of PP' is $(2,\ -2)$. Since the mirror is the perpendicular bisector of PP', it passes through this midpoint.
2. PP' is vertical, so the perpendicular bisector is horizontal. The mirror is the line $y = -2$.
3. Reflect Q using $(x,\ 2k - y)$ with $k = -2$: $Q'(-1,\ 2(-2) - 4) = (-1,-8)$.
4. Check: Q is 6 units above $y = -2$ and Q' is 6 units below it.

ANSWER **Mirror line** $y = -2$**;** $Q'(-1,-8)$**.**

COMMON ERROR · NAMING THE MIRROR, NOT THE COORDINATE

Reflection in the x-axis changes the sign of y, not of x; the name of the axis is the name of the mirror, not of the coordinate that changes. A reflection in $y = x$ swaps the coordinates, while a reflection in $y = -x$ swaps them and changes both signs. Finally, a fold that appears to match two halves is a demonstration, not a proof: paper bends, and printing is not exact.

PRACTICE 6.2 · REFLECTIONS AND MIRROR LINES

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Reflect each point in the x-axis and then in the y-axis. State both images: $A(3, 2)$, $B(-1, 4)$, $C(-2, -3)$, $D(0, 5)$. F
2. Reflect the triangle with vertices $(1, 1)$, $(4, 1)$, $(1, 3)$ in the line $x = 2$. Find the image vertices. F
3. Find the image of $(-3, 2)$ after reflection in (a) $y = x$, (b) $y = -x$, (c) $x = -1$. F
4. A point (a, b) is reflected in the y-axis and then in the x-axis. Show that the final image is $(-a, -b)$. Does the order of the two reflections matter? R
5. A reflection maps $(5, 1)$ to $(5, 9)$. Find the mirror line and the image of $(2, 3)$. R
6. A logo is symmetric about the vertical line $x = 4$. One half passes through $(1, 2)$ and $(3, 5)$. Find the corresponding points on the other half. P
7. A rectangular mirror is placed along the line $x = 3$. A point at $(7, -2)$ appears at the corresponding position behind the mirror. Find that position and state how far each point is from the mirror. P
8. Explain why a reflection can never be equivalent to a non-zero translation. Use orientation and invariant points in your answer. E

6.3 Rotating about a point

CRITERION A · I ◆ CRITERION C · II ◆ SPACE

A rotation turns a shape about a fixed point. The centre stays where it is; every other point travels along a circular arc through the same angle, keeping its distance from the centre.

DEFINITIONS

A **rotation** is described completely by three pieces of information: the **centre of rotation**, the **angle**, and the **direction** (clockwise or anticlockwise). By convention, anticlockwise is positive. The centre is the only invariant point unless the angle is $360°$. A rotation preserves distances and angles, so the image is congruent to the object, and unlike a reflection it preserves orientation.

For rotations about the origin, the coordinates change in a regular way. A quarter-turn anticlockwise takes the horizontal displacement x and turns it into a vertical displacement, and takes the vertical displacement y and turns it into a horizontal displacement pointing left. The image of (x, y) is therefore $(-y, x)$. Applying the rule twice gives a half-turn, and applying it three times gives a three-quarter turn.

Rotation about the origin	Rule	Example: $(2, 1)$
$90°$ anticlockwise	$(x, y) \to (-y, x)$	$(-1, 2)$
$180°$	$(x, y) \to (-x, -y)$	$(-2, -1)$
$270°$ anticlockwise ($90°$ clockwise)	$(x, y) \to (y, -x)$	$(1, -2)$

WORKED EXAMPLE 6.3.1 A QUARTER-TURN ABOUT THE ORIGIN

PROBLEM

Rotate the triangle with vertices $A(2, 1)$, $B(4, 1)$ and $C(2, 3)$ through $90°$ anticlockwise about the origin.

1. Apply $(x, y) \to (-y, x)$ to each vertex: $A'(-1, 2)$, $B'(-1, 4)$, $C'(-3, 2)$.
2. Check the side lengths. AB is horizontal with length 2; $A'B'$ is vertical with length 2. AC is vertical with length 2; $A'C'$ is horizontal with length 2.
3. Check the orientation. Going $A \to B \to C$ is anticlockwise, and going $A' \to B' \to C'$ is also anticlockwise, so the rotation has preserved orientation.
4. The distances from the origin are unchanged: $OA = \sqrt{5}$ and $OA' = \sqrt{5}$, so each vertex has moved along a circular arc centred at the origin.

ANSWER $A'(-1, 2)$, $B'(-1, 4)$, $C'(-3, 2)$. **The image is congruent and has the same orientation.**

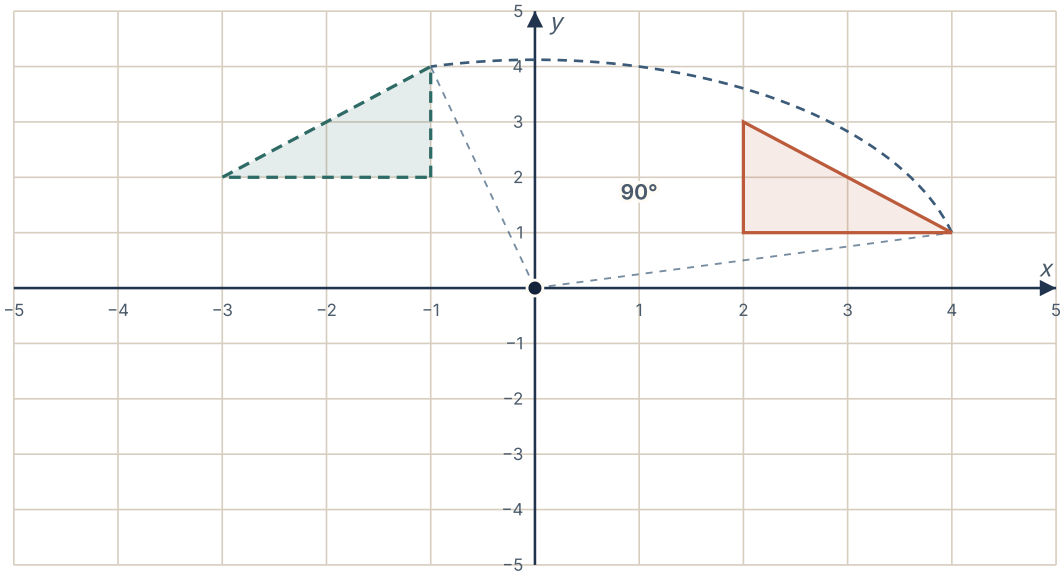


Figure 6.3 A quarter-turn anticlockwise about the origin. The marked $90°$ angle shows the turn; every vertex keeps its distance from the origin.

WORKED EXAMPLE 6.3.2 HALF-TURN AND THREE-QUARTER TURN

PROBLEM

Using the same triangle $A(2,1)$, $B(4,1)$, $C(2,3)$, find the image after (a) a half-turn about the origin, (b) a $90°$ clockwise turn about the origin.

1. (a) A half-turn uses $(x,y)\to(-x,-y)$: $A'(-2,-1)$, $B'(-4,-1)$, $C'(-2,-3)$.
2. (b) A $90°$ clockwise turn is $270°$ anticlockwise, so use $(x,y)\to(y,-x)$: $A'(1,-2)$, $B'(1,-4)$, $C'(3,-2)$.
3. In both cases the image is congruent to the original. Neither turn reverses orientation.
4. A half-turn has the same effect in both directions, so the direction need not be stated for $180°$.

ANSWER **(a)** $A'(-2,-1)$, $B'(-4,-1)$, $C'(-2,-3)$ **(b)** $A'(1,-2)$, $B'(1,-4)$, $C'(3,-2)$

WORKED EXAMPLE 6.3.3 A CENTRE THAT IS NOT THE ORIGIN

PROBLEM

Rotate the triangle $P(4,1)$, $Q(4,3)$, $R(2,1)$ through $90°$ anticlockwise about the point $(2,1)$.

1. Subtract the centre from each vertex to move the centre to the origin: $P\mapsto(2,0)$, $Q\mapsto(2,2)$, $R\mapsto(0,0)$.
2. Apply $(x,y)\to(-y,x)$: $P\mapsto(0,2)$, $Q\mapsto(-2,2)$, $R\mapsto(0,0)$.
3. Add the centre back: $P'(0+2,\,2+1)=(2,3)$; $Q'(-2+2,\,2+1)=(0,3)$; $R'(0+2,\,0+1)=(2,1)$.
4. Check: $PR=2$ and $P'R'=2$; the centre R is invariant, as it must be.

ANSWER $P'(2,3)$, $Q'(0,3)$, $R'(2,1)$.

STRATEGY · DESCRIBING AND PERFORMING A ROTATION

A rotation is not described until the centre, the angle and the direction are all given. To perform one about a centre C, subtract C, apply the rule for a rotation about the origin, then add C back. To check your work, verify that the centre has not moved and that each vertex is the same distance from C as its image.

COMMON ERROR · CLOCKWISE AND ANTICLOCKWISE

$90°$ clockwise is $270°$ anticlockwise, not $90°$ anticlockwise. The rule $(-y,x)$ applies only to an anticlockwise quarter-turn about the origin; using it about a different centre without translating first gives the wrong image. And a half-turn is not a reflection: a half-turn keeps orientation, while a reflection reverses it.

PRACTICE 6.3 · ROTATIONS AND IMAGE COORDINATES

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Rotate each point $90°$ anticlockwise about the origin: $A(3,1)$, $B(-2,4)$, $C(-3,-2)$, $D(0,-5)$. F
2. Rotate the triangle with vertices $(1,2)$, $(4,2)$, $(1,5)$ through $180°$ about the origin. Find the image vertices. F
3. Find the image of $(-2,3)$ under rotations about the origin of (a) $90°$ anticlockwise, (b) $180°$, (c) $90°$ clockwise. F
4. A point (a,b) is rotated $90°$ anticlockwise and then $90°$ anticlockwise again. Show that the result equals a $180°$ rotation. R

5. Rotate the square $(0,0)$, $(2,0)$, $(2,2)$, $(0,2)$ through $90°$ anticlockwise about its centre $(1,1)$ and describe the image. What does this tell you about the rotational symmetry of a square? R
6. The point $(5,2)$ is rotated $90°$ anticlockwise about $(2,2)$. Find the image. P
7. A windmill has four identical blades. Through what angle must it rotate to look unchanged? State the order of its rotational symmetry. P
8. Rotate the triangle $A(0,0)$, $B(4,0)$, $C(0,3)$ through $90°$ anticlockwise about $(2,0)$. Find the image vertices and verify that $AB = A'B'$. E

6.4 Congruence and its criteria

CRITERION A · II ◆ CRITERION C · III, IV ◆ JUSTIFICATION

Two shapes are congruent when one can be moved onto the other without stretching or shrinking. The transformations of the last three sections are exactly the moves that are allowed. Since we cannot always pick a shape up and place it on another, we need tests that decide congruence from measurements.

DEFINITIONS

Two figures are **congruent** if one can be mapped onto the other by a sequence of translations, reflections and rotations. Congruent figures have equal **corresponding parts**: each side and angle matches a partner in the other figure. We write $\triangle ABC \cong \triangle DEF$ to mean that A corresponds to D, B to E and C to F. The standard criteria for triangles are **SSS** (three sides), **SAS** (two sides and the included angle), **ASA** (two angles and the included side), **AAS** (two angles and a corresponding side) and **RHS** (right angle, hypotenuse and one side).

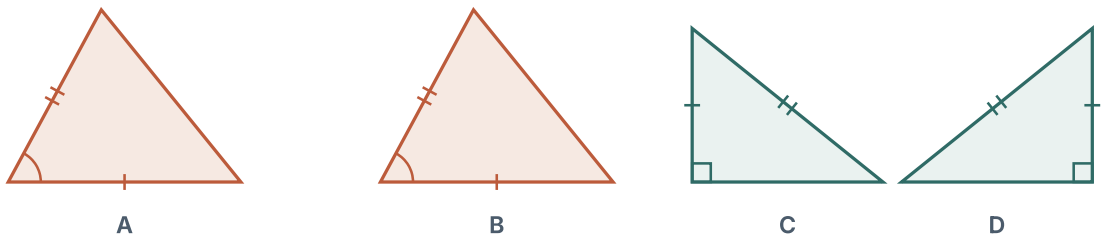


Figure 6.4 Matching marks record equal parts. A and B agree in two sides and the included angle (SAS); C and D agree in the right angle, the hypotenuse and one leg (RHS).

A triangle has six measurements — three sides and three angles — but they are not independent, because the angles always add to $180°$. Three suitably chosen measurements fix the triangle completely. The criteria state which triples are sufficient. The word *included* matters: in SAS the angle must lie between the two named sides; in ASA the side must lie between the two named angles.

WORKED EXAMPLE 6.4.1 — A FORMAL SAS PROOF

PROBLEM

In triangles ABC and DEF, $AB = DE = 7$ cm, $\angle ABC = \angle DEF = 48°$ and $BC = EF = 5$ cm. Prove that the triangles are congruent and list the remaining equal parts.

1. **Given:** $AB = DE = 7$ cm.
2. **Given:** $\angle ABC = \angle DEF = 48°$. In each triangle this angle is between the two sides named in the first and third lines.
3. **Given:** $BC = EF = 5$ cm.
4. **Conclusion:** $\triangle ABC \cong \triangle DEF$ (SAS).
5. Because the triangles are congruent, the remaining corresponding parts are equal: $AC = DF$, $\angle BAC = \angle EDF$ and $\angle ACB = \angle DFE$.

ANSWER $\triangle ABC \cong \triangle DEF$ **by SAS;** $AC = DF$**,** $\angle A = \angle D$**,** $\angle C = \angle F$**.**

WORKED EXAMPLE 6.4.2 — USING A SHARED SIDE: THE KITE

PROBLEM

A kite $ABCD$ has $AB = AD$ and $CB = CD$. The diagonal AC is drawn. Prove that $\triangle ABC \cong \triangle ADC$, and deduce that AC bisects $\angle BAD$.

1. $AB = AD$ (given).
2. $CB = CD$ (given).
3. AC is common to both triangles.
4. Therefore $\triangle ABC \cong \triangle ADC$ (SSS). Note the order: B corresponds to D, because AB matches AD and CB matches CD.
5. Because the triangles are congruent, corresponding angles are equal: $\angle BAC = \angle DAC$. Since these two angles together make $\angle BAD$, the diagonal AC bisects it.

ANSWER $\triangle ABC \cong \triangle ADC$ **by SSS; hence** $\angle BAC = \angle DAC$**, so** AC **bisects** $\angle BAD$**.**

STRATEGY · WRITING A CONGRUENCE PROOF

1. Sketch the two triangles and mark every given equal part. 2. Count the marked parts and check where they sit — is the angle included, is the side included? 3. Name the criterion before you write the conclusion. 4. Write the congruence statement with the vertices in matching order. 5. Extract the corresponding part you were asked to prove. Every line of a proof needs a reason: *given*, *common side*, *proved above*, or the name of the criterion.

WORKED EXAMPLE 6.4.3 ENOUGH INFORMATION, OR NOT?

PROBLEM

For each set of information about triangles ABC and DEF, name the criterion that proves congruence, or explain why the information is insufficient.
(a) $\angle A = \angle D = 90°$, $BC = EF$, $AB = DE$. (b) $AB = DE$, $BC = EF$, $\angle A = \angle D$. (c) $\angle A = \angle D$, $\angle B = \angle E$, $BC = EF$. (d) $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$.

1. (a) The right angles are included between a leg and the hypotenuse in each triangle, and the hypotenuses BC, EF and the legs AB, DE are equal. This is RHS.
2. (b) The equal angles are opposite one pair of equal sides but are not included between the two equal sides. This is SSA, which is not a congruence criterion: a different triangle can satisfy the same three measurements.
3. (c) Two angles and a corresponding side are equal, and the side BC lies opposite the equal angles A and D. This is AAS. (The third angles are equal too, since angles sum to $180°$.)
4. (d) Three angles determine the shape but not the size. Two equilateral triangles of different side lengths have all angles equal yet are not congruent. This is AAA, which proves similarity at most.

ANSWER **(a) RHS (b) insufficient (SSA) (c) AAS (d) insufficient (AAA)**

COMMON ERROR · "IT LOOKS THE SAME"

AAA fixes the shape but not the size; SSA can produce two genuinely different triangles. To see why, fix a side AB and draw a ray from A at a given angle, then mark the second side from B with a compass: the arc can meet the ray in two places. Congruence statements written in the wrong order are also wrong, even when the conclusion is true, because the notation itself asserts which parts correspond.

PRACTICE 6.4 · PROVING CONGRUENCE

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Name the criterion that would prove congruence from: (a) three pairs of equal sides; (b) two pairs of equal sides and the included angles; (c) two pairs of equal angles and a pair of equal sides; (d) a right angle, equal hypotenuses and a pair of equal legs. F
2. In triangles PQR and STU, $PQ = ST = 9$ cm, $\angle P = \angle S = 63°$ and $PR = SU = 4$ cm. Prove the triangles congruent and state the remaining equal parts. F
3. Triangles ABC and ADC share the side AC, with $AB = AD$ and $BC = DC$. Prove $\triangle ABC \cong \triangle ADC$ and deduce that $\angle BCA = \angle DCA$. F
4. Explain why AAA is not enough to prove congruence. Use two equilateral triangles of different sizes in your explanation. R
5. Fix $AB = 8$ cm and draw a ray from A at $30°$ to AB. Side BC must be 6 cm, so C lies where a circle of radius 6 centred at B meets the ray. Explain why this construction gives two different triangles, and what this shows about SSA. R
6. A surveyor measures a triangular plot with one side 40 m and the angles at its ends $65°$ and $42°$. A second plot is claimed to be congruent from the same three measurements. Name the criterion, and find the third angle. P
7. A designer claims two triangular panels are congruent because their side lengths are $30, 40, 50$ cm and $30, 40, 52$ cm. Are they congruent? Explain. P
8. In rectangle $ABCD$, prove that $\triangle ABC \cong \triangle CDA$ and deduce that the diagonal AC divides the rectangle into two congruent triangles. What can you conclude about $\angle BAC$ and $\angle DCA$? E

6.5 Symmetry and tessellation

CRITERION B · II ◆ CRITERION D · II ◆ PATTERNS

A shape has symmetry when a transformation maps it onto itself. Symmetry is congruence applied to a single object, and it explains why a pattern can repeat across a surface without gaps or overlaps.

DEFINITIONS

A shape has **line symmetry** if a reflection in some line maps it onto itself; that line is a **line of symmetry**. It has **rotational symmetry** if a rotation of less than 360° about a point maps it onto itself. The **order** of rotational symmetry is the number of positions, including the starting position, in which the shape looks unchanged during one full turn; the angle of symmetry is $360^\circ \div \text{order}$. A **tessellation** is a covering of the plane by shapes with no gaps and no overlaps. A **regular polygon** has all sides equal and all angles equal.

For a regular polygon with n sides, the two symmetry numbers agree: there are n lines of symmetry and rotational symmetry of order n. For other shapes they can differ. A non-square rectangle has two lines of symmetry and rotational symmetry of order 2; a parallelogram that is not a rectangle has no lines of symmetry but still has rotational symmetry of order 2; an isosceles triangle has one line of symmetry and no rotational symmetry.

COMMON ERROR · COUNTING CORNERS, NOT SYMMETRIES

The diagonals of a rectangle are not lines of symmetry, even though they pass through the centre: folding along a diagonal does not match the rectangle onto itself. Nor is the order of rotational symmetry an angle — an order of 6 means six matching positions, 60° apart. The starting position counts once, not twice.

WORKED EXAMPLE 6.5.1 COUNTING SYMMETRIES

PROBLEM

State the number of lines of symmetry and the order of rotational symmetry for (a) an equilateral triangle, (b) a regular octagon, (c) an isosceles trapezium, (d) a parallelogram that is not a rectangle.

1. (a) An equilateral triangle has 3 lines of symmetry — one through each vertex and the midpoint of the opposite side — and rotational symmetry of order 3, through 120°.
2. (b) A regular octagon has 8 lines of symmetry and rotational symmetry of order 8, through 45°.
3. (c) An isosceles trapezium has 1 line of symmetry, through the midpoints of its parallel sides. Any rotation other than 360° moves it, so its rotational symmetry has order 1.
4. (d) A parallelogram has no line of symmetry, because a fold along either diagonal or either midline fails to match the shape; but a half-turn about the point where the diagonals meet maps it onto itself, so the order is 2.

ANSWER **(a) 3 lines, order 3 (b) 8 lines, order 8 (c) 1 line, order 1 (d) 0 lines, order 2**

A tessellation works because the angles meeting at each interior vertex must add to exactly $360°$. For a regular polygon with n sides, each interior angle is $\frac{180(n-2)}{n}$ degrees. The polygon tiles the plane by itself exactly when this angle divides $360°$. Only three regular polygons pass the test.

Regular polygon	Interior angle	Number meeting at a vertex	Tiles alone?
Triangle, $n = 3$	$60°$	$360 \div 60 = 6$	Yes
Square, $n = 4$	$90°$	$360 \div 90 = 4$	Yes
Pentagon, $n = 5$	$108°$	$360 \div 108 = 3.33\ldots$	No
Hexagon, $n = 6$	$120°$	$360 \div 120 = 3$	Yes
Octagon, $n = 8$	$135°$	$360 \div 135 = 2.67\ldots$	No

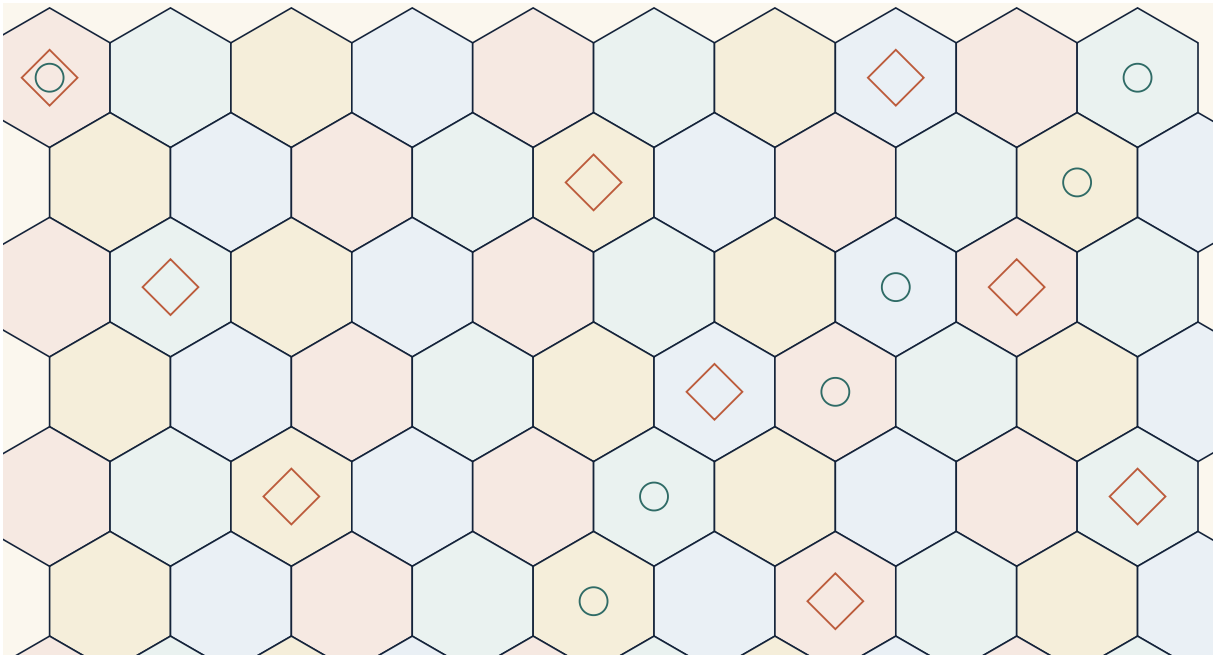


Figure 6.5 A regular hexagonal tiling. Three hexagons meet at every vertex, and $3 \times 120° = 360°$, so the pattern closes exactly with no gap and no overlap.

WORKED EXAMPLE 6.5.2 WHY PENTAGONS LEAVE A GAP

PROBLEM

Show that regular pentagons cannot tessellate by themselves.

1. The interior angle of a regular pentagon is $\frac{180(5-2)}{5} = \frac{540}{5} = 108°$.
2. At a vertex, a whole number of pentagons must meet. Three pentagons give $3 \times 108° = 324°$, leaving a gap of $36°$.
3. Four pentagons give $4 \times 108° = 432°$, which overlaps; more would overlap further.
4. Since no whole number of $108°$ angles makes exactly $360°$, regular pentagons cannot close around a point. A tessellation is therefore impossible.

ANSWER $108°$ **does not divide** $360°$**, so pentagons leave gaps; they cannot tile the plane alone.**

WORKED EXAMPLE 6.5.3 WORKING BACKWARDS FROM AN ANGLE

PROBLEM

A regular polygon has interior angle $150°$. Find its number of sides and decide whether it tessellates by itself.

1. Set $\frac{180(n-2)}{n} = 150$.
2. Multiply by n: $180n - 360 = 150n$.
3. Solve: $30n = 360$, so $n = 12$. The polygon is a regular dodecagon.
4. Test the tiling condition: $360 \div 150 = 2.4$, which is not a whole number. Two tiles leave a $60°$ gap and three exceed $360°$, so it does not tessellate by itself.

ANSWER $n = 12$**; interior angle** $150°$ **does not divide** $360°$**, so it does not tessellate alone.**

PRACTICE 6.5 · SYMMETRY AND TILINGS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. State the number of lines of symmetry and the order of rotational symmetry: (a) a square, (b) a regular hexagon, (c) an isosceles triangle, (d) a parallelogram, (e) a circle. F
2. Find the interior angle of a regular octagon. Does it tessellate by itself? Explain using division. F
3. A regular polygon has interior angle $156°$. Find the number of sides and decide whether it tessellates by itself. F
4. Explain why a regular pentagon cannot tessellate while a regular hexagon can. Support your answer with angle calculations. R
5. A design has rotational symmetry of order 5. Through what angle does it turn onto itself? If it also has 5 lines of symmetry, which familiar shape could it be based on? R
6. Design a square tile with a motif. Describe two different transformations that could be used to repeat the motif across a wall so that the pattern has no gaps. P
7. A floor is tiled with regular octagons and small squares. Show that two octagons and one square fit exactly around a vertex. P
8. Explain why any triangle tessellates, using two congruent copies to form a parallelogram. Extend the argument to any quadrilateral. E

Investigation · The angle that fits

CRITERION B

A regular polygon tiles the plane by itself when copies meet edge to edge with no gaps. The decisive quantity is the interior angle at each vertex. In this investigation you will test the candidates, find the pattern, and prove why only three regular polygons can work.

1. **Explore.** Complete a table for regular polygons with $n = 3$ to 10: the name, the interior angle $\frac{180(n-2)}{n}$, the value of $360 \div \text{angle}$, and whether the polygon tiles alone. Use a calculator and record angles to one decimal place where needed.
2. **Pattern.** Which values of n give a whole number in the third column? Describe the pattern in words.
3. **Verify.** For $n = 5$, calculate the gap left when three pentagons meet. For $n = 8$, calculate the gap left when two octagons meet. Confirm that in each case a third or further tile cannot repair the fit without overlapping.
4. **Generalise.** Show that $360 \div \frac{180(n-2)}{n} = \frac{2n}{n-2} = 2 + \frac{4}{n-2}$. Explain why this is a whole number exactly when $n - 2$ divides 4, and conclude that n can only be 3, 4 or 6.
5. **Combine.** Tilings can mix tile types. Verify that the angles fit exactly at a vertex for: (a) two octagons and one square; (b) two hexagons and two triangles; (c) two squares and three triangles.
6. **Break it.** A student says "a pentagon is close to a hexagon, so with enough pentagons the gaps will disappear." Explain why "enough" cannot help: the gap at each vertex is fixed at $36°$, and adding more pentagons overshoots $360°$.
7. **Create.** Design a semi-regular tiling using two tile types. Draw at least six tiles and verify the angle sum at two different vertices. Label the transformations that carry one tile onto the next.

A neat drawing suggests a tiling; only the angle sum proves that a vertex closes exactly. A single counterexample — one vertex where the angles do not total $360°$ — is enough to refute a proposed tiling. State which claim your evidence supports.

Application · A tile panel for the community centre

CRITERION D · I, II, III, IV ◆ SYMMETRY ◆ DESIGN

A community centre commissions a rectangular frieze 360 cm long and 60 cm high, made from square tiles of side 15 cm. Tiles cost 2.40 units each, and the supplier recommends ordering 8% extra for cutting and breakages. The design repeats a four-tile-wide motif along the length, and each motif is mirrored in its own vertical centre line.

1. Calculate the number of tiles along the length and the number of rows. How many tiles cover the panel exactly?
2. Calculate the cost of the exact coverage, then the cost after adding 8% wastage. Round the number of tiles ordered up to a whole tile and explain why rounding down would be a mistake.
3. How many repeats of the four-tile motif fit along the frieze? Describe the transformation that carries the first motif onto the second, giving the vector in tiles and in centimetres.
4. The mirror line of each motif runs along a grout line between two columns of tiles. How many tiles lie on the mirror line? What would change if the motif were five tiles wide?

5. Each square tile is divided by a diagonal into two triangular pieces for the border motif. Prove that the two triangles are congruent, naming the criterion, and state the angles of each triangle.
6. The designer claims the whole panel looks unchanged after a half-turn about its centre. What must be true of each motif and of their arrangement for this claim to hold? Is the half-turn guaranteed by the design as described?
7. A contrasting border one tile high runs along the top and bottom edges. How many contrast tiles are needed? At 3.10 units each, what is the additional cost, and what is the new total?
8. Grout lines are 2 mm wide. Show that 24 tiles of 15 cm plus 23 grout lines do not fit into 360 cm. Propose two practical solutions and evaluate which is better.
9. Evaluate the design process. Which assumptions (tile size, wastage, grout, labour) are uncertain, and what records should the designer keep so that the panel can be reproduced or repaired later?
10. The motif echoes a geometric tradition from another culture. What should the designer record and acknowledge? Distinguish between mathematical inspiration and direct copying of a specific sacred pattern, and explain why the distinction matters.

CRITERION D FOCUS

Translate a design brief into measurements, justify geometric claims with congruence and symmetry, cost the materials honestly, and evaluate the practical and cultural limits of the model.

Criterion task · The kite window

CRITERIA B AND C
INVESTIGATING AND COMMUNICATING
24 MARKS

A stained-glass studio is making a kite-shaped window $ABCD$, with $AB = AD$ and $CB = CD$. A diagonal bar runs from A to C. The studio must be able to justify every geometric claim in its construction notes.

1. Prove that $\triangle ABC \cong \triangle ADC$, naming the criterion and giving a reason for each equal part. *(5 marks)*
2. Deduce that AC bisects both $\angle BAD$ and $\angle BCD$. *(4 marks)*
3. Let M be the point where AC meets BD. Prove that $BM = DM$. *(3 marks)*
4. Prove that AC is perpendicular to BD. *(4 marks)*
5. State the number of lines of symmetry and the order of rotational symmetry of the kite, justifying each answer from your earlier results. *(4 marks)*
6. A second window is a rhombus that is not a square. Using your results, explain why both diagonals of the rhombus are lines of symmetry, and state the order of its rotational symmetry. *(4 marks)*

PLANNING FRAME

Sketch: copy the given equal lengths onto a diagram and mark the shared side. **Prove:** list three equal parts with reasons under the heading "In $\triangle\ldots$ and $\triangle\ldots$". **Conclude:** name the criterion, then write the congruence statement in matching order. **Use:** extract exactly the corresponding part you need — an angle for bisection, a length for a midpoint. **Communicate:** one reason per line, no unstated assumptions.

BAND	DESCRIPTOR	MARKS
4	Congruence proved with complete reasons; deductions about angles, midpoint and perpendicularity follow logically; symmetry justified from earlier results; notation and correspondence accurate.	19–24
3	Congruence proved with most reasons; most deductions valid; symmetry stated with partial justification; minor gaps in correspondence or notation.	13–18
2	Some equal parts identified; congruence asserted with limited reasoning; one or two deductions attempted; symmetry stated without justification.	7–12
1	Isolated correct statements about the figure; little structure; conclusions do not follow from the given information.	1–6

CRITERIA: B, C (A) **ATL:** THINKING **TIME:** 55 MINUTES **CONDITIONS:** RULER AND PROTRACTOR ALLOWED

MAPPED TO: B strands i–iii and C strands i–iv, with A strand ii. The rubric is a course summary, not an official IB descriptor.

Chapter summary and review

KEY IDEAS

- A transformation maps every point of an object to an image; an isometry preserves all distances and angles.
- A translation moves every point by the same column vector: $(x, y) \to (x + p,\ y + q)$. It preserves orientation.
- A reflection reverses orientation; the mirror line is the perpendicular bisector of every segment joining a point to its image.
- Reflections in the axes: $(x, y) \to (x, -y)$ in the x-axis and $(x, y) \to (-x, y)$ in the y-axis; in $y = x$ the coordinates swap.
- A rotation is fixed by its centre, angle and direction. About the origin: $90°$ anticlockwise $(x, y) \to (-y, x)$, $180°$ $(x, y) \to (-x, -y)$, $90°$ clockwise $(x, y) \to (y, -x)$.
- Congruent figures have equal corresponding sides and angles; the rigid transformations produce congruent images.
- SSS, SAS, ASA, AAS and RHS are sufficient criteria for congruence; AAA and SSA are not.
- Correspondence order matters: $\triangle ABC \cong \triangle DEF$ asserts $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$.
- A regular n-gon has n lines of symmetry, rotational symmetry of order n, and interior angle $\frac{180(n-2)}{n}$.
- A regular polygon tiles the plane alone exactly when its interior angle divides $360°$, which happens only for $n = 3, 4, 6$.

REVISION CHECKLIST

- ☐ I can translate, reflect and rotate a shape and give the image coordinates.
- ☐ I can describe a transformation fully, including centre, angle and direction where needed.
- ☐ I can recover a mirror line or translation vector from a point and its image.
- ☐ I can justify congruence with a named criterion and correct correspondence.
- ☐ I can explain why AAA and SSA do not prove congruence.
- ☐ I can count lines of symmetry and the order of rotational symmetry.
- ☐ I can use interior angles to decide whether a regular polygon tessellates.

VOCABULARY

transformation
A rule mapping each point of a shape to an image point.

image
The shape produced by a transformation.

translation
A slide specified by a column vector.

reflection
A flip across a mirror line.

invariant point
A point that does not move under a transformation.

object (pre-image)
The original shape before a transformation.

isometry
A transformation that preserves all distances and angles.

column vector
Two components giving horizontal and vertical displacement.

mirror line
The line of reflection; it is the perpendicular bisector of each point–image segment.

rotation

A turn about a centre through a given angle and direction.

congruent
Identical in shape and size, with equal corresponding parts.

AAA / SSA
Angle-angle-angle and side-side-angle; neither proves congruence.

rotational symmetry
A rotation below $360°$ that maps a shape onto itself.

tessellation
A gap-free, overlap-free tiling of the plane.

centre of rotation
The fixed point of a rotation.

corresponding parts
Matching sides and angles in two congruent figures.

line symmetry
A reflection that maps a shape onto itself.

order
The number of matching positions in one full turn.

MIXED PRACTICE · CHAPTER 6

Connecting the ideas

FLUENCY Core procedures

1. Find the image of $A(4, -2)$, $B(0, 0)$ and $C(-1, -4)$ under the translation $\begin{pmatrix} -3 \\ 5 \end{pmatrix}$. F
2. For the triangle with vertices $(2, 3)$, $(5, 3)$ and $(2, 6)$, find the image after (a) a reflection in the x-axis, (b) a $180°$ rotation about the origin. F
3. State the number of lines of symmetry and the order of rotational symmetry of (a) a regular decagon, (b) an isosceles trapezium, (c) a parallelogram. F
4. Find the interior angle of a regular nonagon and of a regular dodecagon. Decide whether either tessellates by itself. F

REASONING Justify and connect

1. The point $(3, 1)$ is rotated $90°$ anticlockwise about the origin and then reflected in the y-axis. Find the final image. Now reverse the order of the two transformations and compare. What does this show? R
2. ABC is a triangle and D is a point such that $AB = AD$, $BC = DC$, with AC drawn. Prove $\triangle ABC \cong \triangle ADC$ and state one consequence about the angles at C. R
3. Explain, with one counterexample for each, why SSA and AAA do not guarantee congruence. R
4. The line $y = 2x + 1$ is translated by $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$. Find the equation of the image line and explain why its gradient is unchanged. R

APPLICATION Use and evaluate

1. A logo is a square of side 8 cm divided by a diagonal. Prove that the two triangles formed are congruent, name the criterion, and state the symmetry of the square. P
2. A wall panel is 240 cm by 90 cm and is tiled with 30 cm square tiles. Find the number of tiles needed, add 10% for wastage, and cost the order at 3.50 units per tile. P
3. A frieze has its first motif at $(2, 4)$ and repeats by the vector $(6, 0)$. Find the starting point of the twelfth motif. The motif is also reflected in the line $y = 6$; find the image of $(2, 4)$. State one symmetry the finished frieze definitely has. P
4. A woven band repeats a unit cell through four quarter-turns about a central point. Explain how one quarter specifies the whole design, and why each quarter is congruent to the first. What should the designer record about the tradition that inspired it? E
5. In rectangle $ABCD$, prove that the diagonals AC and BD are equal in length, naming the criterion you use. E

6. Show that no regular polygon with seven or more sides can tessellate by itself. Use the interior angle formula and the fact that at least three tiles must meet at a vertex. E

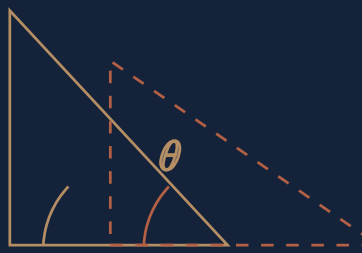
Check your progress

I CAN …	CONFIDENCE	EVIDENCE	NEXT ACTION
Translate, reflect and rotate a shape and describe the transformation completely.	1 / 2 / 3		
Calculate image coordinates and recover a mirror line or vector.	1 / 2 / 3		
Prove congruence using SSS, SAS, ASA, AAS or RHS.	1 / 2 / 3		
Explain why AAA and SSA are insufficient.	1 / 2 / 3		
Analyse symmetry and decide whether a regular polygon tessellates.	1 / 2 / 3		

Reflection

A pattern can be reproduced exactly by applying transformations, but reproducing a tradition's pattern is not the same as understanding it. Where should a designer draw the line between mathematical inspiration and cultural appropriation?

7



GEOMETRY, PROOF AND SPACE

Similarity, proof and trigonometry

A photograph, a map and a shadow all keep a shape while changing its size. Similarity explains why. It also turns the right triangle into an instrument: one measured angle and one measured length can reveal a height no one can reach with a tape.

KEY CONCEPT · RELATIONSHIPS | RELATED · JUSTIFICATION | RELATED · MEASUREMENT | CONTEXT · ORIENTATION IN SPACE AND TIME

INQUIRY

How can an angle and a single length tell you the height of something you cannot reach?

ATL · RESEARCH AND CRITICAL THINKING

Indirect measurement is an argument built from assumptions: level ground, a vertical tower, a sun angle that does not change during the measurement. Record what you assume, choose instruments whose precision matches the claim, and calculate how far an error in one measurement can move the answer. A number is only as trustworthy as the reasoning and the evidence behind it.

MYP UNIT PLANNER

Framing the inquiry

UNIT 7 · APPROX. 16 HOURS
ATL FOCUS · RESEARCH

KEY AND RELATED CONCEPTS

Relationships. Similarity is a relationship between shapes: angles stay equal while lengths change by a common factor. Trigonometric ratios are relationships between an angle and two sides of a right triangle, and they hold for every triangle with that angle.

Justification. AA, SSS and SAS similarity criteria licence a proportion; Pythagoras' theorem and its converse licence a right-angle claim. Each result has conditions that must be checked before it is used.

Measurement. Indirect measurement replaces an inaccessible length with an accessible one plus an angle. Precision, rounding and model assumptions decide how much the answer can be trusted.

GLOBAL CONTEXT AND EXPLORATION

Orientation in space and time. Exploration: how surveyors, navigators and builders have measured inaccessible heights and distances, and how angle-based methods changed maps, buildings and travel.

Before satellites, a surveyor climbed a hill with a theodolite, read an angle and calculated. The mathematics has not changed: a ratio, a right triangle and a careful record. What has changed is the precision of the instruments and the scale of the claims we can justify.

STATEMENT OF INQUIRY

Similar figures preserve shape through proportional change, and right-triangle ratios depend only on the angle; together they make indirect measurement possible and justifiable.

FACTUAL

What does it mean for two figures to be similar? What are the sine, cosine and tangent of an angle? What does Pythagoras' theorem state, and what is its converse?

CONCEPTUAL

Why do trigonometric ratios depend only on the angle and not on the size of the triangle? Why does multiplying every length by k multiply area by k^2? When is a calculated length justified?

DEBATABLE

How precise should a measurement claim be? When a model rests on uncertain assumptions, should its answer be reported as a fact or as an estimate with a range?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii	Identify similar figures; apply scale factors; use Pythagoras and the three trigonometric ratios; select the correct ratio.
B · Investigating patterns	i, ii, iii	Investigate the constancy of height-to-shadow ratios, tabulate results, generalise the shadow rule and test its limits.
C · Communicating	i, ii, iii, iv	Write similarity statements in corresponding order; show each step of a trigonometric solution with units; justify rounding.
D · Applying mathematics	i, ii, iii, iv, v	Measure an inaccessible height; design and cost an accessible ramp; evaluate assumptions, precision and fitness for purpose.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** what similarity preserves and how scale factors affect length and area.
- **Know** and use the AA, SSS and SAS criteria for similar triangles.
- **Be able to** apply Pythagoras' theorem and its converse to lengths, distances and right-angle tests.
- **Be able to** choose and apply sine, cosine or tangent to find a side or an angle in a right triangle.
- **Be able to** use angles of elevation and depression to measure indirectly.
- **Be able to** evaluate the assumptions and precision of a measurement model.

SUCCESS CRITERIA

- [ ] I can find a missing side in similar figures using a scale factor and state the area ratio.
- [ ] I can prove two triangles similar and write the similarity statement in matching order.
- [ ] I can calculate an unknown side or a distance using Pythagoras' theorem.
- [ ] I can use the converse to decide whether a triangle is right-angled.
- [ ] I can solve a right triangle using sin, cos and tan, including finding an angle with an inverse ratio.
- [ ] I can interpret an angle of elevation or depression in a diagram and report an indirect measurement with sensible precision.

PREREQUISITE CHECK · 12 MINUTES, NO CALCULATOR

1. Simplify the ratio $12 : 18$ and write it as a fraction.
2. Solve $\frac{3}{4} = \frac{x}{12}$.
3. Calculate 7^2, 12^2 and 13^2.
4. Round 6.847 to one decimal place and to two significant figures.
5. State the angle sum of a triangle and the size of each angle in an equilateral triangle.
6. In a right triangle, which side is the hypotenuse?

Answers are in the Chapter 7 answer section. If more than two are incorrect, review ratio, rounding and the vocabulary of right triangles before continuing.

BEFORE YEAR 3
Ratio and proportion; fractions and rounding; angle facts; properties of right triangles; area of rectangles and triangles; congruence criteria.

YEAR 3 · THIS CHAPTER
Similarity and scale factors; similarity criteria; Pythagoras' theorem and its converse; trigonometric ratios; indirect measurement.

LATER
The sine and cosine rules; trigonometric graphs and identities; coordinate geometry of circles; vectors; three-dimensional trigonometry.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Surveyors, navigators and builders measure distances they cannot walk.	How can a ratio in a small triangle predict a length in a much larger one?	Similarity, proportional sides, Pythagoras and the trigonometric ratios.	Test whether height-to-shadow ratios stay constant, and use the pattern to measure a tree.	Record the diagram, the ratio selected and the units at every stage.	How should an answer built on measurements communicate its uncertainty?

7.1 Similar figures and scale factors

CRITERION A · I ◆ CRITERION C · I ◆ RELATIONSHIPS

Two figures are similar when one is a scaled copy of the other. Every length changes by the same factor, every angle stays the same, and the shape is not distorted. A photograph enlarged without cropping, a map and a scale drawing are all similar to what they represent.

DEFINITIONS

Two figures are **similar** if their corresponding angles are equal and their corresponding sides are in the same ratio. The **scale factor** k is the ratio $\frac{\text{image length}}{\text{object length}}$. An **enlargement** is a transformation with a centre and a scale factor; it is not an isometry unless $k = 1$. For triangles, the similarity statement $\triangle ABC \sim \triangle DEF$ asserts $A \leftrightarrow D$, $B \leftrightarrow E$ and $C \leftrightarrow F$, in that order.

For a rectangle, it is enough that the ratio of length to width is the same: a 3×2 rectangle is similar to a 6×4 rectangle with $k = 2$. For a general polygon, both conditions are needed. A rectangle 3×2 and a parallelogram with the same angles but sides 4 and 3 have equal angles but different side ratios, so they are not similar.

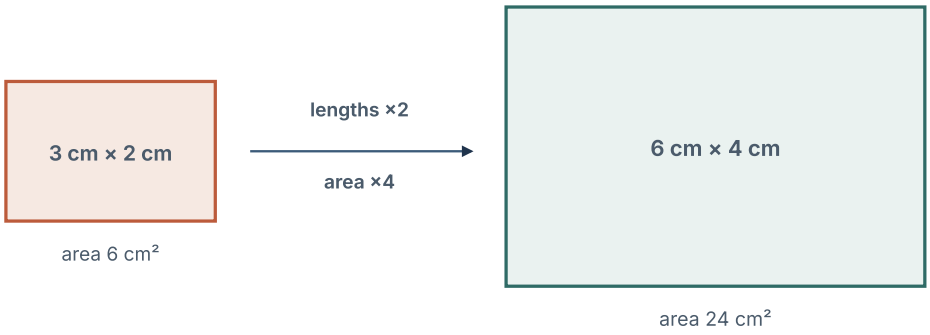


Figure 7.1 Lengths scale by $k = 2$, but area scales by $k^2 = 4$. The larger rectangle is not "twice as big" in every sense.

KEY RELATIONSHIP · LENGTH, AREA AND VOLUME

If two similar figures have scale factor k, then corresponding lengths have ratio k, corresponding areas have ratio k^2, and corresponding volumes have ratio k^3. The area result follows because area is the product of two lengths, each multiplied by k.

WORKED EXAMPLE 7.1.1 ENLARGING FROM A CENTRE

PROBLEM

A rectangle has vertices $(1, 1)$, $(4, 1)$, $(4, 3)$ and $(1, 3)$. It is enlarged by scale factor 3 from the centre $(1, 1)$. Find the image vertices, the dimensions of the image, and the ratio of the areas.

1. The centre is fixed. For each vertex, multiply its displacement from $(1, 1)$ by 3: $(x, y) \to (1 + 3(x - 1),\ 1 + 3(y - 1))$.
2. The vertices map to $(1, 1)$, $(10, 1)$, $(10, 7)$ and $(1, 7)$.
3. The object is 3 by 2, so its area is 6. The image is 9 by 6, so its area is 54.
4. Area ratio $= 54 \div 6 = 9 = 3^2$, which agrees with the scale factor squared.

ANSWER **Images** $(1, 1)$, $(10, 1)$, $(10, 7)$, $(1, 7)$**; dimensions** 9×6**; area ratio** $9 : 1$**.**

WORKED EXAMPLE 7.1.2 FINDING A SCALE FACTOR AND MISSING SIDES

PROBLEM

Triangles ABC and DEF are similar, with $AB = 5$ cm, $BC = 7$ cm and $CA = 9$ cm corresponding to DE, EF and FD. Given $DE = 7.5$ cm, find the scale factor, the other two sides of DEF, and the ratio of their areas.

1. The scale factor is $\frac{DE}{AB} = \frac{7.5}{5} = 1.5$.
2. Multiply each side of ABC by 1.5: $EF = 7 \times 1.5 = 10.5$ cm and $FD = 9 \times 1.5 = 13.5$ cm.
3. Check the ratio: $7.5 : 10.5 : 13.5 = 5 : 7 : 9$, the same ratio as ABC.
4. The area ratio is $k^2 = 1.5^2 = 2.25$. The larger triangle has 2.25 times the area, not 1.5 times.

ANSWER $k = 1.5$**;** $EF = 10.5$ **cm,** $FD = 13.5$ **cm; area ratio** $2.25 : 1$**.**

WORKED EXAMPLE 7.1.3 A MAP AND ITS SCALE

PROBLEM

A map has scale $1 : 25\,000$. (a) A path measures 4.8 cm on the map. Find its real length in kilometres. (b) A lake has area 3.2 cm^2 on the map. Find its real area in square kilometres.

1. The scale means 1 cm on the map represents $25\,000$ cm $= 250$ m $= 0.25$ km.
2. (a) $4.8 \times 0.25 = 1.2$ km.
3. (b) Areas scale by the square of the length scale: 1 cm^2 represents $0.25^2 = 0.0625$ km^2.
4. $3.2 \times 0.0625 = 0.2$ km^2.

ANSWER **(a)** 1.2 **km** **(b)** 0.2 **km^2, which is** $200\,000$ **m^2.**

COMMON ERROR · ADDING THE SAME AMOUNT INSTEAD OF MULTIPLYING

Adding 2 cm to every side does not create a similar figure: a 3×2 rectangle would become 5×4, and $5 : 4 \neq 3 : 2$. Similarity requires a common multiplier. Also do not use k for area: if lengths double, area quadruples. And when finding k, compare corresponding sides, not any two sides.

PRACTICE 7.1 · SIMILARITY AND SCALE

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. These rectangles are similar. Find the scale factor and the missing side: 4×6 and $10 \times x$. F
2. Triangles PQR and STU are similar with $PQ = 8$ cm, $QR = 10$ cm, $RP = 14$ cm and $ST = 12$ cm. Find TU and US. F
3. A photograph 10 cm by 15 cm is enlarged so that its shorter side becomes 25 cm. Find the new longer side and the ratio of the areas. F
4. A plan uses scale $1 : 200$. A room is 7.5 cm by 5 cm on the plan. Find its real dimensions in metres and its real area. R
5. Two similar triangles have areas 36 cm^2 and 81 cm^2. Find the scale factor from the smaller to the larger. Explain your reasoning. R
6. A photograph is enlarged by $k = 1.5$ and then by $k = 2$. What is the single equivalent scale factor? What is the overall area multiplier? Justify both answers. P
7. A map has scale $1 : 50\,000$. A forest on the map has area 18 cm^2. Find its area in square kilometres and in hectares (1 km$^2 = 100$ ha). P
8. Rectangle A is 8 cm by d cm and rectangle B is d cm by 18 cm. Find the value of d that makes the two rectangles similar, and state the scale factor. E

7.2 Similar triangles and indirect measurement

CRITERION A · I, II ◆ CRITERION C · II ◆ JUSTIFICATION

Triangles are the easiest shapes to test for similarity, because three angles and three sides are tightly linked. If two angles match, the third must match too, and the sides are forced into a fixed ratio. That is why a small triangle can predict the dimensions of a large one.

CRITERIA FOR SIMILAR TRIANGLES

AA: two angles of one triangle equal two angles of another. The third pair is then equal because angles sum to $180°$. **SSS:** all three pairs of corresponding sides are in the same ratio. **SAS:** two pairs of corresponding sides are in the same ratio and the included angles are equal. Any one of these is enough to conclude that the triangles are similar and that all corresponding sides share the ratio.

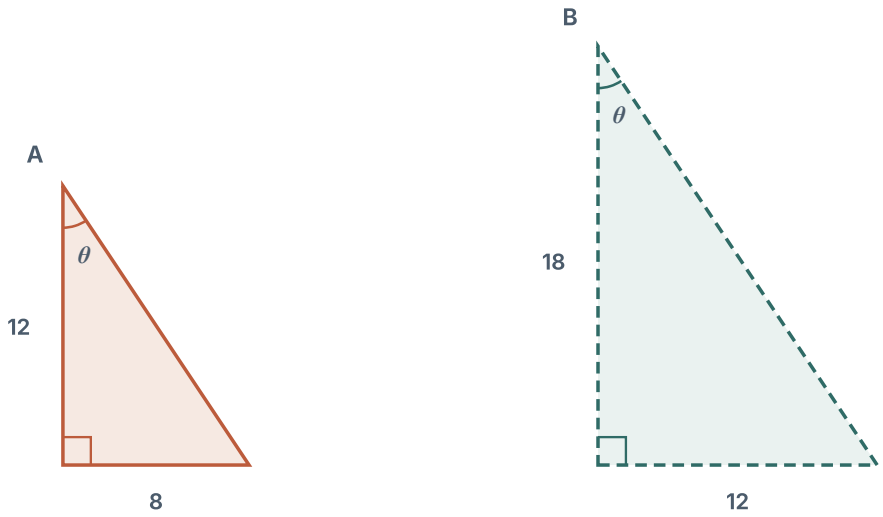


Figure 7.2 Both triangles contain the angle θ, so they are similar by AA. The base ratio $12 : 8 = 1.5$ equals the height ratio $18 : 12 = 1.5$.

Parallel lines are a reliable source of equal angles. If a line parallel to one side of a triangle cuts the other two sides, the corresponding angles it makes with those sides are equal. The small triangle cut off at the top is therefore similar to the whole triangle.

WORKED EXAMPLE 7.2.1 A LINE PARALLEL TO THE BASE

PROBLEM

In triangle ABC, D lies on AB and E lies on AC, with $DE \parallel BC$. Given $AD = 4$ cm, $DB = 2$ cm and $DE = 3$ cm, find BC and the ratio of the area of $\triangle ADE$ to the area of $\triangle ABC$.

1. Because $DE \parallel BC$, $\angle ADE = \angle ABC$ and $\angle AED = \angle ACB$ (corresponding angles). Hence $\triangle ADE \sim \triangle ABC$ by AA.
2. The scale factor from $\triangle ADE$ to $\triangle ABC$ is $\frac{AB}{AD} = \frac{4+2}{4} = 1.5$.
3. So $BC = DE \times 1.5 = 4.5$ cm.
4. The area ratio is $1.5^2 = 2.25$, so the small triangle has area $\frac{1}{2.25} = \frac{4}{9}$ of the whole triangle.

ANSWER $BC = 4.5$ **cm; area ratio** $4 : 9$ **(small : whole).**

WORKED EXAMPLE 7.2.2 THE SHADOW METHOD

PROBLEM

A vertical post 1.8 m high casts a shadow 2.7 m long at the same time that a tree casts a shadow 12 m long. Assume the ground is level and the sun's rays are parallel. Find the height of the tree.

1. The post and its shadow, and the tree and its shadow, form two right triangles.
2. The sun's rays make the same angle with the ground in both triangles, and both have a right angle, so the triangles are similar by AA.
3. Match the vertical sides and the shadows: $\frac{h}{1.8} = \frac{12}{2.7}$.
4. $h = 1.8 \times \frac{12}{2.7} = 8$ m.

ANSWER **The tree is** 8 **m tall.**

WORKED EXAMPLE 7.2.3 THE MIRROR METHOD

PROBLEM

A student's eye is 1.6 m above the ground. She stands 2.4 m from a mirror lying flat on the ground and sees the top of a tree reflected in it. The mirror is 9.6 m from the base of the tree. Find the height of the tree.

1. The angle at which the student looks down equals the angle at which the reflection appears to come from the tree (equal angles at the mirror).
2. Both triangles are right-angled at the ground, so they are similar by AA.
3. Match heights to horizontal distances: $\frac{h}{1.6} = \frac{9.6}{2.4} = 4$.
4. $h = 1.6 \times 4 = 6.4$ m.

ANSWER **The tree is 6.4 m tall.**

COMMON ERROR · WRONG CORRESPONDENCE

In $\triangle ABC \sim \triangle DEF$, side AB corresponds to DE, not to EF. Write the similarity statement before forming any ratio: the order of the letters is the correspondence. Also check that you are comparing like with like — two heights, or a height with a shadow — and never subtract corresponding sides when the relationship is multiplicative.

PRACTICE 7.2 · SIMILAR TRIANGLES SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. State whether each pair is similar, and if so by which criterion: (a) two angles $48°$ and $67°$ in both; (b) sides $3, 4, 5$ and $6, 8, 10$; (c) sides $4, 6$ with included angle $50°$ and sides $6, 9$ with included angle $50°$. F
2. In $\triangle ABC$, $DE \parallel BC$ with $AD = 6$ cm, $DB = 3$ cm and $AE = 8$ cm. Find EC and AC. F
3. Triangles ABC and XYZ are similar with $AB = 9$ cm, $BC = 12$ cm and $XY = 6$ cm. Find YZ and the ratio of their areas. F
4. A 1.5 m post casts a 2.1 m shadow. At the same time a building casts a 28 m shadow. Find the height of the building and explain which assumptions you used. R
5. Explain why two triangles with equal angles are always similar, but two triangles with three proportional sides are not always congruent. Use an example. R
6. A student uses the mirror method and measures 1.7 m, 3.0 m and 12.0 m. Find the height. If the mirror-to-tree distance is measured 0.2 m too short, how much does the answer change? P
7. A ramp is $k = 1.25$ times as long and 1.25 times as high as a model. The model's triangular side has area 0.24 m^2. Find the real side's area. P
8. Two similar triangles have perimeter ratio $3 : 5$ and area of the smaller 27 cm^2. Find the area of the larger. Then find the ratio of their heights. E

7.3 Pythagoras' theorem

CRITERION A · II ◆ CRITERION C · II ◆ RELATIONSHIPS

A right angle turns two lengths into a third. Given any two sides of a right-angled triangle, Pythagoras' theorem fixes the third exactly. The theorem is also reversible: a numerical check can prove that an angle is, or is not, a right angle.

DEFINITIONS

In a right-angled triangle, the side opposite the right angle is the **hypotenuse**. It is always the longest side. If the other two sides (the legs) have lengths a and b, and the hypotenuse has length c, then **Pythagoras' theorem** states that $a^2 + b^2 = c^2$. The **converse** is also true: if a triangle with longest side c satisfies $a^2 + b^2 = c^2$, then the angle opposite c is a right angle. A **Pythagorean triple** is a set of three positive whole numbers that satisfy the equation, such as $3, 4, 5$; $5, 12, 13$; $8, 15, 17$ and $7, 24, 25$.

The theorem is a statement about areas. Draw a square of side $a + b$ and place four copies of the right triangle around its edge. The four triangles leave an uncovered square in the middle, and that square has side c, so its area is c^2. Equating two expressions for the large area gives the theorem:

$$(a+b)^2 = 4\left(\tfrac{1}{2}ab\right) + c^2 \quad \Rightarrow \quad a^2 + 2ab + b^2 = 2ab + c^2 \quad \Rightarrow \quad a^2 + b^2 = c^2.$$

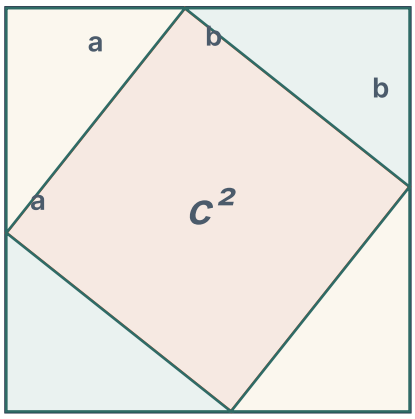


Figure 7.3 Four copies of one triangle leave a central square of area c^2. The algebra holds for every positive a and b, even though the drawing shows one particular pair.

The diagram alone does not prove the theorem: it shows one arrangement, and a drawing can be inaccurate. The proof is the calculation that the two expressions for the large area must be equal. This is a useful pattern — a visual argument supplies insight, and algebra supplies the general justification.

WORKED EXAMPLE 7.3.1 FINDING THE HYPOTENUSE

PROBLEM

A right-angled triangle has legs 9 cm and 12 cm. Find the length of the hypotenuse.

1. The hypotenuse c is opposite the right angle, so $9^2 + 12^2 = c^2$.
2. $81 + 144 = 225$, so $c^2 = 225$.
3. $c = \sqrt{225} = 15$, taking the positive root because a length is positive.
4. Check: 15 is greater than both 9 and 12, as the hypotenuse must be.

ANSWER **The hypotenuse is 15 cm.**

WORKED EXAMPLE 7.3.2 FINDING A LEG, TWICE

PROBLEM

A 2.6 m ladder leans against a vertical wall with its foot 1.0 m from the wall. (a) How far up the wall does it reach? (b) How far from the wall must the foot be placed for the top to reach exactly 2.5 m?

1. The ladder is the hypotenuse. For (a), the height h satisfies $h^2 + 1.0^2 = 2.6^2$.
2. $h^2 = 6.76 - 1.00 = 5.76$, so $h = 2.4$ m.
3. For (b), let the distance be d. Then $d^2 + 2.5^2 = 2.6^2$.
4. $d^2 = 6.76 - 6.25 = 0.51$, so $d = \sqrt{0.51} \approx 0.714$ m.
5. The foot must move closer to the wall, to about 0.71 m, which is consistent with the top rising.

ANSWER **(a) 2.4 m (b) about 0.71 m from the wall.**

WORKED EXAMPLE 7.3.3 DISTANCE AND THE CONVERSE

PROBLEM

(a) Find the distance between $A(-2, 3)$ and $B(4, -5)$. (b) A triangle has sides 8 cm, 15 cm and 17 cm. Show that it is right-angled and state where the right angle is.

1. (a) The horizontal difference is $4 - (-2) = 6$ and the vertical difference is $-5 - 3 = -8$.
2. These are the legs of a right triangle, so $AB^2 = 6^2 + (-8)^2 = 36 + 64 = 100$.
3. $AB = 10$.
4. (b) The longest side is 17, so test $8^2 + 15^2$ against 17^2: $64 + 225 = 289 = 17^2$.
5. By the converse of Pythagoras, the angle opposite the 17 cm side is $90°$.

ANSWER **(a) 10 units (b) right-angled, with the right angle between the 8 cm and 15 cm sides.**

COMMON ERROR · SQUARING THE WRONG SIDE, OR ADDING WHEN YOU SHOULD SUBTRACT

The equation $a^2 + b^2 = c^2$ only works when c is the hypotenuse. To find a leg, subtract the square of the known leg from the square of the hypotenuse; the result must be smaller than the hypotenuse. Do not decide from a sketch which side is longest — identify the right angle first. When using the converse, square the longest side on its own and compare, rather than assuming the triangle is right-angled and forcing the numbers to fit.

PRACTICE 7.3 · PYTHAGORAS AND ITS CONVERSE

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Find the hypotenuse of a right triangle with legs 6 cm and 8 cm. F
2. Find the missing leg when the hypotenuse is 17 m and one leg is 8 m. F
3. Find the diagonal of a rectangle measuring 5 m by 12 m. F
4. An isosceles triangle has equal sides 10 cm and base 12 cm. Find its height and area. R
5. Find the distance between $(1, -2)$ and $(7, 6)$. F
6. Decide whether each triangle is right-angled: (a) $7, 24, 25$; (b) $9, 12, 16$; (c) $10, 24, 26$. Justify each answer. R
7. A 4.5 m ladder has its foot 1.8 m from a wall. How high does it reach? If the foot is moved to 2.7 m from the wall, how far does the top slide down? P

8. A right triangle has hypotenuse 25 cm and one leg 5 cm longer than the other. Find both legs. E

7.4 Trigonometric ratios

CRITERION A · I, II ◆ CRITERION C · II ◆ RELATIONSHIPS

In a right-angled triangle, the ratio of two sides is fixed by the acute angle. That single fact converts an angle measurement into a length, and a length ratio back into an angle. The three standard ratios are called sine, cosine and tangent.

DEFINITIONS

For an acute angle θ in a right-angled triangle, label the side opposite θ as the **opposite**, the side next to θ (but not the hypotenuse) as the **adjacent**, and the longest side as the **hypotenuse**. Then

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

The ratios are properties of the angle, not of the triangle: any two right triangles with the same acute angle are similar by AA, so their corresponding side ratios are equal. The mnemonic **SOH–CAH–TOA** records the three definitions.

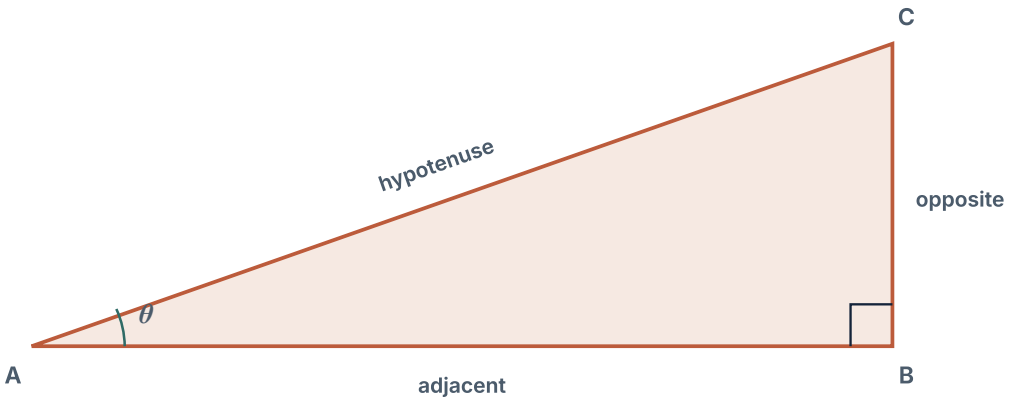


Figure 7.4 The labels "opposite" and "adjacent" move if the other acute angle is chosen; the hypotenuse does not. The three ratios are defined relative to the marked angle θ.

Because the ratios depend only on the angle, a calculator can store them. To recover an angle from a ratio, use the inverse functions $\sin^{-1}$, $\cos^{-1}$ and $\tan^{-1}$. Work in degree mode unless a question states otherwise. Tangent has a direct graphical meaning: in a right triangle whose adjacent side is horizontal, $\tan\theta$ is the gradient of the sloping side, rise divided by run.

EXACT VALUES FOR SPECIAL ANGLES

Two triangles supply the exact values that appear throughout later mathematics. A square of side 1 has diagonal $\sqrt{2}$ by Pythagoras, and its diagonal splits the right angle into two 45° angles. An equilateral triangle of side 2, cut along an altitude, gives a 30°–60°–90° triangle whose altitude is $\sqrt{2^2 - 1^2} = \sqrt{3}$.

EXACT TRIGONOMETRIC VALUES

θ	$\sin\theta$	$\cos\theta$	$\tan\theta$
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$
45°	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$

These values are exact, unlike the decimal approximations a calculator returns. They also give a quick check: $\sin 30^\circ$ must display as 0.5. If it does not, the calculator is in the wrong mode.

WORKED EXAMPLE 7.4.1 — WRITING THE THREE RATIOS

PROBLEM

Triangle PQR is right-angled at Q, with $PQ = 4$ cm, $QR = 3$ cm and $PR = 5$ cm. Write $\sin P$, $\cos P$ and $\tan P$ as fractions and decimals, then find the size of angle P to one decimal place.

1. Relative to angle P, the opposite side is $QR = 3$, the adjacent side is $PQ = 4$ and the hypotenuse is $PR = 5$.
2. $\sin P = \frac{3}{5} = 0.6$, $\cos P = \frac{4}{5} = 0.8$ and $\tan P = \frac{3}{4} = 0.75$.
3. Check with Pythagoras: $3^2 + 4^2 = 5^2$, so the triangle is right-angled as stated.
4. $P = \tan^{-1}(0.75) \approx 36.9^\circ$. Using $\sin^{-1}(0.6)$ or $\cos^{-1}(0.8)$ gives the same angle, which is a useful check.

ANSWER $\sin P = 0.6$, $\cos P = 0.8$, $\tan P = 0.75$; $P \approx 36.9^\circ$.

WORKED EXAMPLE 7.4.2 — FINDING TWO SIDES FROM AN ANGLE

PROBLEM

A right-angled triangle has an acute angle of 32° and its adjacent side is 45 mm. Find the opposite side and the hypotenuse, each to one decimal place.

1. For the opposite side, use tangent: $\tan 32^\circ = \frac{\text{opposite}}{45}$.
2. Opposite $= 45 \tan 32^\circ \approx 45 \times 0.6249 \approx 28.1$ mm.
3. For the hypotenuse, use cosine: $\cos 32^\circ = \frac{45}{\text{hypotenuse}}$, so hypotenuse $= \frac{45}{\cos 32^\circ} \approx 53.1$ mm.
4. Check with Pythagoras: $45^2 + 28.1^2 \approx 2025 + 789.6 = 2814.6$, and $53.1^2 \approx 2819.6$. The small difference comes from rounding 28.1.

ANSWER **Opposite** ≈ 28.1 **mm; hypotenuse** ≈ 53.1 **mm.**

WORKED EXAMPLE 7.4.3 — FROM A GRADIENT TO AN ANGLE

PROBLEM

A ramp rises 1.2 m over a horizontal run of 9.0 m. Find the angle the ramp makes with the horizontal, to one decimal place, and the length of the sloping surface, to two decimal places.

1. The rise is opposite the angle and the run is adjacent, so $\tan\theta = \frac{1.2}{9.0} = 0.1333\ldots$
2. $\theta = \tan^{-1}(0.1333\ldots) \approx 7.6^\circ$.
3. The sloping surface is the hypotenuse: $L = \sqrt{1.2^2 + 9.0^2} = \sqrt{1.44 + 81} = \sqrt{82.44}$.
4. $L \approx 9.08$ m. Equivalently, $L = \frac{9.0}{\cos 7.6^\circ} \approx 9.08$ m.

ANSWER **Angle** $\approx 7.6^\circ$**; sloping length** ≈ 9.08 **m.**

COMMON ERROR · THE WRONG RATIO, OR THE WRONG MODE

"Opposite" and "adjacent" swap when you change the angle you are working with, so mark the angle before choosing a ratio. The hypotenuse is always opposite the right angle. Check that $\sin\theta$ and $\cos\theta$ are less than 1 for an acute angle; a value such as 1.4 means the fraction was inverted. Keep the calculator in degree mode, and do not round an intermediate ratio before the final calculation.

PRACTICE 7.4 · SINE, COSINE AND TANGENT

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. In a right triangle, angle θ has opposite side 7, adjacent side 24 and hypotenuse 25. Write $\sin\theta$, $\cos\theta$ and $\tan\theta$ as fractions. F
2. Find x, the side opposite a 38° angle, when the hypotenuse is 20 cm. F
3. Find y, the side adjacent to a 55° angle, when the hypotenuse is 14 m. F
4. Find θ to one decimal place: (a) $\tan\theta = 0.85$; (b) $\sin\theta = 0.4$; (c) $\cos\theta = 0.25$. F
5. Without a calculator, find the exact value of (a) $\sin 45^\circ + \cos 30^\circ$ and (b) $\tan 60^\circ \times \tan 30^\circ$. R
6. A 14 m tree casts a shadow when the sun's angle of elevation is 58°. Find the length of the shadow to two decimal places. P
7. A kite string 32 m long makes an angle of 41° with the horizontal. Find the height of the kite above the hand, to one decimal place. P
8. A right triangle has an acute angle of 30° and hypotenuse $2x$. Show that the shorter leg is x and the longer leg is $x\sqrt{3}$. E

7.5 Angles of elevation and depression

CRITERION C · II, III ◆ CRITERION D · II ◆ MEASUREMENT

Surveyors, pilots and architects rarely measure a height directly. They measure a horizontal distance and an angle from a horizontal line, then use a trigonometric ratio. The angle above the horizontal is called an angle of elevation; the angle below it is an angle of depression.

Figure 7.5 The horizontal distance is the adjacent side, the height is the opposite side, and the line of sight is the hypotenuse. The angle of elevation is measured upward from the horizontal.

DEFINITIONS

The **angle of elevation** of an object is the angle between the horizontal line through the observer's eye and the line of sight to the object, measured upward. The **angle of depression** is the corresponding angle measured downward from the horizontal. Both are measured from the horizontal, never from the vertical. The **line of sight** is the straight line from the observer to the object. Because the horizontal lines at the two ends of a line of sight are parallel, the angle of elevation from A to B equals the angle of depression from B to A (alternate angles).

The same triangle can be read in two directions. From the ground, the object's height is opposite the elevation angle. From the top, the object on the ground is below the horizontal, and the depression angle is the angle inside the triangle at the top. Drawing both horizontal lines makes the alternate-angle relationship visible and prevents the most common error: placing the depression angle at the bottom of the diagram.

WORKED EXAMPLE 7.5.1 HEIGHT FROM AN ELEVATION ANGLE

PROBLEM

From a point on level ground 60 m from the base of a vertical tower, the angle of elevation of the top is 28°. Find the height of the tower to one decimal place, assuming the ground is horizontal and the tower is vertical.

1. The tower, the horizontal ground and the line of sight form a right triangle. The height is opposite 28° and 60 m is adjacent.
2. $\tan 28^\circ = \frac{h}{60}$, so $h = 60 \tan 28^\circ$.
3. $\tan 28^\circ \approx 0.5317$, so $h \approx 60 \times 0.5317 \approx 31.9$ m.
4. The answer is a model: if the ground slopes or the tower leans, the triangle is not right-angled and the height is only an estimate.

ANSWER **The tower is approximately 31.9 m tall.**

WORKED EXAMPLE 7.5.2 A DEPRESSION ANGLE FROM A CLIFF

PROBLEM

From the top of a cliff 45 m above sea level, the angle of depression of a boat is 34°. Find (a) the horizontal distance from the boat to the foot of the cliff and (b) the direct distance from the observer to the boat, each to one decimal place.

1. Draw a horizontal line at the top of the cliff. The angle between it and the line of sight is 34°, so the angle inside the right triangle at the top is also 34°.
2. The vertical drop 45 m is opposite this angle and the horizontal distance d is adjacent, so $\tan 34^\circ = \frac{45}{d}$.
3. $d = \frac{45}{\tan 34^\circ} \approx \frac{45}{0.6745} \approx 66.7$ m.
4. For the direct distance L, the vertical drop is opposite the 34° angle and L is the hypotenuse: $L = \frac{45}{\sin 34^\circ} \approx \frac{45}{0.5592} \approx 80.5$ m.

ANSWER **(a) 66.7 m (b) 80.5 m.**

WORKED EXAMPLE 7.5.3 ALLOWING FOR EYE HEIGHT

PROBLEM

A student whose eyes are 1.55 m above the ground stands 28 m from a building. She measures the angle of elevation of the top of the building as 36°. Find the height of the building to one decimal place.

1. The right triangle starts at eye level, not at the ground. Let x be the height of the building above eye level.
2. $\tan 36^\circ = \frac{x}{28}$, so $x = 28 \tan 36^\circ \approx 28 \times 0.7265 \approx 20.34$ m.
3. The total height is x plus the eye height: $20.34 + 1.55 = 21.89$ m.
4. Rounding at the end gives 21.9 m. Rounding 20.34 to 20 first would lose almost 0.4 m.

ANSWER **The building is approximately 21.9 m tall.**

COMMON ERROR · MEASURING THE ANGLE FROM THE WRONG LINE

An angle of depression of 34° is not the 56° angle between the line of sight and the vertical, and it is not the angle at the foot of the cliff. Mark the horizontal at the observer and use alternate angles to transfer the angle into the triangle. Remember to add the observer's eye height when the triangle starts at eye level. Finally, the horizontal distance is the adjacent side; do not use the sloping line of sight as the adjacent side.

MAKING CONNECTIONS · GRADIENT AS AN ANGLE

Builders and engineers often quote a slope as a percentage grade rather than an angle. The grade is $100 \tan\theta$, so a 10% grade rises 10 m for every 100 m horizontally, but the angle is only $\tan^{-1}(0.10) \approx 5.7^\circ$. The percentage and the angle are not interchangeable, and the gap grows quickly as the slope steepens.

COMMON GRADIENTS AND THEIR ANGLES

Gradient	Grade	Angle	Typical use
$1:20$	5%	2.9°	railway ruling gradient
$1:12$	8.3%	4.8°	maximum accessible ramp
$1:8$	12.5%	7.1°	steep driveway
$1:4$	25%	14.0°	steep mountain road

Keeping the two measures distinct matters: a ramp described as “8%” is close to the $1:12$ limit, while one described as “8 degrees” is almost twice as steep and would fail the guideline.

PRACTICE 7.5 · ELEVATION AND DEPRESSION

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. From a point 80 m from the base of a tower on level ground, the angle of elevation of the top is 35°. Find the height of the tower. F
2. From the top of a 70 m cliff, the angle of depression of a boat is 42°. Find the horizontal distance to the boat. F
3. An observer's eyes are 1.7 m above the ground. She stands 40 m from a flagpole and measures an angle of elevation of 23°. Find the height of the flagpole. F
4. A 25 m tower stands on level ground. Find the distance from a point where the angle of elevation of the top is 61°. F

5. An aircraft is 900 m above the ground. The angle of depression of the airport runway is $17°$. Find the horizontal distance to the runway, in kilometres. R
6. Two points A and B are 50 m apart on level ground on the same side of a tower. The angles of elevation of the top from A and B are $40°$ and $55°$ respectively, with B closer to the tower. Find the height of the tower. P
7. A ramp rises 0.8 m over a horizontal run of 11 m. Find its angle to the horizontal and the length of its sloping surface. P
8. From the top of a 30 m building, the angle of depression of the top of a shorter building 24 m away is $15°$. Find the height of the shorter building. E

Investigation · The shadow rule

CRITERION B

On a sunny day, every vertical object casts a shadow. The sun's rays are effectively parallel, so all the shadow triangles formed at the same moment have the same angles. In this investigation you will test whether the ratio of height to shadow length is constant, prove why it must be, and use the result to measure something you cannot reach.

MEASUREMENTS TAKEN ON LEVEL GROUND AT THE SAME TIME

Object	Height h (m)	Shadow s (m)	$h \div s$
Metre rule	1.00	0.75	1.333
Pole	1.20	0.90	1.333
Student	1.65	1.24	1.331
Tree	?	9.60	—

1. **Measure.** In a school yard, measure the height and shadow length of at least three vertical objects. Record each to the nearest centimetre and calculate $h \div s$ to three decimal places. Keep all measurements in the same unit.
2. **Pattern.** Compare the ratios. Are they constant within the precision of your measurements? State a conjecture relating height, shadow length and a constant k.
3. **Explain.** Draw one triangle for the metre rule and one for the pole. Mark the right angle at the foot of each object and the angle of the sun's rays. Explain why the two triangles are similar by AA, and why this forces the ratio $h \div s$ to be the same.
4. **Apply.** Use your value of k to estimate the height of the tree from its 9.60 m shadow. Compare with a direct estimate and comment on the difference.
5. **Quantify.** If the tree's shadow is measured 0.10 m too long, by how much does the estimated height change? Which measurement — height of a reference object or shadow length — contributes more error, and why?
6. **Challenge the assumptions.** List three conditions the method requires. For each, describe what happens to the estimate if the condition fails: sloping ground, objects that are not vertical, measurements taken at different times.
7. **Connect.** Show that the constant k equals $\tan\theta$, where θ is the sun's angle of elevation. Use one row of the table to calculate θ to one decimal place.
8. **Evaluate.** Would you trust this method for a 30 m building? Explain what you would change to improve reliability.

A constant ratio in three rows is evidence, not proof. The proof is the AA similarity argument, which explains why the ratio must be constant for every vertical object at that moment. Distinguish clearly between what your data shows and what the geometry guarantees.

Application · The library ramp

CRITERION D · I, II, III, IV ◆ MEASUREMENT ◆ DESIGN

A library must provide step-free access at two entrances. The local accessibility guideline sets a maximum ramp gradient of $1 : 12$ (one unit of rise for every twelve units of horizontal run). A designer surveys both entrances and records the data below. The ramp will be 1.5 m wide, with a 1.5 m square landing at the top.

SURVEY DATA FOR THE TWO ENTRANCES

Entrance	Total rise (m)	Available run (m)	Minimum run at $1 : 12$ (m)	Outcome
Main	0.72	12.00	8.64	fits, with spare length
Side	1.05	11.50	12.60	1.10 m short

1. For the main entrance, verify the minimum run at $1 : 12$. Using the full available run of 12.00 m, calculate the actual gradient as a ratio, the angle of the ramp to the horizontal, and the sloping length of the surface.
2. For the side entrance, calculate the gradient if the full 11.50 m run is used, and its angle. Compare it with the $1 : 12$ limit. Find the greatest rise that a 11.50 m run can accommodate at $1 : 12$, and state the shortfall.
3. The side entrance cannot be lengthened. Propose two compliant solutions and evaluate each in terms of cost, space, user experience and maintenance.
4. One solution is a switchback: two flights, each rising half the total height at $1 : 12$, joined by a 1.5 m square landing. Calculate the run of each flight and the overall length and width the switchback requires. Compare its footprint with the single ramp.
5. The main ramp deck costs 48 units per square metre of sloping surface, and the top landing costs the same rate. Handrails cost 32 units per metre and are fitted on both sides of the sloping surface. Calculate the total cost, then add a 10% contingency and round to the nearest unit.
6. The deck is surfaced with 30 cm by 30 cm non-slip strips. How many strips are needed for the sloping surface, allowing no wastage? Explain why the supplier would round up and add a margin.
7. The surveyor's rise measurement is accurate to ± 2 cm. At $1 : 12$, by how much does the required run change? Evaluate whether the design has enough tolerance at each entrance.
8. Evaluate the model. Which assumptions (level ground, constant gradient, rigid deck, weather, user behaviour) are uncertain? List the records the designer should keep so that the ramp can be inspected, maintained and reproduced.

CRITERION D FOCUS

Convert a guideline into trigonometry, test a design against a physical constraint, cost it transparently, and evaluate the gap between a clean mathematical model and a safe, buildable ramp.

Criterion task · The lighthouse

CRITERIA B AND C
INVESTIGATING AND COMMUNICATING
24 MARKS

A coastal survey team needs the height of a lighthouse lamp above sea level. The lighthouse stands on a cliff that rises straight from the water. From a boat at position A, the angle of elevation of the lamp is $21°$. The boat moves 120 m directly toward the cliff to position B, where the angle of elevation is $35°$. A chart gives the cliff height as 78 m above sea level. The lamp is treated as a point at the top of a vertical tower.

1. Draw a diagram showing the cliff, the tower, the lamp, both boat positions and both angles. Define a variable h for the height of the lamp above sea level and d for the horizontal distance from A to the cliff. State two assumptions of the model. *(3 marks)*
2. Form two equations connecting h, d and the given angles. Eliminate d and solve for h, showing each algebraic step. Give your answer to the nearest metre. *(5 marks)*
3. Find the horizontal distances from each boat position to the cliff. Verify that their difference is 120 m, and explain why this check does not prove that the answer is physically exact. *(3 marks)*
4. Use the chart value to deduce the height of the lighthouse tower above the cliff. Comment on whether the result is plausible. *(3 marks)*
5. A second boat is 250 m horizontally from the cliff. Calculate the angle of elevation of the lamp from this boat, to one decimal place. *(3 marks)*
6. The 120 m baseline is measured to the nearest metre, so it could be 118 m or 122 m. Recalculate h for both values and state the range. Which input — the baseline or the angles — do you think contributes more uncertainty, and why? *(4 marks)*
7. Evaluate the model. Explain how the observer's height above the water, the state of the sea and the shape of the cliff affect the result. Recommend one improvement to the survey procedure and justify it. *(3 marks)*

PLANNING FRAME

Draw: one vertical line for cliff and tower, one horizontal for sea level, two lines of sight from A and B. **Label:** put h and d on the diagram before writing any ratio. **Relate:** in each right triangle, h is opposite and the horizontal distance is adjacent, so use tangent. **Eliminate:** write both distances in terms of h, then subtract. **Check:** substitute back into both triangles. **Communicate:** keep full calculator values until the final rounding, and give units at every answer.

BAND	DESCRIPTOR	MARKS
4	Clear labelled diagram; both equations correct; elimination and solution accurate; distances checked; tower height and second angle correct; sensitivity interpreted; assumptions evaluated with a justified improvement.	19–24
3	Diagram and equations mostly correct; minor algebraic or rounding slip; distances and tower height largely correct; sensitivity attempted; evaluation identifies some assumptions.	13–18
2	Some correct trigonometric ratios; equations may use the wrong side; solution incomplete; limited checking; evaluation generic.	7–12
1	Isolated correct statements or a partially labelled diagram; no coherent model; conclusions unsupported.	1–6

CRITERIA: B, C (D) **ATL:** CRITICAL THINKING **TIME:** 60 MINUTES **CONDITIONS:** SCIENTIFIC CALCULATOR ALLOWED

MAPPED TO: B strands i–iii and C strands i–iv, with D strand iii. The rubric is a course summary, not an official IB descriptor.

Chapter summary and review

KEY IDEAS

- Similar figures have equal corresponding angles and corresponding sides in the same ratio; the scale factor is $k = \frac{\text{image length}}{\text{object length}}$.
- Under a scale factor k, lengths scale by k, areas by k^2 and volumes by k^3.
- Triangles are similar by AA, SSS (all three side ratios equal) or SAS (two side ratios and the included angle).
- In a right triangle, Pythagoras' theorem gives $a^2 + b^2 = c^2$, where c is the hypotenuse; the converse tests for a right angle.
- Pythagorean triples such as $3, 4, 5$ and $5, 12, 13$ generate further triples when multiplied by a whole number.
- For an acute angle θ: $\sin\theta = \frac{\text{opp}}{\text{hyp}}$, $\cos\theta = \frac{\text{adj}}{\text{hyp}}$, $\tan\theta = \frac{\text{opp}}{\text{adj}}$.
- The ratios depend only on θ, because all right triangles with that acute angle are similar.
- The exact values for $30°$, $45°$ and $60°$ come from a halved equilateral triangle and a halved square.
- Inverse trigonometric functions recover an angle from a ratio; keep the calculator in degree mode.
- Angles of elevation and depression are measured from the horizontal; the two are equal when taken between the same two points.

REVISION CHECKLIST

- ☐ I can find a scale factor and use it to calculate lengths, areas and volumes.
- ☐ I can prove two triangles similar by AA, SSS or SAS and write the correspondence correctly.
- ☐ I can use Pythagoras to find either leg or the hypotenuse, and its converse to test a triangle.
- ☐ I can choose sine, cosine or tangent from a labelled diagram.
- ☐ I can find a side or an angle with trigonometry and round appropriately.
- ☐ I can solve elevation and depression problems, including eye height.
- ☐ I can state the assumptions behind an indirect measurement and assess its uncertainty.

VOCABULARY

similar
Same shape, with equal corresponding angles and proportional sides.

enlargement
A transformation with a centre and a scale factor.

AA
Two equal angles; sufficient for similarity.

SAS similarity
Two side ratios equal with equal included angles.

Pythagoras' theorem
$a^2 + b^2 = c^2$ in a right triangle.

converse
The reverse statement: a right angle follows from $a^2 + b^2 = c^2$.

sine
Opposite divided by hypotenuse.

scale factor
The multiplier relating corresponding lengths.

corresponding sides
Sides in matching positions in similar figures.

SSS similarity
All three pairs of sides in the same ratio.

indirect measurement
Finding a length you cannot measure directly, using a model.

hypotenuse
The side opposite the right angle; the longest side.

Pythagorean triple
Three whole numbers satisfying $a^2 + b^2 = c^2$.

cosine
Adjacent divided by hypotenuse.

tangent
Opposite divided by adjacent; the gradient of the sloping side.

angle of depression
Angle measured downward from the horizontal.

angle of elevation
Angle measured upward from the horizontal.

line of sight
The straight line from an observer to an object.

MIXED PRACTICE · CHAPTER 7

Connecting the ideas

FLUENCY Core procedures

1. Triangles ABC and DEF are similar, with $AB = 6$, $BC = 9$, $AC = 12$ and $DE = 8$. Find EF and DF. F
2. Find the hypotenuse of a right triangle with legs 7 cm and 24 cm. F
3. A right triangle has an angle of $42°$ and hypotenuse 15 cm. Find the side opposite the $42°$ angle, to one decimal place. F
4. A 1.8 m person casts a 2.4 m shadow. At the same time, a vertical pole casts a 6.0 m shadow. Find the height of the pole. F

REASONING Justify and connect

1. Two similar cylinders have heights 6 cm and 15 cm. Find the ratio of their curved surface areas and the ratio of their volumes. R
2. Explain why the angle of elevation from A to B equals the angle of depression from B to A. Name the angle relationship you use. R
3. A triangle has sides 9 cm, 12 cm and 15 cm. A second has sides 9 cm, 12 cm and 17 cm. Decide which is right-angled and justify both decisions. R
4. A student calculates $\sin\theta = 1.4$ for an acute angle of a right triangle. Explain why this is impossible. R

APPLICATION Use and evaluate

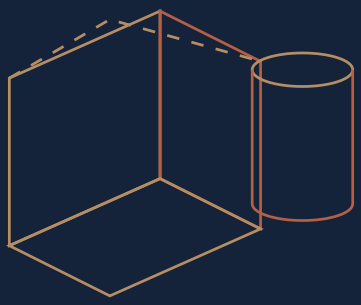
1. A 6 m ladder reaches 5.4 m up a vertical wall. Find the angle it makes with the ground and the distance of its foot from the wall. P
2. From the top of a 40 m tower, the angles of depression of two cars on the same straight road are $28°$ and $42°$. Find the distance between the cars. P
3. A map has scale $1 : 20\,000$. A lake has area 6.5 cm^2 on the map. Find its real area in square kilometres. P
4. A drone hovers 120 m above level ground. The angle of depression of a marker on the ground is $38°$. Find the horizontal distance to the marker and the direct distance from the drone, each to one decimal place. P
5. A ramp must rise 1.1 m at a gradient no steeper than $1 : 12$. Find the minimum horizontal run and the corresponding sloping length. If only 12 m of run is available, show that the design is not compliant and state the shortfall. E
6. A tree is estimated from a 9.6 m shadow. If the shadow is actually 0.3 m shorter, by how much does the estimated height change? Compare the reliability of the shadow method with the angle method, and state one condition under which each fails. E

Check your progress

I CAN …	CONFIDENCE	EVIDENCE	NEXT ACTION
Find and use a scale factor for lengths, areas and volumes.	1 / 2 / 3		
Prove triangles similar and use the correspondence to find a missing side.	1 / 2 / 3		
Apply Pythagoras and its converse, including coordinate distance.	1 / 2 / 3		
Choose and apply sine, cosine or tangent to find a side or an angle.	1 / 2 / 3		
Solve elevation and depression problems and evaluate the assumptions.	1 / 2 / 3		

Reflection

An indirect measurement produces a single number that carries the uncertainty of every input. When should a surveyor report 102 *m, and when should the report say "between* 100 *m and* 104 *m"? How does the purpose of the measurement decide?*



8

GEOMETRY, PROOF AND SPACE

Three-dimensional solids and measurement

A carton, a can and a water tank are all answers to the same question: how much space must be enclosed, and how much material will enclose it? Unfold the surface, slice the solid, fill it with layers, and the answer becomes a calculation.

KEY CONCEPT · FORM | RELATED · MEASUREMENT | RELATED · MODELS | CONTEXT · GLOBALIZATION AND SUSTAINABILITY

INQUIRY

How much material does it take to hold a given volume — and how precisely can we know?

ATL · SELF-MANAGEMENT AND CRITICAL THINKING

A volume problem is rarely one calculation. It is a sequence: choose a model, convert every measurement to one unit, decompose the solid, calculate, check that the answer is plausible, then decide how many figures to report. Organise the work so that a wrong unit or a forgotten face cannot hide inside a long chain of arithmetic.

MYP UNIT PLANNER

Framing the inquiry

UNIT 8 · APPROX. 16 HOURS
ATL FOCUS · SELF-MANAGEMENT

KEY AND RELATED CONCEPTS

Form. A solid is defined by its faces, edges and vertices. A net, a set of views or a cross-section is another representation of the same form, chosen to make a measurement visible.

Measurement. Surface area measures the boundary in square units; volume measures enclosed space in cubic units. Both depend on length, but differently, and must be reported in matching units.

Models and change. A formula models an ideal solid; real cans have seams, rims and varied wall thickness. Scaling changes area and volume at different rates, so large containers use less material per litre.

GLOBAL CONTEXT AND EXPLORATION

Globalization and sustainability. Exploration: packaging and material use — how the shape of a container affects the material consumed, the energy needed to transport it, and the waste left behind.

A drink can is a compromise. A taller, thinner can uses less metal for the same volume, but it may be less stable, harder to hold and more expensive to ship. Mathematical optimisation does not make the decision; it makes the trade-offs visible and quantifiable.

STATEMENT OF INQUIRY

The form of a solid determines both its capacity and its material cost; by modelling solids as nets, layers and measurable parts, we can compare packaging designs and justify choices about efficiency and sustainability.

FACTUAL

What are the faces, edges and vertices of a polyhedron? What is a net? Which formulas measure prisms, cylinders, pyramids, cones and spheres?

CONCEPTUAL

How does decomposing a solid make its enclosed space measurable? Why does doubling every length multiply area by four but volume by eight? Why can equal volumes need different amounts of material?

DEBATABLE

Should a manufacturer minimise material even when another shape sells better or protects the product? How precise should a stated capacity be when every measurement is uncertain?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii	Identify solids and their faces, edges and vertices; interpret nets and views; apply area and volume formulas.
B · Investigating patterns	i, ii, iii	Measure cans, tabulate surface area against radius for a fixed volume, conjecture the most efficient proportion and test it.
C · Communicating	i, ii, iii, iv	Draw and label nets, plans and elevations; present multi-stage calculations with units; explain rounding and tolerance.
D · Applying mathematics	i, ii, iii, iv, v	Specify and cost a water tank, test a packaging claim, and evaluate a design against structural and sustainability constraints.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** the language of solids: faces, edges, vertices, nets, plans and elevations.
- **Know** and use Euler's relationship for convex polyhedra.
- **Be able to** calculate the surface area of prisms, cylinders, pyramids and cones.
- **Be able to** calculate the volume of prisms, cylinders, pyramids, cones and spheres.
- **Be able to** decompose composite solids and apply the correct scale relationships for similar solids.
- **Be able to** work with capacity, density, tolerance and bounds, and report a measurement with appropriate precision.

SUCCESS CRITERIA

- ☐ I can classify a solid and count its faces, edges and vertices without double-counting.
- ☐ I can draw or interpret a net and use it to find a surface area.
- ☐ I can read a plan and elevations and use them to determine a solid's shape and volume.
- ☐ I can choose the correct formula for surface area or volume and keep the units consistent.
- ☐ I can find the volume of a composite solid by adding and subtracting standard parts.
- ☐ I can express a measurement's uncertainty as a range and use it to judge whether a design meets a specification.

PREREQUISITE CHECK · 12 MINUTES, NO CALCULATOR

1. Find the area of a rectangle 8 cm by 5 cm and of a triangle with base 8 cm and height 5 cm.
2. Find the circumference and area of a circle of radius 7 cm, using $\pi \approx \frac{22}{7}$.
3. Calculate 2^2, 2^3, 3^2 and 3^3.
4. Convert 1.5 m to centimetres and 2500 cm^3 to litres.
5. In a right triangle with legs 6 cm and 8 cm, find the hypotenuse.
6. Round 3.14159 to two decimal places and to three significant figures.

Answers are in the Chapter 8 answer section. If more than two are incorrect, review area, circle measurement, unit conversion and Pythagoras before continuing.

BEFORE YEAR 3

Area and perimeter of rectangles, triangles and circles; angle facts; Pythagoras' theorem; metric units of length and capacity; rounding to decimal places.

YEAR 3 · THIS CHAPTER

Nets, views and Euler's relationship; surface area and volume of standard solids; composite solids; similar solids; density, tolerance and bounds.

LATER

Surface area and volume by integration; three-dimensional coordinate geometry; optimisation with calculus; statistical process control and measurement uncertainty.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Containers are designed to hold a volume using as little material as practical.	For a fixed volume, which shape uses the least material, and what other constraints matter?	Nets, surface area, volume, scaling and measurement precision.	Measure cans, compute the material-to-volume ratio and search for the most efficient proportion.	Draw the net, label the dimensions, keep units consistent and justify the rounding.	When should a designer choose the mathematically efficient shape, and when should another constraint win?

8.1 Solids, nets and views

CRITERION A · I, II ◆ CRITERION C · I, II ◆ FORM

A solid is a three-dimensional region, but its surface can be described by counting and naming its parts. Those counts are not bookkeeping: for a whole family of solids they obey a fixed relationship, and that relationship is a first example of a structural theorem.

DEFINITIONS

A **polyhedron** is a closed solid whose faces are polygons. A **face** is a flat polygonal surface, an **edge** is a segment where two faces meet, and a **vertex** is a point where edges meet. A **prism** has two congruent parallel bases joined by rectangular faces; a **pyramid** has one polygonal base and triangular faces meeting at an apex. Cylinders, cones and spheres have curved surfaces, so they are not polyhedra. A **net** is a flat arrangement of faces that folds into a solid. A **plan** is the view from directly above; an **elevation** is a view from the front or the side, looking horizontally.

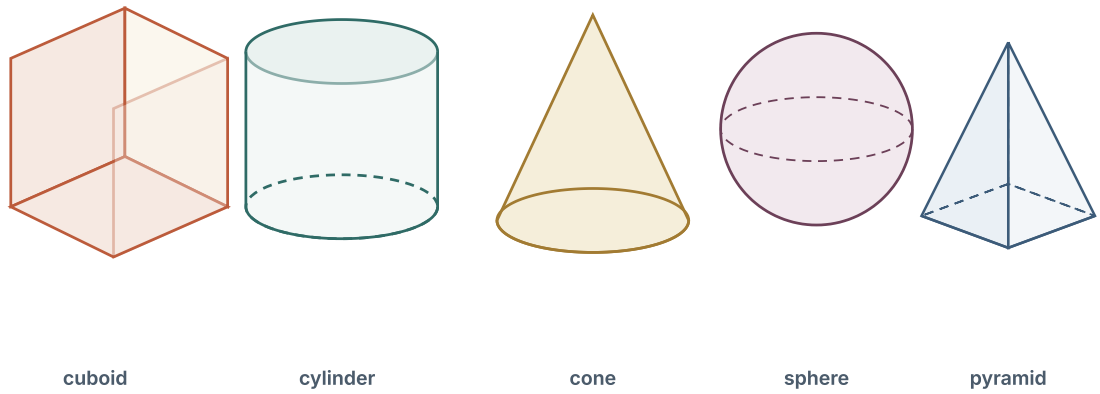


Figure 8.1 Five standard solids. The cuboid and pyramid are polyhedra; the cylinder, cone and sphere have curved surfaces.

The names of a prism and a pyramid come from their bases. A triangular prism has triangular bases; a pentagonal pyramid has a pentagonal base. The orientation of the solid does not change its name: a triangular prism standing on a rectangular face is still a triangular prism, and its triangular faces are still the bases used in the volume formula.

FACES, EDGES AND VERTICES OF FIVE COMMON POLYHEDRA

Solid	Faces F	Edges E	Vertices V
Cuboid	6	12	8
Triangular prism	5	9	6
Square pyramid	5	8	5
Tetrahedron	4	6	4
Pentagonal prism	7	15	10

EULER'S RELATIONSHIP

For any convex polyhedron, $F + V = E + 2$. Check the table: the cuboid gives $6 + 8 = 14$ and $12 + 2 = 14$; the pentagonal prism gives $7 + 10 = 17$ and $15 + 2 = 17$. The relationship does not apply to a cylinder or a sphere, because neither is a polyhedron: a curved surface is not a polygonal face and a circular rim is not a straight edge.

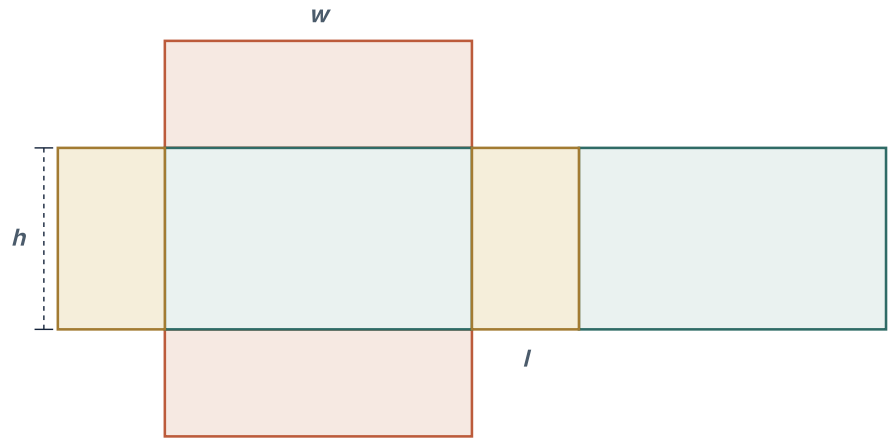


Figure 8.2 A cuboid unfolds into three pairs of congruent rectangles. The area of the net is the surface area of the solid.

A net converts a surface-area problem into a two-dimensional area problem. Every face appears exactly once, and opposite faces appear as congruent pairs. A cube has eleven different nets up to rotation and reflection; all of them contain the same six squares. A net is also a manufacturing document: a cardboard carton is cut from a flat net, folded along the edges and glued along one seam.

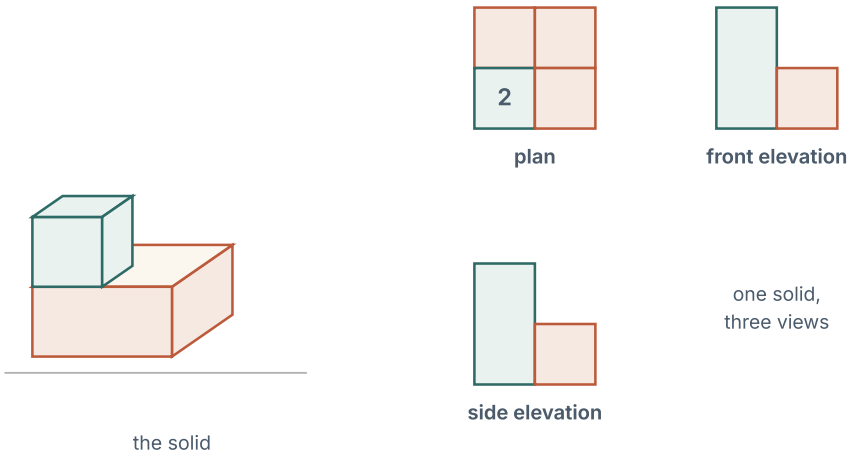


Figure 8.3 One solid, three orthogonal views. The plan gives the footprint; the elevations give the heights.

Plans and elevations are orthogonal projections: each view flattens the solid along one direction. A plan answers “what is directly below?”; a front elevation answers “what do you see from the front?”. No single view is enough. In Figure 8.3 the plan shows four unit squares, but the solid contains five cubes: the marked square is two cubes high, and only the elevations reveal that.

WORKED EXAMPLE 8.1.1 COUNTING A PRISM

PROBLEM

A prism has a regular 12-sided polygon as its base. State the number of faces, edges and vertices, and verify Euler's relationship.

1. The two bases are faces, and each of the 12 sides of the base produces one rectangular lateral face: $F = 12 + 2 = 14$.
2. Each of the 12 base vertices is repeated at the other end: $V = 2 \times 12 = 24$.
3. Each base contributes 12 edges, and there are 12 lateral edges joining corresponding vertices: $E = 12 + 12 + 12 = 36$.
4. Check: $F + V = 14 + 24 = 38$ and $E + 2 = 36 + 2 = 38$. The relationship holds.

ANSWER $F = 14$, $E = 36$, $V = 24$; $14 + 24 = 36 + 2$.

WORKED EXAMPLE 8.1.2 FROM NET TO SOLID

PROBLEM

A net consists of two 12 cm by 8 cm rectangles, two 12 cm by 5 cm rectangles and two 8 cm by 5 cm rectangles. Identify the solid and find the area of the net.

1. Opposite faces of a cuboid are congruent, and there are three pairs of dimensions. The pairs 12×8, 12×5 and 8×5 are exactly the three face types of a cuboid 12 cm by 8 cm by 5 cm.
2. Area of one pair: $2(12 \times 8) = 192$ cm^2.
3. Area of the second pair: $2(12 \times 5) = 120$ cm^2.
4. Area of the third pair: $2(8 \times 5) = 80$ cm^2.
5. Total: $192 + 120 + 80 = 392$ cm^2.

ANSWER **A 12 cm by 8 cm by 5 cm cuboid; net area 392 cm^2.**

WORKED EXAMPLE 8.1.3 READING THREE VIEWS

PROBLEM

The solid in Figure 8.3 is built from unit cubes. Use the plan and elevations to find the number of cubes, the volume and the heights of the columns.

1. The plan shows a 2×2 footprint, so there are four columns of cubes.
2. The plan marks one square with "2", meaning that column is two cubes high; the other three columns are one cube high.
3. Number of cubes: $2 + 1 + 1 + 1 = 5$.
4. Each unit cube has volume 1 cm^3, so the total volume is 5 cm^3.
5. The front elevation shows heights 2 and 1 from left to right; the side elevation shows the same profile because the tall column is at a front corner. Without the elevations, the plan alone would allow anything from four to eight cubes.

ANSWER **Five cubes; volume 5 cm^3; column heights 2, 1, 1, 1.**

COMMON ERROR · COUNTING WHAT YOU CANNOT SEE, OR MISSING WHAT YOU CAN

Count each face, edge and vertex once. On a cuboid, 12 edges is not "4 top + 4 bottom + 4 more"; it is 4 top, 4 bottom and 4 verticals. On a net, faces must not overlap when folded, and a face shared by two parts of the net is one face, not two. In views, hidden edges are usually drawn dashed, but a dashed line is still an edge of the solid.

PRACTICE 8.1 · SOLIDS, NETS AND VIEWS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Complete the table for a triangular prism, a square pyramid, a tetrahedron and a pentagonal prism: faces, edges and vertices. F
2. A convex polyhedron has 9 faces and 14 edges. Find the number of vertices. F
3. A convex polyhedron has 8 vertices and 12 edges. Find the number of faces and name a solid with these counts. F
4. Sketch a net for a cube of side 4 cm and state the total area of the net. F
5. Explain why Euler's relationship does not apply to a sphere, and why a sphere cannot be made from a net of flat polygons without stretching. R
6. A prism has 30 edges. How many sides does its base have? Justify your answer. R
7. An open box, with no lid, measures 20 cm by 15 cm by 10 cm. Find the area of cardboard needed, ignoring overlaps. P
8. A 4 cm by 3 cm by 2 cm cuboid is painted on all faces and then cut into unit cubes. How many unit cubes have paint on exactly two faces? Explain your counting. E

8.2 Surface area of prisms and cylinders

CRITERION A · II ◆ CRITERION C · II, III ◆ MEASUREMENT

Surface area is the total area of the boundary. For a prism, the boundary consists of two congruent bases and a set of rectangles; for a cylinder, the rectangles become one curved rectangle whose width is the circumference.

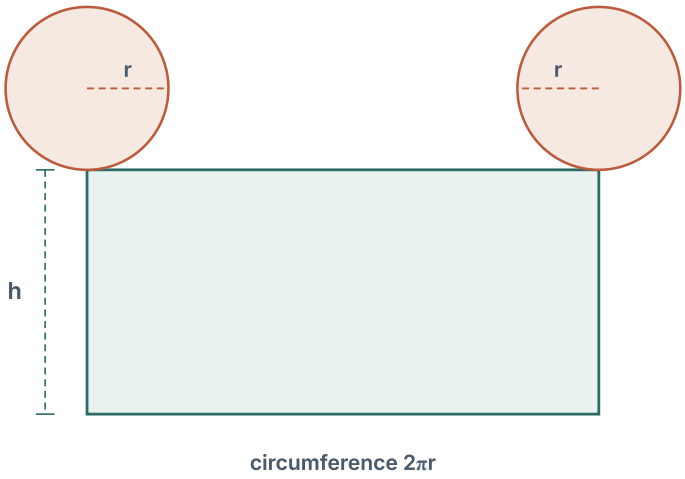


Figure 8.4 The curved surface unrolls into a rectangle. Its width is the circumference $2\pi r$, not the diameter.

FORMULAS · PRISMS AND CYLINDERS

For a prism with base area B and base perimeter P, the lateral surface area is Ph and the total surface area is $SA = 2B + Ph$. For a cylinder of radius r and height h, the curved surface area is $2\pi rh$ and the total surface area is $SA = 2\pi r^2 + 2\pi rh$. The **lateral surface** excludes the bases; the total surface includes them.

The formula $SA = 2B + Ph$ works for every prism, including triangular and trapezoidal prisms, because the lateral faces always form a rectangle of height h and total width P. The formula for a cylinder is the same idea with a circular base: $B = \pi r^2$ and $P = 2\pi r$.

WORKED EXAMPLE 8.2.1 — A CUBOID

PROBLEM

Find the total surface area of a cuboid measuring 15 cm by 9 cm by 6 cm.

1. The three pairs of faces have areas $15 \times 9 = 135$, $15 \times 6 = 90$ and $9 \times 6 = 54$ cm^2.
2. Each pair occurs twice: $2(135 + 90 + 54)$.
3. Inside the bracket: $135 + 90 + 54 = 279$.
4. $SA = 2 \times 279 = 558$ cm^2.

ANSWER 558 **cm^2.**

WORKED EXAMPLE 8.2.2 — A TRIANGULAR PRISM

PROBLEM

A triangular prism has a right-angled triangular base with legs 6 cm and 8 cm and hypotenuse 10 cm. Its length is 15 cm. Find its total surface area.

1. Base area: $B = \frac{1}{2} \times 6 \times 8 = 24$ cm^2.
2. Base perimeter: $P = 6 + 8 + 10 = 24$ cm.
3. Lateral area: $Ph = 24 \times 15 = 360$ cm^2. This is the sum of the three rectangular faces: 6×15, 8×15 and 10×15.
4. Total: $SA = 2(24) + 360 = 48 + 360 = 408$ cm^2.

ANSWER 408 **cm^2.**

WORKED EXAMPLE 8.2.3 — A CYLINDER, CLOSED AND OPEN

PROBLEM

A cylinder has radius 7 cm and height 10 cm. (a) Find its total surface area. (b) Find its surface area if the top is removed. Use $\pi \approx 3.142$.

1. Two bases: $2\pi r^2 = 2\pi(49) = 98\pi$ cm^2.
2. Curved surface: $2\pi rh = 2\pi(7)(10) = 140\pi$ cm^2.
3. (a) Total $= 98\pi + 140\pi = 238\pi \approx 747.7$ cm^2.
4. (b) Remove one base: $238\pi - 49\pi = 189\pi \approx 593.8$ cm^2. The difference, $49\pi \approx 153.9$ cm^2, is the area of the missing disc.

ANSWER **(a)** 747.7 **cm^2 (b)** 593.8 **cm^2.**

COMMON ERROR · TWO BASES, ONE RADIUS

The commonest omission is a base: a closed cylinder has two circular faces, and a closed prism has two polygonal bases. The commonest substitution error is using the diameter in place of the radius: with diameter 10 cm and height 4 cm, the curved area is $2\pi(5)(4) = 40\pi$ cm², not $2\pi(10)(4) = 80\pi$ cm². Keep units squared: a surface area is never in cm³.

PRACTICE 8.2 · SURFACE AREA OF PRISMS AND CYLINDERS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Find the total surface area of a cuboid 10 cm by 4 cm by 3 cm. F
2. Find the total surface area of a cylinder with radius 5 cm and height 12 cm, to one decimal place. F
3. Find the surface area of a cube of side 9 cm. F
4. A right triangular prism has base sides 5 cm, 12 cm and 13 cm and length 20 cm. Find its total surface area. F
5. An open-topped cylinder has radius 3.5 cm and height 8 cm. Find the area of material needed, to one decimal place. R
6. A cube has total surface area 294 cm². Find its side length and its volume. P
7. A closed cylinder has radius 5 cm and total surface area 150π cm². Find its height. P
8. A closed cylinder of radius r and height h has total surface area equal to three times its curved surface area. Find h in terms of r. E

8.3 Surface area of pyramids and cones

CRITERION A · II ◆ CRITERION C · II, III ◆ SPACE

A pyramid has triangular lateral faces; a cone has a curved lateral surface that unrolls into a sector. In both cases the lateral area depends on the *slant* height rather than the vertical height, because the surface is reached by travelling up the sloping side.

FORMULAS · PYRAMIDS AND CONES

For a pyramid with base area B, base perimeter P and slant height l, the lateral surface area is $\frac{1}{2}Pl$ and the total surface area is $SA = B + \frac{1}{2}Pl$. For a cone of radius r and slant height l, the curved surface area is πrl and the total surface area is $SA = \pi r^2 + \pi rl$. The **slant height** is the distance from the apex to a point on the base edge; for a right cone, $l = \sqrt{r^2 + h^2}$.

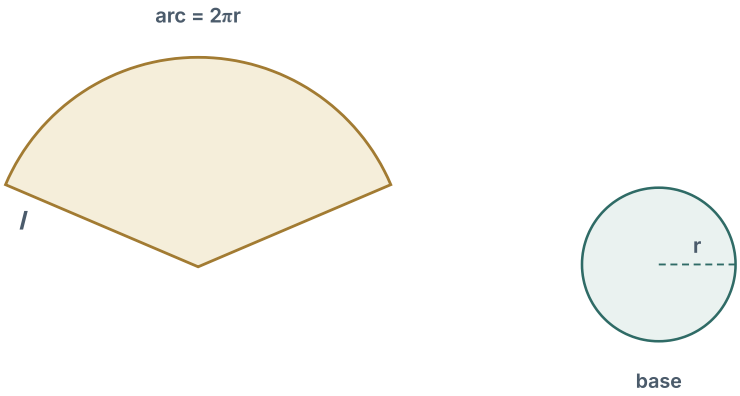


Figure 8.5 A cone unrolls into a sector of radius l, the slant height, plus one circular base. The sector's arc length equals the base circumference, $2\pi r$.

The sector area formula explains the cone result. A sector of radius l and arc length $2\pi r$ has area $\frac{1}{2} \times \text{arc} \times \text{radius} = \frac{1}{2}(2\pi r)l = \pi r l$. For a pyramid, each triangular face has area $\frac{1}{2} \times \text{base edge} \times l$; summing over all edges gives $\frac{1}{2}Pl$.

WORKED EXAMPLE 8.3.1 A SQUARE PYRAMID

PROBLEM

A square pyramid has base side 10 cm and slant height 13 cm. Find its total surface area.

1. Base area: $B = 10^2 = 100$ cm^2.
2. Base perimeter: $P = 4 \times 10 = 40$ cm.
3. Lateral area: $\frac{1}{2}Pl = \frac{1}{2}(40)(13) = 260$ cm^2. This is four triangles, each of area $\frac{1}{2}(10)(13) = 65$ cm^2.
4. Total: $SA = 100 + 260 = 360$ cm^2.

ANSWER 360 **cm^2.**

WORKED EXAMPLE 8.3.2 A CONE FROM RADIUS AND HEIGHT

PROBLEM

A cone has radius 6 cm and vertical height 8 cm. Find its total surface area, to one decimal place.

1. The slant height is not given, so calculate it: $l = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ cm.
2. Base area: $\pi r^2 = 36\pi$ cm^2.
3. Curved area: $\pi r l = \pi(6)(10) = 60\pi$ cm^2.
4. Total: $96\pi \approx 301.6$ cm^2.

ANSWER $96\pi \approx 301.6$ **cm^2.**

WORKED EXAMPLE 8.3.3 A CONE FROM DIAMETER

PROBLEM

A cone has diameter 14 cm and slant height 25 cm. Find its total surface area, to one decimal place.

1. The radius is half the diameter: $r = 7$ cm.
2. Base area: $\pi(7^2) = 49\pi$ cm^2.
3. Curved area: $\pi(7)(25) = 175\pi$ cm^2.
4. Total: $224\pi \approx 703.7$ cm^2.

ANSWER $224\pi \approx 703.7$ **cm^2.**

COMMON ERROR · SLANT HEIGHT IS NOT VERTICAL HEIGHT

The surface of a cone or pyramid slopes, so its area uses the slant height l. Using the vertical height h underestimates every lateral area: with $r = 6$ and $h = 8$, the slant is 10, not 8, and the curved area is 60π, not 48π. The base is included only when the solid is closed. For a pyramid, the slant height is measured from the apex to the midpoint of a base edge for a regular pyramid; it is not the height of the solid.

PRACTICE 8.3 · PYRAMIDS AND CONES SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. A square pyramid has base side 8 cm and slant height 10 cm. Find its total surface area. F
2. A cone has radius 3 cm and slant height 7 cm. Find its total surface area, to one decimal place. F
3. A cone has radius 5 cm and vertical height 12 cm. Find its total surface area, to one decimal place. F
4. A square pyramid has base side 12 cm and vertical height 8 cm. Find its slant height and total surface area. R
5. A cone has diameter 16 cm and slant height 17 cm. Find its total surface area, to one decimal place. R
6. A cone has radius 6 cm and total surface area 90π cm^2. Find its slant height and vertical height. P
7. A square-based tent has base side 3 m and slant height 2.5 m. Canvas costs 12 units per square metre. Find the cost with and without a floor. P
8. A cone has radius and slant height in the ratio $3 : 5$ and total surface area 216π cm^2. Find its radius and slant height. E

8.4 Volume of prisms and cylinders

CRITERION A · II ◆ CRITERION D · I ◆ MEASUREMENT

Volume measures the space inside a solid. For a prism or cylinder, the solid can be built from identical layers: one base-shaped layer stacked h times. That is why the volume is the base area multiplied by the height, whatever the shape of the base.

FORMULAS · VOLUME AND CAPACITY

For a prism or cylinder with base area B and perpendicular height h, $V = Bh$. For a cylinder, $V = \pi r^2 h$. **Capacity** is the volume a container can hold, usually quoted in litres or millilitres. $1\ \text{cm}^3 = 1\ \text{mL}$; $1000\ \text{cm}^3 = 1\ \text{L}$; $1\ \text{m}^3 = 1000\ \text{L} = 1\,000\,000\ \text{cm}^3$.

The height in these formulas is perpendicular to the base, not the length of a sloping edge. A prism whose base is a trapezium still has volume Bh: the base is the trapezium, and the distance between the two parallel trapezium faces is h. For a cylinder, $B = \pi r^2$, so $V = \pi r^2 h$.

WORKED EXAMPLE 8.4.1 — A CUBOID AND ITS CAPACITY

PROBLEM

A tank is a cuboid measuring 25 cm by 18 cm by 12 cm. Find its volume in cm^3 and its capacity in litres.

1. $V = lwh = 25 \times 18 \times 12$.
2. $25 \times 18 = 450$, and $450 \times 12 = 5400$.
3. The volume is $5400\ \text{cm}^3$.
4. Since $1000\ \text{cm}^3 = 1\ \text{L}$, the capacity is $5400 \div 1000 = 5.4\ \text{L}$.

ANSWER $5400\ \textbf{cm}^3 = 5.4\ \textbf{L}$.

WORKED EXAMPLE 8.4.2 — A CYLINDRICAL CONTAINER

PROBLEM

A cylinder has radius 4.5 cm and height 12 cm. Find its volume and its capacity in litres, to three significant figures.

1. $V = \pi r^2 h = \pi(4.5^2)(12)$.
2. $4.5^2 = 20.25$, and $20.25 \times 12 = 243$.
3. $V = 243\pi \approx 763.4\ \text{cm}^3$.
4. Capacity: $763.4 \div 1000 \approx 0.763\ \text{L}$.
5. Check: the radius is about 4.5 cm, so the base is about $64\ \text{cm}^2$, and 64×12 is about $770\ \text{cm}^3$. The answer is plausible.

ANSWER $V \approx 763\ \textbf{cm}^3$**; capacity** $\approx 0.763\ \textbf{L}$.

WORKED EXAMPLE 8.4.3 — A TRAPEZOIDAL PRISM

PROBLEM

A prism has a trapezium as its base, with parallel sides 8 cm and 14 cm and perpendicular distance 6 cm between them. The prism is 20 cm long. Find its volume.

1. Base area: $B = \frac{1}{2}(8 + 14)(6) = \frac{1}{2}(22)(6) = 66\ \text{cm}^2$.
2. The length of the prism is the distance between the two trapezium faces, so $h = 20$ cm.
3. $V = Bh = 66 \times 20 = 1320\ \text{cm}^3$.
4. Capacity: $1320\ \text{cm}^3 = 1.32\ \text{L}$.

ANSWER $1320\ \textbf{cm}^3 = 1.32\ \textbf{L}$.

COMMON ERROR · MIXING LENGTH, AREA AND VOLUME UNITS

Convert every dimension to the same unit before multiplying. $1\text{ m} = 100\text{ cm}$, but $1\text{ m}^2 = 10\,000\text{ cm}^2$ and $1\text{ m}^3 = 1\,000\,000\text{ cm}^3$, because each length factor is applied three times. A volume in cm^3 becomes litres by dividing by 1000, not by 100. Do not use a slant height as the height of a prism or cylinder; if the height is not perpendicular to the base, the simple formula does not apply.

PRACTICE 8.4 · VOLUME OF PRISMS AND CYLINDERS SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Find the volume of a cuboid 9 cm by 7 cm by 4 cm. F
2. Find the volume of a cylinder with radius 6 cm and height 10 cm, to one decimal place. F
3. Find the volume of a cylinder with diameter 10 cm and height 14 cm, to one decimal place. F
4. Convert 3.2 m^3 to litres. F
5. A prism has a triangular base with base 10 cm and height 6 cm, and a length of 25 cm. Find its volume. R
6. A tank measures 1.2 m by 80 cm by 60 cm. Find its capacity in litres. R
7. A cylinder must hold 2 L and has radius 5 cm. Find its height, to one decimal place. P
8. A cuboid 20 cm by 10 cm by 5 cm is melted and recast as a cylinder of radius 5 cm. Find the height of the cylinder, to one decimal place. E

8.5 Pyramids, cones, spheres and composite solids

CRITERION A · II, III ◆ CRITERION C · III ◆ CHANGE

A pyramid or cone tapers to a point, so it fills exactly one third of the prism or cylinder with the same base and height. A sphere has no base at all, and its formulas bring both π and a cube of the radius. Composite solids combine these standard parts — sometimes by addition, sometimes by subtraction.

FORMULAS · TAPERING SOLIDS AND SPHERES

For a pyramid or cone with base area B and perpendicular height h: $V = \frac{1}{3}Bh$. For a cone, $V = \frac{1}{3}\pi r^2 h$. For a sphere of radius r: $V = \frac{4}{3}\pi r^3$ and $SA = 4\pi r^2$. A hemisphere has half the volume and half the curved surface area of the sphere, but its flat circular face is an extra surface if the hemisphere is closed.

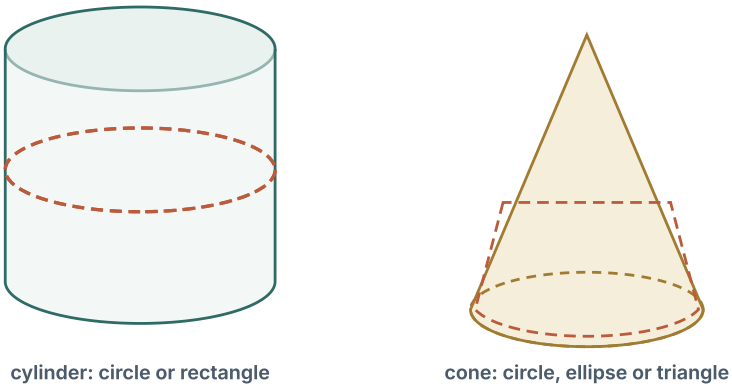


Figure 8.6 A plane cut exposes a cross-section. Slices of the same solid can be circles, rectangles or triangles depending on the direction of the cut.

Cross-sections explain why the volume formulas work. A prism has the same cross-section at every height, so its volume is the cross-sectional area times the height. A pyramid or cone shrinks steadily towards the apex; averaging the base area and the zero area at the top gives the factor $\frac{1}{3}$. A sphere can be sliced into thin discs whose areas follow a curved pattern, which is where the factor $\frac{4}{3}$ comes from.

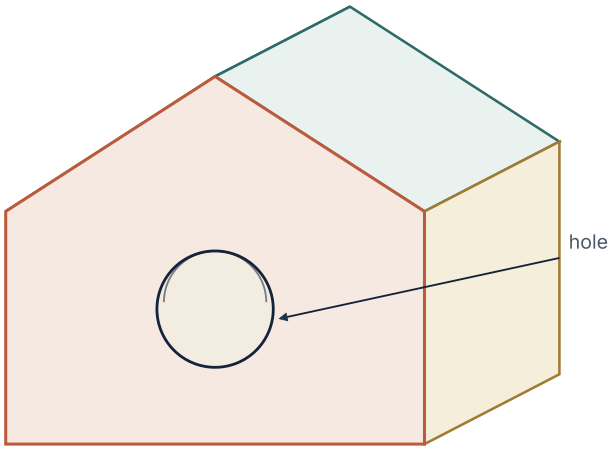


Figure 8.7 A composite solid can be decomposed into a rectangular prism and a triangular prism, then corrected for the cylindrical hole.

WORKED EXAMPLE 8.5.1 — A SQUARE PYRAMID

PROBLEM

Find the volume of a square pyramid with base side 9 cm and vertical height 7 cm.

1. Base area: $B = 9^2 = 81$ cm^2.
2. $V = \frac{1}{3}Bh = \frac{1}{3}(81)(7)$.
3. $81 \div 3 = 27$, and $27 \times 7 = 189$.
4. The volume is 189 cm^3. It is one third of the 567 cm^3 cuboid with the same base and height.

ANSWER 189 **cm^3.**

WORKED EXAMPLE 8.5.2 — A CONE

PROBLEM

A cone has radius 5 cm and vertical height 12 cm. Find its volume, to one decimal place.

1. Base area: $B = \pi(5^2) = 25\pi$ cm^2.
2. $V = \frac{1}{3}(25\pi)(12) = 100\pi$ cm^3.
3. $100\pi \approx 314.2$ cm^3.
4. Check: the enclosing cylinder would have volume $300\pi \approx 942.5$ cm^3, and one third of that is 100π.

ANSWER $100\pi \approx 314.2$ **cm^3.**

WORKED EXAMPLE 8.5.3 — A SPHERE

PROBLEM

A sphere has radius 6 cm. Find its volume and surface area, to one decimal place.

1. Volume: $V = \frac{4}{3}\pi(6^3) = \frac{4}{3}\pi(216) = 288\pi \approx 904.8$ cm^3.
2. Surface area: $SA = 4\pi(6^2) = 144\pi \approx 452.4$ cm^2.
3. Units check: volume in cm^3, surface area in cm^2.
4. Order check: 904.8 cm^3 is roughly the volume of a 10 cm cube, which is plausible for a sphere of diameter 12 cm.

ANSWER $V \approx 904.8$ **cm^3;** $SA \approx 452.4$ **cm^2.**

WORKED EXAMPLE 8.5.4 A COMPOSITE SOLID

PROBLEM

A solid is a rectangular prism 10 cm by 8 cm by 6 cm with a triangular-prism roof of base 10 cm and height 4 cm. A cylindrical hole of radius 1.5 cm is drilled through the 6 cm depth. Find the volume, to one decimal place.

1. Rectangular prism: $10 \times 8 \times 6 = 480$ cm^3.
2. Triangular prism roof: base area $\frac{1}{2}(10)(4) = 20$ cm^2, depth 6 cm, so $V = 20 \times 6 = 120$ cm^3.
3. Total before drilling: $480 + 120 = 600$ cm^3.
4. Cylindrical hole: $\pi(1.5^2)(6) = 13.5\pi \approx 42.4$ cm^3.
5. Remaining volume: $600 - 42.4 = 557.6$ cm^3.

ANSWER $600 - 13.5\pi \approx 557.6$ **cm^3.**

WORKED EXAMPLE 8.5.5 SIMILAR SOLIDS

PROBLEM

Two similar solid cylinders have volumes 64 cm^3 and 216 cm^3. The smaller has surface area 80 cm^2. Find the scale factor, the surface-area ratio and the surface area of the larger cylinder.

1. For similar solids, the volume ratio is the cube of the length scale factor: $k^3 = \frac{216}{64} = \frac{27}{8}$.
2. $k = \sqrt[3]{\frac{27}{8}} = \frac{3}{2} = 1.5$.
3. The surface-area ratio is $k^2 = \left(\frac{3}{2}\right)^2 = \frac{9}{4} = 2.25$.
4. Larger surface area: $80 \times 2.25 = 180$ cm^2.
5. Check: linear dimensions grow by 1.5, areas by 2.25, volumes by 3.375; $64 \times 3.375 = 216$.

ANSWER $k = 1.5$**; area ratio** $9 : 4$**; larger surface area** 180 **cm^2.**

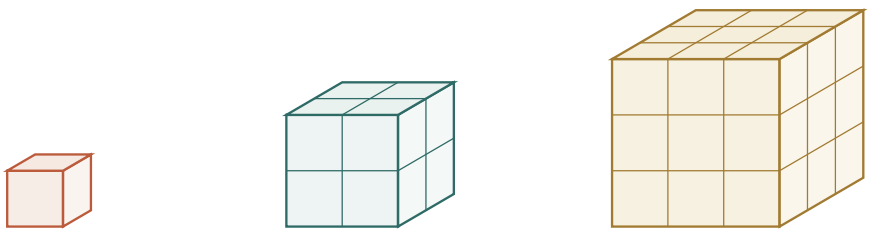


Figure 8.8 Triple the side and the volume becomes $3^3 = 27$ times larger. Surface area grows only $3^2 = 9$ times.

COMMON ERROR · THE MISSING THIRD, AND THE FACES THAT DISAPPEAR

A pyramid or cone is one third of its enclosing prism or cylinder; forgetting $\frac{1}{3}$ overstates the volume by a factor of three. For a sphere, use the radius, not the diameter: a sphere of diameter 10 has radius 5, and its volume is $\frac{500}{3}\pi$, not $\frac{4000}{3}\pi$. In a composite solid, do not count the faces where two parts are glued: they are internal, not surface. When a hole is drilled, subtract its volume but add back its curved surface if the hole is open.

PRACTICE 8.5 · VOLUME AND COMPOSITE SOLIDS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Find the volume of a square pyramid with base side 6 cm and height 10 cm. F
2. Find the volume of a cone with radius 3 cm and height 7 cm, to one decimal place. F
3. Find the volume of a sphere of radius 3 cm, to one decimal place. F
4. Find the volume of a hemisphere of radius 6 cm, to one decimal place. F
5. A cone has diameter 10 cm and height 12 cm. Find its volume, to one decimal place. R
6. Two similar spheres have radii 4 cm and 10 cm. Find the ratios of their surface areas and volumes. R
7. A tank is a cylinder of radius 4 cm and height 15 cm with a hemispherical top. Find its total volume, to one decimal place. P
8. A cone of radius 6 cm and height 8 cm is cut by a plane parallel to its base, halfway up. Find the volume of the small cone removed and the volume of the remaining frustum. E

8.6 Measurement, density and precision

CRITERION C · III, IV ◆ CRITERION D · III, IV ◆ APPROXIMATION

Every measurement is an interval, not a point. A length quoted as 8 cm “to the nearest centimetre” could be anywhere from 7.5 cm to 8.5 cm. When measurements are multiplied, those intervals combine, and a volume can be noticeably less certain than any single length.

DEFINITIONS · DENSITY AND UNCERTAINTY

Density is mass per unit volume: $\rho = \frac{m}{V}$, commonly in g/cm^3 or kg/m^3. **Significant figures** are the digits that carry meaning in a measurement. The **percentage error** of an estimate is $\frac{|\text{measured} - \text{true}|}{\text{true}} \times 100\%$. If a measurement is given to the nearest unit, its **lower bound** is the value minus half a unit and its **upper bound** is the value plus half a unit; the interval is called the **tolerance**.

Density links a solid's geometry to its material. A block of volume 200 cm^3 and mass 540 g has density 2.7 g/cm^3, which identifies aluminium. Rearranging gives $m = \rho V$ for the mass of a known volume, or $V = \frac{m}{\rho}$ for the volume of a known mass. Water has density approximately 1 g/cm^3, so 1 L of water has mass about 1 kg.

WORKED EXAMPLE 8.6.1 — CONVERTING CUBIC UNITS

PROBLEM

Convert $0.75\ \text{m}^3$ to litres and to cubic centimetres.

1. $1\ \text{m}^3 = 1000\ \text{L}$, so $0.75 \times 1000 = 750\ \text{L}$.
2. $1\ \text{m}^3 = 100\ \text{cm} \times 100\ \text{cm} \times 100\ \text{cm} = 1\,000\,000\ \text{cm}^3$.
3. $0.75 \times 1\,000\,000 = 750\,000\ \text{cm}^3$.
4. Check: $750\ \text{L} = 750 \times 1000\ \text{cm}^3 = 750\,000\ \text{cm}^3$. The two conversions agree.

ANSWER $750\ \mathbf{L} = 750\,000\ \mathbf{cm^3}$.

WORKED EXAMPLE 8.6.2 — IDENTIFYING A METAL BY DENSITY

PROBLEM

A rectangular metal block measures 4 cm by 5 cm by 10 cm and has mass 540 g. Find its density and suggest a likely metal.

1. Volume: $4 \times 5 \times 10 = 200\ \text{cm}^3$.
2. Density: $\rho = \frac{540}{200} = 2.7\ \text{g/cm}^3$.
3. Aluminium has density about $2.7\ \text{g/cm}^3$, so the block is likely aluminium.
4. Check: $2.7 \times 200 = 540$ g, as given.

ANSWER $2.7\ \mathbf{g/cm^3}$**; consistent with aluminium.**

WORKED EXAMPLE 8.6.3 — PERCENTAGE ERROR

PROBLEM

A student measures a length as 2.48 m. The accepted value is 2.50 m. Find the absolute error and the percentage error.

1. Absolute error: $|2.48 - 2.50| = 0.02$ m.
2. Percentage error: $\frac{0.02}{2.50} \times 100\%$.
3. $\frac{0.02}{2.50} = 0.008$, so the percentage error is 0.8%.
4. The measurement is within one percent of the accepted value; whether that is acceptable depends on the instrument and the purpose.

ANSWER $0.02\ \mathbf{m}$**;** 0.8%.

WORKED EXAMPLE 8.6.4 BOUNDS OF A VOLUME

PROBLEM

A cuboid is measured as 20 cm by 15 cm by 10 cm, each to the nearest centimetre. Find the greatest and least possible volumes, and compare them with the volume calculated from the stated dimensions.

1. Each dimension lies in an interval of width 1 cm centred on its stated value: $19.5 \le l < 20.5$, $14.5 \le w < 15.5$, $9.5 \le h < 10.5$.
2. Stated volume: $20 \times 15 \times 10 = 3000$ cm^3.
3. Upper bound: $20.5 \times 15.5 \times 10.5 = 3336.375$ cm^3.
4. Lower bound: $19.5 \times 14.5 \times 9.5 = 2686.125$ cm^3.
5. The true volume lies between about 2686 cm^3 and 3336 cm^3. The stated 3000 cm^3 is near the middle, but the interval is about $\pm 11\%$, much wider than any single ± 0.5 cm error.

ANSWER **Between 2686 cm^3 and 3336 cm^3; about $2686 \le V < 3336$.**

COMMON ERROR · MULTIPLYING THE UNIT ERROR AS WELL AS THE MEASUREMENT

If a length is measured to the nearest centimetre, the possible error is 0.5 cm, not 1 cm. In a product of three dimensions, the percentage errors combine approximately by addition, so three measurements each uncertain by about 2.5% can produce a volume uncertain by about 7.5%. Do not write 1 m^3 = 100 cm^3: the correct conversion is 1 m^3 = 1 000 000 cm^3. And keep density units consistent: g/cm^3 with grams and cm^3, or kg/m^3 with kilograms and metres, never a mixture.

PRACTICE 8.6 · MEASUREMENT AND PRECISION

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Convert 2.4 L to cm^3 and to m^3. [F]
2. Convert 5000 cm^3 to litres. [F]
3. A sample has mass 810 g and volume 300 cm^3. Find its density. [F]
4. Find the mass of 2.5 L of water, taking the density as 1 g/cm^3. [F]
5. A metal block measures 2 cm by 5 cm by 8 cm and has mass 624 g. Find its density. [R]
6. A length is estimated as 4.8 m; the true length is 5.0 m. Find the percentage error. [R]
7. A cylinder has radius 3.0 cm and height 10.0 cm, each measured to the nearest 0.1 cm. Find the greatest and least possible volumes, to one decimal place. [P]
8. A cube's volume is stated as 64 cm^3, with the side measured to the nearest centimetre. Find the range of possible side lengths and the range of possible volumes. [E]

Investigation · The 330 mL can

CRITERION B

A drinks can is a cylinder with a fixed volume. The manufacturer chooses its radius and height, and those choices determine how much metal the can requires. In this investigation you will measure real cans, calculate their material-to-volume ratio, and search for the proportion that uses the least material.

FIVE CYLINDERS DESIGNED TO HOLD 330 CM³, WITH HEIGHT ADJUSTED TO KEEP THE VOLUME FIXED

Radius r (cm)	Height h (cm)	$r \div h$	Surface area (cm²)	Area per cm³
3.0	11.67	0.257	276.5	0.838
3.4	9.09	0.374	266.8	0.808
3.7	7.67	0.482	264.4	0.801
4.0	6.57	0.609	265.5	0.805
4.4	5.43	0.811	271.6	0.823

1. **Measure.** Take three cylindrical cans. Measure the circumference with a strip of paper and divide by 2π to find the radius; measure the height with a ruler. Record both to the nearest millimetre, then calculate the volume and total surface area of each can.
2. **Tabulate.** For each can, calculate $r \div h$, the volume, the surface area and the ratio $\text{surface area} \div \text{volume}$. Keep the units consistent and record the calculator value before rounding.
3. **Pattern.** Compare the ratios. Which can uses the least material per millilitre? Does the most efficient can have a particular relationship between radius and height?
4. **Test.** Use the table above, which holds the volume at 330 cm³ while the radius changes. Plot surface area against radius, or compare consecutive rows. Estimate the radius that gives the smallest surface area and the corresponding height.
5. **Conjecture.** State a relationship between r and h at the minimum. Test it against the table: at the smallest area, is h approximately $2r$? Explain what the ratio $r \div h$ would be if $h = 2r$.
6. **Explain.** Write the surface area as $SA = 2\pi r^2 + \frac{2V}{r}$, using $h = \frac{V}{\pi r^2}$. Explain why a very small radius makes the second term large, and why a very large radius makes the first term large. This is why a minimum exists between the extremes.
7. **Apply.** Measure a real 330 mL can and compare its proportions with your optimum. Suggest three reasons why manufacturers might choose a shape that is not mathematically optimal.
8. **Evaluate.** List the assumptions in the model — uniform metal thickness, no seams, no dome or rim, identical cost per square centimetre. Choose the assumption that you think is most unrealistic and describe how removing it would change the conclusion.

A table of five values suggests a pattern; it does not prove that the minimum is exactly at $h = 2r$. The algebra in step 6 explains why a minimum must exist, and a finer table would locate it more precisely. Distinguish between numerical evidence, an algebraic model and a manufacturing decision.

Application · The rainwater tank

CRITERION D · I, II, III, IV ◆ MEASUREMENT ◆ SUSTAINABILITY

A school plans to install a cylindrical rainwater tank beside a classroom. The tank sits on a square concrete base and collects water from the roof. The business manager wants to know the capacity,

the load on the base, the cost of painting the exposed surface, and how long the tank will take to fill.

DESIGN SPECIFICATION FOR THE RAINWATER TANK

Quantity	Value	Note
Tank radius	1.20 m	measured externally
Tank height	2.00 m	flat top, flat base
Concrete base	3.00 m square, 0.15 m thick	tank stands at the centre
Roof catchment	180 m^2	horizontal projected area
Annual rainfall	750 mm	long-term average
Paint cost	8.50 per m^2	two coats included
Concrete cost	240 per m^3	supplied and poured
Inlet flow	12 L per minute	mains top-up

1. Calculate the volume of the tank in cubic metres and its capacity in litres. Use $\pi \approx 3.142$ and round your final answers sensibly.
2. The tank is full. Taking the density of water as 1 kg per litre, find the mass of the water and the load per square metre on the concrete base. Comment on whether the base is likely to need reinforcement.
3. The tank is painted on its curved side and its flat top, but not on the base, which rests on the concrete. Find the area to be painted and the cost at 8.50 per m^2.
4. Find the volume of concrete in the base and its cost. The base extends 0.60 m beyond the tank on every side; verify that the stated 3.00 m square is consistent with this.
5. The roof collects 750 mm of rain over 180 m^2 in a year. Convert the rainfall to metres, calculate the annual volume collected in cubic metres and in litres, and find how many full tanks this represents. Comment on the difference between annual harvest and storage capacity.
6. The tank is empty and is filled from the mains at 12 L per minute. How many minutes and hours will it take to fill? Give your answer to the nearest 0.1 hour.
7. The radius is measured to the nearest 0.05 m and the height to the nearest 0.05 m. Find the greatest and least possible volumes, and express the uncertainty as a percentage of the design volume. Which dimension contributes more to the uncertainty, and why?
8. Evaluate the design. Consider siting, overflow, maintenance, water quality, the weight of a full tank, and the choice between a cylindrical and a cuboid tank. Recommend one improvement and justify it with calculations or estimates.

CRITERION D FOCUS

Turn a specification into volume, mass, area and cost; carry units through every step; then judge the design against structural, practical and sustainability constraints rather than against the arithmetic alone.

Criterion task · The school greenhouse

CRITERIA C AND D
COMMUNICATING AND APPLYING
24 MARKS

A school garden club is building a greenhouse in the shape of a half-cylinder on a rectangular base. The greenhouse is 3.0 m wide and 6.0 m long, with vertical side walls 1.0 m high and a semicircular roof of radius 1.5 m. The curved roof will be covered in polycarbonate; the two end walls, including the semicircular gables, and the two long side walls will be glazed. A door 0.9 m wide and 1.9 m high is cut into one end wall and is not glazed.

1. Draw a labelled cross-section of the greenhouse and a sketch of its three-dimensional form. State two assumptions used in your model. *(3 marks)*
2. Calculate the floor area, the volume of the rectangular part, the volume of the half-cylinder roof and the total internal volume. *(5 marks)*
3. Calculate the area of polycarbonate needed for the curved roof, the area of the two long side walls, and the area of the two end walls including the semicircular gables. Hence find the total glazing area before the door is removed. *(5 marks)*
4. Polycarbonate costs 42 units per m^2 and glass costs 55 units per m^2. The door is deducted from the glazed area. Calculate the total material cost, showing the area and cost of each material separately. *(4 marks)*
5. The frame runs along the base perimeter, the four vertical corners, the two roof edges and the two gable arcs. Calculate the total length of frame required and its cost at 18 units per metre. *(2 marks)*
6. The dimensions are measured to the nearest 0.05 m. Find the greatest and least possible internal volumes and express the range as a percentage of the design volume. *(3 marks)*
7. Evaluate the design for a school garden. Consider condensation, ventilation, wind loading, orientation, maintenance and cost. Recommend one modification and justify it. *(2 marks)*

PLANNING FRAME

Draw: a rectangle 3.0 m by 1.0 m topped by a semicircle of radius 1.5 m; label every dimension before calculating. **Decompose:** rectangular prism plus half-cylinder for volume; curved surface, side walls and end walls for area. **Track:** write each area with its units and keep a running total. **Adjust:** subtract the door once, from the glazed area, not from the polycarbonate. **Bound:** use ± 0.025 m for each measured dimension, not ± 0.05 m. **Evaluate:** connect at least two limitations to a specific design change.

BAND	DESCRIPTOR	MARKS
4	Clear labelled cross-section and sketch; all volumes and areas correct with consistent units; door and gable handled correctly; costs accurate; bounds interpreted; evaluation links limitations to a justified modification.	19–24
3	Diagram and decomposition mostly correct; minor arithmetic or unit slip; most areas and costs correct; bounds attempted; evaluation identifies some limitations.	13–18
2	Some correct formulas; missing component or double-counted surface; costs partly correct; bounds or evaluation incomplete.	7–12
1	Isolated correct statements; no coherent model; little or no justification.	1–6

CRITERIA: C, D **ATL:** SELF-MANAGEMENT **TIME:** 60 MINUTES **CONDITIONS:** SCIENTIFIC CALCULATOR ALLOWED

MAPPED TO: C strands i–iv and D strands i–v. The rubric is a course summary, not an official IB descriptor.

Chapter summary and review

KEY IDEAS

- A polyhedron has flat polygonal faces, straight edges and vertices; a cylinder, cone or sphere has a curved surface and is not a polyhedron.
- For a convex polyhedron, $F + V = E + 2$. Counting carefully is part of understanding the solid.
- A net is a flat pattern that folds into a solid; the area of the net is the surface area of the solid.
- A plan and elevations are orthogonal views; together they determine the footprint and the heights.
- Prism: $SA = 2B + Ph$ and $V = Bh$. Cylinder: $SA = 2\pi r^2 + 2\pi rh$ and $V = \pi r^2 h$.
- Pyramid: $SA = B + \frac{1}{2}Pl$ and $V = \frac{1}{3}Bh$. Cone: $SA = \pi r^2 + \pi rl$ and $V = \frac{1}{3}\pi r^2 h$, with $l = \sqrt{r^2 + h^2}$.
- Sphere: $SA = 4\pi r^2$ and $V = \frac{4}{3}\pi r^3$. A hemisphere has half the curved area and half the volume.
- Composite solids are decomposed into standard parts; volumes of holes are subtracted and internal faces are not counted.
- For similar solids, surface areas scale by k^2 and volumes by k^3.
- Capacity, density and tolerance connect geometry to measurement: $1\ \text{cm}^3 = 1\ \text{mL}$, $\rho = \frac{m}{V}$, and bounds use half the smallest scale division.

REVISION CHECKLIST

- ☐ I can classify a solid and count its faces, edges and vertices.
- ☐ I can use Euler's relationship to find a missing count.
- ☐ I can interpret or draw a net and find the surface area from it.
- ☐ I can read plans and elevations to reconstruct a solid.
- ☐ I can choose and apply the surface-area formula for any standard solid.
- ☐ I can choose and apply the volume formula, including pyramids, cones and spheres.
- ☐ I can find the volume of a composite solid by adding and subtracting parts.
- ☐ I can use the k^2 and k^3 rules for similar solids.
- ☐ I can convert between cm^3, mL, L and m^3, and calculate density.
- ☐ I can express a measurement as a range and use it to evaluate a design.

VOCABULARY

polyhedron
A closed solid whose faces are polygons.

face
A flat polygonal surface of a solid.

edge
A segment where two faces meet.

vertex
A point where edges meet.

net
A flat pattern that folds into a solid.

plan
The view from directly above.

elevation
A view from the front or side, looking horizontally.

surface area
The total area of a solid's boundary.

lateral surface
The curved or side surface, excluding the bases.

slant height
The distance from the apex to the base edge along the surface.

cross-section
The shape exposed by a plane cut through a solid.

capacity
The volume a container can hold, in litres or millilitres.

density
Mass per unit volume, $\rho = \frac{m}{V}$.

tolerance

The permitted range of variation in a measurement or specification.

lower bound
The smallest value a measurement could take.

similar solids
Solids whose corresponding lengths are in the same ratio.

upper bound
The largest value a measurement could take.

composite solid
A solid built from two or more standard solids.

MIXED PRACTICE · CHAPTER 8

Connecting the ideas

FLUENCY Core procedures

1. Find the total surface area and volume of a cuboid 12 cm by 7 cm by 4 cm. F
2. Find the total surface area and volume of a cylinder with radius 6 cm and height 9 cm, each to one decimal place. F
3. Find the total surface area and volume of a cone with radius 4 cm and height 3 cm, each to one decimal place. F
4. Find the surface area and volume of a sphere of radius 5 cm, each to one decimal place. F

REASONING Justify and connect

1. Two similar cones have heights 6 cm and 9 cm. Find the ratios of their surface areas and volumes, and explain how each ratio follows from the scale factor. R
2. Explain why a cylinder with a fixed volume cannot have an arbitrarily small surface area. Refer to both terms of $SA = 2\pi r^2 + \frac{2V}{r}$ in your answer. R
3. A cube and a sphere have the same volume of 1000 cm^3. Which has the smaller surface area? Calculate both and comment on why manufacturers do not make all containers spherical. R
4. A student says that doubling the radius of a cylinder doubles its volume. Explain what actually happens when the height is fixed, and what happens to the surface area. R

APPLICATION Use and evaluate

1. A fish tank measures 60 cm by 30 cm by 35 cm and is open at the top. Find its capacity in litres and the area of glass required. P
2. A cylindrical silo has radius 2.5 m and height 8 m. Its curved side and flat top are painted; the base is not. Find the painted area, the area for two coats, and the number of 20 L drums needed at 0.2 L per m^2. P
3. An ice-cream cone has radius 3 cm and height 12 cm and is filled level with a hemispherical scoop of the same radius. Find the total volume, to one decimal place. P
4. A cylindrical tank has diameter 1.5 m and height 2 m. Find its capacity in litres and the number of days it lasts a household using 180 L per day. P
5. Two boxes each have volume 1200 cm^3. Box A measures 10 cm by 10 cm by 12 cm; Box B measures 8 cm by 10 cm by 15 cm. Calculate both surface areas and decide which uses less cardboard. P
6. A tank is a cylinder of radius 1 m and height 2 m with a hemispherical dome on top; the base is not painted. Find its volume and exposed surface area, to two decimal places. E

Check your progress

I CAN …	CONFIDENCE	EVIDENCE	NEXT ACTION
Classify solids and count faces, edges and vertices.	1 / 2 / 3		
Interpret nets, plans and elevations.	1 / 2 / 3		
Calculate surface area of prisms, cylinders, pyramids and cones.	1 / 2 / 3		
Calculate volume, including pyramids, cones, spheres and composites.	1 / 2 / 3		
Use capacity, density and bounds to evaluate a design.	1 / 2 / 3		

Reflection

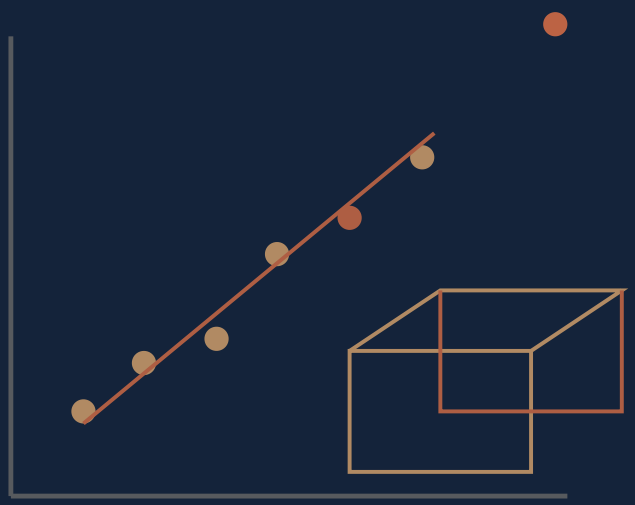
A can is labelled "330 mL", but the true capacity depends on manufacturing tolerances and the shape of the base and lid. Should the label state an exact figure or a guaranteed minimum? How should a manufacturer communicate uncertainty without losing the customer's trust?

PART IV · INTERPRETING EVIDENCE

IV

Data, chance and evidence

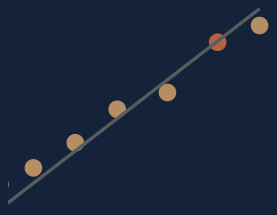
Data do not speak for themselves. A scatter plot can suggest a relationship; a correlation coefficient can measure its strength; neither can establish that one variable causes another. This part develops the tools of bivariate analysis and probability, then turns them on the questions that matter: What does the evidence support, how was the sample chosen, and how should uncertainty be communicated?



09 Bivariate data, correlation and causation

10 Probability, sampling and evidence

9



DATA, CHANCE AND EVIDENCE

Bivariate data, correlation and causation

Two measurements on the same person, place or day form a pair. Plot the pairs and a shape may appear: a rising cloud, a falling cloud, or no pattern at all. The shape is evidence, but it is not yet an explanation.

KEY CONCEPT · RELATIONSHIPS | RELATED · MODELS | RELATED · JUSTIFICATION | CONTEXT · FAIRNESS AND DEVELOPMENT

INQUIRY

When does a pattern in paired data justify a prediction — and when does it justify a claim about cause?

ATL · RESEARCH AND CRITICAL THINKING

A correlation arrives with a history: who was measured, how the variables were defined, which cases were left out and who paid for the study. Before interpreting a coefficient, locate the data's provenance. A number can be arithmetically correct and still support a conclusion that the study design cannot bear.

MYP UNIT PLANNER

Framing the inquiry

UNIT 9 · APPROX. 14 HOURS
ATL FOCUS · RESEARCH

KEY AND RELATED CONCEPTS

Relationships. Bivariate data records two variables on the same unit. A scatter plot shows whether the variables vary together, in what direction and how tightly.

Models. A line of best fit is a simple model of a relationship. Its gradient and intercept have meaning only in the context and range of the data that produced them.

Justification. Association is not causation. A causal claim needs a mechanism, correct temporal order and the elimination of plausible alternative explanations.

GLOBAL CONTEXT AND EXPLORATION

Fairness and development. Exploration: evaluating a public claim — how a statistic about schools, health or income is selected, summarised and presented, and whether the evidence supports the conclusion drawn from it.

Public decisions are often argued with correlations: class size and attainment, screen time and sleep, income and life expectancy. The mathematics cannot settle the policy, but it can show whether the comparison is fair, whether a third variable could explain the pattern, and how much confidence the data warrant.

STATEMENT OF INQUIRY

Relationships between two variables can be displayed, measured and modelled to support prediction; a model becomes evidence for causation only when alternative explanations have been tested and ruled out.

FACTUAL

What is a scatter plot? How is the correlation coefficient calculated and interpreted? What is a line of best fit, and how is it used to predict?

CONCEPTUAL

How does a single number summarise the strength of a linear relationship? Why can a strong association fail to establish cause? How do outliers and subgroups change an interpretation?

DEBATABLE

Should a correlation be enough to justify a public health recommendation? How should a newspaper report a study that shows association but not causation?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii	Identify variables and associations; calculate and interpret a correlation coefficient; use a line of best fit.
B · Investigating patterns	i, ii, iii	Collect paired measurements, describe the scatter, calculate r, fit a line and test the model against new data.
C · Communicating	i, ii, iii, iv	Draw labelled scatter plots and lines; report r and predictions with units; state the limits of a model.
D · Applying mathematics	i, ii, iii, iv, v	Evaluate a public claim by separating association from causation and proposing a stronger study design.

Learning intentions and success criteria

YOU ARE LEARNING TO

— **Understand** the structure of bivariate data and the meaning of association.

— **Be able to** draw, describe and compare scatter plots.

— **Know** how to calculate and interpret Pearson's correlation coefficient r.

— **Be able to** construct and use a line of best fit, including interpolation and extrapolation.

— **Understand** the difference between correlation and causation, and recognise confounding and reverse causation.

— **Be able to** evaluate a public claim built on bivariate data and recommend a stronger design.

SUCCESS CRITERIA

- [ ] I can identify the explanatory and response variables and plot them on correctly labelled axes.
- [ ] I can describe direction, strength, form and unusual features of a scatter plot.
- [ ] I can calculate r from a small table and interpret its sign and size.
- [ ] I can fit a line, state its equation and use it to predict within the data range.
- [ ] I can explain why a correlation may not be causal, naming at least one plausible alternative explanation.
- [ ] I can judge whether a graph or headline overstates the evidence.

PREREQUISITE CHECK · 12 MINUTES, NO CALCULATOR

1. Plot the points $(1, 3)$, $(2, 5)$, $(3, 4)$, $(4, 7)$ on a grid with axes from 0 to 8.
2. Find the mean of $4, 6, 7, 9, 14$.
3. Find the equation of the line through $(1, 5)$ and $(3, 11)$.
4. Calculate the gradient of the line $y = 4x - 7$ and state its intercept.
5. Evaluate $\sqrt{2.5}$ and $\sqrt{0.64}$ to two decimal places.
6. A student scores 62% on a test. The class mean is 70%. Find the difference from the mean.

Answers are in the Chapter 9 answer section. If more than two are incorrect, review plotting points, means and the equation of a straight line before continuing.

BEFORE YEAR 3

Reading and plotting coordinates; mean, median and range; drawing straight-line graphs; calculating a gradient from two points.

YEAR 3 · THIS CHAPTER

Scatter plots; correlation coefficient; line of best fit; prediction; correlation, causation and confounding; critical reading of claims.

LATER

Regression with least squares and residual analysis; Spearman's rank correlation; multiple regression; causal inference and experimental design.

INQUIRY PATHWAY FOR THIS CHAPTER

1 · CONTEXT
Claims about schools, health and behaviour are often based on two variables measured together.

2 · QUESTION
How strong is the relationship, what does it predict, and does it show cause?

3 · CONCEPT
Association, correlation, linear models, confounding and justification.

4 · INVESTIGATION
Measure paired data, calculate the correlation, fit a line and test the model against a hold-out sample.

5 · COMMUNICATION
Plot and label accurately; report the correlation and predictions with units; state the range of valid inference.

6 · REFLECTION
What would it take to turn this association into evidence for a causal claim?

9.1 Two variables at once

CRITERION A · I, II ◆ CRITERION C · I, II ◆ RELATIONSHIPS

A single-variable data set answers “how big?”. A bivariate data set answers “how do two measurements move together?”. Each observation contributes a pair, and the pair is plotted as one point.

DEFINITIONS

Bivariate data consists of two measurements made on the same individual, object or occasion. The **explanatory variable** is the one we use to explain or predict; the **response variable** is the one we measure as an outcome. A **scatter plot** displays each pair as a point, with the explanatory variable on the horizontal axis. An **association** is a pattern in which the response tends to change as the explanatory variable changes. An **outlier** is a point that does not fit the overall pattern. A **cluster** is a separated group of points.

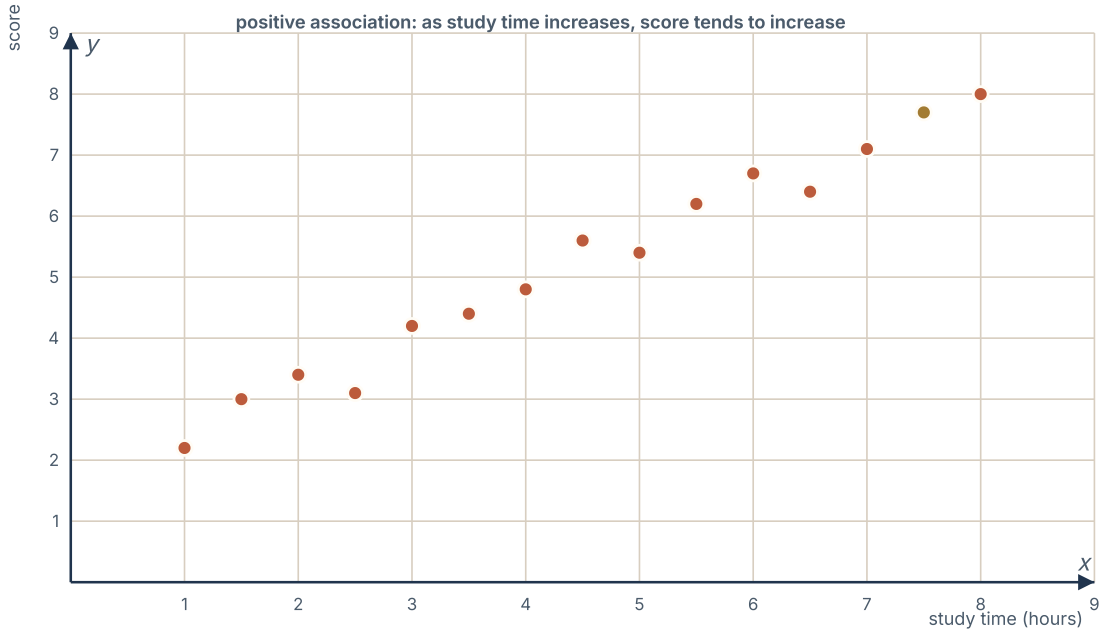


Figure 9.1 Each point is one student: study time and score measured together. The points rise from left to right, so the association is positive. The highlighted point sits above the pattern for its study time.

Label the axes with the quantity and its unit, and choose scales that use the available space without distorting the pattern. The explanatory variable is usually placed on the horizontal axis because we are asking how the response depends on it. When neither variable is clearly prior, either axis may be used, but the choice must be stated and kept consistent when comparing plots.

Describe a scatter plot in four moves: **direction** (positive, negative or none), **form** (linear or curved), **strength** (how closely the points follow the form) and **unusual features** (outliers, clusters or gaps). “Positive, linear and strong, with one high outlier” is a complete description; “it goes up” is not.

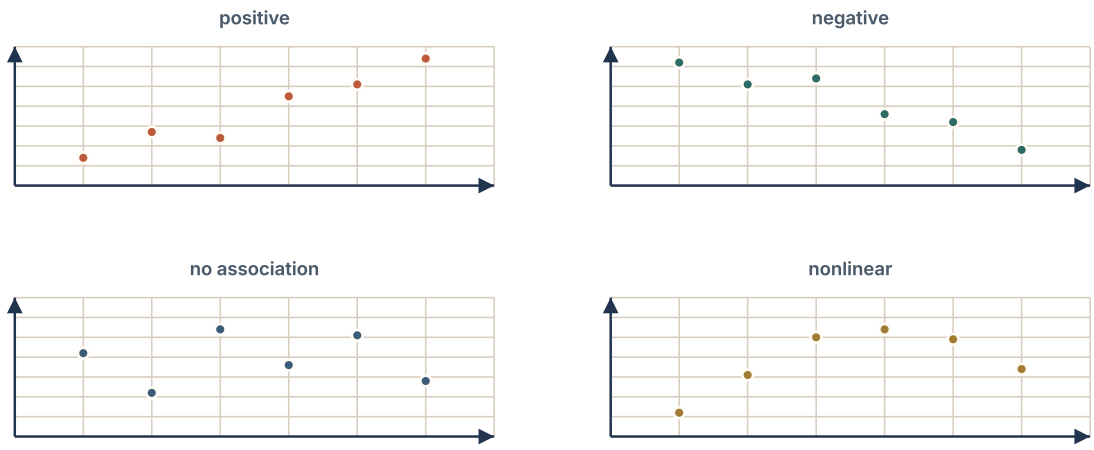


Figure 9.2 Direction and form are different questions. The bottom-right plot has a clear pattern but no overall linear trend: it rises and then falls.

WORKED EXAMPLE 9.1.1 DESCRIBING A SCATTER PLOT

PROBLEM

A researcher records the age in years and the resting heart rate in beats per minute of 20 adults. The scatter plot slopes downwards and the points lie fairly close to a straight line, except for one 22-year-old with a heart rate of 96 bpm. Describe the association and identify any unusual features.

1. **Direction:** as age increases, resting heart rate tends to decrease, so the association is negative.
2. **Form:** the points cluster around a downward-sloping straight line, so the form is approximately linear.
3. **Strength:** the points are fairly close to the line, so the association is moderately strong.
4. **Unusual features:** the point $(22, 96)$ lies well above the downward trend for its age, so it is a possible outlier. Before removing it, check whether it is a recording error or a genuine unusual case.

ANSWER **Moderately strong negative linear association, with one possible outlier at age 22 and heart rate 96 bpm.**

WORKED EXAMPLE 9.1.2 CHOOSING VARIABLES AND AXES

PROBLEM

A shop records, for each of 30 days, the outdoor temperature in °C and the number of cold drinks sold. Identify the explanatory and response variables, state which belongs on each axis, and predict the direction of the association.

1. Temperature is the variable used to explain the number of drinks sold, so temperature is explanatory and drinks sold is the response.
2. Temperature goes on the horizontal axis; drinks sold goes on the vertical axis. Label them "temperature (°C)" and "cold drinks sold".
3. Warmer days are likely to produce more cold-drink sales, so expect a positive association. It need not be perfectly linear: very hot days may reach a practical limit set by stock or the number of customers.
4. Because sales also depend on day of the week, price and weather, the points will be scattered rather than lying exactly on a line.

ANSWER **Explanatory: temperature; response: drinks sold; positive association expected, with scatter from other influences.**

COMMON ERROR · PLOTTING PAIRS THAT DO NOT BELONG TOGETHER

Each point must join two measurements of the *same* individual or occasion. Sorting the two lists separately before plotting destroys the pairing and can create a pattern that does not exist. Also check units: a height recorded in centimetres and an arm span in metres will produce a meaningless cluster near the vertical axis. Convert to a common unit first.

PRACTICE 9.1 · TWO VARIABLES AT ONCE

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. For each pair, identify the explanatory and response variables: (a) hours of revision and test score; (b) age of a car and its resale value; (c) daily rainfall and umbrella sales. [F]
2. Describe the direction of each association: (a) altitude and air pressure; (b) distance from a city centre and house price; (c) practice time and typing speed. [F]
3. Plot these points: $(1, 4)$, $(2, 5)$, $(2.5, 4.6)$, $(3, 6)$, $(4, 6.8)$, $(5, 7.2)$, $(6, 8.1)$. Describe direction, form and strength. [F]
4. Explain why "the points go up" is not a complete description of a scatter plot. [R]
5. A scatter plot of plant height against daily sunlight is strong and positive for the first four weeks, then flattens. Explain why a single straight line would describe the whole period poorly. [R]
6. A data set contains the point $(14, 3)$ while the other points lie between 2 and 6 on the horizontal axis. Explain why this point may be influential as well as unusual. [R]
7. A school records, for 25 students, the number of minutes spent on homework and the score on a test. One student reports 600 minutes, far above the rest. Suggest two checks before treating this as a genuine observation. [P]
8. A shop's sales data show two separate clusters: weekday sales and weekend sales. Explain how plotting a single line of best fit through both clusters could give a misleading summary. [E]

9.2 Measuring linear association

CRITERION A · I, II ◆ CRITERION B · I, II ◆ MODELS

A scatter plot shows strength by eye. The correlation coefficient turns that impression into a number between -1 and 1, so that different data sets can be compared on the same scale.

DEFINITIONS · PEARSON'S CORRELATION COEFFICIENT

For paired data (x_i, y_i) with means $\bar{x}$ and $\bar{y}$, the **correlation coefficient** is

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}.$$

It measures the direction and strength of a *linear* association. $r > 0$ means positive association, $r < 0$ means negative association, and r close to 0 means little linear association. The value is unchanged if either variable is rescaled or if the two variables are swapped.

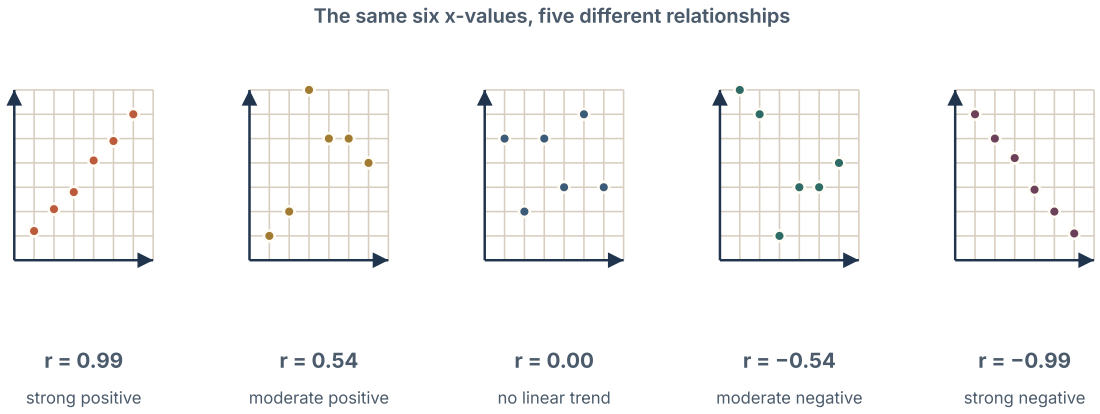


Figure 9.3 The same horizontal values with different vertical values. The sign gives the direction; the distance from 0 gives the linear strength. The middle plot has no linear trend, but that does not mean the two variables are unrelated.

The calculation is easiest in a table. Subtract each mean, multiply the paired deviations, square each deviation, and total. The numerator measures whether the variables deviate in the same direction; the denominator scales the result so that it always lies between -1 and 1.

CALCULATING R FOR FIVE STUDENTS

x	y	Dev. x	Dev. y	Product	Dev. x^2	Dev. y^2
1	52	−2	−12	24	4	144
2	58	−1	−6	6	1	36
3	64	0	0	0	0	0
4	68	1	4	4	1	16
5	78	2	14	28	4	196
Total		0	0	62	10	392

WORKED EXAMPLE 9.2.1 — A POSITIVE CORRELATION

PROBLEM

Five students recorded hours of study x and test score y: $(1, 52)$, $(2, 58)$, $(3, 64)$, $(4, 68)$, $(5, 78)$. Calculate r and interpret it.

1. Means: $\bar{x} = 3$ and $\bar{y} = 64$.
2. From the table, $\sum(x - \bar{x})(y - \bar{y}) = 62$, $\sum(x - \bar{x})^2 = 10$ and $\sum(y - \bar{y})^2 = 392$.
3. $r = \dfrac{62}{\sqrt{10 \times 392}} = \dfrac{62}{\sqrt{3920}} = \dfrac{62}{62.61} \approx 0.990$.
4. The sign is positive and the value is very close to 1: a strong positive linear association. Students who studied more tended to score higher.

ANSWER $r \approx 0.990$**: a very strong positive linear association.**

WORKED EXAMPLE 9.2.2 — A NEGATIVE CORRELATION

PROBLEM

At a café, the mean outdoor temperature x (°C) and hot drinks sold y (hundreds) were recorded on five days: $(5, 9)$, $(10, 8)$, $(15, 6)$, $(20, 4)$, $(25, 3)$. Calculate r and interpret it.

1. Means: $\bar{x} = 15$ and $\bar{y} = 6$.
2. Deviations for x: $-10, -5, 0, 5, 10$; for y: $3, 2, 0, -2, -3$.
3. $\sum(x - \bar{x})(y - \bar{y}) = -30 - 10 + 0 - 10 - 30 = -80$.
4. $\sum(x - \bar{x})^2 = 250$ and $\sum(y - \bar{y})^2 = 26$.
5. $r = \dfrac{-80}{\sqrt{250 \times 26}} = \dfrac{-80}{\sqrt{6500}} = \dfrac{-80}{80.62} \approx -0.992$.
6. The negative sign shows that hotter days tend to have fewer hot-drink sales; the size shows the linear pattern is very strong in these five days.

ANSWER $r \approx -0.992$**: a very strong negative linear association.**

WORKED EXAMPLE 9.2.3 READING THE COEFFICIENT

PROBLEM

Interpret each value of r: (a) 0.92, (b) -0.35, (c) 0.08, (d) -0.99. Then explain how a strong curved relationship could produce $r \approx 0$.

1. (a) 0.92: strong positive linear association; the response tends to rise as the explanatory variable rises.
2. (b) -0.35: weak negative linear association; there is a slight downward tendency but considerable scatter.
3. (c) 0.08: essentially no linear association.
4. (d) -0.99: very strong negative linear association; the points lie close to a downward-sloping line.
5. A curved relationship can rise and then fall symmetrically. A straight line through it has no overall slope, so the positive and negative deviations cancel and r can be near 0 even though the pattern is strong.

ANSWER **Sign gives direction; distance from 0 gives linear strength. $r \approx 0$ rules out a linear trend, not every relationship.**

A related summary is the coefficient of determination r^2, the square of the correlation. It gives the proportion of the variation in the response that the linear model accounts for: $r = 0.8$ gives $r^2 = 0.64$, so about 64% of the variation in y is associated with the linear pattern in x, leaving 36% unexplained. Squaring removes the sign, so r^2 measures fit but not direction.

> **COMMON ERROR · TREATING R AS A PERCENTAGE OR AS PROOF**
>
> An r of 0.8 does not mean "80% of the variation is explained" — r^2 is the proportion, and it is 0.64. Nor does $r = 0.8$ mean the relationship is 80% reliable. r describes how closely the points follow a straight line. It is sensitive to outliers, it can be close to 0 for a strong curve, and it says nothing about causation. Always plot the data before trusting the number.

PRACTICE 9.2 · MEASURING LINEAR ASSOCIATION

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Calculate r for $(1, 2)$, $(2, 4)$, $(3, 5)$, $(4, 8)$. F
2. Calculate r for $(1, 10)$, $(2, 12)$, $(3, 11)$, $(4, 15)$, $(5, 17)$. F
3. Calculate r for $(1, 20)$, $(2, 18)$, $(3, 19)$, $(4, 14)$, $(5, 13)$. F
4. Interpret $r = -0.78$ in context: "hours of television watched per day" and "minutes of moderate exercise per day". R
5. Explain why swapping the two variables in a data set does not change r. R
6. A student says $r = 0.5$ means the relationship is "half as strong as $r = 1$". Criticise this statement and describe a better interpretation. R
7. Add one point to the data in question 1 so that r becomes negative. Explain your choice and calculate the new value. P
8. Two data sets both have $r = 0.85$. In one, the points are tightly grouped around a line; in the other, one extreme point creates most of the correlation. Explain how a scatter plot distinguishes the two cases. E

9.3 The line of best fit

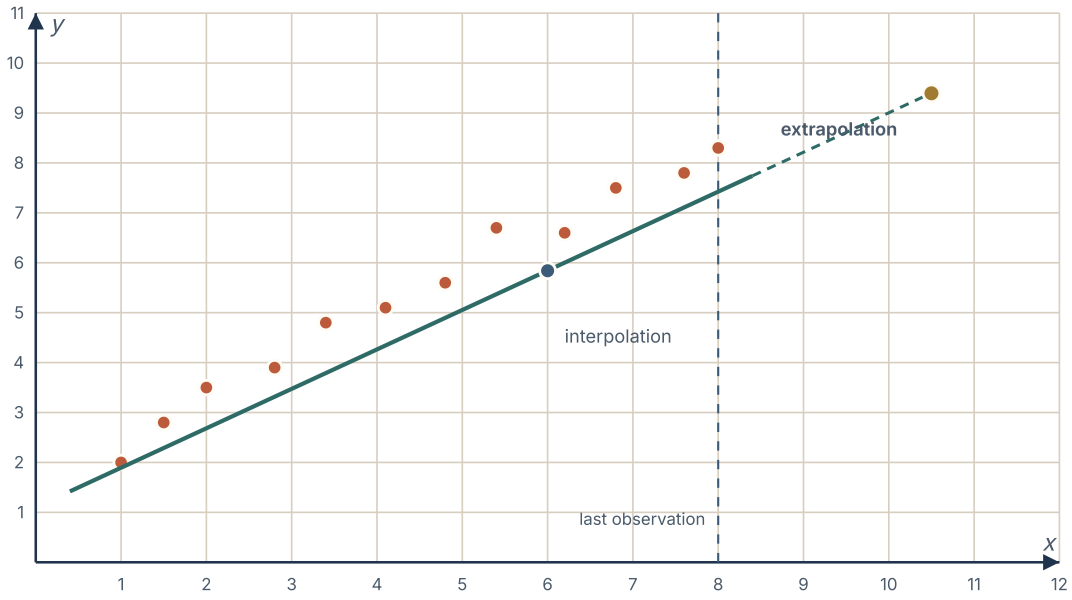
CRITERION A · II ◆ CRITERION C · III, IV ◆ MODELS

A line of best fit is a model of the trend. It smooths the scatter into a single statement: a starting value and a rate of change. Its usefulness depends on how well the points fit and on where the prediction is made.

DEFINITIONS

A **line of best fit** is the straight line that best summarises a linear association on a scatter plot. It passes through the mean point $(\bar{x}, \bar{y})$ and balances the points above and below it. A prediction inside the range of observed x-values is **interpolation**; a prediction outside that range is **extrapolation**. The vertical difference between an observed point and the line is a **residual**: residual $= y_{\text{observed}} - y_{\text{predicted}}$.

To draw a line by eye, plot the mean point, choose two points on the line that are far apart, and check that the number of points above and below the line is roughly balanced. A ruler placed through the mean point can be rotated until the vertical distances of the points from the line look as small as possible. The least-squares line makes the sum of the squared residuals as small as possible; software and spreadsheets compute it directly.



A line of best fit summarises a linear trend; its predictions are only meaningful near the observed range.

Figure 9.4 The line summarises the trend. Interpolation borrows strength from nearby data; extrapolation extends the model into a region where nothing was observed, so it is less reliable.

WORKED EXAMPLE 9.3.1 EQUATION AND PREDICTION

PROBLEM

A line of best fit passes through $(2, 14)$ and $(8, 32)$, where x is hours of practice and y is score. Find the equation of the line, predict the score after 6 hours, and state whether the prediction is interpolation or extrapolation.

1. Gradient: $m = \dfrac{32 - 14}{8 - 2} = \dfrac{18}{6} = 3$. Each extra hour of practice is associated with about 3 more points.
2. Use $y = mx + c$ with $(2, 14)$: $14 = 3(2) + c$, so $c = 8$. Hence $y = 3x + 8$.
3. At $x = 6$: $y = 3(6) + 8 = 26$.
4. 6 lies between 2 and 8, so this is interpolation. A prediction at $x = 11$ would give 41, but it would be extrapolation beyond the observed range.

ANSWER $y = 3x + 8$**; predicted score** 26**; interpolation.**

WORKED EXAMPLE 9.3.2 RESIDUALS

PROBLEM

For the study data in Section 9.2, a line of best fit is $y = 6x + 46$. Calculate the residual for each observation and comment on the fit.

1. At $x = 1$, predicted 52; observed 52; residual 0.
2. At $x = 2$, predicted 58; observed 58; residual 0.
3. At $x = 3$, predicted 64; observed 64; residual 0.
4. At $x = 4$, predicted 70; observed 68; residual -2. The point lies 2 below the line.
5. At $x = 5$, predicted 76; observed 78; residual $+2$. The point lies 2 above the line.
6. The residuals sum to 0, as they should for a line through the mean point. All are small relative to the scores, so the line summarises these five points well.

ANSWER **Residuals** $0, 0, 0, -2, +2$**; small and balanced, so the linear model fits closely.**

WORKED EXAMPLE 9.3.3 INTERPRETING INTERCEPT AND GRADIENT

PROBLEM

A company models the monthly cost C (in units) of producing n items by $C = 1800 + 7.5n$. Interpret the intercept and gradient, find the cost of 400 items, and explain why using the model for 100 000 items may be unreliable.

1. The intercept 1800 is the fixed cost when no items are produced: rent, equipment or salaries that do not depend on output.
2. The gradient 7.5 is the variable cost per item: each additional item adds about 7.5 units to the monthly cost.
3. For $n = 400$: $C = 1800 + 7.5(400) = 1800 + 3000 = 4800$ units.
4. The model assumes the same variable cost per item at every scale, but at very high output, bulk discounts, extra shifts or capacity limits may change the gradient. A 100 000-item prediction is far outside the data used to build the model.

ANSWER **Fixed cost** 1800**;** 7.5 **per item;** $C(400) = 4800$ **units; extrapolation far beyond the data is unsafe.**

COMMON ERROR · SAYING "THE LINE PROVES…"

A line of best fit is a summary, not a law. It does not prove that x causes y, it does not guarantee that a new observation will lie on the line, and it should not be extended far beyond the observed values. The intercept may also be meaningless: a model predicting test scores from hours of study may have a sensible intercept, but a model using age in years to predict height has an intercept at age 0 that lies outside the data and outside the context.

PRACTICE 9.3 · THE LINE OF BEST FIT

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. A line of best fit passes through $(1, 7)$ and $(5, 19)$. Find its equation. F
2. Using $y = 4x + 3$, predict y when $x = 6$ and when $x = 10$. F
3. A model $y = 2.4x + 11$ is built from data with x between 5 and 25. State whether each prediction is interpolation or extrapolation: (a) $x = 12$; (b) $x = 30$; (c) $x = 4$. F
4. A line $y = 3x - 2$ is fitted to five points. Calculate the residuals for $(1, 2)$, $(2, 3)$, $(3, 8)$, $(4, 9)$, $(5, 14)$. R
5. Explain why the line of best fit passes through the mean point, using the idea of balancing residuals. R
6. A student predicts a country's population in 2200 by extending a line fitted to data from 1950 to 2020. Explain two reasons why this prediction is unreliable. R
7. The cost of a taxi is modelled by $C = 4 + 2.1d$, where d is distance in kilometres. Interpret both numbers, find the cost of a 12 km journey, and explain whether the model is valid for a 1 km journey. P
8. Two lines are proposed for the same data: $y = 2x + 5$ and $y = 1.8x + 8$. Explain how residuals could be used to choose between them, without calculating every value. E

9.4 Correlation, cause and consequence

CRITERION A · II ◆ CRITERION D · I, II, IV ◆ JUSTIFICATION

A correlation is a statement about how two variables vary together in a sample. A causal claim is a statement about what would happen if one variable were changed. The second is much stronger than the first, and much harder to establish.

DEFINITIONS

Causation means that a change in one variable produces a change in the other. A **confounding variable** is a third variable that influences both the explanatory and response variables, creating an association that is not due to a direct causal link. **Reverse causation** occurs when the response variable actually influences the explanatory variable. An **observational study** records variables without intervening; a **controlled experiment** deliberately changes the explanatory variable and compares outcomes under otherwise similar conditions.

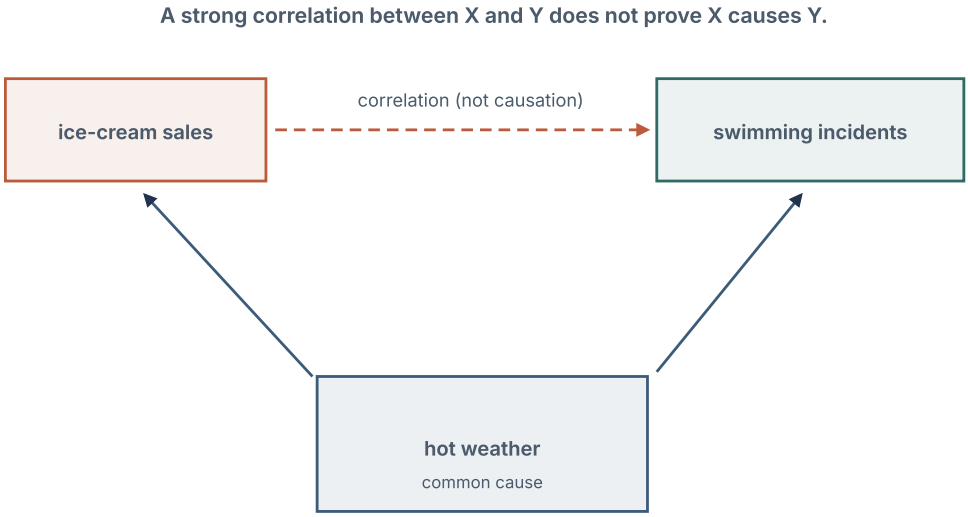


Figure 9.5 Ice-cream sales and swimming incidents rise together, but buying ice cream does not cause swimming incidents. Hot weather is a common cause of both. The correlation is real; the causal direction is not.

The classic alternative explanations are worth checking systematically. Is there a **common cause**, such as temperature, income or age? Could the direction be **reversed**, with the response influencing the explanatory variable? Could the pattern arise from **selection**, because the people in the data chose themselves or were chosen in a non-random way? Could it be a **coincidence**, especially in a small sample or when many possible pairs were examined? Could the variables be **measuring the same underlying quantity**, such as height and arm span?

WORKED EXAMPLE 9.4.1 — A PLAUSIBLE CONFOUNDER

PROBLEM

A study finds that children who own more books score higher on a reading test. A newspaper concludes that buying books raises reading scores. Identify a plausible confounding variable, an alternative explanation involving selection, and a way to test the claim more fairly.

1. A plausible confounder is family income or parental education. Families with more books may also have more resources, more time for reading and more educated parents, all of which could raise scores independently of the books.
2. **Selection:** families that buy many books may already value reading; the children may differ from those with few books before any book is bought.
3. A fairer test would randomly allocate books to some families and not others, then compare reading scores after a fixed period, keeping other support similar.
4. If a controlled trial is impossible, a longitudinal study could track children before and after books arrive, controlling for income and parental education. The conclusion would still be tentative.

ANSWER **Income and parental education confound the association; selection is also plausible; random allocation or a controlled longitudinal design would test the claim.**

WORKED EXAMPLE 9.4.2 DESIGNING A CAUSAL TEST

PROBLEM

A teacher believes that listening to quiet instrumental music during independent work improves concentration. Design an experiment to test the claim, identifying the explanatory and response variables, a control, and two variables to keep constant.

1. **Explanatory variable:** whether music is played (two conditions: music or no music). **Response variable:** a measure of concentration, such as the number of correctly completed problems in 20 minutes or the number of off-task glances recorded by an observer.
2. **Control:** a no-music condition using the same task, time and classroom conditions.
3. **Random allocation:** assign students to the order of conditions at random, or use a crossover design in which each student experiences both conditions in random order. This reduces the chance that one group is already more focused.
4. **Keep constant:** the task, time of day, teacher instructions, seating and noise level apart from the music. Use the same measure and the same observer, or blind the observer to the condition.
5. **Repeat:** a single session cannot rule out coincidence. Repeat across several sessions and compare average performance, reporting variability as well as the mean.

ANSWER **A randomised comparison of music and no music, same task and conditions, repeated over several sessions, with a pre-specified concentration measure.**

WORKED EXAMPLE 9.4.3 EVALUATING A CAUSAL CLAIM

PROBLEM

A report states: "Schools with more library books have higher examination scores, so increasing a school's book stock will raise its results." The correlation is $r = 0.61$ across 200 schools. Evaluate the claim.

1. The association is moderately strong and positive, but it is based on schools, not on individual students. A school-level correlation does not show that individual students who read more books score higher.
2. Wealthier schools may have more books, better facilities, smaller classes and more experienced teachers. These are plausible confounders, so the correlation could reflect advantage rather than books.
3. The claim "increasing stock will raise results" asserts causation and predicts the effect of an intervention. The study is observational, so it cannot establish that changing the stock would change the scores.
4. A stronger design would allocate additional books to a random subset of comparable schools and compare score changes over time, while monitoring how the books are used.
5. A reasonable conclusion is that book stock is associated with results, but the evidence does not justify the causal recommendation.

ANSWER **The correlation is real but moderate; confounding by school resources and the ecological nature of the data undermine the causal claim. A randomised intervention would be stronger evidence.**

COMMON ERROR · "CORRELATION DOES NOT IMPLY CAUSATION" AS A FULL ANSWER

The phrase is a warning, not an explanation. A complete evaluation names the specific alternative explanation that matters in the context — a confounder, reverse causation, selection or chance — and explains how it could produce the observed association. It also says what evidence would distinguish the causal explanation from the alternatives.

PRACTICE 9.4 · CORRELATION AND CAUSE

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. For each association, name a plausible common cause: (a) shoe size and reading age in primary school children; (b) ice-cream sales and drowning incidents; (c) number of firefighters and damage at a fire. F
2. Explain the difference between an observational study and a controlled experiment in one sentence. F
3. Give an example of reverse causation for the association "people who use fitness apps tend to be healthier". R
4. A study finds that students who eat breakfast score higher on average. Suggest two confounders and one way to reduce their influence. R
5. Explain why a correlation calculated from group averages can differ from the correlation calculated from individuals. R
6. A headline says "Playing video games makes teenagers less sociable". Describe the evidence you would need before accepting the causal claim. R
7. Design a simple experiment to test whether a plant food increases tomato yield. Include a control, a response variable and two variables to keep constant. P
8. A city introduces a new tram line and later finds that house prices near the line have risen faster than elsewhere. Explain why this does not prove that the tram caused the rise. E

9.5 Reading claims critically

CRITERION C · I, IV ◆ CRITERION D · III, IV, V ◆ JUSTIFICATION

A graph can be accurate and still mislead. The scale, the range, the summary chosen and the cases omitted all shape what a reader concludes. Critical reading is a mathematical skill, not a suspicion of all numbers.

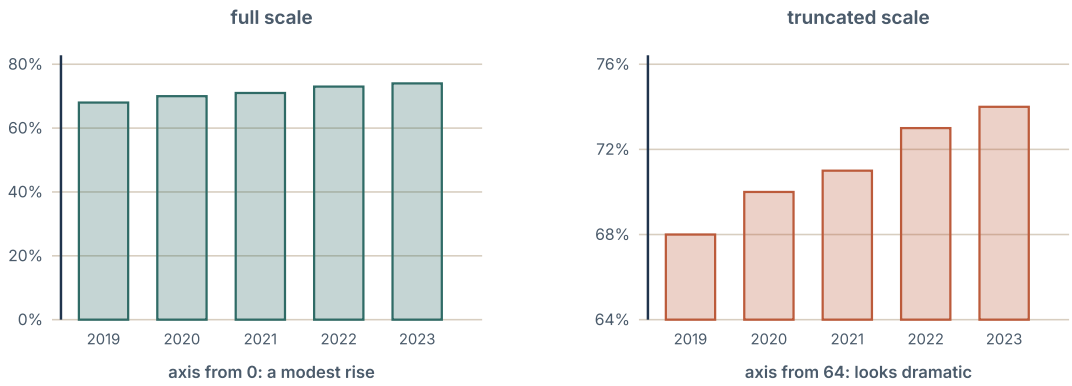


Figure 9.6 The same five values plotted twice. Truncating the vertical axis does not change the data, but it changes the impression: a rise of about 9% looks like a dramatic climb.

WORKED EXAMPLE 9.5.1 A TRUNCATED GRAPH

PROBLEM

A bar chart shows a school's pass rate rising from 68% to 74% over five years, with the vertical axis starting at 64% and ending at 76%. The bars nearly double in height. Evaluate the presentation.

1. The actual increase is $74 - 68 = 6$ percentage points, or $\frac{6}{68} \times 100 \approx 8.8\%$ relative to the starting value.
2. On an axis from 0% to 100%, the same bars would differ by a small amount. Starting at 64% magnifies the visual difference without changing the numbers.
3. Truncating an axis is not always dishonest: it can make small changes visible when the baseline is far from zero. But the axis must be labelled clearly, and the visual impression should not be described as a doubling.
4. A fair presentation would either start at zero or state the change in percentage points and show the truncated axis with an explicit break.

ANSWER **A 6 percentage-point rise (about 8.8%) is presented as an almost-doubling of bar height; the truncation exaggerates the change.**

WORKED EXAMPLE 9.5.2 WHEN SUBGROUPS REVERSE THE SUMMARY

PROBLEM

Two clinics report recovery rates. Clinic A: 81 of 87 minor cases and 192 of 263 major cases recovered. Clinic B: 234 of 270 minor cases and 55 of 80 major cases recovered. Compare the overall rates and the rates within each type of case.

1. Clinic A overall: $\frac{81+192}{87+263} = \frac{273}{350} = 0.780$, or 78.0%.
2. Clinic B overall: $\frac{234+55}{270+80} = \frac{289}{350} = 0.826$, or 82.6%.
3. Minor cases: A $\frac{81}{87} \approx 93.1\%$; B $\frac{234}{270} \approx 86.7\%$. A is higher.
4. Major cases: A $\frac{192}{263} \approx 73.0\%$; B $\frac{55}{80} = 68.8\%$. A is higher again.
5. Clinic A has the higher rate in both subgroups, yet Clinic B has the higher overall rate. Clinic B treated mostly minor cases, which have a higher recovery rate; its case mix raises the overall figure.
6. Comparing overall rates without accounting for case mix is misleading. A fair comparison reports rates within each type of case, or standardises the rates to a common case mix.

ANSWER **A: 78.0% overall and higher in both subgroups; B: 82.6% overall. B's higher overall rate is explained by its larger proportion of minor cases, not by better care.**

WORKED EXAMPLE 9.5.3 RELATIVE RISK AND BASE RATES

PROBLEM

A headline reports that teenagers who drink energy drinks are "twice as likely" to feel tired at school. In the study, 20% of energy-drink users felt tired, compared with 10% of non-users. Interpret the claim and suggest two alternative explanations.

1. The relative risk is $\frac{20}{10} = 2$, so "twice as likely" is arithmetically correct in this sample.
2. The absolute difference is $20 - 10 = 10$ percentage points. Reporting only the ratio hides the size of the effect; reporting only the difference hides the baseline.
3. **Reverse causation:** teenagers who are already tired may use energy drinks to compensate, so tiredness could precede and cause the drink use.
4. **Confounding:** users may sleep less, play more sport, work longer hours or have different health conditions. These could raise tiredness independently of the drinks.
5. The headline also generalises from one sample to "teenagers" without stating who was studied. The causal word "makes" would not be justified by an observational survey.

ANSWER **Relative risk 2, absolute difference 10 percentage points; reverse causation and sleep or lifestyle confounders are plausible, so the causal implication is not established.**

Ask four questions of any quantitative claim. **What is being compared?** A change from 68 to 74 is 6 percentage points or about 8.8%, depending on the comparison; both can be correct, but the wording should say which. **What is the baseline?** "Twice the risk" is alarming if the risk rises from 1 in 10 to 2 in 10, and less alarming if it rises from 1 in 10 000 to 2 in 10 000. **Who is in the data?** A sample of volunteers, a single school or a group with a particular age profile may not represent the population named in the headline. **What is left out?** Outliers, subgroups, non-response and missing data can reverse or erase an apparent pattern.

COMMON ERROR · CONFUSING PERCENTAGE POINTS WITH PER CENT

A rise from 40% to 44% is 4 percentage points, but it is a 10% relative increase, because 4 is 10% of 40. A fall from 50% to 40% is 10 percentage points and a 20% relative decrease. Say which one you mean; a headline that says only "10%" leaves the reader to guess.

PRACTICE 9.5 · READING CLAIMS CRITICALLY

SUPPORT 1–3 · CORE 1–8 · EXTENSION 11–12

1. Convert each change into percentage points and a relative change: (a) 30% to 36%; (b) 80% to 72%; (c) 2.0% to 2.5%. F
2. Explain why a bar chart with a vertical axis starting at 90 can make a change from 92 to 95 look large. F
3. A headline says "Risk of injury doubles on wet days". The risk rises from 0.2% to 0.4%. Explain what "doubles" hides. R
4. Explain how an outlier can create a strong correlation that disappears when the outlier is removed. R
5. In Worked example 9.5.2, calculate the proportion of minor cases at each clinic and explain how the case mix produces the reversal. R
6. A survey of 500 readers of a fitness magazine finds that 90% exercise daily. Explain why this result should not be generalised to all adults. R
7. A remedial programme is offered to the students with the lowest scores in a test. Their mean score rises when they are retested. Explain why this rise alone does not show that the programme worked. R
8. A study reports that people who walk more steps per day have lower blood pressure. Suggest one confounding variable and one possible reverse-causation explanation. R
9. A company reports that its new training programme raised average sales by 12%, but does not report the number of staff or the previous average. List three questions you would ask before accepting the claim. P
10. A dual-axis graph plots mean monthly temperature and ice-cream sales on the same chart; the two lines rise and fall almost together. Explain how the choice of scales could exaggerate the match, and state what you would plot instead. P
11. A small trial of 40 patients reports a 30% improvement, while a large trial of 4,000 reports a 4% improvement. Explain which result is more trustworthy, and why a larger measured effect is not automatically a more important finding. E
12. Find or invent a headline that reports a correlation as a cause. Rewrite it so that it accurately describes the evidence, and explain the changes you made. E

Investigation · Arm span and height

CRITERION B

Leonardo da Vinci observed that a person's arm span is approximately equal to their height. How strong is that relationship, and can one measurement predict the other?

SAMPLE DATA FROM TWELVE YEAR 3 STUDENTS

Student	Arm span x (cm)	Height y (cm)
1	152	155
2	158	159
3	161	163
4	165	166
5	168	168
6	170	172
7	173	174
8	176	178
9	179	180
10	182	183
11	186	187
12	190	191

1. **Measure.** Work in pairs. Measure arm span from fingertip to fingertip with arms outstretched, and height from the floor to the top of the head, both to the nearest centimetre. Keep each student's two values on one row.
2. **Plot.** Put arm span on the horizontal axis and height on the vertical axis. Label both axes with units and choose a scale that uses most of the grid. Describe the direction, form and strength of the scatter.
3. **Calculate.** Estimate the two means and mark the mean point on your plot. Calculate r with a spreadsheet or calculator and state what it says about the strength of the linear relationship.
4. **Model.** Draw a line of best fit through the mean point. Find its equation from two well-separated points on the line, and interpret the gradient and intercept in context.
5. **Predict.** Use your line to predict the height of a student with arm span 170 cm. Is this interpolation or extrapolation for your sample? Then predict for 120 cm and for 210 cm, and comment on reliability.
6. **Test.** Swap data with another pair. Predict three of their students' heights, compare with the measured heights, and calculate the residuals to decide whether your model generalises.
7. **Conjecture.** If arm span equals height on average, what gradient would you expect? Does your fitted gradient support the conjecture? Explain why individual variation and measurement error prevent an exact match.
8. **Evaluate.** Identify two limitations of the data — for example, the age range, the sample size, posture, shoes or who volunteered. Suggest one change that would make the conclusion more reliable, and one claim the data cannot support.

A high r shows only that the two measurements vary together in this sample; both are driven by growth. This is a model-building exercise, not a biological explanation.

Application · The screen-time claim

CRITERION D · I, II, III, IV ◆ MODELS ◆ FAIRNESS AND DEVELOPMENT

A school surveys 40 students about their average nightly screen time before bed and their average hours of sleep. The data produce a correlation of $r = -0.62$ and a line of best fit $s = 9.1 - 0.42t$, where t is screen time in hours and s is sleep in hours. The student newspaper wants to publish the headline "Screens steal your sleep".

SUMMARY OF THE SURVEY

Quantity	Value	Note
Students surveyed	40	volunteers from Years 3–5
Mean screen time	2.6 hours	self-reported, typical night
Mean sleep	8.0 hours	self-reported
Correlation	$r = -0.62$	moderate negative linear association
Line of best fit	$s = 9.1 - 0.42t$	t in hours, s in hours
Range of t	0.5 to 5.5 hours	observed values
Outlier	$(5.5, 5.0)$	one student reported both extremes

1. Interpret the gradient -0.42 in context. Does it mean that one extra hour of screen time causes 0.42 hours less sleep? Explain.
2. Interpret the intercept 9.1. Is a prediction at $t = 0$ meaningful for this data set? Explain.
3. Predict the sleep of a student with 2 hours of screen time and one with 5 hours. State whether each prediction is interpolation or extrapolation.
4. Calculate the residual for the outlier $(5.5, 5.0)$. Does the point lie above or below the line? What effect might it have on the correlation and the line?
5. Calculate r^2. Interpret it as the proportion of variation in sleep associated with the linear model, and explain why it does not prove that screen time causes lost sleep.
6. The survey relied on volunteers and self-report. Describe two ways these choices could bias the result, and suggest a better sampling method.
7. Name two plausible confounders and explain how each could produce the negative association without screen time causing less sleep.
8. Rewrite the headline for the student newspaper so that it is accurate, informative and still interesting. Justify each change in a short editorial note.
9. Design a follow-up study that could provide stronger evidence about causation. State the explanatory and response variables, the comparison, and one thing you would keep constant.

CRITERION D FOCUS

Use the model to quantify the association, then separate three questions: How strong is the pattern? What does it predict inside the data? What would it take to show that changing one variable changes the other? The third question is not answered by r.

Criterion task · The breakfast claim

CRITERIA C AND D
COMMUNICATING AND APPLYING
24 MARKS

A newspaper reports: "Students who eat breakfast score 12 points higher in end-of-year tests. Schools should provide breakfast to raise attainment." The report is based on a survey of 600 students in 12 schools. Students were asked whether they usually eat breakfast; test scores were supplied by the schools. The correlation between breakfast frequency (days per week) and score is $r = 0.31$. Students who ate breakfast on five or more days averaged 68 points; those who ate it on fewer than five days averaged 56 points. The overall average was 64 points.

1. Identify the explanatory and response variables, and state whether the study is observational or experimental. *(2 marks)*
2. Describe the association using the correlation and the group averages. Include the difference in points and comment on its size relative to the overall average. *(4 marks)*
3. Explain why $r = 0.31$ is consistent with a real but modest linear association, and why the headline's "12 points higher" is a comparison of group averages rather than a causal effect. *(4 marks)*
4. Name three plausible confounding variables and explain how each could raise both breakfast consumption and test scores. *(5 marks)*
5. Explain how selection could affect who eats breakfast and who does not. Suggest one way to reduce selection bias. *(3 marks)*
6. Design a stronger study to test whether providing breakfast raises attainment. Include allocation, comparison, response variable and two controls. *(4 marks)*
7. Write a 120-word summary for a school newsletter. Report the evidence accurately, avoid causal language that the design cannot support, and recommend one reasonable action. *(2 marks)*

PLANNING FRAME

Identify: breakfast frequency explains; test score responds; no intervention, so observational. **Describe:** $r = 0.31$ is weak-to-moderate positive; 12 points is 18.75% of the overall mean of 64. **Separate:** group difference is not the same as an individual causal effect. **Challenge:** income, parental support, sleep, prior attainment, school quality. **Design:** random allocation of breakfast provision, compare score change, same test and attendance conditions. **Communicate:** "associated with" not "causes"; state the sample and the limits.

BAND	DESCRIPTOR	MARKS
4	Variables and design identified correctly; association described with r and group difference; three specific confounders and selection explained; strong randomised design; newsletter accurate and appropriately cautious.	19–24
3	Most identification and description correct; some confounders named with partial explanation; design mostly sound; newsletter mostly accurate with minor overstatement.	13–18
2	Basic description; confounders listed without explanation; design weak or incomplete; newsletter repeats the causal claim.	7–12
1	Little correct interpretation; no clear distinction between association and cause; design unusable.	1–6

CRITERIA: C, D **ATL:** RESEARCH **TIME:** 60 MINUTES **CONDITIONS:** CALCULATOR ALLOWED

MAPPED TO: C strands i–iv and D strands i–v. The rubric is a course summary, not an official IB descriptor.

Chapter summary and review

KEY IDEAS

- Bivariate data pairs two measurements on the same unit; a scatter plot displays the pairs.
- Describe a scatter plot by direction, form, strength and unusual features.
- Pearson's r measures the direction and strength of a linear association, from -1 to 1.
- r is unchanged by rescaling or swapping variables, but it is sensitive to outliers and blind to curved patterns.
- A line of best fit passes through the mean point and balances residuals; its gradient and intercept need contextual interpretation.
- Interpolation is safer than extrapolation; a model should not be extended far beyond the observed range.
- Correlation is not causation: common causes, reverse causation, selection and chance are alternative explanations.
- A causal claim is supported by controlled experiments, correct temporal order, a mechanism and the ruling out of confounders.
- Graphs and headlines can mislead through truncated axes, relative versus absolute risk, selection and subgroup reversal.
- Always report the sample, the measure, the range and the uncertainty alongside a correlation.

REVISION CHECKLIST

- ☐ I can identify explanatory and response variables and label axes with units.
- ☐ I can describe direction, form, strength and outliers.
- ☐ I can calculate r from a small table.
- ☐ I can interpret the sign and size of r, including its limits.
- ☐ I can draw and interpret a line of best fit.
- ☐ I can calculate residuals and use them to judge a model.
- ☐ I can distinguish interpolation from extrapolation.
- ☐ I can name a plausible confounder and explain reverse causation.
- ☐ I can design a simple controlled experiment.
- ☐ I can evaluate a graph or headline for exaggeration and missing context.

VOCABULARY

bivariate data
Two measurements made on the same unit.

response variable
The outcome measured in a study.

association
A tendency for two variables to vary together.

negative association
As one variable increases, the other tends to decrease.

cluster
A separated group of points.

line of best fit
The line that best summarises a linear trend.

interpolation
Prediction inside the observed range.

causation
A change in one variable produces a change in another.

explanatory variable
The variable used to explain or predict.

scatter plot
A graph of paired data as points.

positive association
As one variable increases, the other tends to increase.

outlier
A point that does not fit the overall pattern.

correlation coefficient
A number from -1 to 1 measuring linear association.

residual
Observed value minus predicted value.

extrapolation
Prediction outside the observed range.

confounding variable
A third variable influencing both variables in an association.

reverse causation
The response variable influences the explanatory variable.

observational study
A study that records variables without intervening.

controlled experiment
A study that deliberately changes one variable and compares outcomes.

relative risk
The ratio of two risks or probabilities.

MIXED PRACTICE · CHAPTER 9

Connecting the ideas

FLUENCY Core procedures

1. Calculate r for $(2, 5)$, $(4, 8)$, $(6, 10)$, $(8, 14)$, $(10, 15)$. F
2. A line of best fit passes through $(3, 11)$ and $(7, 23)$. Find its equation and predict y at $x = 9$. F
3. For the line $y = 2.5x + 6$, calculate the residual at $(4, 17)$. F
4. Convert a rise from 24% to 30% into percentage points and a relative increase. F

REASONING Justify and connect

1. A data set has $r = 0.88$. Explain why this does not mean that 88% of the variation is explained by the linear model. R
2. Explain how the same increase in a pass rate can be described as "5%" or "5 percentage points", and why the distinction matters. R
3. A strong correlation is found between the number of books in a home and children's test scores. Explain why a randomised experiment with books would be difficult to run, and what observational evidence could still strengthen a causal argument. R
4. Explain why a line of best fit should not be used to predict beyond the range of the data, using an example where the model eventually becomes impossible. R

APPLICATION Use and evaluate

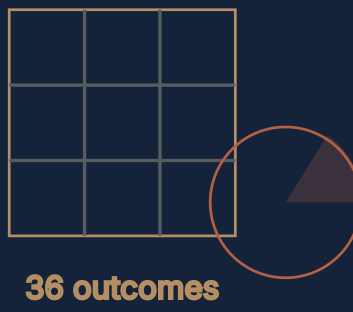
1. The age of a used car x (years) and its price y (thousands) give $r = -0.84$ and $y = 18 - 2.1x$. Interpret both numbers, predict the price of a 5-year-old car, and state one reason the model may fail for a classic car. P
2. A study of 50 students finds a correlation of 0.45 between hours of sport per week and mathematics score. A second study of 500 students finds 0.12. Explain why the larger study is more trustworthy and what the difference suggests about the first sample. P
3. A health campaign reports that a disease rate fell from 0.8% to 0.5%. Calculate the absolute and relative reductions, and write one accurate sentence for a poster. P
4. A school wants to know whether a new revision app improves scores. Describe an observational comparison and a randomised experiment, and explain which provides stronger evidence for causation. P
5. A scatter plot of 20 countries shows a positive association between average income and life expectancy, with $r = 0.72$. Explain why this does not show that raising an individual's income by a fixed amount adds a fixed number of years of life. E
6. A data set has a strong correlation created almost entirely by one point far from the rest. Describe how you would detect this and how you would report the result responsibly. E

Check your progress

I CAN …	CONFIDENCE	EVIDENCE	NEXT ACTION
Identify variables and describe a scatter plot.	1 / 2 / 3		
Calculate and interpret the correlation coefficient.	1 / 2 / 3		
Fit and use a line of best fit, including residuals.	1 / 2 / 3		
Distinguish correlation from causation and name confounders.	1 / 2 / 3		
Evaluate a graph or headline built on bivariate data.	1 / 2 / 3		

Reflection

A correlation can be genuine, statistically strong and still useless for a decision — or weak and still important. What should a reader ask first: how strong the association is, or how the data were produced? Write two sentences explaining your order of priorities.



10

DATA, CHANCE AND EVIDENCE

Probability, sampling and evidence

Chance has structure. Outcomes can be listed, events compared and long-run frequencies predicted. A sample adds a second question: not only what is likely, but who was asked, and what their answers can reasonably support.

KEY CONCEPT · LOGIC | RELATED · QUANTITY | RELATED · MODELS | CONTEXT · IDENTITIES AND RELATIONSHIPS

INQUIRY

How can a limited sample support a claim about a whole population — and how should the uncertainty in that claim be communicated?

ATL · COMMUNICATION

A probability is only as good as the assumptions behind it, and a survey is only as good as the people it reached. Both have to be explained in plain language: what was counted, what was assumed, who was included and how large the uncertainty is. “One in four” and “25%” can describe the same number yet leave a reader with different impressions.

MYP UNIT PLANNER

Framing the inquiry

UNIT 10 · APPROX. 13 HOURS
ATL FOCUS · COMMUNICATION

KEY AND RELATED CONCEPTS

Logic. Probability reasons from a clearly defined set of outcomes to a quantified conclusion. The conclusion is only valid while the assumptions — a fair die, a representative sample — remain true.

Quantity. Probabilities are numbers between 0 and 1. Expected frequencies scale with the number of trials, and a sample proportion is an estimate rather than the population value itself.

Models and justification. A theoretical probability and a sampling method are both models of a real process. Evidence for a claim about a population depends on how the sample was selected and how the result is reported.

GLOBAL CONTEXT AND EXPLORATION

Identities and relationships. Exploration: how surveys about young people's habits, health risks and school decisions are sampled, summarised and used — and who is left out when a sample is chosen badly.

Statistical claims shape how groups are seen: "most teenagers", "one in four students", "a 90% accurate test". The mathematics cannot decide the policy, but it can show whether the sample represents the group, whether a risk is stated fairly, and how much confidence a number deserves.

STATEMENT OF INQUIRY

Probability models uncertainty, and a sample supports a claim about a population only when the method of selection, the size of the sample and the limits of the evidence are communicated honestly.

FACTUAL

How is a probability calculated from a sample space? How are compound events represented? What is a random sample, and how is it selected?

CONCEPTUAL

Why does relative frequency approach theoretical probability as the number of trials grows? Why can a larger sample be more reliable yet still biased? How does the framing of a risk change a decision?

DEBATABLE

Should a school act on a survey of volunteers? When is a "90% accurate" test good enough to use for screening? Who should decide how a risk is reported?

CRITERION	STRANDS DEVELOPED	EVIDENCE IN THIS CHAPTER
A · Knowing and understanding	i, ii	Use sample spaces, probability rules and sampling vocabulary correctly; interpret risk statements.
B · Investigating patterns	i, ii, iii	Collect trial data, compare relative frequencies with theory, design a simulation and plan a sample survey.
C · Communicating	i, ii, iii, iv	Represent compound events with tables, Venn diagrams and trees; report estimates, methods and limitations.
D · Applying mathematics	i, ii, iii, iv, v	Evaluate a screening claim and a sampling claim; recommend a fairer method and a clearer form of words.

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** outcomes, events and sample spaces, and calculate theoretical probabilities.
- **Be able to** combine events using tables, Venn diagrams and tree diagrams.
- **Know** how relative frequency estimates probability and how simulation tests a model.
- **Understand** populations, samples, random selection and the main sources of bias.
- **Be able to** interpret absolute and relative risk and recognise base-rate errors.
- **Be able to** design a small sample survey and report its estimate with appropriate caution.

SUCCESS CRITERIA

- ☐ I can list a sample space systematically and use it to calculate probabilities.
- ☐ I can use the complement rule and the addition rule correctly.
- ☐ I can draw a tree or Venn diagram for a two-stage or two-event problem.
- ☐ I can compare experimental relative frequency with theoretical probability and explain the difference.
- ☐ I can identify the population and sample, and name a specific source of bias.
- ☐ I can report a risk or survey estimate with the absolute numbers and the method behind it.

PREREQUISITE CHECK · 12 MINUTES, NO CALCULATOR

1. Write $\frac{3}{8}$ as a decimal and as a percentage.
2. A bag contains 6 red and 4 blue counters. Find the probability that a counter drawn at random is red.
3. Write 0.07 as a fraction and as a percentage.
4. Find 15% of 240.
5. Order 0.6, $\frac{2}{3}$, 0.65 and $\frac{3}{5}$ from smallest to largest.
6. Find the mean of $12, 18, 15, 20, 15$.

Answers are in the Chapter 10 answer section. If more than two are incorrect, review fraction–decimal–percentage conversion, simple probability and the mean before continuing.

BEFORE YEAR 3

Simple probability as a fraction; listing outcomes; conversion between fractions, decimals and percentages; the mean of a small data set.

YEAR 3 · THIS CHAPTER

Sample spaces; compound events; tree and Venn diagrams; relative frequency; simulation; random sampling and bias; absolute and relative risk.

LATER

Conditional probability and Bayes' theorem; probability distributions; confidence intervals and significance tests; large-scale survey design and causal inference.

INQUIRY PATHWAY FOR THIS CHAPTER

1	2	3	4	5	6
CONTEXT	**QUESTION**	**CONCEPT**	**INVESTIGATION**	**COMMUNICATION**	**REFLECTION**
Public claims use percentages, probabilities and survey results to describe groups and risks.	What is likely, who was asked, and what can the evidence support?	Probability, sample space, relative frequency, sampling, bias and risk.	Design and carry out a small random sample; compare the estimate with a census and report the limits.	Use fractions, decimals and percentages accurately; report absolute and relative risk; state the sample and method.	When does a probability become evidence for a decision, and what would change your mind?

10.1 Outcomes, events and probability

CRITERION A · I, II ◆ CRITERION C · I, II ◆ QUANTITY

Probability begins with a clear question: what could happen? Once the possible outcomes are listed, likelihood can be counted rather than guessed.

DEFINITIONS

A **probability experiment** is a process with an uncertain result; each repetition is a **trial**. An **outcome** is one possible result, and the **sample space** is the set of all possible outcomes. An **event** is a set of outcomes. Outcomes are **equally likely** when each has the same chance of occurring. For a finite sample space with equally likely outcomes,

$$P(A) = \frac{\text{number of outcomes in } A}{\text{total number of outcomes}}.$$

The **complement** of A, written A', consists of all outcomes not in A. A coin, die or spinner is **fair** when its outcomes are equally likely.

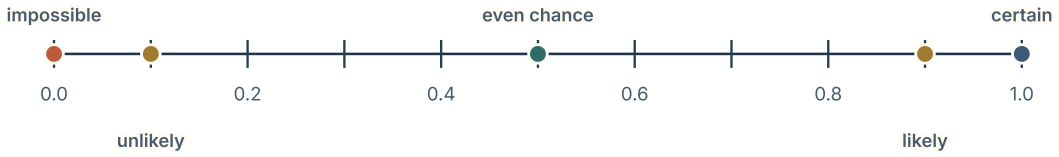


Figure 10.1 Probability is measured on a scale from 0 to 1. A probability of 0.5 is an even chance, not a guarantee that one of each outcome will occur in the next two trials.

Three rules follow from the definition. First, every probability lies between 0 and 1, because an event cannot contain fewer than no outcomes or more than all of them. Second, the probabilities of all the outcomes in a sample space add to 1. Third, an event either happens or does not, so

$$P(A) + P(A') = 1.$$

The complement rule is often the quickest route: if “at least one” is awkward to count, count “none” instead and subtract from 1.

WORKED EXAMPLE 10.1.1 — ONE FAIR DIE

PROBLEM

A fair six-sided die is rolled once. Find (a) $P(\text{even})$, (b) $P(\text{greater than } 4)$, (c) $P(\text{not prime})$ and (d) $P(\text{even or prime})$.

1. The sample space is $\{1, 2, 3, 4, 5, 6\}$, with 6 equally likely outcomes.
2. (a) The even outcomes are $2, 4, 6$, so $P(\text{even}) = \frac{3}{6} = \frac{1}{2}$.
3. (b) The outcomes greater than 4 are $5, 6$, so $P(\text{greater than } 4) = \frac{2}{6} = \frac{1}{3}$.
4. (c) The primes are $2, 3, 5$, so the non-primes are $1, 4, 6$. Hence $P(\text{not prime}) = \frac{3}{6} = \frac{1}{2}$.
5. (d) "Even or prime" contains $2, 3, 4, 5, 6$. The outcome 2 is in both events but must be counted once, so the probability is $\frac{5}{6}$, not $\frac{3}{6} + \frac{3}{6} = \frac{6}{6}$.

ANSWER **(a)** $\frac{1}{2}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{5}{6}$.

WORKED EXAMPLE 10.1.2 — A BAG OF COUNTERS

PROBLEM

A bag contains 5 red, 3 blue and 2 green counters, all the same size. One counter is drawn at random. Find (a) $P(\text{red})$, (b) $P(\text{blue or green})$, (c) $P(\text{not blue})$, and (d) the number of red counters expected in 40 draws if each counter is replaced after it is drawn.

1. There are $5 + 3 + 2 = 10$ equally likely counters.
2. (a) $P(\text{red}) = \frac{5}{10} = \frac{1}{2}$.
3. (b) The blue and green counters are $3 + 2 = 5$, so $P(\text{blue or green}) = \frac{5}{10} = \frac{1}{2}$.
4. (c) The complement of blue contains the 7 red and green counters, so $P(\text{not blue}) = \frac{7}{10}$.
5. (d) In 40 draws, the expected number of reds is $40 \times \frac{5}{10} = 20$. This is a long-run average, not a promise that exactly 20 of the next 40 draws will be red.

ANSWER **(a)** $\frac{1}{2}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{7}{10}$ **(d)** 20.

When an experiment has two stages, a table lists the sample space systematically and prevents omissions. The figure below shows every result of rolling two fair dice; each cell is one of 36 equally likely outcomes.

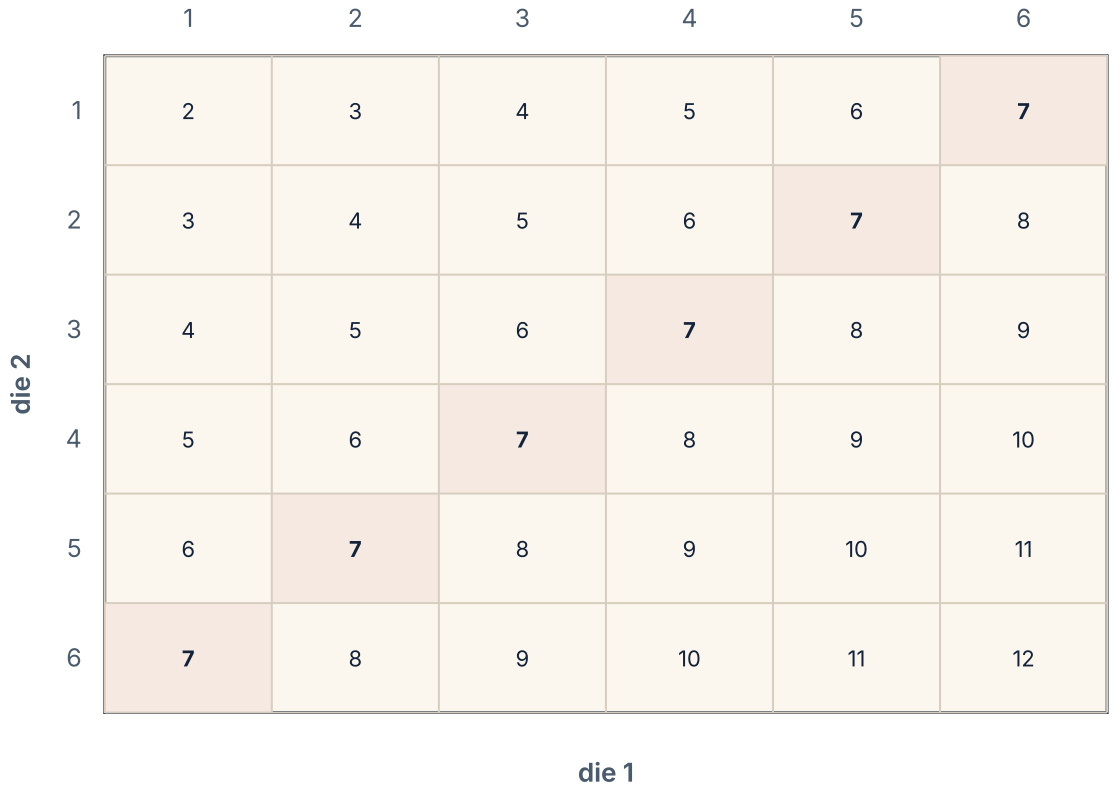


Figure 10.2 The sample space for the sum of two fair dice. Each of the 36 cells is equally likely, even though the possible sums are not: there is one way to make 2 and six ways to make 7.

WORKED EXAMPLE 10.1.3 TWO FAIR DICE

PROBLEM

Two fair dice are rolled and the sum of the faces is recorded. Find (a) $P(\text{sum } 7)$, (b) $P(\text{prime sum})$, (c) $P(\text{at least one } 6)$ and (d) $P(\text{sum greater than } 9)$.

1. There are $6 \times 6 = 36$ equally likely outcomes. Using the table or an ordered list avoids counting the same roll twice.
2. (a) The sums equal to 7 are $(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)$, six outcomes. So $P(\text{sum } 7) = \dfrac{6}{36} = \dfrac{1}{6}$.
3. (b) The prime sums are $2, 3, 5, 7, 11$, with $1 + 2 + 4 + 6 + 2 = 15$ outcomes. So $P(\text{prime sum}) = \dfrac{15}{36} = \dfrac{5}{12}$.
4. (c) The complement of "at least one 6" is "no 6", which has $5 \times 5 = 25$ outcomes. Hence $P(\text{at least one } 6) = 1 - \dfrac{25}{36} = \dfrac{11}{36}$.
5. (d) Sums greater than 9 are $10, 11, 12$, with $3 + 2 + 1 = 6$ outcomes, so the probability is $\dfrac{6}{36} = \dfrac{1}{6}$.

ANSWER **(a)** $\dfrac{1}{6}$ **(b)** $\dfrac{5}{12}$ **(c)** $\dfrac{11}{36}$ **(d)** $\dfrac{1}{6}$.

GUIDED PRACTICE · USE THE DICE TABLE

Using Figure 10.2, find (a) $P(\text{sum } 5)$, (b) $P(\text{sum } 12)$, (c) $P(\text{sum not } 7)$ and (d) $P(\text{at least one } 3)$. For each, first count the favourable outcomes, then divide by 36. Check (c) with the complement rule.

COMMON ERROR · ASSUMING OUTCOMES ARE EQUALLY LIKELY

"It either happens or it does not" does not make the probability $\frac{1}{2}$. A biased coin can land heads with probability 0.8. Equally likely outcomes must be justified by the physical setup — a fair die, a well-balanced spinner, a random draw from a thoroughly mixed bag — not by the number of descriptions we can invent.

PRACTICE 10.1 · SAMPLE SPACES AND PROBABILITY SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–9

1. A fair coin is tossed once. List the sample space and find $P(\text{head})$. F
2. A fair eight-sector spinner is numbered 1 to 8. Find $P(\text{prime})$, $P(\text{multiple of } 4)$ and $P(\text{not } 3)$. F
3. A bag contains 4 red, 6 yellow and 2 black counters. One counter is drawn at random. Find $P(\text{red})$, $P(\text{not black})$ and $P(\text{red or yellow})$. F
4. Two fair dice are rolled. Find $P(\text{sum } 9)$, $P(\text{sum at most } 3)$ and $P(\text{both faces odd})$. R
5. Explain why a forecast of $P(\text{rain}) = 0.3$ does not mean that it will rain for 30% of the day. R
6. A student says: "There are two outcomes, win or lose, so the probability of winning is $\frac{1}{2}$." Critique the statement and give a counterexample. R
7. A bag contains 3 red counters and some blue counters. The probability of drawing red is $\frac{1}{4}$. Find the number of blue counters. P
8. Design a twelve-sector spinner so that $P(\text{red}) = \frac{1}{3}$, $P(\text{blue}) = \frac{1}{4}$ and the remaining sectors are green. State the number of sectors of each colour. P
9. Two fair dice are rolled. Find one event with probability $\frac{1}{2}$ and one with probability $\frac{1}{3}$, other than the events in Worked example 10.1.3. Justify each by counting outcomes. E

10.2 Compound events: tables, trees and Venn diagrams

CRITERION A · I, II ◆ CRITERION C · I, II, III ◆ LOGIC

Most real questions combine events: rain and a delayed bus, a red counter and then another red, a student who studies both languages. A representation that makes the overlap visible is the safest way to count.

DEFINITIONS

A **compound event** combines two or more events. Two events are **mutually exclusive** when they cannot occur together. They are **independent** when the occurrence of one does not change the probability of the other. A **dependent** event's probability does change, as when a counter is drawn and not replaced. **With replacement** means the item is returned before the next draw; **without replacement** means it is not. The **union** $A \cup B$ means A or B or both; the **intersection** $A \cap B$ means both.

Venn diagrams are useful when the same individuals can belong to more than one group. Count the overlap first, then fill in the "only" regions, and check that all the counts add to the total.

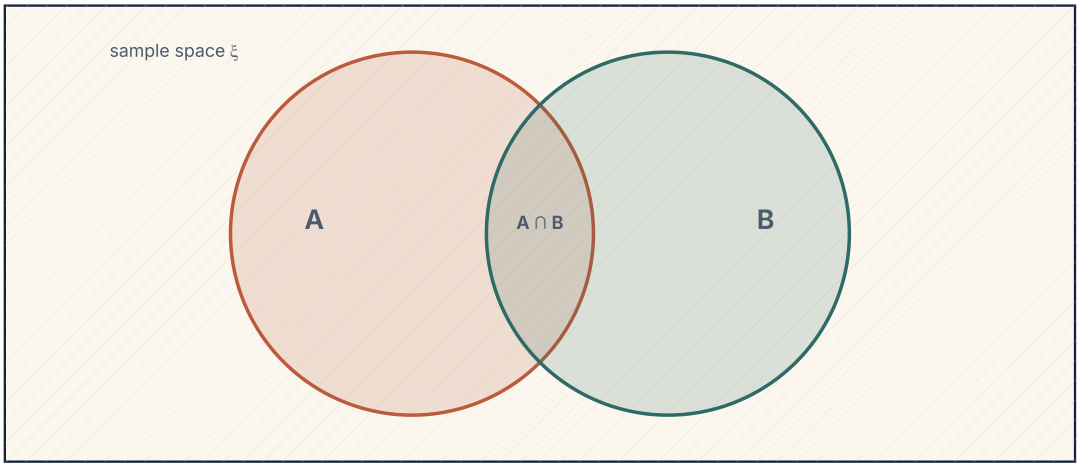


Figure 10.3 Two events A and B inside a sample space. Adding $P(A)$ and $P(B)$ counts the overlap twice, so the addition rule subtracts it once.

If A and B are events in the same sample space, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

The rule is a counting statement: $A \cup B$ consists of the outcomes in A only, in B only, and in both. Adding the counts of A and B includes the overlap twice, so one copy is subtracted. When A and B are mutually exclusive, the intersection is empty and the rule reduces to $P(A \cup B) = P(A) + P(B)$.

WORKED EXAMPLE 10.2.1 A TWO-LANGUAGE CLASS

PROBLEM

In a class of 30 students, 18 study French, 12 study Spanish and 5 study both. A student is chosen at random. Find $P(\text{French})$, $P(\text{Spanish})$, $P(\text{both})$, $P(\text{French or Spanish})$ and $P(\text{neither})$.

1. Start with the overlap: 5 students study both languages.
2. French only: $18 - 5 = 13$. Spanish only: $12 - 5 = 7$.
3. Students studying at least one language: $13 + 7 + 5 = 25$, leaving $30 - 25 = 5$ who study neither.
4. $P(\text{French}) = \dfrac{18}{30} = \dfrac{3}{5}$ and $P(\text{Spanish}) = \dfrac{12}{30} = \dfrac{2}{5}$.
5. $P(\text{both}) = \dfrac{5}{30} = \dfrac{1}{6}$.
6. Using the addition rule, $P(\text{French or Spanish}) = \dfrac{18}{30} + \dfrac{12}{30} - \dfrac{5}{30} = \dfrac{25}{30} = \dfrac{5}{6}$.
7. The complement gives $P(\text{neither}) = 1 - \dfrac{25}{30} = \dfrac{1}{6}$, which agrees with the count of 5 students.

ANSWER $\dfrac{3}{5}, \dfrac{2}{5}, \dfrac{1}{6}, \dfrac{5}{6}, \dfrac{1}{6}$.

Two events are independent when one outcome tells you nothing about the other. For independent events, the multiplication rule applies:

$$P(A \cap B) = P(A) \times P(B).$$

This is why a die and a coin are convenient: whatever the die shows, the coin is still equally likely to land heads. Drawing without replacement is different, because the first draw changes the contents of the bag and therefore the probability on the second draw.

WORKED EXAMPLE 10.2.2 INDEPENDENT EVENTS

PROBLEM

A fair die is rolled and a fair coin is tossed. Find (a) $P(\text{a 6 and a tail})$, (b) $P(\text{an even number and a head})$ and (c) $P(\text{a 6 or a head})$.

1. The die and the coin are independent, so multiply the probabilities for "and".
2. (a) $P(6) = \frac{1}{6}$ and $P(\text{tail}) = \frac{1}{2}$, so $P(6 \text{ and tail}) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}$.
3. (b) $P(\text{even}) = \frac{1}{2}$, so $P(\text{even and head}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.
4. (c) For "or", use the addition rule: $P(6 \text{ or head}) = \frac{1}{6} + \frac{1}{2} - \frac{1}{12} = \frac{2}{12} + \frac{6}{12} - \frac{1}{12} = \frac{7}{12}$.

ANSWER **(a)** $\frac{1}{12}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{7}{12}$.

A tree diagram shows the stages of an experiment in order. Multiply along a branch to find the probability of that sequence, and add the probabilities of all complete branches that satisfy the event.

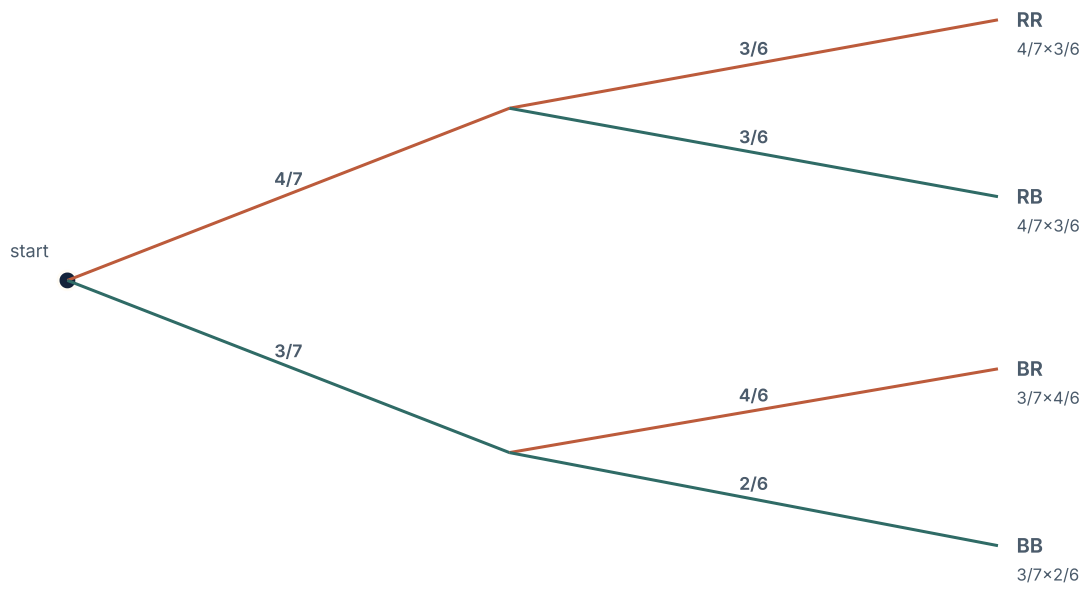


Figure 10.4 Drawing twice without replacement from 4 red and 3 blue counters. The second-stage denominator is 6 because one counter has already been removed.

WORKED EXAMPLE 10.2.3 TWO DRAWS WITHOUT REPLACEMENT

PROBLEM

A bag contains 4 red and 3 blue counters. Two counters are drawn without replacement. Find (a) $P(\text{both red})$, (b) $P(\text{one of each colour})$ and (c) $P(\text{at least one red})$.

1. First draw: $P(\text{red}) = \frac{4}{7}$ and $P(\text{blue}) = \frac{3}{7}$.
2. If the first is red, 3 red and 3 blue remain: $P(\text{red second}) = \frac{3}{6}$ and $P(\text{blue second}) = \frac{3}{6}$.
3. If the first is blue, 4 red and 2 blue remain: $P(\text{red second}) = \frac{4}{6}$ and $P(\text{blue second}) = \frac{2}{6}$.
4. (a) $P(\text{RR}) = \frac{4}{7} \times \frac{3}{6} = \frac{12}{42} = \frac{2}{7}$.
5. (b) One of each colour is RB or BR: $\frac{4}{7} \times \frac{3}{6} + \frac{3}{7} \times \frac{4}{6} = \frac{12}{42} + \frac{12}{42} = \frac{24}{42} = \frac{4}{7}$.
6. (c) The complement of "at least one red" is BB: $P(\text{BB}) = \frac{3}{7} \times \frac{2}{6} = \frac{6}{42} = \frac{1}{7}$. Hence $P(\text{at least one red}) = 1 - \frac{1}{7} = \frac{6}{7}$.

ANSWER **(a)** $\frac{2}{7}$ **(b)** $\frac{4}{7}$ **(c)** $\frac{6}{7}$.

COMMON ERROR · CONFUSING "MUTUALLY EXCLUSIVE" WITH "INDEPENDENT"

Mutually exclusive events cannot happen together; independent events do not influence each other. Drawing a red card and drawing a black card in a single draw are mutually exclusive, but not independent: if you draw red, the probability of black is 0. Rolling a 6 and tossing a head are independent, but not mutually exclusive: both can happen. For independent events, use multiplication; for "or" with an overlap, use the addition rule.

PRACTICE 10.2 · COMPOUND EVENTS

SUPPORT 1–3 · CORE 1–7 · EXTENSION 8–10

1. Two fair coins are tossed. List the sample space and find $P(\text{two heads})$ and $P(\text{exactly one head})$. F
2. A fair die is rolled and a fair coin is tossed. Find $P(\text{odd and tail})$ and $P(\text{odd or tail})$. F
3. $P(A) = 0.5$, $P(B) = 0.3$, and A and B are mutually exclusive. Find $P(A \cup B)$ and $P(\text{neither})$. F
4. $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$. Find $P(A \cup B)$ and decide whether A and B are independent. R
5. In a group of 80 students, 30 play football, 25 play basketball and 10 play both. Find the probability that a student chosen at random plays football, plays basketball, plays both, plays at least one, and plays neither. R
6. A bag contains 5 red and 4 green counters. Two are drawn without replacement. Find $P(\text{both green})$ and $P(\text{at least one red})$. R
7. Explain why $P(A \cap B) = P(A) \times P(B)$ applies to independent events but not to draws without replacement. Give one example of each. R
8. A bag contains 3 red and 2 white counters. Draw a tree diagram for two draws without replacement and find $P(\text{same colour})$ and $P(\text{different colours})$. P
9. A game uses two fair dice. A player wins if the sum is 7 or 11. Find the probability of winning. In 180 games, how many wins would you expect? P
10. $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \cup B) = 0.9$. Find $P(A \cap B)$ and determine whether A and B are independent. Justify your answer. E

10.3 From theory to experiment

CRITERION A · I, II ◆ CRITERION B · I, II ◆ MODELS

A theoretical probability is a prediction from assumptions. An experiment gives a measurement. Comparing the two is how we test whether the assumptions — and the model — are reasonable.

DEFINITIONS

The **relative frequency** of an outcome is $\frac{\text{number of times it occurred}}{\text{number of trials}}$. It is also called the **experimental probability**. A **simulation** imitates a probability experiment using a model such as random digits, a spreadsheet or a die. The **law of large numbers** says, informally, that as the number of trials increases, relative frequencies tend to settle near the theoretical probabilities, provided the trials are independent and the model is correct.

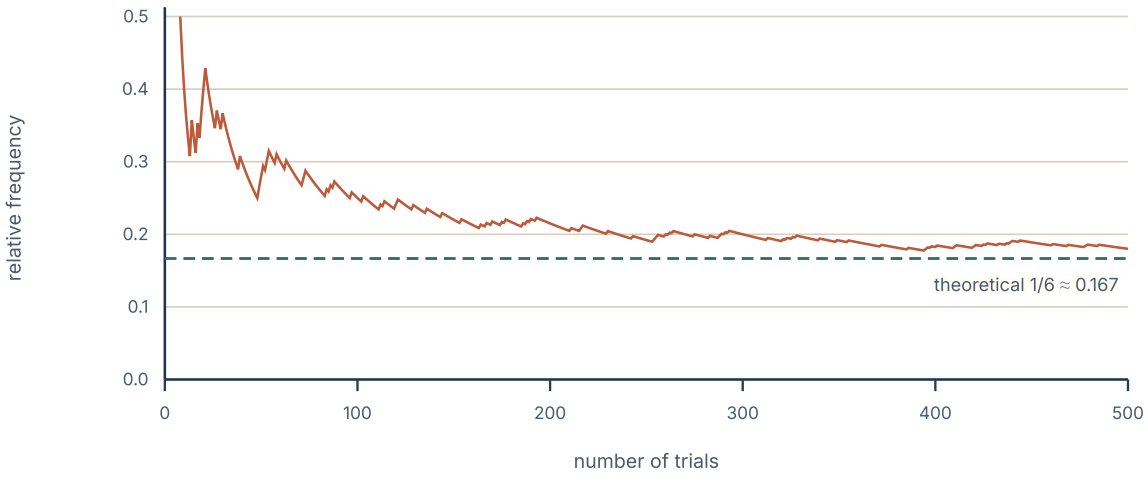


Figure 10.5 Relative frequencies from repeated trials. Early trials scatter widely; with more trials the values cluster more tightly around the theoretical probability.

WORKED EXAMPLE 10.3.1 TWO HUNDRED SPINS

PROBLEM

A spinner has a green half and a quarter each of red and blue. In 200 spins, it lands on red 46 times, blue 58 times and green 96 times. Calculate the relative frequency of each colour, compare with the theoretical probability, and comment.

1. Relative frequencies: red $\frac{46}{200} = 0.23$; blue $\frac{58}{200} = 0.29$; green $\frac{96}{200} = 0.48$.
2. Theoretical probabilities: red 0.25, blue 0.25, green 0.50.
3. All three experimental values are close to theory. Blue is 0.04 above its theoretical value and green 0.02 below.
4. These differences are consistent with ordinary sampling variation. They do not, by themselves, show that the spinner is biased; more trials would show whether the gap persists.

ANSWER **Red 0.23, blue 0.29, green 0.48; close to $0.25, 0.25, 0.50$. The differences are small and plausible for 200 trials.**

WORKED EXAMPLE 10.3.2 SIMULATING FREE THROWS

PROBLEM

A basketball player scores about 70% of free throws. Simulate 20 throws using the random digits

$4\ 8\ 1\ 0\ 6\ 6\ 9\ 3\ 5\ 0\ 7\ 2\ 1\ 4\ 8\ 3\ 9\ 5\ 0\ 6$

taking 0–6 as a score and 7–9 as a miss. Estimate the probability from the simulation and compare with 0.7.

1. There are 7 scoring digits out of 10, so each digit scores with probability 0.7.
2. Scoring digits in the list: $4, 1, 0, 6, 6, 3, 5, 0, 2, 1, 4, 3, 5, 0, 6$ — 15 scores.
3. Misses: $8, 9, 7, 8, 9$ — 5 misses. Check: $15 + 5 = 20$.
4. The simulated relative frequency is $\frac{15}{20} = 0.75$, above 0.70 but not unreasonable for only 20 trials.
5. A second run would give a different value. Averaging over many runs, or using 200 trials, would produce an estimate closer to 0.70.

ANSWER **15 scores in 20 trials gives 0.75; a small-sample estimate of the 0.70 model.**

WORKED EXAMPLE 10.3.3 IS THE DIE FAIR?

PROBLEM

A die is rolled 120 times with the frequencies shown. Calculate the relative frequency of each face, compare with $\frac{1}{6}$, and decide what can and cannot be concluded.

Face	1	2	3	4	5	6
Frequency	18	22	19	21	17	23
Relative frequency	0.150	0.183	0.158	0.175	0.142	0.192

1. The expected frequency of each face in 120 rolls is $120 \times \frac{1}{6} = 20$.
2. Each observed frequency is within 3 of 20. The relative frequencies lie between 0.142 and 0.192, around 0.167.
3. These deviations are small and balanced: three faces are above 20 and three below.
4. The data are consistent with a fair die, but they do not prove fairness. A larger number of rolls, or a formal test, would be needed to detect a small bias.

ANSWER **Relative frequencies $0.150, 0.183, 0.158, 0.175, 0.142, 0.192$, all close to $\frac{1}{6} \approx 0.167$. No evidence of bias, but fairness is not proved.**

COMMON ERROR · THE GAMBLER'S FALLACY

After five heads in a row, some people say a tail is "due". If the coin is fair, the next toss is still $P(\text{head}) = 0.5$. The coin has no memory, and the law of large numbers does not schedule compensation; it says that in a very long sequence the proportion of heads will settle near one half, not that the next trial must correct the past.

PRACTICE 10.3 · EXPERIMENTAL PROBABILITY

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. A coin is tossed 80 times and lands heads 36 times. Find the relative frequency and compare it with the theoretical probability. F
2. A spinner is expected to land red with probability 0.2. In 250 spins it lands red 60 times. Find the relative frequency and the expected number of reds. F
3. A simulation uses digits 0–9, and success is a digit from 0 to 3. State the modelled probability. In 15 trials the successes are $1, 4, 0, 2, 9, 3, 3, 7, 0, 5, 2, 8, 1, 3, 4$. Estimate the probability and compare. R
4. Explain why rolling a die 10 times is not enough to decide whether it is biased. R
5. A fair coin lands heads six times in a row. Find the probability that the next toss is a head, and explain your reasoning. R
6. Design a simulation for the sexes of three children in a family, assuming $P(\text{boy}) = P(\text{girl}) = 0.5$. Explain how to use random digits and how to estimate $P(\text{all three the same sex})$. P
7. A game website claims a "70% win rate" from 20 plays. Explain why this is weak evidence and describe a better way to test the claim. P
8. Two students simulate 20 trials each and obtain relative frequencies of 0.30 and 0.45. One says their results prove the true probability is between them. Critique this reasoning and explain what combining the trials would show. E

10.4 Populations, samples and bias

CRITERION A · I, II ◆ CRITERION B · I, II, III ◆ JUSTIFICATION

A probability describes a model. A survey describes people. The bridge between them is the sample: a small group chosen so that what is true of the group is likely to be roughly true of the whole.

DEFINITIONS

The **population** is the whole group about which we want information. A **sample** is the part we actually observe, and the **sample size** is the number of individuals in it. A **census** measures every member of the population. A **random sample** gives every member a known, non-zero chance of being chosen. The **sampling frame** is the list from which the sample is drawn. **Bias** is a systematic tendency for a sample or measurement to differ from the population in a particular direction. A **representative** sample has roughly the same composition as the population on the characteristics that matter. A **stratified sample** selects from each subgroup in proportion to its size.

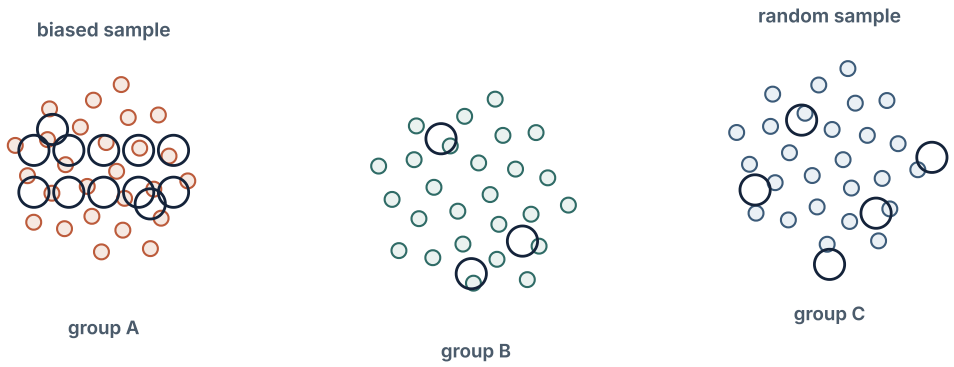


Figure 10.6 A biased sample over-represents one group, while a random sample gives every group a chance. Sample size alone cannot repair a method that systematically misses part of the population.

Random selection is the strongest defence against bias, but it is not the only decision. A sample also needs a complete frame: if the list omits students who are absent, or people without a phone, the "random" draw is random only from the part that could be reached. Three common methods are:

— **Simple random sample.** Number the frame and use a random-number generator to choose individuals. Every member has the same chance.

— **Systematic sample.** Choose every kth name on the frame after a random start. Convenient when the list is long, provided the list has no repeating pattern that matches k.

— **Stratified sample.** Divide the population into groups such as year levels or regions, then sample within each group in proportion to its size. This guarantees representation of every group.

Convenience samples — asking the first 50 people in the queue — and voluntary-response samples — inviting anyone to vote online — are not random. They are cheap, but they systematically favour people who are easy to reach or strongly motivated, and no increase in size removes that direction of error.

WORKED EXAMPLE 10.4.1 A WEBSITE POLL

PROBLEM

A school asks visitors to its website to vote on a later start time. Of 1,200 responses, 78% support the change. The head teacher says: "Most students want a later start." Evaluate the sample and the claim.

1. The population is all students at the school; the sample is the 1,200 people who chose to vote.
2. The sample is self-selected. Students with a strong opinion, easy internet access or an interest in the website are more likely to respond; students who are indifferent or offline are under-represented.
3. Nothing guarantees that respondents are students at all. The poll measures support among website visitors, not among the student body.
4. A better design would draw a random sample from the school roll, contact each selected student, and follow up non-responses. Stratifying by year group would ensure each year is represented.
5. The claim "most students" is not supported by this poll, even though 78% is a large percentage of a large number of responses.

ANSWER **The sample is voluntary and not representative. A random sample from the roll, with follow-up, is needed before the claim about all students can be made.**

WORKED EXAMPLE 10.4.2 A STRATIFIED SAMPLE

PROBLEM

A school has 720 students: 300 in Year 7, 240 in Year 8 and 180 in Year 9. A sample of 60 students is to be selected, stratified by year group. Find the number selected from each year.

1. Each year group's share of the school is its count divided by 720.
2. Year 7: $\frac{300}{720} = \frac{5}{12}$, so $60 \times \frac{5}{12} = 25$ students.
3. Year 8: $\frac{240}{720} = \frac{1}{3}$, so $60 \times \frac{1}{3} = 20$ students.
4. Year 9: $\frac{180}{720} = \frac{1}{4}$, so $60 \times \frac{1}{4} = 15$ students.
5. Check: $25 + 20 + 15 = 60$. Within each year group, students should still be chosen at random from the roll.

ANSWER **25 from Year 7, 20 from Year 8 and 15 from Year 9.**

WORKED EXAMPLE 10.4.3 WRITING A FAIR QUESTION

PROBLEM

A student writes: "Don't you agree that the canteen should stop selling unhealthy food? Yes / No." Identify the problems and rewrite the item as two neutral questions.

1. The phrase "Don't you agree" invites a yes; it presents one side as the reasonable position.
2. "Unhealthy food" is undefined. Different respondents will classify different items as unhealthy, so their answers are not comparable.
3. "Yes / No" forces a choice and offers no way to say "don't know" or "no opinion".
4. A neutral version asks about behaviour and preferences separately: "How often do you buy food from the canteen?" with options Never / Less than once a week / 1–3 times a week / 4 or more times a week. Then: "Which one change to the canteen would you most like?" with several balanced options, including "no change".
5. Assure anonymity and avoid collecting names with the responses; respondents are more likely to answer honestly when they cannot be identified.

ANSWER **The original is leading and ambiguous. Ask about frequency with balanced options, then about preferred changes with a "no change" option, and keep responses anonymous.**

COMMON ERROR · "A BIGGER SAMPLE IS ALWAYS BETTER"

A larger sample reduces random variation, but it does not remove bias. A voluntary online poll with 50,000 responses can be far less representative than a random sample of 200 students drawn from a complete roll. Ask first "Who could not be selected?", then "How many were selected?".

PRACTICE 10.4 · SAMPLING AND BIAS

SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. A school surveys 50 of its 800 students. Identify the population, the sample and the sample size. F
2. Give one advantage and one disadvantage of a census compared with a sample. F
3. A year group of 200 students is numbered 1 to 200. Describe how to choose a simple random sample of 10 using random numbers. F
4. Explain why an online poll on a news website is likely to be biased, even if tens of thousands of people respond. R
5. A school has 600 students: 240 in Year 7, 200 in Year 8 and 160 in Year 9. A stratified sample of 90 is needed. Find the number from each year. R
6. A researcher compares a sample of 20 with a sample of 80, both randomly chosen from the same population. Which estimate is likely to be more accurate, and why? R
7. A questionnaire asks: "How much do you love sport? A lot / A little." Critique the item and rewrite it with clearer, balanced options. P
8. Design a sample to estimate the average travel time from home to school. State the population, sampling frame, method, sample size and one source of bias you would monitor. E

10.5 Risk, base rates and decisions

CRITERION A · I, II ◆ CRITERION D · I, II, III ◆ JUSTIFICATION

A risk is a probability attached to a consequence. It can be described with different denominators, and the choice of words changes what a reader believes. Natural frequencies and absolute risks make comparisons fairer.

DEFINITIONS

Risk is the probability of an unwanted outcome. **Absolute risk** is stated as a number out of a group, such as "3 in 1,000". **Relative risk** compares two risks as a ratio or percentage change, such as "twice as likely". **Prevalence** is the proportion of a population that has a condition at a given time. In a test, a **false positive** is a positive result for someone without the condition, and a **false negative** is a negative result for someone with it.

Relative risk alone can exaggerate a small danger. A rise from 1 in 10,000 to 2 in 10,000 is a doubling, but the absolute increase is only 1 extra case in 10,000 people. A rise from 20 in 1,000 to 40 in 1,000 is also a doubling, but it affects 20 extra people per thousand. Both statements are true; only both together are informative.

WORKED EXAMPLE 10.5.1 RELATIVE AND ABSOLUTE RISK

PROBLEM

A report says a new food additive "doubles the risk" of a rare illness. The risk rises from 1 in 10,000 to 2 in 10,000. Express the change as an absolute risk and as a percentage-point change. Then describe a doubling from 2% to 4% in the same two ways and compare the messages.

1. Absolute risk before: $\frac{1}{10{,}000} = 0.0001 = 0.01\%$. After: $\frac{2}{10{,}000} = 0.02\%$.
2. The absolute increase is 0.01 percentage points, or 1 extra case per 10,000 people.
3. The relative risk is $\frac{0.02}{0.01} = 2$, so "doubles" is correct in relative terms.
4. For 2% to 4%, the relative risk is also 2, but the absolute increase is 2 percentage points, or 20 extra cases per 1,000 people.
5. The same word "doubles" describes a negligible absolute change in the first case and a substantial one in the second. A responsible report gives both the relative and the absolute comparison.

ANSWER **First: 0.01% to 0.02%, an increase of 0.01 percentage points. Second: 2% to 4%, an increase of 2 percentage points. Both double, but the absolute consequences differ greatly.**

Screening tests are a particularly rich source of confusion because a positive result is not the same as having the condition. The table below records 1,000 people tested in a clinic. The test detects 90% of true cases, and it also gives a positive result for 10% of people who do not have the condition.

SCREENING 1,000 PEOPLE: TEST RESULT BY TRUE CONDITION

	Test positive	Test negative	Total
Has condition	45	5	50
No condition	95	855	950
Total	140	860	1,000

WORKED EXAMPLE 10.5.2 WHAT A POSITIVE TEST MEANS

PROBLEM

Using the table, find (a) the probability that a randomly chosen person tests positive, (b) the probability that a person has the condition given a positive test, and (c) explain why "90% accurate" is an incomplete description.

1. (a) 140 of 1,000 people test positive, so $P(\text{positive}) = \dfrac{140}{1{,}000} = 0.14 = 14\%$.
2. (b) Of the 140 positive results, only 45 belong to people with the condition. So $P(\text{condition} \mid \text{positive}) = \dfrac{45}{140} \approx 0.321$, about 32%.
3. (c) The test is 90% sensitive: it finds $\dfrac{45}{50} = 90\%$ of true cases. It is also 90% specific: it correctly clears $\dfrac{855}{950} = 90\%$ of healthy people. But because the condition is rare (50 in 1,000), the 95 false positives outnumber the 45 true positives.
4. Most positive results therefore come from people without the condition. A positive result should lead to further testing, not a diagnosis.

ANSWER **(a) 14% (b) about 32% (c) "90% accurate" hides the low prevalence and the false positives; the test detects 90% of cases but only about 32% of positives are true cases.**

The prevalence matters. If the same test were used where 10% of people have the condition, 100 of 1,000 would be true cases. With 90% sensitivity, 90 would test positive; with 10% false positives among the 900 healthy, another 90 would test positive. Then $P(\text{condition} \mid \text{positive}) = \dfrac{90}{180} = 0.5$, so half of the positives would be true cases. The test has not improved; the population has changed.

WORKED EXAMPLE 10.5.3 RISK IN A SCHOOL POPULATION

PROBLEM

A leaflet states that "1 in 4 people will experience a mental health problem". A school of 800 students uses this to predict that 200 students are currently affected. Identify two errors in the interpretation and suggest a more careful statement.

1. The "1 in 4" figure usually refers to people who experience a problem at some point in their lives, not the proportion affected at this moment.
2. The figure may come from a different population, age range or country. Prevalence in Year 3 students is not necessarily the same as in the general adult population.
3. Even if the prevalence applied, 1 in 4 of 800 is 200 as an expected number, not a count of identified students. Many students would not have disclosed a problem.
4. A more careful statement: "National surveys suggest that about one in four people experience a mental health problem at some point; the number among our students is not known, and support should be available without requiring disclosure."

ANSWER **The lifetime figure is being treated as a current rate, and the population is different. Report the source, the time frame and the uncertainty rather than converting it into a precise count.**

COMMON ERROR · IGNORING THE BASE RATE

A test with a 1% false-positive rate sounds excellent until the condition is rare. Among 10,000 healthy people, 1% is 100 false alarms. If only 100 people have the condition, even a perfect test would produce as many true positives as false ones. Always ask "Out of how many?" before interpreting a positive result or a risk increase.

PRACTICE 10.5 · RISK AND EVIDENCE

SUPPORT 1–3 · CORE 1–8 · EXTENSION 9–12

1. Write "3 in 1,000" as a percentage and as a decimal. F
2. A risk rises from 4 in 1,000 to 6 in 1,000. Find the absolute increase in percentage points and the relative increase. F
3. A condition affects 3% of a population of 5,000. Find the expected number of cases. F
4. A headline says a risk "doubles" from 1 in 100,000 to 2 in 100,000. Explain why the headline may mislead and how it could be written more fairly. R
5. In 1,000 people, 20 have a condition. A test detects 90% of cases and has a 5% false-positive rate. Complete a two-way table and find $P(\text{condition} \mid \text{positive})$. R
6. A test is 99% specific. In 10,000 healthy people, how many false positives are expected? Explain why this matters when the condition is rare. R
7. A school proposes screening every student for a rare condition. List two benefits and two harms, and state what data you would need before deciding. P
8. Design a one-paragraph poster statement that reports a doubled risk honestly. Include the absolute risk before and after, the relative risk and one reason the change matters. E
9. A campaign reports that the risk of injury "fell by 25%", from 8 in 1,000 to 6 in 1,000. Verify the relative fall, state the absolute change in percentage points, and write a fairer headline. P
10. Two clinics use the same test, with 95% sensitivity and a 4% false-positive rate. Clinic A tests 2,000 people when prevalence is 2%; Clinic B tests 500 people when prevalence is 12%. Find $P(\text{condition} \mid \text{positive})$ for each clinic and explain the difference. P

11. A leaflet claims that "90% survive this disease for five years" and concludes that treatment works for nine out of ten patients. Explain two ways this statistic could mislead, even if the number is correct. E

12. A test has 90% sensitivity and a 5% false-positive rate, and the condition affects 1% of people tested. Build a table for 10,000 people, find $P(\text{condition} \mid \text{positive})$, and write two sentences a patient could understand. State one harm of leaving the base rate out. E

Investigation · How representative is a sample?

CRITERIA B AND C

A school wants to know how long students spend on homework on a typical school night. Surveying all 900 students is impractical, so the council will use a sample. Your task is to design the sample, carry it out on a small scale and report what the result can and cannot support.

PILOT SAMPLE: HOMEWORK MINUTES REPORTED BY TWELVE STUDENTS

Student	1	2	3	4	5	6	7	8	9	10	11	12
Minutes	30	45	40	60	35	50	75	20	45	55	60	85

1. **Define.** State the population, the variable and the exact wording of the question. Explain why a census is impractical and why the definition of "typical school night" matters.
2. **Frame.** Obtain the school roll and number every student. Describe how you would use a random-number generator to select 30 students without duplicates.
3. **Ask.** Write a neutral question with response options, such as 0–30, 31–60, 61–90 and more than 90 minutes. Pilot it on two students and revise any wording that suggests a preferred answer.
4. **Collect.** Record refusals and absentees as well as responses. Compare the achieved sample with the school by year group and check whether any group is under-represented.
5. **Calculate.** Use the pilot table to find the mean, the median and the proportion reporting more than 60 minutes. Then calculate the same quantities for your own sample.
6. **Compare.** Exchange results with another pair. Explain why two random samples from the same school can produce different means, and whether either is "wrong".
7. **Evaluate.** Identify three sources of bias or error — non-response, self-report, ambiguous wording, or who is absent — and suggest a remedy for each.
8. **Report.** Write a 100-word report for the school council. State the estimate, the method, the sample size, the limitations and one recommendation. Say clearly what the data cannot tell you.

A random sample reduces bias but does not guarantee a representative sample; sample-to-sample variation remains. Report an estimate with its method and limits, not a single number presented as exact.

Application · The screening test

CRITERION D · I, II, III, IV ◆ MODELS ◆ IDENTITIES AND RELATIONSHIPS

A clinic evaluates a rapid test for a condition that affects about 5% of the people who attend. In a study of 1,000 attendees, 50 have the condition. The test is positive for 45 of them. Among the 950 people without the condition, 95 also test positive. The clinic's leaflet calls the test "90% accurate".

STUDY RESULTS FOR 1,000 CLINIC ATTENDEES

Quantity	Value	Note
People tested	1,000	clinic attendees
With condition	50	confirmed by a second test
True positives	45	test positive and have condition
False positives	95	test positive, no condition
False negatives	5	test negative, have condition
Sensitivity	90%	45 of 50 true cases detected
Specificity	90%	855 of 950 healthy people cleared

1. Complete a two-way table with rows “has condition” and “no condition” and columns “test positive” and “test negative”. Include the totals.
2. Find the probability that a randomly chosen attendee tests positive.
3. Find $P(\text{condition} \mid \text{positive})$. Explain in one sentence what this probability means for a person who receives a positive result.
4. The leaflet says the test is “90% accurate”. Explain why this description is incomplete, using the sensitivity and specificity from the table.
5. Suppose the test were used in a group where 10% have the condition, with the same sensitivity and specificity. Calculate the new $P(\text{condition} \mid \text{positive})$ and explain why it differs.
6. Clinic attendees are not a random sample of the general population. Describe two ways this could affect the estimated prevalence and the usefulness of the test.
7. Who might be harmed by a false positive? Who might be harmed by a false negative? Use these consequences to recommend whether the test should be used for screening everyone, only for people with symptoms, or not at all.
8. Write a 100-word explanation for patients that reports the test’s real performance honestly. Include the natural frequencies 45 true positives and 95 false positives out of 1,000, and state what a positive result does and does not mean.

CRITERION D FOCUS

Translate the percentages back into people. A 90% sensitive test used on a rare condition can still produce more false positives than true ones. The quality of the test, the prevalence in the group tested and the consequences of each type of error all belong in the decision.

Criterion task · The travel survey

CRITERIA B AND C
INVESTIGATING AND COMMUNICATING
24 MARKS

The student council wants to know how students travel to school. A group surveys 60 students outside the main gate at 8:30 one morning and finds that 45 walk or cycle. The council tweets: "75% of students travel actively — we need more bike racks than car spaces." The school has 900 students: 320 in Year 7, 300 in Year 8 and 280 in Year 9.

1. Identify the population, the sample, the variable and the claim. *(3 marks)*
2. Explain two reasons why the gate survey may not represent the whole school. Refer to specific groups who could be missed. *(3 marks)*
3. Propose a sampling frame and a random method for selecting 90 students. Calculate how many should come from each year group in a stratified sample. *(6 marks)*
4. Write two questionnaire items for the survey: one categorical question about usual travel mode with balanced options, and one numerical question about journey time. Avoid leading wording. *(3 marks)*
5. Explain how non-response and self-report could affect the results, and suggest one remedy for each. *(3 marks)*
6. A random sample of 90 students finds that 54 travel actively. Estimate the number of active travellers in the school and compare this with the gate survey's 75%. Which estimate is more trustworthy, and why is neither exact? *(4 marks)*
7. Write a 120-word report for the student council. Include the estimate, the method, two limitations and one recommendation, using language that does not overstate the evidence. *(2 marks)*

PLANNING FRAME

Identify: population 900 students; sample 60 at the gate; variable usual travel mode; claim that 75% travel actively. **Challenge:** the gate sample misses bus and car arrivals, late students, absentees and students who enter by other gates; walkers are over-represented. **Design:** school roll as frame; random numbers within year groups; 90 split 32, 30, 28 in proportion to 320, 300, 280. **Compare:** $54/90 = 0.6$, so about 540 of 900; the gate figure would suggest 675, a difference of 135. **Communicate:** random selection and the larger, structured sample support 60%; report a range and the limits rather than "exactly 540".

CRITERIA: B, C **ATL:** COMMUNICATION **TIME:** 60 MINUTES **CONDITIONS:** CALCULATOR ALLOWED

BAND	DESCRIPTOR	MARKS
4	Population and sample precise; biases linked to specific missed groups; frame, random method and stratified numbers correct; questionnaire neutral and well structured; non-response and self-report explained with remedies; comparison accurate and caution justified; clear, honest report.	19–24
3	Most identification and bias analysis correct; sampling method workable with a minor numerical slip; questionnaire largely neutral; comparison mostly accurate; report conveys the estimate with some caution.	13–18
2	Basic identification; bias described vaguely; sampling method not genuinely random or numbers incorrect; questionnaire leading or incomplete; comparison repeats the claim without evaluating it.	7–12
1	Little correct vocabulary; no clear distinction between sample and population; no usable sampling method; report overstates the evidence.	1–6

MAPPED TO: B strands i–iii and C strands i–iv. The rubric is a course summary, not an official IB descriptor.

Chapter summary and review

KEY IDEAS

- A sample space lists all possible outcomes; for equally likely outcomes, probability is a ratio of counts.
- Probabilities lie between 0 and 1, and $P(A) + P(A') = 1$.
- For two events, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; the overlap is counted once.
- For independent events, $P(A \cap B) = P(A) \times P(B)$; without replacement, probabilities change from stage to stage.
- Relative frequency estimates probability and tends to settle near it as trials increase, if the model and trials are sound.
- Simulation replaces a real experiment with a random model; the assumptions must match the situation.
- A random sample gives every member a known chance of selection; bias is a systematic direction of error that a larger sample does not fix.
- Stratified sampling selects within subgroups in proportion to their size.
- Absolute risk, relative risk and natural frequencies describe the same data; reporting only one can mislead.
- A positive test result depends on prevalence as well as test accuracy; base rates cannot be ignored.

REVISION CHECKLIST

- ☐ I can list a sample space systematically.
- ☐ I can calculate probabilities for one-stage and two-stage experiments.
- ☐ I can use the complement rule and the addition rule.
- ☐ I can draw and read a Venn diagram or tree diagram.
- ☐ I can calculate relative frequency and compare it with theory.
- ☐ I can design a simple simulation.
- ☐ I can identify a population, sample and sampling frame.
- ☐ I can explain selection bias, non-response and leading questions.
- ☐ I can convert between absolute and relative risk.
- ☐ I can interpret a test result using natural frequencies.

VOCABULARY

trial One repetition of a probability experiment.

outcome One possible result of a trial.

sample space The set of all possible outcomes.

event A set of outcomes.

equally likely Outcomes with the same probability.

complement All outcomes not in the event.

mutually exclusive Events that cannot occur together.

independent Events that do not affect each other's probability.

relative frequency Frequency divided by the number of trials.

simulation A random model used to imitate an experiment.

population The whole group being studied.

sample The part of the population observed.

census A measurement of every member of the population.

random sample A sample in which every member has a known chance of selection.

bias A systematic tendency to differ from the population.

sampling frame The list from which a sample is drawn.

stratified sample A sample drawn proportionally from each subgroup.

absolute risk A risk stated as a number out of a group.

relative risk A ratio comparing two risks.

false positive A positive result for someone without the condition.

MIXED PRACTICE · CHAPTER 10

Connecting the ideas

FLUENCY Core procedures

1. A bag contains 7 red, 5 blue and 3 green counters. Find $P(\text{red})$, $P(\text{not blue})$ and $P(\text{red or green})$. F
2. Two fair dice are rolled. Find $P(\text{sum } 4)$, $P(\text{sum } 8)$ and $P(\text{even sum})$. F
3. $P(A) = 0.35$ and $P(B) = 0.4$, with A and B mutually exclusive. Find $P(A \cup B)$ and $P(\text{neither})$. F
4. An experiment has 36 successes in 240 trials. Find the relative frequency and the expected number of successes in 800 trials at the same rate. F

REASONING Justify and connect

1. A fair coin has landed heads ten times. Explain why the probability of a head on the next toss is still 0.5, and why this does not contradict the law of large numbers. R
2. $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$. Show that A and B are independent and find $P(A \cup B)$. R
3. A gardening magazine surveys 500 of its readers and finds that 80% grow vegetables. Explain why this should not be used to estimate the proportion of all adults who grow vegetables. R
4. A headline says a risk has "doubled" from 1 in 20,000 to 2 in 20,000. Explain how the same information could be reported more fairly. R

APPLICATION Use and evaluate

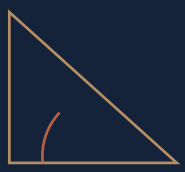
1. A school has 840 students, equally divided among Years 7, 8 and 9. A stratified sample of 84 is chosen. Find the number from each year. If 35 of the 84 walk to school, estimate the number of walkers in the school. P
2. In 2,000 people, 100 have a condition. A test detects 95% of true cases and has a 10% false-positive rate. Complete a two-way table and find $P(\text{condition} \mid \text{positive})$. P
3. A simulation models a 30% chance of success with random digits, using $0, 1, 2$ as success. Explain why this is appropriate, state the expected number of successes in 40 trials, and describe how to estimate the probability from the results. P
4. Class A (20 students) improves its mean score by 6 marks. Class B (5 students) improves by 10 marks. Explain which result is stronger evidence of a real improvement, and why a larger effect is not automatically more reliable. P
5. Explain why increasing a biased sample from 500 to 50,000 responses does not make it representative. Use a specific example. E
6. A condition affects 1% of a population. A test has 99% sensitivity and 99% specificity. Starting with 10,000 people, find the number of true positives, false positives and $P(\text{condition} \mid \text{positive})$. Explain the result. E

Check your progress

I CAN …	CONFIDENCE	EVIDENCE	NEXT ACTION
List a sample space and calculate simple probabilities.	1 / 2 / 3		
Combine events with tables, trees and Venn diagrams.	1 / 2 / 3		
Compare experimental relative frequency with theory.	1 / 2 / 3		
Design a random sample and identify bias.	1 / 2 / 3		
Interpret absolute and relative risk, including a test result.	1 / 2 / 3		

Reflection

A probability can be correct and still be poor evidence for a decision: the sample may be biased, the risk may be framed unfairly, or the outcome may not be the one that matters. Which is more dangerous — an inaccurate number or an accurately calculated number presented without its context? Write three sentences defending your view.



FOUR CRITERION-STYLE PAPERS AND AN EXIT TASK

Cumulative review and examination practice

A course is not a list of chapters but a set of connected tools. These papers ask you to choose a method when nobody names it for you.

HOW TO USE THESE PAPERS

Work each paper under timed conditions, then mark it against the answers. Paper A checks knowledge, B investigates a pattern, C communicates a proof and D applies mathematics; the exit task is a capstone.

ATL · THINKING AND COMMUNICATION

Selecting a method, checking a result for sense, and writing an argument another person can follow are the skills these papers reward. Marks are earned for reasoning made visible, not for a bare number.

Paper A · Knowing and understanding

CRITERION A
SHOW ALL WORKING
37 MARKS · 45 MIN

Answer every question. Write down the method, not only the answer: a correct result with no working can earn partial credit only. A non-calculator question is marked "NC".

Short responses across the course

CRITERION A · I, II, III
RECALL AND PROCEDURE
37 MARKS

1. Write 4.7×10^{-3} as an ordinary number. *(1)*
2. Simplify $\sqrt{98}$. *(2)*
3. Evaluate $(3 \times 10^{4}) \times (2 \times 10^{-6})$, giving your answer in scientific notation. *(2)*
4. Write $2^{-2} \times 2^{5} \div 2^{-1}$ as a single power, then evaluate it. *(2)*
5. Expand and simplify $(x - 4)(x + 7)$. *(2)*
6. Factorise $x^{2} - 9$. *(1)*
7. Solve $4x - 7 = 2x + 5$. *(2)*
8. Solve the simultaneous equations $2x + y = 11$ and $x - y = 1$. *(3)*
9. Find the equation of the line with gradient -3 that passes through $(2, -1)$. *(2)*
10. A line has equation $2y = 6x - 8$. State its gradient and its y-intercept. *(2)*
11. Solve the inequality $3(x - 2) \leq x + 8$. *(2)*
12. A triangle has sides 7 cm, 24 cm and 25 cm. Show that it is right-angled. *(2)*
13. In a right triangle the hypotenuse is 20 cm and one angle is $35°$. Find the side opposite the $35°$ angle, to three significant figures. *(2)*
14. A cylinder has radius 5 cm and height 12 cm. Calculate its volume, to three significant figures. *(2)*
15. Express $135°$ as a fraction of a full turn, in lowest terms. *(1)*
16. For the data set $4,\ 6,\ 6,\ 7,\ 12$, find the mean and the median. *(2)*
17. A scatter plot of two variables has correlation coefficient $r = -0.92$. Describe the correlation in words. *(1)*
18. A fair die is rolled. Find the probability of obtaining a prime number. *(2)*
19. A bag holds 4 red and 6 blue counters. One counter is taken at random. Find $P(\text{red})$ and $P(\text{not red})$. *(2)*
20. A value x satisfies $|x - 8| \leq 3$. Solve for x. *(2)*

CRITERION: A **ATL:** RECALL AND TRANSFER

CONDITIONS: SCIENTIFIC CALCULATOR ALLOWED EXCEPT WHERE MARKED NC

MAPPED TO: Criterion A strands i–iii across all ten chapters. The rubric is a course summary, not an official IB descriptor.

Paper B · Investigating patterns

CRITERION B
INVESTIGATE AND GENERALISE
24 MARKS · 45 MIN

This paper rewards a clear table, a general rule you can verify, and honest comment on what your model does not capture. Draw your own diagrams; neat sketches count as working.

Criterion task · The staircase pattern

CRITERIA B · I, II, III
DESCRIBE AND GENERALISE
24 MARKS

A designer tiles a wall with a staircase pattern built from unit squares. Figure 1 is a single square. Figure 2 sets a column two squares high beside the first; Figure 3 adds a column three squares high, and the pattern continues. In Figure n the columns have heights $1,\ 2,\ 3,\ \ldots,\ n$, placed side by side from left to right.

1. Draw Figures 1 to 4. Copy and complete a table giving the number of tiles T and the perimeter P (measured in unit lengths) for Figures $n = 1$ to 5. *(4 marks)*
2. Find the first differences of the T values, and then the second difference. Explain what the constant second difference tells you about the type of rule connecting T and n. *(4 marks)*
3. Find a rule for T in terms of n. Show that your rule gives $T = 21$ when $n = 6$, and verify it by counting. *(4 marks)*
4. The perimeter follows a simpler pattern. Find a rule for P in terms of n, and verify it for $n = 4$ by counting edges on your diagram. *(4 marks)*
5. Find the least figure whose tile count exceeds 100. Show your method clearly. *(3 marks)*
6. A second designer claims that the perimeter is always an even number of units. Use your rule for P to decide whether the claim is true, and justify your answer. *(3 marks)*
7. State one limitation of this model and describe how you would check your generalisation for a much larger figure. *(2 marks)*

PLANNING FRAME

Tabulate: put n, T and P in three columns before looking for a rule. **Compare:** differences of a linear rule are constant; a constant second difference signals a quadratic rule. **Guess and test:** try $T = \frac{n(n+1)}{2}$ against your table. **Verify:** substitute $n = 6$ and count the squares. **Count edges:** for the perimeter, add the bottom edge, the top steps and the two sides. **Conclude:** state the rule, the evidence and its limitation.

CRITERION: B (I–III) **ATL:** PATTERN-SEEKING **TIME:** 45 MINUTES **CONDITIONS:** CALCULATOR ALLOWED

MAPPED TO: Criterion B strands i–iii. The rubric is a course summary, not an official IB descriptor.

Paper C · Communicating

CRITERION C
REASONING AND PROOF
24 MARKS · 45 MIN

Write in full sentences. A proof is a sequence of statements, each of which follows from the one before, ending in the result you were asked to establish.

Criterion task · The angle in a semicircle

CRITERIA C · I, II, III, IV
JUSTIFY AND COMMUNICATE
24 MARKS

AB is a diameter of a circle with centre O. The point C lies on the circumference, and C is neither A nor B. You are to prove that $\angle ACB = 90°$.

1. Draw the circle, the diameter AB, the point C and the chords AC and BC. Add the radius OC. State which segments are equal in length and explain why each equality holds. *(4 marks)*
2. Name the two triangles formed by the radius and state why each is isosceles. *(3 marks)*
3. Let $\angle OAC = \angle OCA = \alpha$ and $\angle OBC = \angle OCB = \beta$. Write the sum of the angles of triangle ABC in terms of α and β, and deduce that $\alpha + \beta = 90°$. *(5 marks)*
4. Explain how the previous step proves that $\angle ACB = 90°$. *(3 marks)*
5. Prove that the sum of any three consecutive integers is divisible by 3. Begin by writing the three integers in terms of a single variable. *(4 marks)*
6. A student writes the following “proof”. Identify the single invalid step and explain why it is not allowed, using numerical values to illustrate your point.

   STUDENT WORKING

   Let $a = b$. Then $a^2 = ab$, so $a^2 - b^2 = ab - b^2$. Hence $(a - b)(a + b) = b(a - b)$. Dividing both sides by $(a - b)$ gives $a + b = b$, so $2b = b$, and therefore $2 = 1$.

   (5 marks)

PLANNING FRAME

Name: define every point and label every equal length before you argue. **Use a theorem:** every radius of a circle has the same length. **Link:** equal sides in a triangle give equal base angles. **Sum:** the interior angles of a triangle add to $180°$. **Algebra:** for consecutive integers use n, $n + 1$, $n + 2$ and factorise the total. **Check:** a step that divides by an expression is only valid if that expression is never zero.

CRITERION: C (I–IV) **ATL:** COMMUNICATION AND REASONING **TIME:** 45 MINUTES
CONDITIONS: CALCULATOR NOT REQUIRED

MAPPED TO: Criterion C strands i–iv. The rubric is a course summary, not an official IB descriptor.

Paper D · Applying mathematics

CRITERION D
REAL-LIFE CONTEXTS
33 MARKS · 60 MIN

Two contexts are modelled with the same tools. State your assumptions explicitly; a reasonable answer built on clearly stated assumptions scores better than a number with no reasoning.

Criterion task · Two weekly tariffs

CRITERION D · I–V
MODEL AND EVALUATE
33 MARKS

A small delivery business compares two annual tariffs from a courier firm. Tariff X charges a fixed €8 per week plus €0.90 for every kilometre driven. Tariff Y charges a fixed €20 per week plus €0.40 for every kilometre driven. Let k stand for the number of kilometres driven in a week.

1. Write a formula for the weekly cost C_X under Tariff X and C_Y under Tariff Y in terms of k. *(2 marks)*
2. Copy and complete a table of costs for $k = 0,\ 10,\ 20,\ 30,\ 40$ km. *(3 marks)*
3. Draw both graphs on the same axes. Find the distance at which the two costs are equal, both by reading the graph and by solving the two equations. *(5 marks)*
4. Interpret the value you found in words: for which distances is each tariff the cheaper one? *(3 marks)*
5. The business expects to drive 900 km per week. Calculate the weekly cost under each tariff and recommend one, stating the weekly saving. *(3 marks)*
6. Fuel prices rise, so the per-kilometre charge under Tariff Y becomes €0.60 while its fixed charge is unchanged. Recalculate the break-even distance and explain how the recommendation for 900 km is affected. *(4 marks)*
7. State two assumptions of your model and give one reason a real quotation might differ from your calculation. *(4 marks)*

A second problem uses the same habit of modelling. A gardener has 40 m of fencing and wants to fence a rectangular vegetable bed on three sides, using an existing straight wall as the fourth side.

8. Let each of the two equal fenced sides have length x m. Write the length of the third fenced side in terms of x, and hence write an expression for the area A of the bed. *(3 marks)*
9. Use a table of values for $x = 4,\ 6,\ 8,\ 10,\ 12,\ 14,\ 16$ to estimate the value of x that gives the greatest area, and state that greatest area. *(4 marks)*
10. Explain why your answer is only an estimate, and describe one way to improve it. *(2 marks)*

PLANNING FRAME

Define: write what each letter means and its unit. **Relate:** each cost is a fixed charge plus a rate times distance, a linear function. **Compare:** the break-even point is where the two expressions are equal. **Read off:** the cheaper tariff is the lower line on the graph. **Optimise:** tabulate the area and look for the turning point. **Evaluate:** name the assumptions and say how the answer would change if they failed.

CRITERION: D (I–V) **ATL:** MODELLING AND TRANSFER **TIME:** 60 MINUTES **CONDITIONS:** CALCULATOR ALLOWED

MAPPED TO: Criterion D strands i–v. The rubric is a course summary, not an official IB descriptor.

Year 3 exit task · Does the revision app work?

INTERDISCIPLINARY
DATA, CHANCE AND ALGEBRA
30 MARKS · 90 MIN

This capstone draws on bivariate data, probability and algebra from the whole course. Work through it as a single investigation: collect the evidence, model it, then decide how much the evidence really supports.

Criterion task · Evidence for a claim

CRITERIA D AND B, WITH C
MODEL, DECIDE AND JUSTIFY
30 MARKS

A school is considering buying a revision app. Twelve students recorded the number of hours they used a trial version in one week and their score on a test out of 100. Your calculator gives a correlation coefficient of $r \approx 0.985$.

HOURS USING THE APP	1	2	3	4	5	6	7	8	9	10	11	12
TEST SCORE	52	58	55	64	68	71	69	78	82	85	88	90

1. Plot a scatter plot of test score against hours using the app. Describe the correlation, giving its direction and its strength, and link your description to the value of r. *(4 marks)*
2. Draw a line of best fit through the points and estimate its equation in the form $y = mx + c$, where y is the test score and x is the number of hours. *(4 marks)*
3. Use your line to predict the score of a student who uses the app for 14 hours. Comment on how reliable this prediction is. *(4 marks)*
4. A student claims "a student who uses the app for 12 hours will score about 12 marks more than a student who uses it for 7 hours". Use your equation to check the claim and correct it if necessary. *(3 marks)*
5. The school concludes that the app causes higher scores. Explain why the data do not prove causation. Give one alternative explanation for the pattern and describe one way to test causation properly. *(5 marks)*
6. The app is offered free to 200 students. Records from another school show that about 30% of students buy such an app at full price. Estimate how many of the 200 students would take the free offer, and give one reason this estimate might be too high. *(4 marks)*
7. Write a short recommendation to the school. State one strength of your analysis, one limitation, and what further information you would request before committing to the purchase. *(6 marks)*

PLANNING FRAME

Plot: hours on the horizontal axis, score on the vertical axis; label both axes. **Describe:** direction, form and strength, then quote r. **Fit:** a line that passes through the middle of the cloud, not through the extremes. **Predict:** substitute into your equation, then ask whether 14 hours lies inside the range of the data. **Compare:** a difference of 5 hours changes the score by $5m$, with m the gradient. **Question:** high correlation does not establish cause; name a confounding variable such as motivation. **Conclude:** balance the strength against the limitation.

CRITERIA: D, B (C) **ATL:** CRITICAL THINKING AND COMMUNICATION **TIME:** 90 MINUTES

CONDITIONS: CALCULATOR ALLOWED

MAPPED TO: Criterion D strand iii with B strands i–ii and C strand iv. The rubric is a course summary, not an official IB descriptor.

Assessment rubric for the cumulative papers

CRITERIA A–D
COURSE SUMMARY
148 MARKS IN TOTAL

The bands below describe one route through the whole set. A paper may show a higher band in one criterion than another; record the band per criterion rather than averaging across the four.

CRITERION	BAND 4 — 7 TO 8	BAND 3 — 5 TO 6	BAND 2 — 3 TO 4	BAND 1 — 1 TO 2
A · Knowing and understanding	Accurate recall and procedure across every topic; correct notation and units; errors, if any, are slips corrected on review.	Sound recall with minor errors in one or two topics; mostly correct notation.	Partial recall; procedures attempted but with repeated errors.	Isolated correct facts; most procedures incomplete.
B · Investigating patterns	Complete table; general rule found, verified and justified; limitations identified precisely.	Table and rule largely correct; verification sound; some comment on limits.	Pattern partly described; rule stated without full verification.	Few correct values; no usable generalisation.
C · Communicating	Proof complete and logically ordered; notation consistent; assumptions and the flawed step identified clearly.	Proof mostly correct with a small gap; explanation generally clear.	Some correct reasoning; structure and notation inconsistent.	Statements unconnected; conclusion unsupported.
D · Applying mathematics	Accurate model with stated assumptions; correct calculations; recommendation justified and limitations evaluated.	Model largely correct; calculations mostly accurate; evaluation attempted.	Model partly correct; interpretation limited; evaluation generic.	Little correct modelling; answer unsupported by reasoning.

TOTAL: 148 MARKS ACROSS PAPERS A–D AND THE EXIT TASK **MARKING:** PER CRITERION, NOT AVERAGED

REVIEW: RETURN TO THE CHAPTER SECTIONS NAMED IN EACH ANSWER

MAPPED TO: the four MYP mathematics criteria as paraphrased for this course. These bands are a course summary, not official IB descriptors, and a response may sit in different bands for different strands.

Self-assessment and examination technique

Marks are lost more often to presentation than to knowledge. Before you mark your own work, read the three habits below; then record, paper by paper, where the evidence of your understanding actually sits.

READ THE COMMAND WORD

"Show", "explain", "justify" and "evaluate" ask for different amounts of writing. Answer the verb you were given.

MAKE THE METHOD VISIBLE

Write the formula, substitute, then calculate. A bare answer cannot earn method marks, even when it is correct.

CHECK FOR SENSE

Ask whether the size, sign and unit of the result are plausible. A negative length or a probability above 1 signals an error to find.

PAPER	TARGET BAND	EVIDENCE	NEXT ACTION
A · Knowing and understanding	1 / 2 / 3 / 4		
B · Investigating patterns	1 / 2 / 3 / 4		
C · Communicating	1 / 2 / 3 / 4		
D · Applying mathematics	1 / 2 / 3 / 4		
Exit task · Evidence for a claim	1 / 2 / 3 / 4		

Reflection

Which paper revealed the largest gap between what you understand and what you can explain? A high score in Paper A but a low one in Paper C means the mathematics is secure but the communication needs practice. Choose one habit from above and describe, in two sentences, how you will apply it in your next assessment.

Answers and worked solutions

COMPLETE SOLUTIONS FOR EVERY EXERCISE

An answer ends a calculation; a solution records a decision. The entries below show the reasoning behind each result, so a difference between your work and mine becomes a precise question rather than a guess about what went wrong.

HOW EACH ENTRY IS ARRANGED

Exercise numbers match the chapter. Part letters follow the number, for example **3c**. Working is shown wherever it helps you locate a disagreement; routine arithmetic is compressed to a single line. Units and rounding are stated with each result.

Some questions invite more than one valid route. The key gives one efficient method and, where useful, a note beginning *Alternatively*. A different correct method is not an error. Validity, accuracy and a conclusion that answers the question are what count.

Each chapter closes with a **mixed practice** set and a criterion task. Mixed-practice answers are labelled **F** (fluency), **R** (reasoning), **A** (application) and **E** (extension) to match the question tags. A marking guide follows the criterion task in most chapters.

INVESTIGATIONS AND CRITERION TASKS

Open tasks receive *sample responses*, not a single script: representative data, a conjecture, a verification and a limitation. Your data may differ. Use the sample to judge the completeness of your table, the quality of your generalisation and the care of your conclusion.

Academic honesty. These solutions support your attempt; they do not replace it. If you use a sample response, rewrite the reasoning in your own words and explain the choice of method. That is the evidence a teacher can assess.

ACCURACY AND NOTATION

Unless a question states otherwise, exact answers are given for integers, fractions, surds and terminating decimals. Measurements are rounded to a sensible degree of accuracy, generally three significant figures. Never round intermediate values; rounding early is a common source of “correct method, wrong answer”.

- **Surds** in simplest form with a rationalised denominator, for example $\frac{\sqrt{2}}{2}$ rather than $\frac{1}{\sqrt{2}}$.
- **Fractions** in lowest terms; **percentages** with the % symbol and the denominator stated where it matters.
- **Powers** in index form unless a value is asked for; **scientific notation** as $a \times 10^{n}$ with $1 \le a < 10$.
- **Sets** in roster or set-builder form; elements listed once. **Intervals** in either bracket form, $[-3, 5)$, or as an inequality.
- **Equations of lines** in the form $y = mx + c$ unless a context asks otherwise; gradients as exact fractions where possible.
- **Angles** with the degree symbol; lengths, areas and volumes with units, for example cm^2 and m^3.
- **Trigonometric ratios** to three significant figures unless an exact value such as $\frac{\sqrt{3}}{2}$ is available.
- **Probability** as a fraction, decimal or percentage; write $P(\text{event})$ and keep sample spaces explicit. Risk may be absolute or relative.
- **Data summaries** with the unit and number of values; the correlation coefficient r to three decimal places.
- **Money** to two decimal places in currency contexts.

Equivalent fractions, surds, ratios and expressions can look unlike each other. Test numerical equivalence before assuming an error.

Criterion tasks use course-specific rubrics that paraphrase the MYP criteria; they are not official IB descriptors, and sample responses show one level of performance. Criterion D answers depend on the assumptions you state, so a different but well-justified model can score full marks.

Chapter 1 · The real number system

PREREQUISITE CHECK

1 $\frac{18}{24} = \frac{3}{4}$, dividing numerator and denominator by 6.
2 $0.35 = \frac{35}{100} = \frac{7}{20}$.
3 $7^2 = 49$ and $(-7)^2 = 49$; the bracket includes the negative sign in the base.
4 $\frac{3}{4} = \frac{21}{28}$ and $\frac{5}{7} = \frac{20}{28}$, so $\frac{3}{4} > \frac{5}{7}$.
5 $2.5 \times 4 = 10$.
6 $1, 4, 9, 16, 25$.

PRACTICE 1.1 · RATIONAL FORM AND DECIMAL EXPANSION

1 (a) integer and rational; (b) rational; (c) rational; (d) natural, integer and rational.
2 (a) $\frac{7}{20} = 0.35$; (b) $\frac{13}{8} = 1.625$; (c) $\frac{9}{50} = 0.18$.
3 (a) $0.\overline{6}$; (b) $0.\overline{45}$; (c) $0.58\overline{3}$.
4 (a) repeats, since $14 = 2 \times 7$; (b) terminates, since $40 = 2^3 \times 5$; (c) repeats, since $30 = 2 \times 3 \times 5$; (d) terminates, since $64 = 2^6$. In lowest terms, the decimal terminates exactly when the denominator has no prime factor except 2 and 5.
5 (a) $\frac{5}{9}$; (b) $\frac{27}{99} = \frac{3}{11}$; (c) $x = 0.4\overline{1}$, so $10x = 4.1\overline{1}$ and $100x = 41.\overline{1}$; subtracting gives $90x = 37$, so $x = \frac{37}{90}$.
6 The midpoint $\frac{1}{2}\left(\frac{1}{3} + \frac{1}{2}\right) = \frac{5}{12}$ lies strictly between. Between $\frac{1}{3}$ and $\frac{5}{12}$, the midpoint is $\frac{1}{2}\left(\frac{4}{12} + \frac{5}{12}\right) = \frac{3}{8}$. Check: $0.333\ldots < 0.375 < 0.4166\ldots$.
7 Let $x = 0.\overline{9}$. Then $10x = 9.\overline{9}$, so $9x = 9$ and $x = 1$. There is no positive gap between $0.\overline{9}$ and 1, so "just less than 1" is impossible.
8 Counterexample: $\frac{1}{6} = 0.1\overline{6}$ repeats. The denominator 6 has prime factor 3, so a denominator below 10 does not guarantee only factors 2 and 5; $\frac{1}{7}$ and $\frac{1}{9}$ also repeat.

PRACTICE 1.2 · IRRATIONAL NUMBERS AND SURDS

1 (a) rational, since $\sqrt{9} = 3$; (b) irrational; (c) rational, since $0.\overline{4} = \frac{4}{9}$; (d) irrational.
2 (a) 4 and 5, since $16 < 20 < 25$; (b) 8 and 9, since $64 < 75 < 81$; (c) 10 and 11, since $100 < 103 < 121$.
3 (a) $3\sqrt{2}$; (b) $4\sqrt{3}$; (c) $10\sqrt{2}$; (d) $3\sqrt{5}$.
4 $\sqrt{7} < 2.65 < \sqrt{8} < 2.9$. Since $2.65^2 = 7.0225 > 7$, $2.65 > \sqrt{7}$; also $2.65 < 2.8$. Since $2.8^2 = 7.84 < 8 < 8.41 = 2.9^2$ and $2.83^2 = 8.0089 > 8$, $2.8 < \sqrt{8} < 2.83 < 2.9$.
5 $\sqrt{25} = 5 = \frac{5}{1}$, a quotient of integers; 25 is a perfect square. But 26 is not a perfect square. If $\sqrt{26} = \frac{a}{b}$ in lowest terms, then $a^2 = 26b^2$. The factor 2 divides a^2, so a is even; writing $a = 2k$ gives $2k^2 = 13b^2$, so 2 divides b^2 and hence b. Both are even, contradicting lowest terms.
6 Assume $\sqrt{3} = \frac{a}{b}$ in lowest terms. Squaring gives $3b^2 = a^2$, so 3 divides a^2. Here we use the fact that if 3 divides a^2, then 3 divides a (true because 3 is prime). Write $a = 3k$. Then $3b^2 = 9k^2$, so $b^2 = 3k^2$; hence 3 divides b^2 and therefore b. This contradicts the assumption that a and b have no common factor, so $\sqrt{3}$ is irrational.
7 $\sqrt{2}$ and $\sqrt{8}$: both are irrational because 2 and 8 are not perfect squares, yet $\sqrt{2} \times \sqrt{8} = \sqrt{16} = 4$, which is rational. The product rule $\sqrt{a}\sqrt{b} = \sqrt{ab}$ turns the product into a perfect square.
8 A rational decimal must terminate or eventually repeat with a fixed block. Here the zero blocks grow without limit, so no fixed block can repeat forever. The pattern is describable, but describable is not the same as periodic; hence the number is irrational.

PRACTICE 1.3 · ORDER, DISTANCE AND INTERVALS

1 (a) $-6 < -1$; (b) $|-9| = 9$; (c) $\sqrt{17} > 4.1$, since $4.1^2 = 16.81 < 17$.
2 (a) $|7 - (-4)| = 11$; (b) $|-2.5 - (-8)| = 5.5$; (c) $|3 - 11| = 8$.
3 (a) $-3 \le x < 5$, or $[-3, 5)$; (b) $x > 4$, or $(4, \infty)$; (c) $|x - 10| \le 1.5$, so $8.5 \le x \le 11.5$, or $[8.5, 11.5]$.
4 $-3.5 < -\sqrt{10} < -\frac{22}{7} < -3$. Now $-\sqrt{10} \approx -3.162$ and $-\frac{22}{7} \approx -3.143$, so -3.5 is furthest left, then -3.162, then -3.143, then -3.
5 $|L - 45| \le 0.2$ gives $44.8 \le L \le 45.2$ mm. The length 44.75 mm is 0.25 mm below nominal, so reject; 45.15 mm is 0.15 mm above, so accept.
6 $|x - 5| \le 2$ says the distance from x to 5 is at most 2, so x runs from $5 - 2 = 3$ to $5 + 2 = 7$. Conversely, $3 \le x \le 7$ gives $-2 \le x - 5 \le 2$, hence $|x - 5| \le 2$. The two descriptions therefore name the same interval.
7 $\sqrt{10}$ and $\sqrt{11}$. Since $9 < 10 < 11 < 16$, $3 < \sqrt{10} < \sqrt{11} < 4$. Neither 10 nor 11 is a perfect square, so each square root is irrational.
8 False as stated. Counterexample: $-5 < -2$, but $(-5)^2 = 25 > 4 = (-2)^2$. Squaring reverses order when both numbers are negative; the statement is true only when $0 \le a < b$.

PRACTICE 1.4 · SURD ARITHMETIC

1 (a) $2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$; (b) $5\sqrt{2} - 2\sqrt{2} = 3\sqrt{2}$; (c) $5\sqrt{2} + 3\sqrt{2} = 8\sqrt{2}$.
2 (a) $\sqrt{36} = 6$; (b) $6\sqrt{50} = 6 \times 5\sqrt{2} = 30\sqrt{2}$; (c) $\sqrt{16} = 4$.
3 (a) $\frac{4\sqrt{2}}{2} = 2\sqrt{2}$; (b) $\frac{9\sqrt{6}}{6} = \frac{3\sqrt{6}}{2}$; (c) $\frac{7\sqrt{5}}{3\times5} = \frac{7\sqrt{5}}{15}$.
4 (a) $3 - 4 = -1$; (b) $4 + 4\sqrt{5} + 5 = 9 + 4\sqrt{5}$.
5 Incorrect. $\sqrt{8} = 2\sqrt{2}$, so $\sqrt{8} + \sqrt{2} = 3\sqrt{2} \approx 4.243$, not 4. Also $(3\sqrt{2})^2 = 18$, whereas $4^2 = 16$.
6 $\frac{\sqrt{20}}{\sqrt{5}} = \sqrt{\frac{20}{5}} = \sqrt{4} = 2$, an integer. This uses the quotient rule $\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$ for $b \neq 0$.
7 $\frac{1}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1} = \frac{\sqrt{5}+1}{(\sqrt{5})^2-1^2} = \frac{\sqrt{5}+1}{5-1} = \frac{\sqrt{5}+1}{4}$.
8 Area $= (3 + \sqrt{2})^2 = 9 + 6\sqrt{2} + 2 = 11 + 6\sqrt{2}$ cm^2. For area $14 + 6\sqrt{5}$, test $(3 + \sqrt{5})^2 = 9 + 6\sqrt{5} + 5 = 14 + 6\sqrt{5}$, so the side is $3 + \sqrt{5}$ cm; take the positive root because a length is positive.

PRACTICE 1.5 · CLASSIFICATION, COUNTEREXAMPLE AND PROOF

1 (a) integer (also rational); (b) rational; (c) natural, since $\sqrt{16} = 4$; (d) irrational.

2 (a) always true: the integers are closed under multiplication. (b) sometimes true: $6 \div 3 = 2$, but $5 \div 2 = 2.5$; the quotient is an integer only when the divisor is nonzero and divides exactly. (c) sometimes true: $\sqrt{2} + (-\sqrt{2}) = 0$ is rational, while $\sqrt{2} + \sqrt{3}$ is irrational.

3 "Not all rational numbers have terminating decimals", equivalently "there is at least one rational number whose decimal does not terminate". The negation is true: $\frac{1}{3} = 0.\overline{3}$.

4 Let $x = \frac{a}{b}$ and $y = \frac{c}{d}$ with integers a, b, c, d, $b, d \neq 0$. Then $xy = \frac{ac}{bd}$. Both ac and bd are integers and $bd \neq 0$, so xy is a quotient of integers and is rational.

5 Let r be rational and s irrational. Suppose $r + s = q$ were rational. Then $s = q - r$. A difference of two rational numbers is rational, so s would be rational, a contradiction. Hence $r + s$ is irrational.

6 Suppose $\sqrt{2} + \sqrt{3} = \sqrt{n}$ for an integer n. Squaring gives $5 + 2\sqrt{6} = n$, so $\sqrt{6} = \frac{n-5}{2}$, which would be rational. But 6 is not a perfect square, so $\sqrt{6}$ is irrational. Contradiction; no such integer n exists.

7 Take $x = \sqrt{2}$. Then $x^2 = 2$ is rational, but x is irrational. This shows that the converse of the true statement "if x is rational then x^2 is rational" is false; a true implication does not guarantee its converse.

8 Counterexamples: $\sqrt{2} + (-\sqrt{2}) = 0$ and $(2 + \sqrt{2}) + (2 - \sqrt{2}) = 4$. Both are sums of irrationals with rational totals. Revised claim: the sum of two irrational numbers may be rational or irrational; however, the sum of a rational number and an irrational number is always irrational.

INVESTIGATION 1 · CHASING $\sqrt{2}$ WITH FRACTIONS

1 $x_2 = \frac{1}{2}\left(\frac{3}{2} + \frac{4}{3}\right) = \frac{1}{2} \times \frac{17}{6} = \frac{17}{12} = 1.41666667$. Then $x_3 = \frac{1}{2}\left(\frac{17}{12} + \frac{24}{17}\right) = \frac{1}{2} \times \frac{577}{204} = \frac{577}{408} = 1.41421569$, matching the table.

2 The errors are 0.0858, 0.00245, 0.00000212, then 0.00000000. The decrease is not constant: each error is roughly the square of the previous one, so the number of correct digits roughly doubles each step. The final zero reflects the display; the true error is about 1.6×10^{-12}.

3 Since $x_n \times \frac{2}{x_n} = 2 = (\sqrt{2})^2$, if $x_n < \sqrt{2}$ then $\frac{2}{x_n} > \sqrt{2}$, and if $x_n > \sqrt{2}$ then $\frac{2}{x_n} < \sqrt{2}$. One value lies below $\sqrt{2}$ and the other above, so their average lies between them. Also, $(x - \sqrt{2})^2 > 0$ rearranges to $x + \frac{2}{x} > 2\sqrt{2}$, so the average is always at least $\sqrt{2}$, with equality only at $x = \sqrt{2}$. The next estimate is therefore close to $\sqrt{2}$, and the process converges.

4 Use $x_{n+1} = \frac{1}{2}\left(x_n + \frac{7}{x_n}\right)$ with $x_1 = 3$:
$x_2 = \frac{1}{2}\left(3 + \frac{7}{3}\right) = \frac{8}{3} = 2.6666667$;
$x_3 = \frac{1}{2}\left(\frac{8}{3} + \frac{21}{8}\right) = \frac{127}{48} = 2.6458333$;
$x_4 = \frac{1}{2}\left(\frac{127}{48} + \frac{336}{127}\right) = \frac{32257}{12192} = 2.6457513$. This is already $\sqrt{7} = 2.64575131$ to 8 d.p.

5 For $\sqrt{10}$, $x_{n+1} = \frac{1}{2}\left(x_n + \frac{10}{x_n}\right)$. From $x_1 = 3$:
$x_2 = \frac{19}{6} = 3.1666667$; $x_3 = \frac{721}{228} = 3.1622807$;
$x_4 = \frac{1039681}{328776} = 3.1622776602$. A calculator gives 3.1622776602, so after three iterations the error is below 2×10^{-12}.

6 Every term is rational: the starting value is rational and the formula uses only addition and division of rationals, so rationality is preserved. No term can equal $\sqrt{2}$, because then $\sqrt{2}$ would be rational, contradicting the proof in Section 1.2. The terms approach $\sqrt{2}$ as closely as we wish, which shows that the rationals are dense but not complete: there is a "hole" in the rational line at $\sqrt{2}$, filled only when the real numbers are introduced. The method gives approximations, never the exact value in finite form.

APPLICATION · TOLERANCE IS AN INTERVAL

1 $27.2 - 0.05 \leq d \leq 27.2 + 0.05$, so $27.15 \leq d \leq 27.25$ mm; equivalently $|d - 27.2| \leq 0.05$.

2 27.24 mm is definitely acceptable. 27.16 mm is 0.04 mm below nominal, inside $[27.15, 27.25]$, so it is also acceptable. The recorded 27.25 and 27.15 lie on the limits and are borderline because the displayed value could represent a true length just outside. No recorded entry is a definite reject, though the unrounded values 27.252 and 27.148 lie outside.

3 A device reading to 0.01 mm rounds, so a display of 27.25 represents any true length in $[27.245, 27.255)$. Part of that range exceeds 27.25, so the workshop cannot be certain. A finer device (0.001 mm), recording more digits, or a guard band at the limits would remove the ambiguity.

4 The second interval is $[27.16, 27.24]$, strictly inside $[27.15, 27.25]$, so it is stricter. Using displayed values, all four pass ± 0.05, but only 27.24 and 27.16 pass ± 0.04. Using unrounded values, only those two pass either specification, since 27.252 and 27.148 fail both; rounding creates the apparent extra passes.

5 A wider tolerance reduces machining cost and scrap, because a small band requires slower, expensive processes. It may still work functionally: a seat post with clearance can fit and grip across a range. The tolerance describes functional limits; device precision describes how finely we measure. The device must be finer than the tolerance, but not every part needs the finest precision.

CRITERION TASK · WHY $\sqrt{2}$ CANNOT BE WRITTEN AS A FRACTION

Sample response. A unit square has diagonal d with $d^2 = 1^2 + 1^2 = 2$, so $d = \sqrt{2}$. A rational number is one that can be written as $\frac{p}{q}$ with integers p, q and $q \neq 0$. Suppose $\sqrt{2}$ were rational, so $\sqrt{2} = \frac{a}{b}$ in lowest terms. Squaring gives $a^2 = 2b^2$, so a^2 is even. An odd number squared is odd, so a is even; write $a = 2k$. Then $4k^2 = 2b^2$, so $b^2 = 2k^2$, making b even too. This contradicts the assumption that a and b are not both even, so $\sqrt{2}$ is irrational. Every finite decimal such as 1.414 is a fraction with denominator a power of 10, hence rational, so no finite decimal can equal $\sqrt{2}$. Thus $\sqrt{2} \approx 1.414$ reports a rounded value, while $\sqrt{2} = 1.414$ is simply false.

1 The diagram should show the unit square, one diagonal, and the right triangle with legs 1 and 1. By Pythagoras, $d^2 = 1^2 + 1^2 = 2$. A length is positive, so $d = \sqrt{2}$. The diagonal is not a measured approximation; its square is exactly 2, which forces the length $\sqrt{2}$.

2 A rational number has the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$. To say $\sqrt{2}$ is rational would mean there are integers a, b, with $b \neq 0$, such that $\sqrt{2} = \frac{a}{b}$. We may also choose the fraction in lowest terms, so a and b are not both even.

3 Assume, for contradiction, that $\sqrt{2} = \frac{a}{b}$ with a, b integers, $b \neq 0$, and no common factor. Squaring both sides: $2 = \frac{a^2}{b^2}$, hence $a^2 = 2b^2$. Therefore a^2 is even. Since the square of an odd integer is odd, a must be even; write $a = 2k$ for an integer k. Substituting: $(2k)^2 = 2b^2$, so $4k^2 = 2b^2$, hence $b^2 = 2k^2$. Thus b^2 is even, so b is even. But a and b cannot both be even if the fraction is in lowest terms. This is the contradiction; therefore the assumption is false and $\sqrt{2}$ is irrational.

4 Any finite decimal, such as 1.414, equals $\frac{1414}{1000}$, a quotient of integers, so it is rational. If a finite decimal equalled $\sqrt{2}$, then $\sqrt{2}$ would be rational, contradicting the proof. A finite decimal can only truncate or round the expansion; it cannot capture all infinitely many non-repeating digits.

5 The symbol $\approx$ means "is approximately equal to", so $\sqrt{2} \approx 1.414$ states that 1.414 is a rounded value, with error less than 0.0005. The symbol $=$ asserts exact equality, and $\sqrt{2} = 1.414$ is false because $1.414^2 = 1.999396 \neq 2$. The exact form $\sqrt{2}$ preserves the value; the decimal is useful for measuring and comparing.

Marking guide. Part 1 (4 marks): diagram, Pythagoras, $d^2 = 2$, positive root. Part 2 (4 marks): definition $\frac{p}{q}$; $q \neq 0$; translation to $\sqrt{2} = \frac{a}{b}$; lowest terms. Part 3 (8 marks): assumption; squaring; $a^2 = 2b^2$; a even by parity; $a = 2k$; $b^2 = 2k^2$; b even; explicit contradiction and conclusion. Part 4 (4 marks): finite decimal over 10^n; hence rational; contradiction with irrationality; truncation example. Part 5 (4 marks): $\approx$ as approximation; $=$ as exact and false; meaning of the error; appropriate use of each form. Criterion C i–iv: audience, structure, notation and integration of diagram and conclusion. Bands: 19–24 complete and coherent; 13–18 sound with minor gaps; 7–12 outline with significant gaps; 1–6 fragments or bare result.

MIXED PRACTICE 1

F1 $\frac{11}{8} = 1.375$; $\frac{7}{12} = 0.58\overline{3}$.

F2 $0.\overline{45} = \frac{45}{99} = \frac{5}{11}$.

F3 (a) $7\sqrt{2}$; (b) $6\sqrt{2} - 2\sqrt{2} = 4\sqrt{2}$; (c) $\frac{10\sqrt{5}}{5} = 2\sqrt{5}$.

F4 $-4 \leq x < 9$.

R1 Compare squares: $6.3^2 = 39.69$ and $6.4^2 = 40.96$. Since 40 is 0.31 above 39.69 but 0.96 below 40.96, $\sqrt{40}$ is closer to 6.3 (about 6.32).

R2 $-\sqrt{3} \approx -1.732$, $-\frac{5}{3} \approx -1.667$, so the order is $-\sqrt{3} < -1.7 < -\frac{5}{3} < -1.65$.

R3 Let $r \neq 0$ be rational and x irrational. If $rx = q$ were rational, then $x = \frac{q}{r}$, a quotient of rationals with $r \neq 0$, so x would be rational. Contradiction; the product is irrational.

R4 The student truncated: $\frac{1}{3} = 0.\overline{3}$, not 0.333. The decimal repeats forever, so it does not terminate. Also $3 \times 0.333 = 0.999 \neq 1$, confirming that 0.333 is only an approximation.

A1 $|d - 8| \leq 0.03$ gives $7.97 \leq d \leq 8.03$ mm. A pin of 7.96 mm is below 7.97 mm, so reject it.

A2 Area $= \sqrt{5} \times \sqrt{20} = \sqrt{100} = 10$ cm^2 exactly; to one decimal place, 10.0 cm^2.

A3 Rounded to 0.1 kg, a display of 2.4 kg represents $2.35 \leq w < 2.45$ kg. The band "above 2.35 kg" is not certain: a true weight could be exactly 2.35 and still display as 2.4, yet it is not above 2.35. A more precise weight is needed.

A4 Rational: $\frac{3}{2}$, since $2 < \left(\frac{3}{2}\right)^2 = 2.25 < 3$. Irrational: $\sqrt{2.5} = \frac{\sqrt{10}}{2} \approx 1.581$, since $2 < 2.5 < 3$ places it in the interval and $\sqrt{10}$ is irrational.

Common errors. Treating a repeating decimal as terminating (writing $0.\overline{36} = \frac{36}{100}$); assuming $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$; squaring an inequality without checking signs; giving a tolerance as one endpoint instead of a two-sided interval; and offering examples as proof of a universal claim. A surd left in the denominator or a lost negative sign is an incomplete answer.

Chapter 2 · Powers, roots and scientific notation

PREREQUISITE CHECK

1 $3^4 = 3 \times 3 \times 3 \times 3 = 81$.

2 $\sqrt{81} = 9$, since $9^2 = 81$.

3 $10^5 = 100{,}000$.

4 $\frac{8}{12} = \frac{2}{3}$.

5 $2.5 \times 10^2 = 2.5 \times 100 = 250$.

6 7.684 rounds to 7.7, because the hundredths digit is 8.

PRACTICE 2.1 · EXPONENT LAWS

1 (a) a^{10}; (b) b^8; (c) c^{10}; (d) $9d^2$.

2 (a) 125; (b) $\frac{1}{16}$; (c) 81; (d) -32.

3 (a) $3m^5$; (b) $2p^2q^3$; (c) $8x^6y^3$, since $2^3 = 8$ and each exponent is multiplied by 3.

4 $2^3 \times 2^4 = 8 \times 16 = 128 = 2^7$. But $4^7 = (2^2)^7 = 2^{14} = 16{,}384$. The bases differ: the addition law applies only to powers of the same base, and 4^7 has fourteen factors of 2, not seven.

5 By the division law, $\frac{5^8}{5^8} = 5^{8-8} = 5^0$. The same quotient is 1 because any nonzero number divided by itself is 1. Hence $5^0 = 1$; the definition $a^0 = 1$ for $a \neq 0$ is forced by the requirement that the exponent laws stay consistent.

6 (a) $81 = 3^4$, so $n = 4$; (b) $32 = 2^5$, so $n + 1 = 5$ and $n = 4$; (c) $625 = 5^4$, so $2n = 4$ and $n = 2$.

7 $\left(\frac{2a^3}{b^2}\right)^2 \times \frac{b^6}{a^2} = \frac{4a^6}{b^4} \times \frac{b^6}{a^2} = 4a^{6-2}b^{6-4} = 4a^4b^2$.

8 $(-2)^4 = 16$, but $-2^4 = -(2^4) = -16$. In the first, the bracket makes the base -2; in the second, the exponent applies only to 2. The distinction matters when substituting negative values, for example $-x^4$ versus $(-x)^4$, and when reading calculator or spreadsheet output.

PRACTICE 2.2 · ROOTS AND RATIONAL EXPONENTS

1 (a) 11; (b) 3; (c) -4, since $(-4)^3 = -64$; (d) 3, since $3^4 = 81$.

2 (a) $7^{1/2}$; (b) $5^{1/3}$; (c) $3^{-1/2}$; (d) $x^{2/5}$.

3 (a) 8, since $8^2 = 64$; (b) 5; (c) 2; (d) $9^{3/2} = (\sqrt{9})^3 = 3^3 = 27$.

4 (a) $32^{2/5} = (2^5)^{2/5} = 2^2 = 4$; (b) $64^{-1/3} = \frac{1}{\sqrt[3]{64}} = \frac{1}{4}$; (c) $\left(\frac{8}{27}\right)^{2/3} = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$.

5 (a) $x = \pm 12$; (b) $x = 5$, since $5^3 = 125$; (c) $x = 81$, since $81^{1/2} = 9$.

6 $(-8)^{1/3} = \sqrt[3]{-8} = -2$, because $(-2)^3 = -8$; an odd root of a negative number is real. But $(-8)^{1/2} = \sqrt{-8}$ would require a real number whose square is -8, and every real square is non-negative. So the even root is not real.

7 $a^{m/n} = a^{m\times(1/n)}$. The power-of-a-power law gives $a^{m\times(1/n)} = (a^m)^{1/n}$, and the definition of a rational exponent gives $(a^m)^{1/n} = \sqrt[n]{a^m}$. The same law also gives $(a^{1/n})^m = (\sqrt[n]{a})^m$.

8 Edge $= \sqrt[3]{343} = 7$ cm. One face has area $7^2 = 49$ cm^2, so the square's side is $\sqrt{49} = 7$ cm. The numbers coincide because $343 = 7^3$ and $49 = 7^2$; the units are different (cm for length, cm^2 for area).

PRACTICE 2.3 · SCIENTIFIC NOTATION

1 (a) 8.25×10^3; (b) 4.6×10^{-4}; (c) 9.1×10^7; (d) 3.05×10^{-7}.

2 (a) 5,700; (b) 0.0000902; (c) 160,000,000; (d) 0.45.

3 (a) $2.5 \times 4 = 10$, so $10 \times 10^9 = 1.0 \times 10^{10}$; (b) $6.4 \times 2.5 = 16$, so $16 \times 10^3 = 1.6 \times 10^4$.

4 (a) $\frac{9.6}{3.2} \times 10^{8-2} = 3.0 \times 10^6$; (b) $\frac{4.5}{9} \times 10^{-5-(-9)} = 0.5 \times 10^4 = 5.0 \times 10^3$.

5 Align the powers: $3.8 \times 10^5 = 0.38 \times 10^6$. Then $4.7 \times 10^6 + 0.38 \times 10^6 = 5.08 \times 10^6$. Check: $4{,}700{,}000 + 380{,}000 = 5{,}080{,}000$.

6 $3.3 \times 10^{-4} < 4.1 \times 10^{-4} < 3.4 \times 10^{-3} < 2.9 \times 10^{-2}$. In standard form: $0.00033 < 0.00041 < 0.0034 < 0.029$.

7 $7.5 \times 10^{-9} \times 40 = 300 \times 10^{-9} = 3.0 \times 10^{-7}$ kg.

8 $\frac{2.0\times10^{12}}{4.0\times10^{8}} = 0.5 \times 10^4 = 5.0 \times 10^3$, so 5,000 files.

PRACTICE 2.4 · ORDERS OF MAGNITUDE AND ESTIMATION

1 (a) 10^4, since $4.8 \geq 3.16$; (b) 10^7; (c) 10^{-4}, since $2.2 < 3.16$; (d) 10^{-1}, since $5.0 \geq 3.16$.

2 $\frac{6\times10^8}{3\times10^5} = 2 \times 10^3 = 2{,}000$ times larger.

3 $24 \times 60 \times 60 = 86{,}400 \approx 8.6 \times 10^4$ seconds; a one-significant-figure estimate is 10^5 s.

4 1 kg = 1,000 g, so the number is about $\frac{1000}{0.02} = 50{,}000 = 5 \times 10^4$ grains. Assumptions: every grain has mass 0.02 g, the bag contains only rice, and the total mass is exact.

5 1 cm $= 10^{-2}$ m, so $\frac{10^{-2}}{10^{-9}} = 10^7$ molecules could fit across the gap.

6 $\frac{9.1\times10^{-31}}{1.67\times10^{-27}} \approx 5.45 \times 10^{-4}$. The proton is heavier, by a factor of about 1.8×10^3 (roughly 1,835).

7 $\frac{3.2\times10^{15}}{1.28\times10^{11}} = 2.5 \times 10^4 = 25{,}000$ smartphones. This is reasonable: $25{,}000 \times 128$ GB ≈ 3.2 million GB, or about 3.2 PB.

8 Volume $= 25 \times 10 \times 2 = 500$ m^3 $= 5 \times 10^5$ L. At 0.5 L per bottle, that is 10^6 bottles. A school of about 10^3 people is three orders of magnitude smaller, so the bottle count is roughly 1,000 times the school population.

PRACTICE 2.5 · SIGNIFICANT FIGURES AND PRECISION

1 (a) 2; (b) 4; (c) ambiguous, usually 2 unless the trailing zeros are declared significant (then 3 or 4); (d) 3.

2 (a) 4.57; (b) 0.00988; (c) 12,300; (d) 8.99, since the next digit is 4.

3 7.85 rounds to 7.9, but 7.849 rounds to 7.8. Rounding depends on the digit immediately after the retained place: 5 rounds up, 4 does not. The values differ by 0.001, but they fall on opposite sides of the 7.85 threshold.

4 The true length lies in $12.35 \leq L < 12.45$ cm; a value of exactly 12.45 would round to 12.5, so the upper endpoint is excluded.

5 Relative error $= \frac{2}{250} \times 100\% = 0.8\%$.

6 Area $= 4.2 \times 7.5 = 31.5$ cm^2. Both lengths have 2 significant figures, so the answer should be given to 2 s.f.: 32 cm^2.

7 The inputs 2.5 and 4.0 each have 2 s.f., so the product has 2 s.f. The value 10.0 already claims 3 s.f., and 10.000 claims 5. The correct report is 10 (ambiguous) or preferably 1.0×10^1, not a string of zeros that invents precision.

8 3.84×10^8 m has 3 s.f., so the uncertainty is about $\pm 0.005 \times 10^8 = \pm 5 \times 10^5$ m. The true value could lie roughly between 383.5 million and 384.5 million metres. The trailing digits in 384,000,000 are not known, so "exactly" is not justified.

INVESTIGATION 2 · FORTY-TWO FOLDS TO THE MOON

1 For $n = 10$: $0.1 \times 2^{10} = 0.1 \times 1024 = 102.4$ mm $= 0.1024$ m $= 1.024 \times 10^{-1}$ m. For $n = 20$: $0.1 \times 2^{20} = 0.1 \times 1{,}048{,}576 = 104{,}857.6$ mm $= 104.8576$ m $= 1.048576 \times 10^2$ m.

2 The Moon is 3.84×10^8 m $= 3.84 \times 10^{11}$ mm. Require $0.1 \times 2^n > 3.84 \times 10^{11}$, so $2^n > 3.84 \times 10^{12}$. Now $2^{41} = 2.199 \times 10^{12}$ is too small, while $2^{42} = 4.398 \times 10^{12}$ is large enough. The smallest whole number is $n = 42$, giving 4.398×10^{11} mm $= 4.398 \times 10^8$ m.

3 For 300 m $= 3 \times 10^5$ mm, require $2^n > 3 \times 10^6$. Since $2^{21} = 2.10 \times 10^6$ is too small and $2^{22} = 4.19 \times 10^6$, $n = 22$ gives 419.4 m. For 8,849 m $= 8.849 \times 10^6$ mm, $2^{26} = 6.71 \times 10^7$ is too small and $2^{27} = 1.34 \times 10^8$, so $n = 27$ gives 13.4 km. Table: tower 22 folds, 419 m; Everest 27 folds, 13.4 km.

4 A height H metres is $1000H$ mm, so $0.1 \times 2^n > 1000H$, i.e. $2^n > 10^4 H$. Thus $n > \log_2(10^4 H)$. Using $2^{10} \approx 10^3$, estimate $n \approx \frac{10}{3}\log_{10}(10^4 H)$ and round up. Checks: $H = 300$ gives about 21.6, so 22; $H = 8849$ gives about 26.5, so 27; $H = 3.84 \times 10^8$ gives about 42.

5 The model assumes each fold exactly doubles the thickness and that the sheet can keep folding. In reality the area halves each time: after 42 folds it would be about 2×10^{-13} of the original, a microscopic patch, while the thickness would be 4.4×10^8 m. The sheet would be far longer than it is wide long before that. Paper stiffness, the finite radius of a crease, trapped air and imperfect alignment all break the doubling rule. The mathematics correctly predicts the exponential growth, but the physical procedure is impossible; a model can be mathematically sound and still not describe a feasible action.

APPLICATION · A SCALE MODEL OF THE SOLAR SYSTEM

1 Neptune is 4.50×10^9 km $= 4.50 \times 10^{12}$ m. The scale is $450 : 4.50 \times 10^{12} = 1 : 1.0 \times 10^{10}$, or 1 : 10,000,000,000.

2 Distances: Mercury 5.8 m, Earth 15.0 m, Mars 22.7 m, Jupiter 77.8 m, Neptune 450 m. Diameters: Mercury 0.049 cm, Earth 0.127 cm, Mars 0.068 cm, Jupiter 1.4 cm, Neptune 0.49 cm.

3 Earth's model diameter, 0.13 cm, is only just visible as a dot; Jupiter's 1.4 cm is clearly visible. At one true scale the planets are minute compared with their separations, so a model cannot show both distance and size clearly. Museums usually exaggerate planet sizes (or use separate inset models) while keeping distances to scale, and must say so.

4 Jupiter's diameter is 1.40×10^5 km $= 1.40 \times 10^8$ m. A 0.30 m model gives scale $\frac{0.30}{1.40\times10^8} \approx 2.1 \times 10^{-9}$, or about $1 : 4.7 \times 10^8$. Neptune would then be $4.50 \times 10^{12} \times 2.1 \times 10^{-9} \approx 9.6 \times 10^3$ m, about 9.6 km from the Sun. The 500 m path is far too short.

5 Sample label: "This model uses a scale of about $1 : 10^{10}$: one metre here represents ten billion metres in space. Distances are to scale, but the planets are enlarged so that they can be seen; their sizes are therefore not to the same scale as their separations. A model that kept sizes to scale would make the planets almost invisible, so one quantity must be exaggerated. Neptune orbits 450 m from the Sun, while Earth is only 15 m away and just 1.3 mm across; the Sun itself would be 14 cm wide. Scientific notation keeps the design readable: 4.5×10^{12} m becomes 450 m. We quote two significant figures because the scale and the orbital distances are approximate."

CRITERION TASK · A PUBLIC BRIEFING ON VERY SMALL PARTICLES

Sample response. A PM2.5 particle is at most 2.5×10^{-6} m across, while a human hair is about 7×10^{-5} m across. The ratio is $\frac{7\times10^{-5}}{2.5\times10^{-6}} = 28$, so about 28 particles would fit across one hair. The measured annual mean is 8.3×10^{-9} kg/m^3, or 8.3 μg/m^3, compared with the limit of 5.0×10^{-9} kg/m^3, or 5.0 μg/m^3. The measurement is 66% above the limit. Its uncertainty is $\pm 6\%$, which is about $\pm 0.50 \times 10^{-9}$ kg/m^3, giving a range from 7.8×10^{-9} to 8.8×10^{-9} kg/m^3. Even the lowest value is above 5.0×10^{-9}, so the site definitely exceeds the limit. The comparison is reported to two significant figures, matching the input data.

1 $\frac{7\times10^{-5}}{2.5\times10^{-6}} = 2.8 \times 10^{1} = 28$. Because the hair diameter is given as "about 7×10^{-5}" (1 s.f.), a cautious report is "about 30 particles", i.e. 1 s.f.; 28 is acceptable if the data are treated as 2 s.f.

2 Since 1 μg $= 10^{-9}$ kg, multiply kg/m^3 by 10^9: measured 8.3×10^{-9} kg/m^3 = 8.3 μg/m^3; limit 5.0×10^{-9} kg/m^3 = 5.0 μg/m^3.

3 $\frac{8.3-5.0}{5.0} \times 100\% = 66\%$. Equivalently, the measured value is 1.66 times the limit, so it is 66% above it.

4 6% of 8.3×10^{-9} is $0.498 \times 10^{-9} \approx 0.50 \times 10^{-9}$. The interval is $7.8 \times 10^{-9} \le c \le 8.8 \times 10^{-9}$ kg/m^3. The lower bound 7.8×10^{-9} exceeds 5.0×10^{-9}, so the site definitely exceeds the limit even after allowing for uncertainty.

5 Sample briefing: "The particles we are measuring are very small: up to 28 of them would fit across the width of a human hair. Because they are invisible, the hair comparison is the clearest way to show their scale, and it is a whole-number estimate because the hair diameter is itself approximate. The annual average at the tested site was 8.3×10^{-9} kilograms per cubic metre, or 8.3 micrograms per cubic metre. The health-based limit is 5.0 micrograms per cubic metre, so the site is about 66% above the limit. The measurement has an uncertainty of $\pm 6\%$, which means the true annual average is likely between 7.8 and 8.8 micrograms per cubic metre. Even the lowest estimate is well above 5.0, so the site exceeds the limit. We report the comparison to two significant figures, because the measurement and its uncertainty do not justify more; the figures are rounded so that they inform without pretending to be exact."

Marking guide. Part 1 (4 marks): correct ratio; division of mantissas and subtraction of indices; answer 28; appropriate significant-figure comment. Part 2 (3 marks): correct conversion factor 10^9; both values converted; units μg/m^3. Part 3 (4 marks): correct difference; division by the limit; multiplication by 100; answer 66%. Part 4 (5 marks): uncertainty 0.50×10^{-9}; correct interval; comparison of lower bound with the limit; justified "definitely exceeds"; clear units. Part 5 (8 marks): particle–hair comparison; measured value and limit; uncertainty; conclusion; scientific notation used; precision of each number justified; readable for a non-specialist. Criterion D strands i–v are addressed through identifying quantities, selecting ratios and conversions, justifying accuracy, and evaluating the message. Bands: 19–24 correct and well justified; 13–18 mostly correct with minor uncertainty errors; 7–12 some relevant work; 1–6 little correct mathematics.

MIXED PRACTICE 2

F1 (a) $x^{7-3} = x^4$; (b) $3^3a^6 = 27a^6$; (c) $\frac{8}{2}m^{6-(-2)} = 4m^8$.

F2 (a) $81^{3/4} = (3^4)^{3/4} = 3^3 = 27$; (b) $27^{-2/3} = \frac{1}{(\sqrt[3]{27})^2} = \frac{1}{9}$; (c) $\sqrt[3]{-125} = -5$.

F3 $0.000903 = 9.03 \times 10^{-4}$; $28{,}400{,}000 = 2.84 \times 10^7$.

F4 $2.4 \times 5.0 = 12$ and $10^{-4+7} = 10^3$, so $12 \times 10^3 = 1.2 \times 10^4$.

R1 $\frac{2^1}{2^4} = 2^{1-4} = 2^{-3}$. Directly, $\frac{2}{16} = \frac{1}{8}$, so $2^{-3} = \frac{1}{8}$; a negative exponent records division by a positive power.

R2 No: 4.9×10^5 is larger. Its power of 10 is ten times 10^4, and the mantissas 5.2 and 4.9 are close, so the difference in exponent dominates; the ratio is about 9.4.

R3 The student added mantissas and exponents as if multiplying. Addition needs a common exponent: $4.0 \times 10^4 = 0.40 \times 10^5$, so $3.6 \times 10^5 + 0.40 \times 10^5 = 4.00 \times 10^5$. Check: $360{,}000 + 40{,}000 = 400{,}000$.

R4 Four significant figures: 7, 0, 8 and the final 0. The leading zeros are not significant; the zero between 7 and 8 is; the trailing zero after the decimal point is a deliberate precision claim.

A1 10 cm $= 10^{-1}$ m, so $\frac{10^{-1}}{2\times10^{-5}} = 0.5 \times 10^4 = 5 \times 10^3$ cells, about 5,000.

A2 $\frac{4.0\times10^{12}}{3.2\times10^{6}} = 1.25 \times 10^6$, about 1.3×10^6 or 1.25 million photos.

A3 $5.0 \times 10^3 \times 2^8 = 5.0 \times 10^3 \times 256 = 1.28 \times 10^6$ bacteria.

A4 Relative error $= \frac{0.1}{8.4} \times 100\% \approx 1.2\%$. For an 8.4 g measurement this is moderately precise, but for a 100 g sample the same ± 0.1 g is only 0.1%; the smaller sample carries the larger relative uncertainty, so 8.4 g is not a precise way to represent a 100 g quantity.

Common errors. Multiplying exponents when powers are multiplied instead of adding them; writing $(2x)^3 = 2x^3$ rather than $8x^3$; losing the negative sign in a^{-n}; adding exponents when adding numbers in scientific notation; leaving a mantissa outside $1 \le a < 10$; and counting leading zeros as significant. A related error is reporting every calculator digit as if it were measured.

Chapter 3 · Functions, expressions and algebraic reasoning

PREREQUISITE CHECK

1 $4(5) + 3 = 20 + 3 = \mathbf{23}$.

2 $(7 - 2 + 3)a = \mathbf{8a}$.

3 $3(2x + 4) = \mathbf{6x + 12}$.

4 $(-3)^2 = (-3) \times (-3) = \mathbf{9}$.

5 $2x = 17 - 5 = 12$, so $x = \mathbf{6}$.

6 The terms increase by 3: **16**, **19**.

PRACTICE 3.1 · EXPRESSIONS AND SUBSTITUTION

1 (a) Coefficient 7, constant -4. (b) Coefficient -2, constant 9. (c) Coefficient of x is -5, constant 1; the coefficient of x^2 is 1.

2 (a) $(8-3+2)y = \mathbf{7y}$. (b) $(4p-p)+(7q-3q) = \mathbf{3p+4q}$. (c) $(6-1)x^2+2x = \mathbf{5x^2+2x}$.

3 (a) $6a+8+5a = \mathbf{11a+8}$. (b) $10x-5-3x = \mathbf{7x-5}$. (c) $12m+8-2m+10 = \mathbf{10m+18}$.

4 (a) $2(4)+3(2)-4 = 8+6-4 = \mathbf{10}$. (b) $2(1)+3(-1)-4 = 2-3-4 = \mathbf{-5}$. (c) $2(0.25)+1.5-4 = 0.5+1.5-4 = \mathbf{-2}$.

5 $5x$ and $2x$ have the same variable part, so the distributive law gives $(5+2)x = 7x$. The 2 in $5x+2$ is a constant, not a like term, so no common factor of x exists; the expression is already as simple as possible.

6 Writing x^2 as -3^2 squares 3 first and then applies the minus sign, giving -9. The input is the whole number -3, so brackets are needed: $(-3)^2 = 9$. Correct value **9**.

7 $P = 2\big((3x+1)+(x-2)\big) = 2(4x-1) = \mathbf{8x-2}$ (the same unit as x).

8 For example x^2 and $4x$: at $x=2$ both equal 4, but at $x=3$ they give 9 and 12, so they are not identical. One matching input tests only one case; equivalence requires equal values for *every* permitted input, which a single substitution cannot establish.

PRACTICE 3.2 · FUNCTIONS, DOMAIN AND RANGE

1 (a) $3(2)-5 = \mathbf{1}$. (b) $3(0)-5 = \mathbf{-5}$. (c) $3(-4)-5 = -12-5 = \mathbf{-17}$.

2 (a) $3^2+1 = \mathbf{10}$. (b) $(-3)^2+1 = \mathbf{10}$. (c) $0.5^2+1 = 0.25+1 = \mathbf{1.25}$.

3 $10-2x = 4$, so $2x = 6$ and $x = \mathbf{3}$. Check: $h(3) = 10-6 = 4$.

4 (a) All real x except $x = 0$ (division by zero). (b) $x \geq 3$, so that $x-3 \geq 0$ and the root is real. (c) All real x.

5 For $-5 < x < 5$ the equation gives two outputs, $y = \sqrt{25-x^2}$ and $y = -\sqrt{25-x^2}$; for example $x = 0$ gives $y = 5$ and $y = -5$. A vertical line therefore meets the circle twice, so one input has two outputs and the relation is not a function.

6 $f(x) = 2x-3$. Solving $2x-3 = 11$ gives $2x = 14$, so the input is $x = \mathbf{7}$.

7 A sensible domain is $n = 0, 1, 2, 3, \ldots$ (whole numbers of posters), since a fraction of a poster cannot be printed. $C(40) = 25+1.5(40) = 25+60 = \mathbf{85}$ units.

8 $f(x) = \dfrac{2x-4}{x-4}$ is one example. The denominator $x-4$ excludes 4 from the domain; the numerator is chosen so that $f(0) = \dfrac{-4}{-4} = 1$. Any rule of the form $1+\dfrac{kx}{x-4}$ also works.

PRACTICE 3.3 · EXPANDING AND FACTORISING

1 (a) $\mathbf{8x+12}$. (b) $\mathbf{-3x+15}$. (c) $2x(3x)+2x(4) = \mathbf{6x^2+8x}$.

2 (a) $x^2+7x+2x+14 = \mathbf{x^2+9x+14}$. (b) $x^2+5x-4x-20 = \mathbf{x^2+x-20}$. (c) $x^2-2x-6x+12 = \mathbf{x^2-8x+12}$.

3 (a) $3x^2-6x+x-2 = \mathbf{3x^2-5x-2}$. (b) $4x^2+10x-10x-25 = \mathbf{4x^2-25}$. (c) $x^2+8x+8x+64 = \mathbf{x^2+16x+64}$.

4 (a) $5x(x+3)$. (b) $(x-7)(x+7)$. (c) $(x+4)(x+5)$. (d) $(x-4)(x+3)$.

5 $x^2+6x+9 = x^2+2(3)x+3^2 = (x+3)^2$, since $3+3=6$ and $3\times 3 = 9$. For x^2+9 the constants would need sum 0 and product 9; no integers satisfy both (3 and 3 give sum 6). It is a sum, not a difference, of squares, so it does not factorise over the integers.

6 Expanding: $x^2+3x+2x+6 = x^2+5x+6$. Substituting $x = 1$: left side $3\times 4 = 12$, right side $1+5+6 = 12$. Both checks agree, so the factorisation is correct.

7 $(x-7)(x+2) = x^2+2x-7x-14 = \mathbf{x^2-5x-14}$, so $b = \mathbf{-5}$ and $c = \mathbf{-14}$.

8 Integer factors of 7 are 1 and 7, or -1 and -7, with sums 8 and -8; neither sum is 4. So no integer a, b exist. Since $x^2+4x+7 = (x+2)^2+3$, the graph is a parabola with minimum $(-2, 3)$ and no x-intercepts: the equation $y = 0$ has no real solutions.

PRACTICE 3.4 · REARRANGING FORMULAS

1 (a) $x = y-5$. (b) $x = \dfrac{y}{4}$. (c) $y+3 = 2x$, so $x = \dfrac{y+3}{2}$.

2 $v-u = at$, so $t = \dfrac{v-u}{a}$ (for $a \neq 0$).

3 $2A = bh$, so $h = \dfrac{2A}{b}$.

4 $r = \dfrac{C}{2\pi}$.

5 Divide by 5: $x-2 = \dfrac{y}{5}$, so $x = \dfrac{y}{5}+2 = \dfrac{y+10}{5}$. Check with $y = 15$: $x = 5$, and $5(5-2) = 15$.

6 $2A = (a+b)h$, then $a+b = \dfrac{2A}{h}$, so $b = \dfrac{2A}{h}-a$.

7 $y = x(a+b)$, so $x = \dfrac{y}{a+b}$, provided $a+b \neq 0$; if $a+b = 0$ then $y = 0$ for every x and no unique subject exists.

8 $S-2\pi r^2 = 2\pi rh$, so $h = \dfrac{S-2\pi r^2}{2\pi r}$. With $S = 300$, $r = 4$: $h = \dfrac{300-32\pi}{8\pi} = \dfrac{199.47}{25.13} \approx 7.9366$, so $h = \mathbf{7.9\ cm}$ (2 s.f.).

PRACTICE 3.5 · SEQUENCES AND GENERALISATION

1 (a) Add 5: **21**, **26**. (b) Subtract 4: **18**, **14**. (c) Multiply by 3: **54**, **162**.

2 (a) $5n-1$. (b) $15-3n$. (c) $2n-1$.

3 $n = 1$: $5-2 = \mathbf{3}$; $n = 5$: $25-2 = \mathbf{23}$; $n = 20$: $100-2 = \mathbf{98}$.

4 Solve $4n+3 = 71$: $4n = 68$, $n = 17$. Since 17 is a positive whole number, **yes** : 71 is the 17th term.

5 First differences are $3, 5, 7, \ldots$, which change, so there is no constant difference and the rule is not of the form $an+b$. The second differences are constant, so the sequence is quadratic, with nth term n^2.

6 Differences are 4, so the rule is $4n+2$: $4(1)+2 = 6$, $4(2)+2 = 10$, $4(3)+2 = 14$. Structurally, each stage has 2 central tiles and 4 arms of n tiles, giving $2+4n$.

7 $T(n) = 2^n$ (double each time) gives 2, 4, 8, 16. A second rule, $T(n) = 2^n+(n-1)(n-2)(n-3)(n-4)$, also gives 2, 4, 8, 16 but gives 56 for $n = 5$. Four terms therefore do not determine a unique pattern: a rule needs a structural reason or a proof that it continues.

8 First differences $5, 7, 9, 11$; second differences $2, 2, 2$, constant, so the rule is quadratic. $T(n) = n^2+2n$, since $1+2 = 3$, $4+4 = 8$, $9+6 = 15$. Equivalently $T(n) = (n+1)^2-1$.

INVESTIGATION 3 · THE MATCHSTICK TRIANGLES

1 Stage 5 needs 11 matchsticks and stage 6 needs 13. The first differences remain 2, 2, 2, 2, so each new triangle still adds two matchsticks.

2 $M(n) = 2n+1$. The first triangle uses 3 matchsticks. Each additional triangle shares one side with the row, so only two of its three sides are new: $M = 3+2(n-1) = 2n+1$. Check: $n = 4$ gives 9, matching the table. The rule is structural, not merely fitted to the numbers.

3 $M(20) = 2(20) + 1 = \mathbf{41}$; $M(100) = 2(100) + 1 = \mathbf{201}$.

4 Solve $2n + 1 = 61$: $2n = 60$, so $n = \mathbf{30}$. Exactly 30 complete triangles can be made and **no matchsticks remain**, because 61 is odd and every value of $2n + 1$ is odd.

5 In a ring every triangle has one free outer side and two shared sides, so there are n free sides and n shared sides: $M = n + n = 2n$. The domain is $n \geq 3$, since a ring must close and $n = 1$ or $n = 2$ cannot form one. Compared with the row, the ring always uses one fewer matchstick ($2n$ against $2n + 1$): the two end triangles of the row, each with an extra outer side, are joined. The difference is constant, so the ring is still linear.

6 The zigzag has the same count, $M = 2n + 1$. Each new triangle still shares exactly one complete side with the previous triangle, so exactly two of its three sides are new; placing it above or below the previous triangle changes the shape but not the count. Diagram: draw stages 1–3 alternating up and down and mark the shared side; each stage adds two sticks.

APPLICATION · PAINTING A COMMUNITY HALL

1 Four walls have area $2LH + 2WH = 2(L + W)H$. Subtract the door and four windows: $A = 2(L + W)H - 2.0 - 4(1.5) = \mathbf{2(L + W)H - 8.0}$ square metres.

2 Two coats cover $2A$ square metres, and one litre covers 10, so $N = \dfrac{2A}{10} = \dfrac{A}{5} = \dfrac{2(L + W)H - 8}{5}$ litres. Round up because paint is bought in whole litres (and whole tins): a fractional litre cannot be purchased, and running short would leave a coat unfinished.

3 $A = 2(18 + 12)(4) - 8 = 240 - 8 = \mathbf{232\ m^2}$. $N = \dfrac{2(232)}{10} = 46.4$, so buy **47 L**. Tins: $47 \div 5 = 9.4$, so **10** five-litre tins (50 L, with 3.6 L spare).

4 Paint: $10 \times 42 = \mathbf{420}$ units. Extras: $0.08 \times 420 = 33.6$ units. Total $= 420 + 33.6 = \mathbf{453.6}$ units.

5 From $5N = 2(L + W)H - 8$: $2(L + W)H = 5N + 8$, so $H = \dfrac{5N + 8}{2(L + W)}$. With $L = 15$, $W = 9$, $N = 44$: $H = \dfrac{220 + 8}{48} = 4.75$, so $H = \mathbf{4.75\ m}$.

6 The model ignores the ceiling, paint absorbed unevenly by the surface, spillage, masking and trim, and it assumes every litre covers exactly $10\ \text{m}^2$. The opening areas are also fixed, although a real hall may have different windows. It could be improved without measuring every wall by sampling one wall to estimate absorption, using the manufacturer's coverage range for the paint type, and adding a wastage allowance of about 5–10%. Rounding up litres suits a purchasing decision because tins are indivisible and a small surplus is cheaper than an unfinished job; the cost is driven by whole tins, so the answer should report both 47 L and 10 tins. Since 44 L was itself rounded, the height 4.75 m is approximate.

CRITERION TASK · THE GROWING SQUARE FRAME

Sample response. Frames 1 to 5 contain 4, 8, 12, 16 and 20 tiles, with constant first difference 4, so $T = 4n$. The structure gives the same result: an outer square of side $n + 1$ loses an inner square of side $n - 1$, and $(n + 1)^2 - (n - 1)^2 = 4n$. Frame 25 therefore has 100 tiles, and 100 tiles make exactly Frame 25. A rectangular frame of width $n + 1$ and height $n + 2$, with the centre removed, has $T = (n + 1)(n + 2) - n(n - 1) = 4n + 2$, which is still linear.

1 Frame 1: a 2×2 block, 4 tiles. Frame 2: 3×3 with a 1-tile hole, 8 tiles. Frame 3: 4×4 with a 2×2 hole, 12 tiles. Table: $n = 1, 2, 3, 4, 5$ gives $T = 4, 8, 12, 16, 20$.

2 First differences: $8 - 4 = 4$, $12 - 8 = 4$, $16 - 12 = 4$, $20 - 16 = 4$. They are constant, so the pattern is **linear** with gradient 4.

3 $T = (n + 1)^2 - (n - 1)^2 = (n^2 + 2n + 1) - (n^2 - 2n + 1) = 4n$. Structurally, the four sides each contain n tiles when each corner is counted once, so $4n$; equivalently, two opposite sides have $n + 1$ tiles and the other two have $n - 1$, giving $2(n + 1) + 2(n - 1) = 4n$. Verification: $T(5) = 20$, which matches the drawing.

4 $4n = 4(n - 1) + 4$ because $4n - 4 + 4 = 4n$. The form $4n$ shows four arms of n tiles with every corner counted exactly once; the form $4(n - 1) + 4$ separates the four corner tiles from the $n - 1$ non-corner tiles along each side. Both describe the same frame, which is why the expressions are equivalent.

5 $T(25) = 4(25) = \mathbf{100}$ tiles. For 100 tiles, $4n = 100$ gives $n = 25$, so the largest frame is **Frame 25**, using all 100 tiles with none left over.

6 Outer area $(n + 1)(n + 2)$, hole $n(n - 1)$, so $T = n^2 + 3n + 2 - (n^2 - n) = \mathbf{4n + 2}$. Values 6, 10, 14, 18, 22 have constant difference 4, so the relationship is linear; the extra 2 is the two tiles by which the height exceeds the width.

Marking guide. Q1 (4): two for a complete, correct table for $n = 1$ to 4, one for $n = 5$, one for the three labelled drawings. Q2 (4): two for the four first differences, two for the conclusion and reason. Q3 (6): two for $T = 4n$, three for a structural explanation (outer square minus hole, or sides with corners counted once), one for verifying $n = 5$. Q4 (4): two for showing both forms equal $4n$, two for the meaning of each form. Q5 (4): two for 100, two for $n = 25$ with remainder 0. Q6 (2): one for $4n + 2$, one for the linear justification. Criterion B strands i–iii are addressed by the table and drawings (i), the difference analysis (ii), and the generalisation with verification (iii). Criterion C strands ii–iii are credited within the same marks for correct notation $T(n)$, labelled table and diagrams, and a coherent chain from pattern to rule.

MIXED PRACTICE 3

F1 $(6 - 4)x^2 + (-2 + 7)x + (3 - 8) = \mathbf{2x^2 + 5x - 5}$.

F2 $f(0) = \mathbf{4}$; $f(2) = 4 - 6 = \mathbf{-2}$; $f(-1) = 4 + 3 = \mathbf{7}$.

F3 $2x^2 + 12x - x - 6 = \mathbf{2x^2 + 11x - 6}$.

F4 $x^2 + 11x + 24 = \mathbf{(x + 3)(x + 8)}$ (sum 11, product 24); $x^2 - 64 = \mathbf{(x - 8)(x + 8)}$.

F5 $3x = 12 - 2y$, so $x = \dfrac{12 - 2y}{3} = \mathbf{4 - \dfrac{2}{3}y}$.

F6 Difference 4, first term 9, so $T(n) = \mathbf{4n + 5}$.

R1 At $x = 1$: $(1 + 5)^2 = 36$, while $1^2 + 25 = 26$. Since $36 \neq 26$, one counterexample shows the expressions are not equivalent; the square must also produce the middle term $10x$.

R2 $f(3)$ is the y-coordinate of the point on the graph with $x = 3$: move vertically from 3 on the x-axis to the graph, then read across to the y-axis. To solve $f(x) = 2$, draw the horizontal line $y = 2$ and read the x-coordinates of its intersections with the graph.

R3 At $x = -1$ the denominator is zero, and division by zero is undefined, so -1 must be excluded. Domain: all real x except -1, i.e. $x \neq -1$.

R4 $T(n + 1) - T(n) = 7 - 2(n + 1) - (7 - 2n) = -2$, a constant negative difference, so the sequence is **decreasing**. Its terms are $5, 3, 1, -1, \ldots$

A1 $C(m) = 12 + 0.08m$, in units, for m megabytes. $C(500) = 12 + 0.08(500) = 12 + 40 = \mathbf{52}$ units.

A2 Area $= (x + 3)(x - 1) = x^2 + 3x - x - 3 = \mathbf{x^2 + 2x - 3}$. At $x = 6$: $36 + 12 - 3 = \mathbf{45}$ square units.

A3 Sum $= 30^2 = \mathbf{900}$. Table: $n = 1, 2, 3, 4, 5$ gives sums 1, 4, 9, 16, 25, that is n^2; each new odd number adds the next odd difference.

A4 $s^2 = \dfrac{A}{6}$, so $s = \sqrt{\dfrac{A}{6}}$ (positive root for a length). $A = 150$: $s^2 = 25$, so $s = \mathbf{5\ cm}$.

Common errors. Expanding $(x+5)^2$ as x^2+25, which drops the middle term $10x$; substituting a negative without brackets, so that $(-3)^2$ becomes -9; combining unlike terms such as $3x+4$ into $7x$; dividing only the first term when rearranging, as in $2y=6x-8$ giving $y=3x-8$; and treating three terms of a sequence as proof of a rule. In $23-3n$, the negative gradient means the sequence falls, so the tenth term is -7, not 7.

Chapter 4 · Linear relationships and graphs

PREREQUISITE CHECK

1 $(0,0)$ is the origin; $(2,3)$ is in quadrant 1; $(-1,4)$ is in quadrant 2 (1 left, 4 up).

2 $3(4)+2=12+2=\mathbf{14}$.

3 $2x=6$, so $x=\mathbf{3}$.

4 Add 4 each time: **21**, **25**.

5 $\frac{8-2}{5-1}=\frac{6}{4}=\mathbf{1.5}$.

6 $90\div 60=\mathbf{1.5}$ hours.

PRACTICE 4.1 · COORDINATES, TABLES AND GRAPHS

1 $A(3,2)$: quadrant 1. $B(-2,4)$: quadrant 2. $C(-3,-1)$: quadrant 3. $D(4,-3)$: quadrant 4. $E(0,-2)$: on the negative y-axis, not in a quadrant.

2 $y=3x-2$ gives $-5,-2,1,4,7$. The points lie on a straight line through $(0,-2)$, rising 3 for each step of 1 right.

3 First differences $3,3,3,3$ are constant, so the relationship is **linear**. At $x=0$, $y=5$, so $y=3x+5$.

4 (a) Differences $3,3,3$: linear, $y=3x+4$. (b) Differences $2,4,8$: not linear. (c) Differences $-2,-2,-2$: linear, $y=11-2x$. (d) Differences $2,3,4$: not linear (triangular numbers).

5 $w=0,1,2,3,4$ gives $h=12,14.5,17,19.5,22$ cm. The rule is $h=12+2.5w$; at $w=10$, $h=12+25=\mathbf{37\ cm}$.

6 $y=3$ is horizontal: every point has height 3, so the vertical change is 0. $x=3$ is vertical: the input $x=3$ has infinitely many y-values, so it fails the vertical line test; its gradient is undefined.

7 Gradient $=\frac{9.00-4.50}{2-0}=2.25$ units per km; intercept 4.50 units. So $C=4.50+2.25d$, and at $d=11$, $C=4.50+24.75=\mathbf{29.25}$ units.

8 Example: a repair shop charges a 15-unit call-out fee plus 0.40 units per GB recovered; x is GB, C is units. The model fails for very large jobs if the shop caps its charge or charges separately for parts.

PRACTICE 4.2 · GRADIENT AND RATE OF CHANGE

1 (a) $\frac{5-1}{6-2}=1$. (b) $\frac{-4-4}{3-(-1)}=\frac{-8}{4}=-2$. (c) $\frac{5-(-3)}{2-(-2)}=\frac{8}{4}=2$. (d) $\frac{0-0}{7-4}=0$. (e) Run $5-5=0$, so the gradient is **undefined**.

2 $m=\frac{11-2}{3-1}=4.5$. From $x=7$ to $x=9$ is a step of 2, so $y=29+2(4.5)=\mathbf{38}$.

3 Rate $=\frac{340-500}{8-0}=-20$ L/min: the tank loses 20 litres each minute. $500-20t=0$ gives $t=\mathbf{25}$ minutes.

4 Sketches: a rising line $y=x+1$ (positive: y increases with x); a falling line $y=-x+3$ (negative: y decreases as x increases); a horizontal line $y=2$ (zero: no change); a vertical line $x=1$ (undefined: no horizontal change).

5 Gradient 0 means the vertical change is 0, so the line is horizontal, for example $y=2$. An undefined gradient means the horizontal change is 0, so the formula divides by zero; the line is vertical, for example $x=3$, and is not a function. Zero is a number; undefined is not.

6 The x-coordinates were subtracted in the opposite order to the y-coordinates: the denominator should be $5-2$, not $2-5$. Correct: $m=\frac{-1-3}{5-2}=-\frac{\mathbf{4}}{\mathbf{3}}$.

7 A: $12\div 40=0.3$ km/min $=18$ km/h. B: $9\div 30=0.3$ km/min $=18$ km/h, so the gradients are equal. B's 2 km start raises its intercept by 2 km, shifting the line up without changing its steepness.

8 Gradient $=\frac{0.6}{7.2}=\frac{1}{12}\approx 0.0833$. This is exactly the guideline limit, so the ramp is **acceptable but has no margin**. At $\frac{1}{12}$, a rise of 0.6 m needs a run of $0.6\times 12=\mathbf{7.2\ m}$, the same run.

PRACTICE 4.3 · EQUATIONS OF LINES

1 (a) $m=4$, $c=-7$. (b) $m=-0.5$, $c=2$. (c) $y=-2x+5$, so $m=-2$, $c=5$. (d) $y=5-3x$, so $m=-3$, $c=5$. (e) $y=3x-4$, so $m=3$, $c=-4$.

2 (a) $y=3x-2$. (b) $y=-\frac{2}{3}x+4$. (c) $y=-3$. (d) $x=5$.

3 (a) $m=\frac{10-2}{4-0}=2$, $c=2$: $y=2x+2$. (b) $m=\frac{1-5}{3-1}=-2$; $5=-2+c$, so $c=7$: $y=7-2x$. (c) $m=\frac{0-4}{2-(-2)}=-1$; $4=2+c$, so $c=2$: $y=2-x$. (d) $m=0$: $y=-1$.

4 Gradient $=\frac{4-6}{1-0}=-2$, intercept 6: $y=6-2x$.

5 $(1,-1)$: $2-3=-1$ ✓. $(2,1)$: $4-3=1$ ✓. $(3,4)$: $6-3=3\neq 4$ ✗. $(0,-3)$: $0-3=-3$ ✓. So $(1,-1)$, $(2,1)$ and $(0,-3)$ lie on the line; $(3,4)$ does not.

6 Parallel: $m=-1$, $5=-2+c$, so $c=7$: $y=-x+7$. Perpendicular: $m=1$, $5=2+c$, so $c=3$: $y=x+3$.

7 Gradient -0.15 m/min: height falls 0.15 m each minute. Intercept 2.4 m: initial height. $h(8)=2.4-1.2=\mathbf{1.2\ m}$. Empty when $2.4-0.15t=0$, so $t=\mathbf{16}$ min; domain $0\le t\le 16$.

8 Infinitely many, one for each gradient: $y=x+3$, $y=2x+1$, $y=-x+7$. Gradient 0 gives $y=5$; undefined gradient gives $x=2$. Both pass through $(2,5)$, since a point fixes the intercept once the gradient is chosen.

PRACTICE 4.4 · DIRECT PROPORTION

1 (a) **Yes**: $C=4n$ is $y=kx$ with $k=4$. (b) **No**: intercept 10. (c) **Yes**: $k=0.5$. (d) **No**: $y=\frac{2}{x}$ is not linear; xy, not y/x, is constant. (e) **No**: intercept 3 and negative gradient, so y/x changes.

2 $y=7(8)=\mathbf{56}$. $63=7x$, so $x=\mathbf{9}$.

3 $45=9k$, so $k=5$ and $y=5x$. At $x=12$, $y=\mathbf{60}$; at $y=100$, $x=\mathbf{20}$.

4 $3=12k$, so $k=0.25$ and $c=0.25M$. For 20 muffins, $c=\mathbf{5}$ cups; for 7.5 cups, $M=7.5\div 0.25=\mathbf{30}$ muffins.

5 $E=0.93D$. 250 dollars gives 232.50 euros; 500 euros gives $D=\frac{500}{0.93}\approx\mathbf{537.63}$ dollars. Domain $D\ge 0$. Commission, fees and daily rate changes make the model approximate.

6 Table $x = 1, 2, 3$ gives $y = 5, 7, 9$; ratios $5, 3.5, 3$ are not constant, and $y = 3$ at $x = 0$, so the graph does not pass through the origin.

7 $d = 80t$ is $y = kx$ with $k = 80$, and $t = 0$ gives $d = 0$. For 300 km, $t =$ **3.75 h**; domain $t \geq 0$. Starting 20 km from home gives $d = 20 + 80t$, which is linear but **not proportional**, because the intercept is 20, not 0.

8 If $y = kx$, replacing x by $2x$ gives $k(2x) = 2kx = 2y$. For $y = 2x + 3$: $x = 1$ gives 5, $x = 2$ gives 7, not 10. The fixed 3 breaks the doubling, so proportional reasoning needs a zero intercept.

PRACTICE 4.5 · BUILDING AND EVALUATING MODELS

1 (a) $C = 45 + 30h$ (units; h hours). (b) $C = 3.50 + 2.20d$ (units; d km). (c) $C = 18 + 0.40g$ (units per month; g GB).

2 (a) 30 units per hour; 45 units call-out. (b) 2.20 units per km; 3.50 units fixed. (c) 0.40 units per GB; 18 units monthly fee.

3 $m = 0, 3, 6, 9, 12$: A costs $10, 22, 34, 46, 58$; B costs $25, 31, 37, 43, 49$. A is cheaper at 0, 3 and 6 months, B at 9 and 12. $10 + 4m = 25 + 2m$ gives $m = 7.5$, so A is cheaper below 7.5 months (up to 7 whole months) and B above.

4 $15 + 3n = 30 + 1.5n$ gives $n = 10$, and both cost 45 units at $(10, 45)$. Below 10, the first plan is cheaper; above 10, the second.

5 $M = 120$ and $P = 3v$. $3v = 120$ gives $v =$ **40** visits. The gradient is 0 because the annual cost does not depend on visits.

6 Gradient 5: 5 more marks per hour of study. Intercept 40: score with no study. $S(12) =$ **100**. $h = 40$ is extrapolation outside $0 \leq h \leq 20$; it predicts 240, above any plausible maximum, and ignores diminishing returns.

7 5 GB: A costs 25, B 45, so A is cheaper by 20. 40 GB: A costs 60, B 45, so B is cheaper by 15. A suits light, low-income users; B protects heavy users but is regressive for light ones. Assumptions: steady use, truly unlimited data, equal reliability.

8 $C = 20 + 3x$: at $x = 10$ both charge 50. Below 10 the 10-unit lower intercept outweighs the extra 1 per item; above 10 the higher gradient makes it dearer. It benefits light users.

INVESTIGATION 4 · THE CANDLE CLOCK

1 The points form a falling straight line; the first differences are all -1.6 cm, so height falls 1.6 cm each hour.

2 Using $(0, 18.0)$ and $(5, 10.0)$: $m = \dfrac{10.0 - 18.0}{5 - 0} = -1.6$. Using $(1, 16.4)$ and $(4, 11.6)$: $m = \dfrac{-4.8}{3} = -1.6$. Units are cm per hour, since the rise is in cm and the run in hours; the gradient is negative because height falls as time increases.

3 $h = 18 - 1.6t$. The intercept 18.0 cm is the initial height. At $t = 3$, $h = 18 - 4.8 = 13.2$ cm, matching the table.

4 $h(2.5) = 18 - 4 =$ **14.0 cm**. Burn-out: $t = 18 \div 1.6 =$ **11.25 h**. Domain $0 \leq t \leq 11.25$ for the model; the data support only $0 \leq t \leq 5$.

5 $18 - 1.6t = 7.2$, so $t =$ **6.75 h** (6 h 45 min). This is inside the model domain but beyond the measured 5 hours, so it is extrapolation.

6 Draughts make one side burn faster; the melting wax pool and changing wick length alter the rate; the candle may soften or change diameter, and parallax adds error. Improve by shielding the candle, trimming the wick to a fixed length, measuring every 15 minutes with a clamped ruler, repeating with identical candles and controlling temperature.

7 $h(t) = H - rt$ and $T = \dfrac{H}{r}$. H is initial height in cm, r the rate in cm per hour, t the time in hours, T the burning time. The formulas assume constant r and $0 \leq t \leq T$.

8 $24 - 2.4t = 18 - 1.6t$, so $0.8t = 6$ and $t =$ **7.5 h**; the common height is **6 cm** (check: $24 - 18 = 6$). The answer is exact for the models but approximate in reality, and 7.5 h is beyond the 5 h of data.

APPLICATION · A LIBRARY OF THINGS

1 $C_A = 36 + 3h$, $C_B = 9h$. A: 36 units membership before any hire; 3 units per hire. B: no fixed charge; 9 units per hire.

2 $h = 0, 2, 4, 6, 8, 10, 12$: $C_A = 36, 42, 48, 54, 60, 66, 72$; $C_B = 0, 18, 36, 54, 72, 90, 108$.

3 $36 + 3h = 9h$ gives $h =$ **6**, both costing 54 units. The lines cross at $(6, 54)$: six hires a year cost the same; B is cheaper below 6, A above 6.

4 3 hires: A = 45, B = 27, so **B** (saving 18). 15 hires: A = 81, B = 135, so **A** (saving 54).

5 A demands 36 units before any tool is borrowed, a barrier for low-income households even though A is cheaper over a year of frequent use. B is cheaper for light users but not necessarily fairer: its high per-hire charge penalises heavy users. Fairness depends on access and need, not only the annual total.

6 $C_C = 12 + 6h$: $12, 24, 36, 48, 60, 72, 84$ for $h = 0, 2, 4, 6, 8, 10, 12$. B is cheapest below 4 hires, C from 4 to 8, A above 8. At $h = 4$, $C_B = C_C = 36$; at $h = 8$, $C_A = C_C = 60$. C benefits moderate users and those who can afford a small fee but not 36 units upfront.

7 Assumptions: every hire costs the same, membership is paid for a full year, and there is no limit on hires. The models ignore staff time, maintenance and replacement, insurance, late fees and seasonal demand. Collect anonymised hire records by household, a survey of income band and willingness to pay, and the cost of servicing one hire; compare predicted and actual annual totals.

8 **Recommendation:** offer C, with the 12-unit fee waived for low-income households, and keep A for frequent users. At 2 hires C costs 24 against B's 18; at 6 hires C costs 48 against 54 for both A and B; at 12 hires C costs 84 against A's 72. C covers moderate use at a lower entry cost than A, while A rewards very frequent use and gives reliable income. The waiver protects occasional users who cannot pay a joining fee; widespread use of it would need a subsidy or a higher per-hire charge.

CRITERION TASK · TWO WAYS TO PAY FOR THE SCHOOL BUS

Sample response. A costs 240 units whatever the number of trips; B costs $40 + 2t$. B is cheaper until 100 trips, where both cost 240 units; A is cheaper beyond 100. A student making 30 trips pays 100 units under B, not 240; one making 110 trips pays 240 under A, not 260. The break-even is 100 trips, so the recommendation depends on expected use and on ability to pay upfront.

1 $C_A = 240$ and $C_B = 40 + 2t$. A: gradient 0 units per trip, intercept 240 units per year. B: gradient 2 units per trip, intercept 40 units.

2 $t = 0, 20, 40, 60, 80, 100, 120$: $C_A = 240$ throughout; $C_B = 40, 80, 120, 160, 200, 240, 280$.

3 A is horizontal at 240, cutting the axis at $(0, 240)$; B rises from $(0, 40)$ with gradient 2, meeting A at $(100, 240)$. B lies below A until the intersection, then above it.

4 $40 + 2t = 240$ gives $t =$ **100** trips. The table shows both costs 240 at $t = 100$, and the graphs intersect there. In context, 100 trips is the break-even: below it B is cheaper, above it A is cheaper.

5 30 trips: A = 240, B = 100; recommend **B**, saving 140 units. 110 trips: A = 240, B = 260; recommend **A**, saving 20 units. B's low intercept wins for light use, but its gradient of 2 overtakes 240 after 100 trips.

6 B is fairer for rare travellers, who pay a small registration fee and then only for trips made; A favours frequent travellers and large families, since 240 per child is fixed. Three children making 110 trips each cost 720 under A and 780 under B, so A is better; at 60 trips each, B costs 480 and A 720, so B is better. A's 240-unit upfront payment may exclude students who cannot pay a lump sum, even when A would be cheaper. Assumptions: each trip costs 2 units, there is no cap or concession, and the pass is unlimited and non-transferable. Modification: allow instalments, or cap B's annual charge at 240 units.

Marking guide. Q1 (3): one per equation, one for units. Q2 (4): one for each pair of columns at $t = 0, 40, 80$, one for the row at 120. Q3 (5): two for accurate lines and labelled intercepts, two for the horizontal and rising behaviour, one for $(100, 240)$. Q4 (4): two for $t = 100$, two for interpreting the break-even and which option is cheaper. Q5 (4): two for each student's comparison and recommendation. Q6 (4): two for rare travellers, families and upfront payment, one for assumptions, one for a modification. D strands i–v: defining (i), representing (ii), applying (iii), interpreting (iv), evaluating (v); C strands i–iv: notation, tables and graphs, units, coherent justification.

MIXED PRACTICE 4

F1 $y = -2x + 5$, so $m = \mathbf{-2}$, $c = \mathbf{5}$.

F2 $m = \dfrac{10 - 2}{5 - (-3)} = \dfrac{8}{8} = \mathbf{1}$.

F3 $y = 3x - 4$.

F4 $m = \dfrac{9 - 1}{6 - 2} = 2$; $1 = 4 + c$, so $c = -3$: $y = 2x - 3$.

F5 $k = 7 \div 2 = 3.5$; $y = 3.5x$, so at $x = 8$, $y = \mathbf{28}$.

F6 Gradient 0.4 units per unit of use; intercept 25 units. $C(60) = 25 + 24 = \mathbf{49}$ units.

R1 Table $x = 1, 2, 3$ gives $y = 5, 9, 13$; ratios $5, 4.5, 4.33$ change, and $y = 1$ at $x = 0$, so the graph misses the origin.

R2 Equal gradients 3, different intercepts -2 and 5: the lines are parallel and never meet, the second being 7 units above the first at every x.

R3 The horizontal change is missing: $4 - 1 = 3$. $m = \dfrac{8 - 2}{4 - 1} = \dfrac{6}{3} = \mathbf{2}$.

R4 The object is stationary; distance is constant, so the gradient is 0. It has not disappeared: the same distance is held for 10 minutes.

A1 $C(250) = 15 + 20 = \mathbf{35}$ units. $31 = 15 + 0.08m$ gives $m = \mathbf{200}$ MB. Gradient 0.08: each extra MB costs 0.08 units.

A2 $n = 4$: X = 24, Y = 30. $n = 8$: both 36. $n = 12$: X = 48, Y = 42. X is cheaper for $n < 8$, Y for $n > 8$.

A3 Initial height 20 cm; burning rate 2.5 cm per hour. $5 = 20 - 2.5t$ gives $t = \mathbf{6}$ h; domain $0 \le t \le 8$.

A4 $S(30) = 800 + 1200 = \mathbf{2000}$. This is extrapolation far beyond the 6-week range. Request data beyond week 6, seasonal information, promotions and capacity limits.

Common errors. Subtracting coordinates in different orders, which changes the sign of the gradient; reading the intercept of $y = 5 - 2x$ as -2 instead of 5; dividing only one term when rewriting $2y = 6x - 8$; confusing a gradient of 0 with an undefined gradient; treating every linear relationship as proportional; and quoting a gradient without units. Predictions outside the stated domain should be labelled extrapolation.

Chapter 5 · Equations, inequalities and systems

PREREQUISITE CHECK

1 $2x = 12$, $x = 6$. Check: $2(6) + 5 = 17$.

2 $3x - 6$.

3 $-2x = -4$, $x = 2$. Check: $5 - 2(2) = 1$.

4 $(0, -1)$, $(1, 1)$, $(2, 3)$; the points lie on one straight line.

5 $3(4) + 2(1) = 14$.

6 19, 23; add 4 each time.

PRACTICE 5.1 · LINEAR EQUATIONS

1 (a) $4x = 32$, $x = 8$. (b) $-2x = 12$, $x = -6$. (c) $x + 2 = 7$, $x = 5$.

2 (a) $14 = 2x$, $x = 7$. (b) $5x - 5 = 3x + 7$, $2x = 12$, $x = 6$. (c) $1 - 3x = 10$, $-3x = 9$, $x = -3$.

3 Multiply by 12: $4(2x + 1) = 3(x - 2)$, $8x + 4 = 3x - 6$, $5x = -10$, $x = -2$. Check: $\frac{-4+1}{3} = -1 = \frac{-4}{4}$.

4 $0.4x = 1.6$, $x = 4$. Check: $0.4(4) + 1.2 = 2.8$.

5 The first expands to $3x + 6 = 3x + 6$, then $6 = 6$, true for every x: an identity, infinitely many solutions. The second gives $3x + 6 = 3x + 1$, then $6 = 1$, false for every x: no solution. The variable terms cancel; the constants decide.

6 The -2 multiplies both terms: $-2(x - 1) = -2x + 2$, so $5 - 2x + 2 = 9$. Then $7 - 2x = 9$, $-2x = 2$, $x = -1$. Check: $5 - 2(-2) = 9$.

7 Width w, length $w + 3$: $2(2w + 3) = 38$, $4w + 6 = 38$, $w = 8$. The rectangle is 11 cm by 8 cm. Check: $2(11 + 8) = 38$.

8 Sample: $-3(x - 2) = 2x - 14$. Expanding: $-3x + 6 = 2x - 14$, $-5x = -20$, $x = 4$. Check: $-3(2) = -6 = 2(4) - 14$.

PRACTICE 5.2 · INEQUALITIES

1 (a) $x < 5$: open circle 5, shade left. (b) $x \ge 4$: filled 4, shade right. (c) $-x > -3$, $x < 3$: open 3, shade left. (d) $x \ge -4$: filled -4, shade right.

2 (a) $2x > 10$, $x > 5$. (b) $-3x \le 6$, $x \ge -2$. (c) $-3x < -6$, $x > 2$.

3 (a) $-3 < 3x < 9$, so $-1 < x < 3$, or $(-1, 3)$. (b) $6 \le 2x \le 12$, so $3 \le x \le 6$, or $[3, 6]$.

4 $x < 5$ and $5 > x$ are the same statement read from opposite ends: both say 5 is greater than x. Multiplying by -1 reflects the number line about 0 and reverses order: $2 < 5$ but $-2 > -5$. Test $x = 4$: both true; $x = 6$: both false.

5 Dividing by -3 must reverse the sign: $x > -4$, not $x < -4$. Test $x = 0$: $0 < 12$ is true and $0 > -4$ is true, but $0 < -4$ is false, so the student's set loses a solution.

6 $12 + 2n \le 40$, $2n \le 28$, $n \le 14$: at most 14 people. Check: $n = 14$ costs 40; $n = 15$ costs 42.

7 $1.8 + 0.3t > 6$, $0.3t > 4.2$, $t > 14$. The drone is above 6 m after 14 minutes. The model starts at $t = 0$, so negative times are excluded; with $t \ge 0$, the contextual solution is $t > 14$.

8 Sample: $-2 \le x < 5$ is equivalent to $-1 \le x + 1 < 6$. No solution: $x < 2$ and $x \ge 2$; no number is both strictly below 2 and at least 2.

PRACTICE 5.3 · GRAPHICAL AND SUBSTITUTION METHODS

1 (a) $x+1=-x+5$, $x=2$, $y=3$: $(2,3)$. Check: $3=2+1=-2+5$. (b) $2x-3=-x+3$, $3x=6$, $x=2$, $y=1$: $(2,1)$. Check: $1=4-3=-2+3$.

2 (a) $3x+x+4=16$, $x=3$, $y=7$: $(3,7)$. (b) $x+3(2x-1)=11$, $7x=14$, $x=2$, $y=3$: $(2,3)$. (c) $x+2x-5=7$, $x=4$, $y=3$: $(4,3)$.

3 (a) Add: $3x=12$, $x=4$, $y=1$: $(4,1)$. Check: $8+1=9$, $4-1=3$. (b) $3x+2(x+3)=16$, $5x=10$, $x=2$, $y=5$: $(2,5)$.

4 The intersection lies on both lines, so its coordinates satisfy both equations at once. Parallel lines never meet, so there is no common point: no solution. Coincident lines give infinitely many.

5 $3x+2x+1=11$, $5x=10$, $x=2$; then $y=2(2)+1=5$, so $(2,5)$. The student must substitute $x=2$ to find y, then check both originals: $3(2)+5=11$, $5=2(2)+1$.

6 $2a+3c=46$, $a+2c=27$. From the second $a=27-2c$, so $54-c=46$, $c=8$, $a=11$. Adult 11 units, child 8 units. Check: $22+24=46$, $11+16=27$.

7 (a) Substitute: y is isolated, so $2x+4x-1=11$, $x=2$, $y=7$: $(2,7)$. (b) Eliminate by adding: the y-coefficients are opposites, so $2x=6$, $x=3$, $y=2$: $(3,2)$. Choose the method that avoids rearranging.

8 Sample: $x+y=1$, $x-y=5$; both hold at $(3,-2)$: $3+(-2)=1$, $3-(-2)=5$. Construct two different lines through $(3,-2)$, e.g. $y=x-5$ and $y=-2x+4$. A second system: $2x+y=4$, $x+2y=-1$. Infinitely many systems work: any two distinct lines through the point.

PRACTICE 5.4 · ELIMINATION AND SPECIAL SYSTEMS

1 (a) Add: $2x=14$, $x=7$, $y=3$. (b) Add: $5x=15$, $x=3$, $y=5$. (c) Add: $6x=24$, $x=4$; $16+3y=25$, $y=3$. Check: $8-9=-1$.

2 (a) Subtract: $2x=8$, $x=4$; $4+2y=8$, $y=2$. (b) Double the first: $10x-4y=38$; add: $13x=39$, $x=3$; $15-2y=19$, $y=-2$. Check: $9-8=1$.

3 Subtract: $2x=8$, $x=4$; $4+2y=8$, $y=2$. Check: $12+4=16$, $4+4=8$.

4 $4x+2y=12$ is $2(2x+y)=2(6)$, the same line as $2x+y=6$. Every point on it satisfies both equations: infinitely many solutions.

5 Dividing $4x+2y=10$ by 2 gives $2x+y=5$, contradicting $2x+y=6$; eliminating gives $0=2$, false. The lines have equal gradient -2 and different intercepts, so they are parallel: no solution.

6 $3a+2c=74$, $2a+3c=66$. First $\times 3$, second $\times 2$: $9a+6c=222$, $4a+6c=132$; subtract: $5a=90$, $a=18$; $54+2c=74$, $c=10$. Adult 18, child 10. Check: $36+30=66$.

7 (a) Substitute, since y is isolated: $3x+2(2x-3)=8$, $7x=14$, $x=2$, $y=1$: $(2,1)$. (b) Add, since the y-terms cancel: $6x=24$, $x=4$; $16+3y=18$, $y=\frac{2}{3}$: $(4,\frac{2}{3})$. Check: $8-2=6$.

8 No solution: $y=2x+1$, $y=2x-3$; equal gradients, different intercepts, so parallel. Infinitely many: $y=3x-2$, $2y=6x-4$; the second is twice the first, the same line with gradient 3 and intercept -2.

PRACTICE 5.5 · FEASIBLE REGIONS AND OBJECTIVES

1 The triangle with vertices $(0,0)$, $(6,0)$, $(0,6)$, bounded by the solid line $x+y=6$ and the axes.

2 $(2,3)$: $3\geq 3$, $3<4$, feasible (on the boundary). $(0,0)$: $0\geq -1$, $0<4$, feasible. $(3,8)$: $8\geq 5$ but $8<4$ is false, not feasible. The region is above the solid $y=2x-1$ and below the dashed $y=4$.

3 $P=0, 12, 9+4=13, 6$. Maximum 13 at $(3,2)$; minimum 0 at $(0,0)$.

4 $x+y\leq 120$, $y\geq 30$, $x\geq 0$, $y\geq 0$. The triangle with vertices $(0,30)$, $(90,30)$, $(0,120)$: left of $x+y=120$, above $y=30$, right of the y-axis; lattice points only.

5 $x+y\leq 8$, $3x+y\leq 18$, $x\geq 0$, $y\geq 0$. Vertices $(0,0)$, $(6,0)$, $(5,3)$, $(0,8)$. Two allocations: $(4,4)$ uses 8 plots and 16 water units; $(3,5)$ uses 8 plots and 14 water units.

6 $Q=0, 24, 18+10=28, 15$. Maximum 28 at $(3,2)$, the same vertex as for P; both objectives increase with x and y, so the ranking is unchanged, though the margins differ.

7 At $(0,0)$, $0\geq 5$ is false, so the origin is not in the set; shading its side is the error. The correct region is the other half-plane, with a solid boundary: above and right of $x+y=5$.

8 The least total is $10+10=20>15$, so the region is empty. Change: raise the total to at least 20, or lower a minimum, e.g. $x\geq 5$, $y\geq 10$. Trade-off: more plots use more land and labour; lowering a minimum cuts the allocation guaranteed to that group.

INVESTIGATION 5 · THE THREE WEIGHINGS

1 From $2c+s=14$: $c=0,1,\ldots,7$ with $s=14,12,10,8,6,4,2,0$. Testing in $c+2s=13$, only $(5,4)$ works: $5+8=13$. Verify: $10+4=14$ and $5+8=13$.

2 $2c+s=14$, $c+2s=13$. From the first $s=14-2c$; then $c+28-4c=13$, $-3c=-15$, $c=5$, $s=4$. Agrees with the table and is exact.

3 Each equation is a line in the (c,s)-plane. $2c+s=14$ has intercepts $(7,0)$, $(0,14)$; $c+2s=13$ has $(13,0)$, $(0,6.5)$. Gradients -2 and $-\frac{1}{2}$ differ, so the lines cross once at $(5,4)$; mark it.

4 $3c+3s=27$ is the sum of the first two equations: $(2c+s)+(c+2s)=3c+3s=27$. Every solution of the first two satisfies it, so it adds no new information; the solution remains $(5,4)$.

5 $3c+3s=28$ gives $c+s=\frac{28}{3}\approx 9.33$, but the first two give $c+s=9$. No pair satisfies all three: the third weighing is inconsistent. This suggests misreading, limited precision or non-identical objects.

6 For $y=m_1x+c_1$ and $y=m_2x+c_2$: different gradients, one solution; equal gradients with different intercepts, parallel, no solution; equal gradient and intercept, coincident, infinitely many solutions.

7 Sample: $c+s=5$, $2c-s=4$ gives $c=3$, $s=2$. No solution: $c+s=5$, $c+s=7$. Exchange: the partner checks by substitution and by comparing gradients and intercepts.

> **Limitation.** The table is one valid data set. The algebra is exact only if the objects are identical and the scale readings are exact; a balance measures to limited precision.

APPLICATION · THE WATER REFILL STATION

1 x = 5-litre refills, y = 12-litre refills: $x+y=80$, $5x+12y=680$.

2 $x=80-y$: $400+7y=680$, $7y=280$, $y=40$, $x=40$.

3 $40+40=80$, $200+480=680$. Revenue $=40(2)+40(5)=280$ units.

4 $280\div 680=0.4117\ldots\approx 0.41$ units per litre. The 12-litre refill costs $5\div 12=0.4167$ per litre, the 5-litre $2\div 5=0.40$: the larger container is better value (about 0.42 against 0.40).

5 Revenue $=40(3)+40(5)=320$ units, up 40 (14.3%). The 5-litre buyers are hit hardest: their price rises 50%, while 12-litre buyers pay the same. If small-container buyers are poorer, the change is regressive.

6 $680 \times 1.1 = 748$ litres exceeds 700 by 48, so the rise cannot be met from the same mix; 700 litres is the cap whatever the mix. With smaller containers, 80 refills (42 large, 38 small) total $504 + 190 = 694$ litres, but that is less than 748. If refills rise to 88, at most 37 can be large: $444 + 255 = 699$. More, smaller refills are possible; the 748-litre volume is not.

7 $40 \times 3 = 120$ bottles. Assumption: each 5-litre refill displaces exactly three 1.5-litre bottles that would otherwise be bought; 12-litre refills are excluded and no extra bottles are bought.

8 The model ignores leakage, free refills, container reuse (which avoids more bottles), customers buying both sizes, price elasticity, queueing and the water source. "Replaced" bottles are an estimate, not a count. Better data: metered volume, counts by size, receipts, a reuse survey and a spill log.

CRITERION TASK · PLANNING A YOUTH CENTRE PROGRAMME

Sample response. x art workshops, y music sessions: $x + y \le 8$, $2x + y \le 16$, $x + 2y \le 14$, $x, y \ge 0$. Vertices $(0,0)$, $(8,0)$, $(2,6)$, $(0,7)$; $P = 0, 64, 70, 63$, so $(2,6)$ is optimal. Requiring $x \ge 3$ gives $(3,5)$, $P = 69$, a fairer balance at a cost of one point.

1 $x + y \le 8$: at most 8 venue slots. $2x + y \le 16$: art uses 2 tutor hours, music 1, with 16 available. $x + 2y \le 14$: art uses 1 equipment unit, music 2, with 14 available. $x \ge 0$, $y \ge 0$: negative sessions are impossible.

2 Boundaries $x + y = 8$, $2x + y = 16$, $x + 2y = 14$ and the axes. Vertices $(0,0)$; $(8,0)$; $(2,6)$, where $x + y = 8$ meets $x + 2y = 14$ (tutor $4 + 6 = 10 \le 16$); $(0,7)$, where $x + 2y = 14$ meets the y-axis. The quadrilateral is bounded because $x + y \le 8$ with $x, y \ge 0$ confines it to $0 \le x \le 8$, $0 \le y \le 8$.

3 $(4,4)$: $8 \le 8$, $12 \le 16$, $12 \le 14$, feasible. $(6,3)$: $9 > 8$, so it fails the venue limit, although $15 \le 16$ and $12 \le 14$. This matches the graph: $(4,4)$ is inside, $(6,3)$ beyond the venue line.

4 $P = 0$, 64, $16 + 54 = 70$, 63. Maximum 70 at $(2,6)$: 2 art, 6 music. A linear objective attains its maximum at a vertex, and 70 is the largest vertex value.

5 The optimum maximises total participation, not art provision. Music scores 9 against 8 and uses less equipment, so the model favours music; 2 art workshops may fail the centre's duty to art demand. Change: $x \ge 3$. New vertices $(3,0)$, $(8,0)$, $(3,5)$, with $P = 24, 64, 69$; the optimum is $(3,5)$, 3 art and 5 music, costing one point. Raising the art weight instead is blunt: $10x + 9y$ sends the optimum to $(8,0)$.

6 With 10 slots, $x + y \le 10$. The line $x + y = 10$ meets both $2x + y = 16$ and $x + 2y = 14$ at the single point $(6,4)$; the earlier vertex $(8,0)$ remains, while $(10,0)$ is infeasible because $2(10) + 0 = 20 > 16$. Vertices $(0,0)$, $(8,0)$, $(6,4)$, $(0,7)$; $P = 0, 64, 84, 63$; new optimum $(6,4)$, $P = 84$. The original $(2,6)$ remains feasible ($8 \le 10$, $10 \le 16$, $14 \le 14$), with $P = 70$, but is no longer optimal.

Marking guide. Criterion A: constraints, graph, vertices, tests and objective values correct. Criterion D: units and interpretation; fairness discussed with demand evidence; one specific change with its calculated effect; correct treatment of the 10-slot limit. Criterion C: labels, notation and a conclusion that answers the question. 24 marks: Q1 5, Q2 5, Q3 4, Q4 4, Q5 4, Q6 2.

MIXED PRACTICE 5

1 $3x = 15$, $x = 5$.

2 $4x - 4 = 3x + 5$, $x = 9$.

3 $-2x \ge -6$, $x \le 3$; filled 3, shade left.

4 $-3 \le x < 3$.

5 Add: $2x = 16$, $x = 8$, $y = 4$: $(8,4)$.

6 Add: $4x = 20$, $x = 5$; $15 + 2y = 19$, $y = 2$: $(5,2)$.

7 $x = 7 - y$: $14 + y = 16$, $y = 2$, $x = 5$: $(5,2)$.

8 $3 = 2(2) - 1$ is true, but $2 + 3 = 5 \ne 6$, so $(2,3)$ satisfies only the first equation; not a solution.

9 Dividing by -3 requires reversing: $x > -3$. Test 0: $4 < 13$ and $0 > -3$ are true; $0 < -3$ is false.

10 Equal gradients 3, different intercepts 1 and -4: parallel. Algebraically $3x + 1 = 3x - 4$ gives $1 = -4$, false; no solution.

11 Add, since the y-coefficients are opposites: $3x = 12$, $x = 4$, $y = 5$: $(4,5)$.

12 $P = 0$, 20, 21, 12; maximum 21 at $(3,3)$.

13 $g \le 120$ (chairs) and $t \ge 2$ (minimum tables); also $0 \le t \le 30$, $g \ge 0$.

14 $n + p = 35$, $4n + 2p = 100$. Doubling the first and subtracting: $2n = 30$, $n = 15$, $p = 20$. Check: $60 + 40 = 100$.

15 The optimum maximises a total score, which may weight groups unequally, so a group can fall below its request; the objective does not measure fairness. Add a minimum constraint, e.g. $x \ge 4$, and report the loss in total output.

Common errors. Reverse the inequality only when multiplying or dividing by a negative. A negative sign before a bracket multiplies every term inside. A system solution is an ordered pair: find both variables and check both originals. Elimination needs equal or opposite coefficients first. Test a half-plane with a point off the boundary; $\le$ and $\ge$ use a solid line. Evaluate a linear objective at the vertices, then check integer constraints in context.

Chapter 6 · Transformations, congruence and symmetry

PREREQUISITE CHECK

1 $(2,1)$ is in the first quadrant; $(-3,4)$ in the second; $(0,-2)$ on the negative y-axis.

2 $(4,2)$.

3 $2x = 8$, $x = 4$.

4 Triangle $180°$; quadrilateral $360°$.

5 $360 \div 6 = 60$; $360 \div 5 = 72$.

6 Perimeter $4(4) = 16$ cm; area $4^2 = 16$ cm^2.

PRACTICE 6.1 · TRANSLATIONS AND VECTORS

1 $A'(5,1)$, $B'(2,-4)$, $C'(3,-7)$, $D'(-1,-6)$.

2 $P'(-1,5)$, $Q'(2,5)$, $R'(-1,8)$: 2 units left, 3 up. Check: $P'Q' = 3 = PQ$, lengths preserved.

3 $\begin{pmatrix} -7 \\ 5 \end{pmatrix}$; $B'(-7,5)$. Check: $A' - A = (-7,5)$.

4 Vector $\begin{pmatrix}4\\3\end{pmatrix}$; images $(4,3)$, $(6,3)$, $(6,5)$, $(4,5)$. Area 4 square units; a translation preserves lengths, so the side remains 2.

5 $\begin{pmatrix}3\\3\end{pmatrix}$. It is unique: a translation is fixed by the image of one point, and both endpoints give the same vector. A different vector would not send $(-2,1)$ to $(1,4)$.

6 $(2,3)$, $(4,6)$, $(6,9)$, $(8,12)$; they are collinear with the origin, on a line of gradient $\frac{3}{2} = 1.5$.

7 $(14,9) + 4(-5,-3) = (-6,-3)$. Single vector $4\begin{pmatrix}-5\\-3\end{pmatrix} = \begin{pmatrix}-20\\-12\end{pmatrix}$.

8 Vector $\begin{pmatrix}3\\-2\end{pmatrix}$; reverse it: $(0,0) - \begin{pmatrix}3\\-2\end{pmatrix} = (-3,2)$. Check: $(-3+3, 2-2) = (0,0)$.

PRACTICE 6.2 · REFLECTIONS AND MIRROR LINES

1 A: $(3,-2)$, then $(-3,-2)$. B: $(-1,-4)$, then $(1,-4)$. C: $(-2,3)$, then $(2,3)$. D: $(0,-5)$, then $(0,-5)$; D lies on the y-axis, so the second reflection fixes it.

2 $(x,y) \to (4-x,y)$: $(3,1)$, $(0,1)$, $(3,3)$.

3 (a) $(2,-3)$. (b) $(-2,3)$. (c) $2(-1) - (-3), 2) = (1,2)$.

4 y-axis: $(a,b) \to (-a,b)$; then x-axis: $(-a,b) \to (-a,-b)$. Reversing gives $(a,-b)$, then $(-a,-b)$. Order does not matter; the composite is a half-turn about the origin.

5 Midpoint $(5,5)$, PP' vertical, so the mirror is $y = 5$. Image of $(2,3)$: $(2, 2(5) - 3) = (2,7)$.

6 $(x,y) \to (8-x,y)$: $(1,2) \to (7,2)$; $(3,5) \to (5,5)$.

7 $(2(3) - 7, -2) = (-1,-2)$. Each point is 4 units from $x = 3$: $7 - 3 = 4$ and $3 - (-1) = 4$.

8 A reflection reverses orientation; a non-zero translation preserves it. A reflection fixes every point of its mirror line, while a translation fixes none. A reflection also moves different points by different vectors; a translation moves all points by one vector.

PRACTICE 6.3 · ROTATIONS AND IMAGE COORDINATES

1 $A'(-1,3)$, $B'(-4,-2)$, $C'(2,-3)$, $D'(5,0)$.

2 $(-1,-2)$, $(-4,-2)$, $(-1,-5)$.

3 (a) $(-3,-2)$. (b) $(2,-3)$. (c) $(3,2)$.

4 $(a,b) \to (-b,a) \to (-a,-b)$, which is the half-turn rule; two quarter-turns make a half-turn.

5 Subtract $(1,1)$: $(-1,-1)$, $(1,-1)$, $(1,1)$, $(-1,1)$; rotate: $(1,-1)$, $(1,1)$, $(-1,1)$, $(-1,-1)$; add $(1,1)$: $(2,0)$, $(2,2)$, $(0,2)$, $(0,0)$. The image is the same square with its vertices cycled, so a $90°$ turn maps it onto itself: rotational symmetry of order 4.

6 Subtract $(2,2)$: $(3,0)$; rotate: $(0,3)$; add: $(2,5)$.

7 $90°$ (also $180°$ and $270°$); order 4.

8 Subtract $(2,0)$: $(-2,0)$, $(2,0)$, $(-2,3)$; rotate: $(0,-2)$, $(0,2)$, $(-3,-2)$; add: $A'(2,-2)$, $B'(2,2)$, $C'(-1,-2)$. $AB = 4 = A'B'$ and $AC = 3 = A'C'$.

PRACTICE 6.4 · PROVING CONGRUENCE

1 (a) SSS. (b) SAS. (c) ASA if the side is included between the equal angles; otherwise AAS. (d) RHS.

2 $PQ = ST$, the equal angle at P is between PQ and PR (and at S between ST and SU), and $PR = SU$, so SAS. $\triangle PQR \cong \triangle STU$; $QR = TU$, $\angle Q = \angle T$, $\angle R = \angle U$.

3 $AB = AD$, $BC = DC$, AC common: SSS. Hence $\triangle ABC \cong \triangle ADC$, so $\angle BCA = \angle DCA$ and AC bisects $\angle BCD$.

4 AAA fixes shape but not size. Equilateral triangles of side 1 cm and 2 cm all have $60°$ angles but different sides, so they are not congruent; a length is needed.

5 The perpendicular distance from B to the ray is $8\sin 30° = 4$ cm, less than 6 cm, so the circle of radius 6 meets the ray twice, at $AC = 8\cos 30° \pm \sqrt{36-16} \approx 6.93 \pm 4.47$, about 11.4 cm and 2.5 cm. Both satisfy the data but differ, so SSA is ambiguous.

6 ASA, because the 40 m side is included between the $65°$ and $42°$ angles. Third angle: $180 - 65 - 42 = 73°$.

7 Not congruent: SSS needs all three corresponding sides equal, and $50 \neq 52$. The first is right-angled; the second is not.

8 $AB = DC$ and $BC = AD$ (opposite sides), AC common: SSS, so $\triangle ABC \cong \triangle CDA$. Alternatively RHS: right angles at B and D, common hypotenuse AC, $AB = CD$. Hence $\angle BAC = \angle DCA$; the diagonal divides the rectangle into two congruent triangles.

PRACTICE 6.5 · SYMMETRY AND TILINGS

1 (a) 4 lines, order 4. (b) 6 lines, order 6. (c) 1 line, order 1. (d) 0 lines, order 2. (e) infinitely many lines, rotational symmetry of every angle (infinite order).

2 $\frac{180(8-2)}{8} = 135°$. $360 \div 135 = 2.67$, not a whole number; two octagons leave $90°$ and three overlap by $45°$, so no.

3 $\frac{180(n-2)}{n} = 156$, $180n - 360 = 156n$, $24n = 360$, $n = 15$. $360 \div 156 = 2.31$, not whole; no tessellation.

4 Pentagon: $108°$; $3(108) = 324$, gap $36°$; $4(108) = 432$, overlap $72°$. Hexagon: $120°$; $3(120) = 360$ exactly, so it tiles.

5 $360 \div 5 = 72°$. A regular pentagon (5 lines, order 5).

6 Translate the tile by one tile side horizontally, and by one side vertically for the next row; reflect it in a shared edge (or rotate $180°$ about an edge midpoint) to make mirror pairs. Four $90°$ corners meet at each vertex, so squares leave no gaps; the motif must meet the tile edges consistently.

7 $2(135) + 90 = 270 + 90 = 360$, so two octagons and one square close exactly around a vertex.

8 Two congruent copies of a triangle, one rotated $180°$ about the midpoint of a side, form a parallelogram; parallelograms tile by translation, so the triangle tiles. For a quadrilateral, the four angles sum to $360°$, so four copies fit around a point; half-turns about side midpoints make adjacent edges coincide.

INVESTIGATION 6 · THE ANGLE THAT FITS

1 $n = 3$: triangle, $60°$, $360 \div 60 = 6$, yes. $n = 4$: square, $90°$, 4, yes. $n = 5$: pentagon, $108°$, 3.33, no. $n = 6$: hexagon, $120°$, 3, yes. $n = 7$: heptagon, $128.6°$, 2.8, no. $n = 8$: octagon, $135°$, 2.67, no. $n = 9$: nonagon, $140°$, 2.57, no. $n = 10$: decagon, $144°$, 2.5, no.

2 Only $n = 3, 4, 6$ give whole numbers. The quotient falls as n rises (6, 4, 3.33, 3, 2.8, ...), and no later value returns to an integer.

3 $n = 5$: $3(108) = 324$, gap $36°$; four give 432, overlap $72°$. $n = 8$: $2(135) = 270$, gap $90°$; three give 405, overlap $45°$. A further tile overshoots, and part of a tile cannot fill the gap.

4 $360 \div \frac{180(n-2)}{n} = \frac{360n}{180(n-2)} = \frac{2n}{n-2} = 2 + \frac{4}{n-2}$. This is whole exactly when $n - 2$ divides 4; with $n \geq 3$, $n - 2 = 1, 2, 4$, so $n = 3, 4, 6$.

5 (a) $270 + 90 = 360$. (b) $240 + 120 = 360$. (c) $180 + 180 = 360$. All three vertices close exactly.

6 The number of pentagons at a vertex is whole: three leave $36°$, four exceed by $72°$, and more increase the overshoot. The gap is fixed and cannot be filled by a partial pentagon, so "enough" pentagons never help.

7 Sample: a 4.8.8 tiling (one square, two octagons), $90 + 135 + 135 = 360$; or 3.3.3.4.4 (three triangles, two squares), $180 + 180 = 360$. Draw at least six tiles; label translations by edge vectors and reflections or rotations carrying one tile to the next.

> **Limitation.** A neat drawing is evidence, not proof; only the angle sum at every vertex proves that a tiling closes. A single vertex whose angles miss 360° refutes the proposed tiling.

APPLICATION · A TILE PANEL FOR THE COMMUNITY CENTRE

1 $360 \div 15 = 24$ along the length; $60 \div 15 = 4$ rows; $24 \times 4 = 96$ tiles.

2 Exact: $96(2.40) = 230.40$ units. With 8%: $96(1.08) = 103.68$, so order 104 tiles, cost $104(2.40) = 249.60$. Rounding down to 103 leaves only 7 spare (7.3%), below the allowance; cutting and breakages could leave the panel short, and tiles are sold whole.

3 $24 \div 4 = 6$ repeats. Translation by 4 tiles $= 60$ cm: $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$ tiles or $\begin{pmatrix} 60 \\ 0 \end{pmatrix}$ cm.

4 None: the mirror is a grout line between columns, so it cuts no tile; the 24 columns form 12 mirror pairs. A five-wide motif would put the mirror through the middle of the third column, so 4 tiles would lie on it; also $24 \div 5 = 4.8$, so whole motifs would not fit.

5 Diagonal AC: $AB = CD$, $BC = DA$, AC common, so SSS (or RHS with right angles and equal legs). Each triangle has angles $45^\circ, 45^\circ, 90^\circ$, legs 15 cm and hypotenuse $15\sqrt{2} \approx 21.2$ cm.

6 Not guaranteed. With 6 columns and 4 rows, the half-turn sends motif positions to motif positions, but each motif must also map onto itself under 180°, or be paired with a half-turned copy. A motif with only a vertical mirror line need not have half-turn symmetry, and the border or any asymmetric detail must also match. The claim needs an extra condition, such as symmetry about both centre lines or motifs arranged in half-turn pairs.

7 $2 \times 24 = 48$ contrast tiles; $48(3.10) = 148.80$; total $249.60 + 148.80 = 398.40$ if the border is additional. If it replaces the top and bottom rows, 48 standard tiles are saved: extra $48(0.70) = 33.60$, exact total 264.00.

8 $24(15) + 23(0.2) = 360 + 4.6 = 364.6$ cm, which exceeds 360 cm. Solutions: use 23 columns, $23(15) + 22(0.2) = 349.4$ cm, leaving 10.6 cm for a feature border or wider joints; or specify 14.81 cm tiles, since $24s + 4.6 = 360$; or trim the two edge columns. The 23-column plus border option is best: standard tiles, no cut edges, and the 10.6 cm becomes deliberate.

9 Uncertain assumptions: exact 15 cm tiles, 2 mm grout, 8% wastage, no further breakage, labour time and a true wall. Records: a dimensioned drawing, batch and shade codes, grout width, adhesive, cutting plan, supplier, photographs, cost log and maintenance notes.

10 Record the tradition, place, makers, permission or protocol, the source motif, what was changed and what is original; acknowledge it in the brief and label. Mathematical inspiration uses symmetry and tessellation ideas; direct copying reproduces a specific sacred pattern, which may be restricted or misrepresented. The distinction matters for respect, accuracy and credit.

CRITERION TASK · THE KITE WINDOW

> **Sample response.** $AB = AD$ and $CB = CD$ are given and AC is common, so $\triangle ABC \cong \triangle ADC$ by SSS. Hence AC bisects the angles at A and C; congruent triangles ABM and ADM give $BM = DM$, and the adjacent angles at M are both 90°, so $AC \perp BD$. The kite has one line of symmetry, AC, and rotational order 1; a non-square rhombus has both diagonals as mirror lines and order 2.

1 $AB = AD$ (given); $CB = CD$ (given); $AC = AC$ (common). Therefore $\triangle ABC \cong \triangle ADC$ by SSS, with $A \leftrightarrow A$, $B \leftrightarrow D$, $C \leftrightarrow C$.

2 From the congruence, $\angle BAC = \angle DAC$, so AC bisects $\angle BAD$. Also $\angle BCA = \angle DCA$, so AC bisects $\angle BCD$.

3 In $\triangle ABM$ and $\triangle ADM$: $AB = AD$ (given), $\angle BAM = \angle DAM$ (proved in Q2), AM common. SAS gives $\triangle ABM \cong \triangle ADM$, so $BM = DM$.

4 From Q3, $\angle AMB = \angle AMD$. Since B, M, D are collinear, these adjacent angles sum to 180°, so each is 90°; hence $AC \perp BD$.

5 One line of symmetry, AC: reflection in AC fixes A and C and swaps B and D, and the kite maps onto itself. Rotational order 1: a half-turn would require $A \leftrightarrow C$, so M would be the midpoint of AC, which is not true for a general kite. No second mirror line exists, since only AC bisects both angles at A and C.

6 A rhombus has all sides equal, so $AB = AD$ and $CB = CD$: the kite results apply. AC bisects the angles at A and C and is perpendicular to BD; applying the same argument to diagonal BD shows it bisects the angles at B and D and is perpendicular to AC. Hence both diagonals are lines of symmetry. The order is 2: a 180° turn about M swaps $A \leftrightarrow C$ and $B \leftrightarrow D$, while 90° does not (the rhombus is not a square).

> **Marking guide.** Criterion B: a complete chain from the three equal parts to the congruence, then to angle bisection, midpoint and perpendicularity; each deduction justified. Criterion C: correct correspondence and notation, one reason per line, and a clear distinction between given, common and proved. Criterion A: symmetry counts and the rhombus extension justified from earlier results. 24 marks: Q1 5, Q2 4, Q3 3, Q4 4, Q5 4, Q6 4.

MIXED PRACTICE 6

1 $A'(1, 3)$, $B'(-3, 5)$, $C'(-4, 1)$.

2 (a) $(2, -3)$, $(5, -3)$, $(2, -6)$. (b) $(-2, -3)$, $(-5, -3)$, $(-2, -6)$.

3 (a) 10 lines, order 10. (b) 1 line, order 1. (c) 0 lines, order 2.

4 Nonagon 140°; dodecagon 150°. $360 \div 140 = 2.57$ and $360 \div 150 = 2.4$, neither whole, so neither tessellates.

5 Rotate: $(-1, 3)$; reflect in y-axis: $(1, 3)$. Reverse: reflect $(-3, 1)$, rotate $(-1, -3)$. Different results, so the order matters; these transformations do not commute.

6 $AB = AD$, $BC = DC$, AC common: SSS. $\triangle ABC \cong \triangle ADC$, so $\angle BCA = \angle DCA$; AC bisects $\angle BCD$.

7 SSA: $AB = 8$, $BC = 6$, $\angle A = 30^\circ$ gives two triangles, because the circle of radius 6 meets the ray twice. AAA: equilateral triangles of side 1 and 2.

8 A point (x, y) maps to $(x + 3, y - 2)$. Substituting gives $y + 2 = 2(x - 3) + 1$, so the image line is $y = 2x - 7$. The gradient stays 2 because every point shifts by the same vector, preserving the slope.

9 Sides 8, 8 and the diagonal: SSS (or RHS with the right angles). Each triangle is right isosceles, $45^\circ, 45^\circ, 90^\circ$. The square has 4 lines of symmetry and rotational order 4.

10 $8 \times 3 = 24$ tiles; $24(1.10) = 26.4$, so order 27; $27(3.50) = 94.50$ units.

11 Twelfth motif: $(2 + 11(6), 4) = (68, 4)$. Reflection in $y = 6$: $(2, 2(6) - 4) = (2, 8)$. The frieze has reflection symmetry in $y = 6$ (and translations by $(6, 0)$).

12 One quarter, rotated 90° three times about the centre, generates the other three; each quarter is congruent because rotation is an isometry. Record the tradition, permission, attribution and the changes made.

13 In rectangle $ABCD$, AB is common, $BC = AD$ and $\angle ABC = \angle BAD = 90^\circ$, so SAS gives $\triangle ABC \cong \triangle BAD$; hence $AC = BD$.

14 Interior angle $= 180 - \frac{360}{n}$. For $n \geq 7$, it is at least 128.6°: two tiles are below 360°, three exceed it, so no whole number fits. Equivalently $360 \div \text{angle} = 2 + \frac{4}{n-2}$ lies between 2 and 3, never an integer.

Common errors. Reflection in the x-axis changes y, not x; in $y = x$ the coordinates swap. A 90° clockwise turn is 270° anticlockwise. Reverse a translation by subtracting its vector. Congruence statements must list vertices in corresponding order, and SSA and AAA are not criteria. A rectangle's diagonals are not lines of symmetry. Count the starting position once when finding the order, and test tessellation with $360 \div$ interior angle.

Chapter 7 · Similarity, proof and trigonometry

PREREQUISITE CHECK

1 $12 : 18 = 2 : 3$; as a fraction $\frac{2}{3}$.

2 $\frac{3}{4} = \frac{x}{12}$, so $4x = 36$ and $x = 9$.

3 $7^2 = 49$, $12^2 = 144$, $13^2 = 169$.

4 6.847 to one decimal place is 6.8; to two significant figures it is also 6.8, because the third digit is 4 and does not round up.

5 The angle sum is 180°; each angle of an equilateral triangle is 60°.

6 The hypotenuse is the side opposite the right angle; it is the longest side.

PRACTICE 7.1 · SIMILARITY AND SCALE

1 Taking the 4 cm and 10 cm sides as corresponding, $k = \frac{10}{4} = 2.5$, so $x = 6 \times 2.5 = 15$ cm. Check: $4 : 6 = 10 : 15 = 2 : 3$.

2 $k = \frac{ST}{PQ} = \frac{12}{8} = 1.5$, so $TU = 10 \times 1.5 = 15$ cm and $US = 14 \times 1.5 = 21$ cm.

3 $k = \frac{25}{10} = 2.5$; the longer side becomes $15 \times 2.5 = 37.5$ cm. Area ratio $= 2.5^2 = 6.25$, that is $25 : 4$ (new : old).

4 $7.5 \times 200 = 1500$ cm $= 15$ m and $5 \times 200 = 1000$ cm $= 10$ m. Real area $= 15 \times 10 = 150$ m^2.

5 The area ratio is $k^2 = \frac{81}{36} = \frac{9}{4}$, so $k = \frac{3}{2} = 1.5$. Area is the product of two lengths, so it scales by the square of the length factor.

6 The single factor is $1.5 \times 2 = 3$, because lengths multiply. The area multiplier is $3^2 = 9$; equivalently $1.5^2 \times 2^2 = 2.25 \times 4 = 9$, since area depends on two lengths.

7 1 cm represents 0.5 km, so 1 cm^2 represents $0.5^2 = 0.25$ km^2. Area $= 18 \times 0.25 = 4.5$ km^2 $= 450$ ha.

8 $\frac{8}{d} = \frac{d}{18}$, so $d^2 = 144$ and $d = 12$ cm. Then A is 8×12 and B is 12×18; $k = \frac{12}{8} = 1.5$ (A to B). Both have side ratio $2 : 3$.

PRACTICE 7.2 · SIMILAR TRIANGLES

1 (a) Similar by AA; the third angles are both 65°. (b) Similar by SSS; $3 : 6 = 4 : 8 = 5 : 10 = 1 : 2$. (c) Similar by SAS; $\frac{4}{6} = \frac{6}{9} = \frac{2}{3}$ and the included angles are both 50°.

2 By the intercept theorem, $\frac{AD}{DB} = \frac{AE}{EC}$: $\frac{6}{3} = \frac{8}{EC}$, so $EC = 4$ cm and $AC = 8 + 4 = 12$ cm.

3 $k = \frac{XY}{AB} = \frac{6}{9} = \frac{2}{3}$; $YZ = 12 \times \frac{2}{3} = 8$ cm. Area ratio $XYZ : ABC = \left(\frac{2}{3}\right)^2 = 4 : 9$.

4 $\frac{h}{1.5} = \frac{28}{2.1}$, so $h = \frac{1.5 \times 28}{2.1} = 20$ m. Assumptions: level ground, post and building vertical, parallel sun's rays giving equal elevation angles, shadows measured together.

5 Equal angles force the third pair to be equal, so AA gives similarity and one common side ratio. Three proportional sides fix shape but not size: $3, 4, 5$ and $6, 8, 10$ are similar with $k = 2$, yet not congruent. Congruence needs $k = 1$.

6 $h = \frac{1.7 \times 12.0}{3.0} = 6.8$ m. If the distance is 11.8 m, $h = \frac{1.7 \times 11.8}{3.0} = 6.69$ m; the estimate falls by 0.11 m, about 1.7%.

7 Area ratio $= 1.25^2 = 1.5625$, so the real area is $0.24 \times 1.5625 = 0.375$ m^2.

8 The perimeter ratio equals the length ratio, so $k = \frac{5}{3}$. Larger area $= 27 \times \left(\frac{5}{3}\right)^2 = 27 \times \frac{25}{9} = 75$ cm^2. The heights are in the ratio $3 : 5$.

PRACTICE 7.3 · PYTHAGORAS AND ITS CONVERSE

1 $\sqrt{6^2 + 8^2} = \sqrt{100} = 10$ cm.

2 $\sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225} = 15$ m.

3 $\sqrt{5^2 + 12^2} = \sqrt{169} = 13$ m.

4 The height bisects the base, giving 6 cm: $h = \sqrt{10^2 - 6^2} = 8$ cm. Area $= \frac{1}{2} \times 12 \times 8 = 48$ cm^2.

5 Horizontal difference 6, vertical difference 8: distance $= \sqrt{6^2 + 8^2} = 10$ units.

6 (a) $7^2 + 24^2 = 49 + 576 = 625 = 25^2$: right-angled. (b) $9^2 + 12^2 = 225$, but $16^2 = 256$: not right-angled; the angle opposite 16 is obtuse. (c) $10^2 + 24^2 = 100 + 576 = 676 = 26^2$: right-angled.

7 First height $= \sqrt{4.5^2 - 1.8^2} = \sqrt{17.01} = 4.12$ m. Second height $= \sqrt{4.5^2 - 2.7^2} = \sqrt{12.96} = 3.60$ m. The top slides down by $4.12 - 3.60 = 0.52$ m.

8 Let the legs be x and $x + 5$. Then $x^2 + (x + 5)^2 = 25^2$, so $2x^2 + 10x + 25 = 625$, $x^2 + 5x - 300 = 0$, $(x - 15)(x + 20) = 0$. Hence $x = 15$ and the legs are 15 cm and 20 cm.

PRACTICE 7.4 · SINE, COSINE AND TANGENT

1 $\sin\theta = \frac{7}{25}$, $\cos\theta = \frac{24}{25}$, $\tan\theta = \frac{7}{24}$.

2 $x = 20 \sin 38^\circ = 20 \times 0.6157 = 12.3$ cm.

3 $y = 14 \cos 55^\circ = 14 \times 0.5736 = 8.03$ m.

4 (a) $\theta = \tan^{-1} 0.85 = 40.4^\circ$. (b) $\theta = \sin^{-1} 0.4 = 23.6^\circ$. (c) $\theta = \cos^{-1} 0.25 = 75.5^\circ$.

5 (a) $\frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2} = \frac{\sqrt{2}+\sqrt{3}}{2} \approx 1.573$. (b) $\sqrt{3} \times \frac{1}{\sqrt{3}} = 1$.

6 $\tan 58^\circ = \frac{14}{s}$, so $s = \frac{14}{\tan 58^\circ} = 8.75$ m.

7 Height $= 32 \sin 41^\circ = 32 \times 0.6561 = 21.0$ m.

8 The side opposite 30° is $2x \sin 30^\circ = 2x \times \frac{1}{2} = x$, the shorter leg. The adjacent side is $2x \cos 30^\circ = 2x \times \frac{\sqrt{3}}{2} = x\sqrt{3}$, the longer leg.

PRACTICE 7.5 · ELEVATION AND DEPRESSION

1 $h = 80\tan 35^\circ = 80 \times 0.7002 = 56.0$ m.

2 $d = \frac{70}{\tan 42^\circ} = \frac{70}{0.9004} = 77.7$ m.

3 Height above eye level $= 40\tan 23^\circ = 16.98$ m, so the flagpole is $16.98 + 1.7 = 18.7$ m.

4 $d = \frac{25}{\tan 61^\circ} = \frac{25}{1.8040} = 13.9$ m.

5 $d = \frac{900}{\tan 17^\circ} = 2943.8$ m $= 2.94$ km (about 2.9 km).

6 Let the height be h. Then $\frac{h}{\tan 40^\circ} - \frac{h}{\tan 55^\circ} = 50$, so $h(1.1918 - 0.7002) = 50$ and $h = \frac{50}{0.4915} = 101.7$ m.

7 $\tan\theta = \frac{0.8}{11}$, so $\theta = 4.2^\circ$. Sloping length $= \sqrt{0.8^2 + 11^2} = \sqrt{121.64} = 11.03$ m.

8 The vertical drop is $24\tan 15^\circ = 24 \times 0.2679 = 6.43$ m, so the shorter building is $30 - 6.43 = 23.6$ m tall.

INVESTIGATION 7 · THE SHADOW RULE

1 Sample data at one moment on level ground, to the nearest centimetre: door 2.00 m with shadow 1.50 m; post 1.20 m with 0.90 m; student 1.65 m with 1.24 m. Ratios 1.333, 1.333, 1.331; mean $k = 1.332$.

2 The ratios agree to within 0.002, about 0.15%, which is within the precision of the measurements. Conjecture: for every vertical object at that moment, $h = ks$ with $k \approx 1.33$; equivalently, height and shadow length are proportional.

3 Each object and shadow forms a right triangle: the object is vertical and the shadow lies on level ground, so the foot angle is 90°. Parallel sun's rays make the same angle θ with the ground in both triangles. Two equal angles give AA similarity, so corresponding sides share one ratio, $h_1/s_1 = h_2/s_2$.

4 $h = 1.332 \times 9.60 = 12.79$, about 12.8 m. A clinometer estimate might give 12.7 m; the 0.1 m difference is smaller than the uncertainty in the shadow tip, so the methods agree within precision.

5 An extra 0.10 m of shadow changes the estimate by 0.13 m. The reference measurements dominate: ± 0.5 cm is 0.56% of the 0.90 m pole shadow and 0.42% of its 1.20 m height, but only 0.05% of the 9.60 m tree shadow. Small lengths carry the largest relative error.

6 Conditions: level ground, vertical objects, one fixed sun angle (same time). Sloping ground makes the adjacent side non-horizontal; a leaning object changes the angle at its foot and is not similar to a vertical one; a different time changes θ and k. The shadow must also be straight and unobstructed.

7 In the right triangle, $\tan\theta = \frac{\text{height}}{\text{shadow}} = \frac{h}{s} = k$. Using the metre rule, $\theta = \tan^{-1} 1.333 = 53.1^\circ$, so the sun's elevation was about 53°.

8 A 30 m building casts a 22.5 m shadow. Percentage precision is unchanged, but 1% of 30 m is 0.3 m, so absolute error grows. Trust it only if the ground is level and clear for 22.5 m and the shadow tip is visible. Improve reliability with a longer reference pole, several objects averaged, measurements near midday, and a clinometer cross-check. Limitation: one constant sun angle is assumed, which fails if cloud or time changes the light.

APPLICATION · THE LIBRARY RAMP

1 $12 \times 0.72 = 8.64$ m, so the minimum run is verified. With 12.00 m, the gradient is $0.72 : 12 = 1 : 16.7$; $\theta = \tan^{-1}\frac{0.72}{12} = 3.4^\circ$; sloping length $= \sqrt{0.72^2 + 12^2} = 12.02$ m.

2 Gradient $= 1.05 : 11.50 = 1 : 10.95$, steeper than $1 : 12$; $\theta = \tan^{-1}\frac{1.05}{11.5} = 5.2^\circ$. At 1 : 12, 11.50 m of run allows a rise of $\frac{11.50}{12} = 0.958$ m, so the rise is 0.092 m (about 9.2 cm) too great; equivalently the run is 1.10 m short.

3 (i) Reduce the rise to 0.958 m or less by regrading the approach or lowering the threshold: cheap and a single straight flight, but it may conflict with drainage and door heights. (ii) Use a switchback with two flights and a landing: it fits the length, but costs more, occupies more width, needs turning space and more maintenance.

4 Each flight rises $1.05 \div 2 = 0.525$ m, so its run is $12 \times 0.525 = 6.30$ m. With a 1.5 m square turning landing, the switchback is about $6.30 + 1.5 = 7.80$ m long and $2 \times 1.5 = 3.00$ m wide: a 23.4 m^2 footprint. The single ramp is $11.5 \times 1.5 = 17.25$ m^2 (plus a 2.25 m^2 top landing), so the switchback uses about 20% more ground but fits the 11.5 m site length, which the single ramp does not.

5 Deck area $= 18.03$ m^2, cost 865.55. Landing $= 2.25$ m^2, cost 108.00. Handrails $= 24.04$ m, cost 769.38. Total $= 1742.93$; with 10% contingency, 1917.23, about 1917 units.

6 Each strip covers $0.30 \times 0.30 = 0.09$ m^2. $18.03 \div 0.09 = 200.4$, so 201 strips are needed. The supplier rounds up and adds a margin because edge strips must be cut, some break, and later batches may differ in colour; ordering exactly 200 would leave the deck short.

7 A rise error of ± 0.02 m changes the required run by $12 \times 0.02 = 0.24$ m. Main entrance: required 8.64 ± 0.24, that is 8.40 to 8.88 m; the 12.00 m available leaves 3.12 m spare, so it has ample tolerance. Side entrance: required 12.36 to 12.84 m; only 11.50 m is available, so it is between 0.86 m and 1.34 m short and has no tolerance. The side design fails at every possible measurement.

8 Uncertain: level ground, constant gradient, a rigid deck, dry weather, and users keeping to the ramp. Records: survey dates, instruments and readings with precision; calculations and chosen gradient; material and handrail specifications; inspection and maintenance dates; as-built drawings and photographs.

CRITERION TASK · THE LIGHTHOUSE

Sample response. The diagram is a vertical line for the cliff and tower, a horizontal line for sea level, the lamp as a point at height h, position A at horizontal distance d, and position B 120 m nearer the cliff. The angles of elevation 21° and 35° are marked at A and B, each between the horizontal and the line of sight to the lamp.

1 With h the lamp height above sea level and d the horizontal distance from A to the cliff: both lines of sight lie in one vertical plane, the cliff face is vertical, the lamp is a point, sea level is horizontal, and the eye is at sea level. The diagram labels the 78 m cliff, the tower, h, d, the 120 m baseline and both angles.

2 In the triangle from A: $\tan 21^\circ = \frac{h}{d}$, so $d = h\cot 21^\circ$. From B: $\tan 35^\circ = \frac{h}{d-120}$, so $d - 120 = h\cot 35^\circ$. Subtract: $120 = h(\cot 21^\circ - \cot 35^\circ) \approx 1.1769h$. Hence $h = \frac{120}{1.1769} = 101.96$ m, or 102 m to the nearest metre.

3 From the unrounded h, $d = 265.6$ m and $d_B = 145.6$ m; their difference is 120.0 m. With $h = 102$, the distances round to 266 m and 146 m, again 120 m apart. This is an internal consistency check: both distances come from the same h and angles, so the subtraction must return the baseline. It does not test the physical assumptions (vertical cliff, horizontal sea level, correct angles), so it does not prove the answer exact.

4 Tower height $= h - 78 = 102 - 78 = 24$ m above the cliff. A 24 m tower on a 78 m cliff is plausible. Because h is rounded, the tower is between about 23.5 m and 24.5 m.

5 From 250 m horizontally, $\tan\theta = \frac{102}{250} = 0.408$, so $\theta = \tan^{-1} 0.408 = 22.2^\circ$.

6 For 118 m, $h = 100.3$ m; for 122 m, $h = 103.7$ m: about 100 m to 104 m. The angles contribute more: a half-degree error changes $\cot 21^\circ$ by 0.068 and $\cot 35^\circ$ by 0.027, so the difference changes by 0.095, about 8% or 8 m. The baseline changes h by 2/120, about 1.7%. A small angle error at 21° moves the intersection a long way.

7 Eye height matters: ignoring it makes h a height above eye level, so about 2 m must be added. A rough sea makes the boat rise, fall and tilt, so angles vary; a sloping or overhanging cliff means the distance to the face is not the distance to the vertical below the lamp. Improve by measuring from a fixed shore station of known height with a theodolite, using a longer baseline, repeating and averaging readings, and surveying the baseline rather than estimating it from boat movement.

Marking guide. Criterion B: a labelled model with variables and both tangent equations; correct elimination and solution; a distance check; sensitivity to baseline and angles; assumptions evaluated. Criterion C: correct notation, units at every answer, logical sequence, and a conclusion separating the mathematical result from the physical measurement. Criterion D strand iii: precision and assumptions assessed, with one justified improvement. The 24 marks are spread across the parts (3, 5, 3, 3, 3, 4, 3), with method marks for correct ratios.

MIXED PRACTICE 7

Numbering. The chapter restarts numbering in each section, so items are labelled F (Fluency), R (Reasoning) and A (Application) with their section numbers.

F1 $k = \frac{8}{6} = \frac{4}{3}$, so $EF = 9 \times \frac{4}{3} = 12$ and $DF = 12 \times \frac{4}{3} = 16$.

F2 $\sqrt{7^2 + 24^2} = \sqrt{625} = 25$ cm.

F3 $15 \sin 42^\circ = 15 \times 0.6691 = 10.0$ cm.

F4 $\frac{h}{1.8} = \frac{6.0}{2.4}$, so $h = 1.8 \times 2.5 = 4.5$ m.

R1 $k = \frac{15}{6} = 2.5$. Curved surface areas: $k^2 = 6.25$, ratio $25 : 4$. Volumes: $k^3 = 15.625$, ratio $125 : 8$.

R2 The horizontal lines through A and B are parallel, and the line of sight is a transversal. The two angles are alternate angles, so they are equal. The relationship is "alternate angles are equal".

R3 $9, 12, 15$: $81 + 144 = 225 = 15^2$, so right-angled. $9, 12, 17$: $81 + 144 = 225$, but $17^2 = 289$; not right-angled, and since $289 > 225$ the angle opposite 17 cm is obtuse.

R4 $\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$. For an acute angle the opposite side is shorter than the hypotenuse, so the ratio is less than 1. A value of 1.4 would require the opposite side to be longer than the hypotenuse, which is impossible; the fraction was probably inverted.

A1 $\sin\theta = \frac{5.4}{6} = 0.9$, so $\theta = 64.2^\circ$. Foot distance $= \sqrt{6^2 - 5.4^2} = \sqrt{6.84} = 2.62$ m.

A2 Distances $= \frac{40}{\tan 28^\circ} = 75.23$ m and $\frac{40}{\tan 42^\circ} = 44.42$ m; the cars are $75.23 - 44.42 = 30.8$ m apart.

A3 1 cm represents 0.2 km, so 1 cm^2 represents 0.04 km^2. Area $= 6.5 \times 0.04 = 0.26$ km^2.

A4 Horizontal $= \frac{120}{\tan 38^\circ} = 153.6$ m; direct $= \frac{120}{\sin 38^\circ} = 194.9$ m.

A5 Minimum run $= 12 \times 1.1 = 13.2$ m; sloping length $= \sqrt{1.1^2 + 13.2^2} = 13.25$ m. With 12 m, the gradient is $1.1 : 12 = 1 : 10.9$, steeper than $1 : 12$; the run is 1.2 m short.

A6 With $k = \frac{4}{3}$, the estimates are 12.8 m and 12.4 m, a change of 0.4 m. The shadow method needs level ground, vertical objects and steady sun; it fails on a cloudy day or uneven ground. The angle method needs a clear line of sight and a measurable horizontal distance; it fails when the base is inaccessible or the angle is too small.

Common errors. Using k instead of k^2 for area and k^3 for volume; mismatching sides in a similarity statement; adding a fixed length instead of multiplying by k; choosing the hypotenuse by appearance rather than by the right angle; adding squares when a leg is required; placing a depression angle at the foot of the diagram; forgetting eye height; and leaving the calculator in radian mode. Round only at the end and give units.

Chapter 8 · Three-dimensional solids and measurement

PREREQUISITE CHECK

1 Rectangle $= 8 \times 5 = 40$ cm^2; triangle $= \frac{1}{2} \times 8 \times 5 = 20$ cm^2.

2 Circumference $= 2 \times \frac{22}{7} \times 7 = 44$ cm; area $= \frac{22}{7} \times 7^2 = 154$ cm^2.

3 $2^2 = 4$, $2^3 = 8$, $3^2 = 9$, $3^3 = 27$.

4 1.5 m $= 150$ cm; 2500 cm$^3 = 2.5$ L.

5 $\sqrt{6^2 + 8^2} = \sqrt{100} = 10$ cm.

6 3.14159 to two decimal places is 3.14; to three significant figures it is also 3.14.

PRACTICE 8.1 · SOLIDS, NETS AND VIEWS

1 Triangular prism $F = 5, E = 9, V = 6$; square pyramid $F = 5, E = 8, V = 5$; tetrahedron $F = 4, E = 6, V = 4$; pentagonal prism $F = 7, E = 15, V = 10$.

2 $V = E + 2 - F = 14 + 2 - 9 = 7$.

3 $F = E + 2 - V = 12 + 2 - 8 = 6$; a cuboid (rectangular prism).

4 A cross of six 4×4 squares folds into the cube; total area $= 6 \times 16 = 96$ cm^2.

5 A sphere has no flat polygonal faces, no straight edges and no vertices, so F, E and V are undefined and Euler's relationship does not apply. A net of flat polygons cannot wrap a sphere without stretching or overlapping, because the sphere is curved in two directions; any flat map of it distorts distances.

6 For an n-sided prism, $E = 3n$ (two bases plus the lateral edges). $30 = 3n$, so $n = 10$: the base is a decagon.

7 Base $20 \times 15 = 300$; four sides $2(20 \times 10) + 2(15 \times 10) = 400 + 300 = 700$. Cardboard $= 1000$ cm^2.

8 12 cubes. An edge cube that is not a corner has exactly two painted faces. The four edges of length 4 contribute $4 \times (4 - 2) = 8$; the four edges of length 3 contribute $4 \times (3 - 2) = 4$; the four edges of length 2 contribute 0. Total $8 + 4 = 12$.

PRACTICE 8.2 · SURFACE AREA OF PRISMS AND CYLINDERS

1 $SA = 2(10 \times 4 + 10 \times 3 + 4 \times 3) = 2(40 + 30 + 12) = 164$ cm^2.

2 $SA = 2\pi(5^2) + 2\pi(5)(12) = 50\pi + 120\pi = 170\pi = 534.1$ cm^2.

3 $SA = 6 \times 9^2 = 6 \times 81 = 486$ cm^2.

4 $B = \frac{1}{2}(5)(12) = 30$, $P = 5 + 12 + 13 = 30$. $SA = 2(30) + 30(20) = 60 + 600 = 660$ cm^2.

5 One base plus curved surface: $\pi(3.5^2) + 2\pi(3.5)(8) = 12.25\pi + 56\pi = 68.25\pi = 214.4$ cm^2.

6 $6s^2 = 294$, so $s^2 = 49$, $s = 7$ cm; $V = 7^3 = 343$ cm^3.

7 $2\pi(25) + 2\pi(5)h = 150\pi$, so $50 + 10h = 150$ and $h = 10$ cm.

8 $2\pi r^2 + 2\pi rh = 3(2\pi rh)$, so $2r^2 + 2rh = 6rh$, $2r^2 = 4rh$, hence $h = \frac{r}{2}$. Check: $SA = 3\pi r^2$ and curved area $= \pi r^2$.

PRACTICE 8.3 · PYRAMIDS AND CONES

1 $B = 8^2 = 64$, lateral $= \frac{1}{2}(32)(10) = 160$; $SA = 64 + 160 = 224$ cm^2.

2 $SA = \pi(3^2) + \pi(3)(7) = 9\pi + 21\pi = 30\pi = 94.2$ cm^2.

3 $l = \sqrt{5^2 + 12^2} = 13$; $SA = 25\pi + 65\pi = 90\pi = 282.7$ cm^2.

4 Slant height $l = \sqrt{8^2 + 6^2} = 10$ cm; $SA = 144 + \frac{1}{2}(48)(10) = 144 + 240 = 384$ cm^2.

5 $r = 8$; $SA = 64\pi + 136\pi = 200\pi = 628.3$ cm^2.

6 $36\pi + 6\pi l = 90\pi$, so $l = 9$ cm; $h = \sqrt{9^2 - 6^2} = \sqrt{45} = 6.7$ cm.

7 Lateral $= \frac{1}{2}(12)(2.5) = 15$ m^2, cost $15 \times 12 = 180$ units. With floor: $15 + 9 = 24$ m^2, cost 288 units.

8 Let $r = 3x$, $l = 5x$. Then $9x^2 + 15x^2 = 216$, so $24x^2 = 216$, $x = 3$. Hence $r = 9$ cm, $l = 15$ cm (and $h = 12$ cm).

PRACTICE 8.4 · VOLUME OF PRISMS AND CYLINDERS

1 $V = 9 \times 7 \times 4 = 252$ cm^3.

2 $V = \pi(6^2)(10) = 360\pi = 1131.0$ cm^3.

3 $r = 5$; $V = \pi(25)(14) = 350\pi = 1099.6$ cm^3.

4 3.2 m$^3 = 3.2 \times 1000 = 3200$ L.

5 $B = \frac{1}{2}(10)(6) = 30$ cm^2; $V = 30 \times 25 = 750$ cm^3.

6 $120 \times 80 \times 60 = 576\,000$ cm$^3 = 576$ L.

7 $2000 = \pi(25)h$, so $h = \frac{2000}{25\pi} = 25.5$ cm.

8 Volume $= 20 \times 10 \times 5 = 1000$ cm^3; $h = \frac{1000}{25\pi} = 12.7$ cm.

PRACTICE 8.5 · VOLUME AND COMPOSITE SOLIDS

1 $V = \frac{1}{3}(36)(10) = 120$ cm^3.

2 $V = \frac{1}{3}\pi(9)(7) = 21\pi = 66.0$ cm^3.

3 $V = \frac{4}{3}\pi(27) = 36\pi = 113.1$ cm^3.

4 $V = \frac{1}{2} \times \frac{4}{3}\pi(216) = 144\pi = 452.4$ cm^3.

5 $r = 5$; $V = \frac{1}{3}\pi(25)(12) = 100\pi = 314.2$ cm^3.

6 $k = \frac{10}{4} = 2.5$. Surface areas $k^2 = 6.25$, ratio $25 : 4$; volumes $k^3 = 15.625$, ratio $125 : 8$.

7 Cylinder $= \pi(16)(15) = 240\pi$; hemisphere $= \frac{1}{2} \times \frac{4}{3}\pi(64) = \frac{128\pi}{3}$. Total $= \frac{848\pi}{3} = 888.0$ cm^3.

8 The cut is halfway up, so the small cone has $k = \frac{1}{2}$: radius 3 cm, height 4 cm, $V = 12\pi = 37.7$ cm^3. Whole cone $= 96\pi$, so the frustum is $96\pi - 12\pi = 84\pi = 263.9$ cm^3.

PRACTICE 8.6 · MEASUREMENT AND PRECISION

1 2.4 L $= 2400$ cm$^3 = 0.0024$ m^3.

2 $5000 \div 1000 = 5$ L.

3 $\rho = \frac{810}{300} = 2.7$ g/cm^3.

4 2.5 L $= 2500$ cm^3, so mass $= 2500 \times 1 = 2500$ g $= 2.5$ kg.

5 $V = 2 \times 5 \times 8 = 80$ cm^3; $\rho = \frac{624}{80} = 7.8$ g/cm^3.

6 $\frac{|4.8 - 5.0|}{5.0} \times 100\% = 4\%$.

7 Greatest $= \pi(3.05^2)(10.05) = 293.7$ cm^3; least $= \pi(2.95^2)(9.95) = 272.0$ cm^3. The design value is $90\pi = 282.7$ cm^3.

8 The side is 4 cm to the nearest centimetre, so $3.5 \le s < 4.5$ cm. Cubing: $42.875 \le V < 91.125$ cm^3. The stated 64 cm^3 is only the central estimate; the rounding of the side allows this wide range.

INVESTIGATION 8 · THE 330 ML CAN

1 Sample cans, measured with a paper strip and a ruler to the nearest millimetre. Can A (330 mL): circumference 20.7 cm, so $r = 3.30$ cm, $h = 9.6$ cm. Can B (500 mL): $C = 21.4$ cm, $r = 3.40$ cm, $h = 13.8$ cm. Can C (250 mL): $C = 18.2$ cm, $r = 2.90$ cm, $h = 9.5$ cm. Volumes: 328, 501 and 251 cm^3; surface areas 267.5, 367.4 and 225.9 cm^2.

2 Completed ratios: A $r/h = 0.344$, $SA/V = 0.814$; B $r/h = 0.247$, $SA/V = 0.733$; C $r/h = 0.305$, $SA/V = 0.900$. The larger can has the lowest material per millilitre, but it also has the largest volume, so this does not yet isolate the effect of shape.

3 Comparing cans of different volume is unfair: a bigger container has a smaller surface-to-volume ratio whatever its shape. To compare shapes, fix the volume. At 330 cm^3, the table shows the least material near $r = 3.7$ cm, where $SA/V = 0.801$, and the most at the extremes of the table.

4 Surface area falls from 276.5 to 264.4 and then rises to 271.6, so the minimum lies between $r = 3.4$ and $r = 4.0$, close to $r = 3.7$ cm with $h = 7.67$ cm. A finer search gives $r \approx 3.75$ cm, $h \approx 7.49$ cm, $SA \approx 264.4$ cm^2. Consecutive rows confirm that the change of sign occurs around the 3.7 cm row.

5 Conjecture: at the minimum, $h = 2r$, so $r/h = 0.5$. The table's minimum row has $r/h = 0.482$, which is 3.6% below 0.5; with a 0.3 cm grid this is consistent with the conjecture, but the table alone cannot prove it.

6 Since $h = \frac{V}{\pi r^2}$, $SA = 2\pi r^2 + 2\pi r\left(\frac{V}{\pi r^2}\right) = 2\pi r^2 + \frac{2V}{r}$. For very small r, the term $\frac{2V}{r}$ is huge (a tall, thin can with a large curved surface); for very large r, $2\pi r^2$ is huge (a flat, wide can with large discs). One term falls and the other rises, so a minimum lies between.

7 A real 330 mL can typically has $r \approx 3.3$ cm and $h \approx 9.6$ cm, giving $r/h = 0.344$ and $SA/V = 0.814$, about 1.6% more material than the optimum. Manufacturers choose this shape for grip and stability, for vending machines and shelves, for the domed ends and seams that the model ignores, and for marketing.

8 Assumptions: uniform metal thickness, no seams, no domed ends or rim, and equal cost per square centimetre. The most unrealistic is uniform thickness with no domed ends: real cans have thicker rims and shaped ends, and the body and ends are made and printed separately. Removing that assumption changes the position of the optimum and makes a taller can relatively more expensive, so the true best shape is not exactly $h = 2r$.

APPLICATION · THE RAINWATER TANK

1 $V = \pi r^2 h = 3.142 \times 1.20^2 \times 2.00 = 9.049$ m^3, capacity 9049 L, about 9.05 kL.

2 Mass $= 9049$ kg, about 9.05 tonnes. Spread over the $3.00 \times 3.00 = 9.00$ m^2 base, the average load is about 1005 kg/m^2 (roughly 10 kN/m^2); directly under the tank footprint it is about 2000 kg/m^2. The average is modest for soil, but the concentrated load and the 0.15 m slab make mesh reinforcement advisable.

3 Curved side $= 2\pi rh = 2 \times 3.142 \times 1.2 \times 2 = 15.08$ m^2; top $= \pi r^2 = 3.142 \times 1.44 = 4.52$ m^2. Painted area $= 19.61$ m^2; cost $= 19.61 \times 8.50 = 166.65$ units, about 167 units.

4 Base volume $= 3.00^2 \times 0.15 = 1.35$ m^3; cost $= 1.35 \times 240 = 324$ units. But a 0.60 m margin on every side would need a base of $2.40 + 2(0.60) = 3.60$ m, not 3.00 m. The stated 3.00 m square gives only 0.30 m margin, so the specification is inconsistent. At 3.60 m the volume would be 1.944 m^3 and the cost 466.56 units.

5 750 mm $= 0.75$ m, so the annual harvest is $0.75 \times 180 = 135$ m$^3 = 135\,000$ L. This is $135 \div 9.049 = 14.9$ full tanks. The tank stores only about one-fifteenth of the annual harvest, so overflow and timing, not rainfall, limit the supply; most water will be lost unless it is used between storms.

6 $9049 \div 12 = 754.1$ minutes, which is $754.1 \div 60 = 12.57$ hours, or 12.6 hours to the nearest 0.1 hour (12 h 34 min).

7 The radius is 1.20 ± 0.025 m and the height 2.00 ± 0.025 m. Greatest $= \pi(1.225^2)(2.025) = 9.548$ m^3; least $= \pi(1.175^2)(1.975) = 8.567$ m^3. As percentages of 9.049, this is $+5.5\%$ and -5.3%, about $\pm 5.4\%$. The radius contributes more because it is squared: its 2.08% relative error doubles to about 4.2%, while the height contributes only 1.25%.

8 Uncertainties: the ground may not be level, the tank may not be exactly cylindrical, the base may settle, and the rainfall and demand vary. Siting must be shaded, level and accessible, with overflow directed away from the building. Maintenance needs an inlet screen, a first-flush diverter and periodic cleaning; standing water raises quality and mosquito concerns. A full tank weighs about 9 tonnes, so the base and ground must be checked. A cylinder uses less material per litre than a cuboid, but a cuboid may fit a corner better. Recommend a first-flush diverter and leaf screen, with the overflow led to a garden bed; this improves water quality and reduces waste at modest cost.

CRITERION TASK · THE SCHOOL GREENHOUSE

Sample response. The cross-section is a 3.0 m by 1.0 m rectangle with a semicircle of radius 1.5 m on top; the length is 6.0 m. The door is a 0.9 m by 1.9 m rectangle in one end wall.

1 Labelled cross-section: width 3.0 m, side walls 1.0 m, semicircular roof of radius 1.5 m. The sketch shows a half-cylinder on a rectangular prism, with the door in one end. Assumptions: the roof is an exact half-cylinder, the walls are vertical and the dimensions are internal, the door is the only opening, and the frame has negligible width.

2 Floor $= 3.0 \times 6.0 = 18\ \text{m}^2$. Rectangular part $= 3.0 \times 1.0 \times 6.0 = 18\ \text{m}^3$. Half-cylinder $= \frac{1}{2}\pi(1.5^2)(6.0) = 6.75\pi = 21.21\ \text{m}^3$. Total $= 39.21\ \text{m}^3$.

3 Curved roof $= \frac{1}{2}(2\pi rh) = \pi(1.5)(6.0) = 9\pi = 28.27\ \text{m}^2$. Two long side walls $= 2(1.0 \times 6.0) = 12.00\ \text{m}^2$. Each end $= 3.0 \times 1.0 + \frac{1}{2}\pi(1.5^2) = 3.00 + 3.53 = 6.53\ \text{m}^2$, so two ends $= 13.07\ \text{m}^2$. Glazing before the door $= 12.00 + 13.07 = 25.07\ \text{m}^2$; total covered surface $= 28.27 + 25.07 = 53.34\ \text{m}^2$.

4 Door area $= 0.9 \times 1.9 = 1.71\ \text{m}^2$, deducted from the glazed area: glass $= 25.07 - 1.71 = 23.36\ \text{m}^2$. Polycarbonate cost $= 28.27 \times 42 = 1187.52$; glass cost $= 23.36 \times 55 = 1284.72$. Total $= 2472.24$, about 2472 units.

5 Base perimeter $= 2(3.0 + 6.0) = 18.0$ m; four vertical corners $= 4 \times 1.0 = 4.0$ m; two roof edges $= 2 \times 6.0 = 12.0$ m; two gable arcs $= 2\pi(1.5) = 9.42$ m. Total $= 43.42$ m; cost $= 43.42 \times 18 = 781.65$, about 782 units.

6 Use ± 0.025 m for each dimension. The width lies from 2.975 to 3.025, the height from 0.975 to 1.025, the length from 5.975 to 6.025, and the roof radius is half the width, from 1.4875 to 1.5125. Greatest volume $= 3.025 \times 1.025 \times 6.025 + \frac{1}{2}\pi(1.5125^2)(6.025) = 40.33\ \text{m}^3$; least $= 2.975 \times 0.975 \times 5.975 + \frac{1}{2}\pi(1.4875^2)(5.975) = 38.10\ \text{m}^3$. The design $39.21\ \text{m}^3$ lies between, about 2.8% either side, a total range of 5.7%.

7 Condensation will form on single-layer polycarbonate, so vents or a slight slope are needed; ventilation must avoid overheating in summer; the large curved surface catches wind, so the frame must be anchored; orientation should admit winter light without overheating; glazing needs cleaning and panel replacement. Recommend opening ridge vents at both gable ends with closable flaps. They remove humid air in summer, reduce condensation, and are cheaper than mechanical ventilation; the cost is small compared with the glazing and the benefit is direct.

Marking guide. Criterion C: a clear labelled cross-section and sketch, correct formulas, units carried through, and a logical sequence from decomposition to totals. Criterion D: a valid model of the real greenhouse, correct areas, volumes and costs, the door deducted once from the glazing, bounds interpreted as a range, and an evaluation that links at least two limitations to a specific, justified modification. Award the 24 marks across the seven parts (3, 5, 5, 4, 2, 3, 2), with follow-through marks for consistent earlier errors.

MIXED PRACTICE 8

Numbering. The chapter restarts numbering in each section, so items are labelled F (Fluency), R (Reasoning) and A (Application) with their section numbers.

F1 $SA = 2(12 \times 7 + 12 \times 4 + 7 \times 4) = 2(84 + 48 + 28) = 320\ \text{cm}^2$; $V = 12 \times 7 \times 4 = 336\ \text{cm}^3$.

F2 $SA = 2\pi(36) + 2\pi(6)(9) = 72\pi + 108\pi = 180\pi = 565.5\ \text{cm}^2$; $V = \pi(36)(9) = 324\pi = 1017.9\ \text{cm}^3$.

F3 $l = \sqrt{4^2 + 3^2} = 5$; $SA = 16\pi + 20\pi = 36\pi = 113.1\ \text{cm}^2$; $V = \frac{1}{3}\pi(16)(3) = 16\pi = 50.3\ \text{cm}^3$.

F4 $SA = 4\pi(25) = 100\pi = 314.2\ \text{cm}^2$; $V = \frac{4}{3}\pi(125) = \frac{500\pi}{3} = 523.6\ \text{cm}^3$.

R1 $k = \frac{9}{6} = 1.5$. Surface areas: $k^2 = 2.25$, ratio $9 : 4$. Volumes: $k^3 = 3.375$, ratio $27 : 8$. Area depends on two lengths and volume on three, so the factors are squared and cubed.

R2 $SA = 2\pi r^2 + \frac{2V}{r}$. As r becomes very small, $\frac{2V}{r}$ grows without limit; as r becomes very large, $2\pi r^2$ grows without limit. Between the two extremes the sum has a minimum, so the surface area cannot be arbitrarily small.

R3 Cube: $s = 10$, $SA = 600\ \text{cm}^2$. Sphere: $r^3 = \frac{750}{\pi} = 238.7$, $r = 6.20$ cm, $SA = 4\pi r^2 = 483.6\ \text{cm}^2$. The sphere uses less material. Containers are not spherical because spheres roll, stack poorly, waste shipping space and are awkward to fill, label and open.

R4 $V = \pi r^2 h$ becomes $\pi(2r)^2 h = 4\pi r^2 h$, so the volume is four times larger, not double. The curved area doubles ($2\pi rh \to 4\pi rh$), but the total surface area becomes $8\pi r^2 + 4\pi rh$, so it is not simply doubled either.

A1 $V = 60 \times 30 \times 35 = 63\,000\ \text{cm}^3 = 63$ L. Glass $= 60 \times 30 + 2(60 \times 35) + 2(30 \times 35) = 1800 + 4200 + 2100 = 8100\ \text{cm}^2 = 0.81\ \text{m}^2$.

A2 Painted area $= 2\pi(2.5)(8) + \pi(2.5^2) = 40\pi + 6.25\pi = 46.25\pi = 145.30\ \text{m}^2$. Two coats $= 290.60\ \text{m}^2$; paint $= 290.60 \times 0.2 = 58.12$ L, so 3 drums of 20 L.

A3 Cone $= \frac{1}{3}\pi(9)(12) = 36\pi = 113.10\ \text{cm}^3$; hemisphere $= \frac{2}{3}\pi(27) = 18\pi = 56.55\ \text{cm}^3$. Total $= 54\pi = 169.6\ \text{cm}^3$.

A4 $r = 0.75$, $V = \pi(0.5625)(2) = 1.125\pi = 3.534\ \text{m}^3 = 3534$ L. At 180 L per day this lasts $3534 \div 180 = 19.6$ days.

A5 Box A $= 2(100 + 120 + 120) = 680\ \text{cm}^2$; box B $= 2(80 + 120 + 150) = 700\ \text{cm}^2$. Box A uses less cardboard for the same volume.

A6 $V = \pi(1^2)(2) + \frac{2}{3}\pi(1^3) = 2\pi + \frac{2\pi}{3} = \frac{8\pi}{3} = 8.38\ \text{m}^3$. Exposed area $= 2\pi(1)(2) + 2\pi(1^2) = 4\pi + 2\pi = 6\pi = 18.85\ \text{m}^2$.

Common errors. Using the diameter where the radius is required; omitting one or both bases; using vertical height instead of slant height for a cone or pyramid; forgetting the factor $\frac{1}{3}$ for pyramids and cones; cubing the diameter in the sphere formula; counting the glued faces of a composite solid; mixing cm^2 with cm^3; converting cubic units by 100 instead of 1 000 000; and using ± 0.05 rather than ± 0.025 for a measurement to the nearest 0.05.

Chapter 9 · Bivariate data, correlation and causation

PREREQUISITE CHECK

1 Points plotted at $(1,3)$, $(2,5)$, $(3,4)$, $(4,7)$, with both axes labelled and scaled from 0 to 8.

2 $\dfrac{4+6+7+9+14}{5} = \dfrac{40}{5} = 8$.

3 Gradient $\dfrac{11-5}{3-1} = 3$; $5 = 3(1) + c$, so $c = 2$ and $y = 3x + 2$.

4 Gradient 4; intercept -7, so the line crosses the y-axis at $(0, -7)$.

5 $\sqrt{2.5} \approx 1.58$; $\sqrt{0.64} = 0.80$.

6 $62 - 70 = -8$, so the score is 8 percentage points below the mean.

PRACTICE 9.1 · TWO VARIABLES AT ONCE

1 (a) Explanatory: hours of revision; response: test score. (b) Explanatory: age of car; response: resale value. (c) Explanatory: daily rainfall; response: umbrella sales.

2 (a) Negative: pressure falls as altitude rises. (b) Negative in most cities: price tends to fall with distance from the centre. (c) Positive: more practice is associated with faster typing.

3 The points rise from left to right and lie close to a straight line, so the association is positive, linear and strong. The point $(2.5, 4.6)$ sits slightly below the trend but is not an outlier; there are no clusters or gaps.

4 "Going up" gives only the direction. A complete description also states the form (linear or curved), the strength (how closely the points follow the form), any outliers or clusters, and the variables and units on the axes.

5 The form changes: growth is roughly linear at first, then levels off. A single straight line cannot follow both parts, so it will under-predict the early heights or over-predict the later ones, and its residuals will show a pattern rather than random scatter.

6 The point is far from the other x-values, so it has high leverage: the line is pulled towards it. Because it is also far vertically from the trend, removing it could change the gradient substantially. Being unusual and being influential are related but not identical; always check both.

7 Check the unit: 600 minutes is 10 hours per day, which may be a weekly total entered in the daily column. Check the source with the student or teacher, and compare with the rest of the distribution. Only after confirming the value should it be treated as genuine; if it is genuine, report it and consider analysing the data with and without it.

8 A single line through both clusters would mainly describe the difference between weekdays and weekends, not the trend within either. It could show a steep slope that no individual day follows, and it would hide two distinct patterns. Plot the clusters separately or include a group variable.

PRACTICE 9.2 · MEASURING LINEAR ASSOCIATION

1 Means $\bar{x} = 2.5$, $\bar{y} = 4.75$. $\sum(x-\bar{x})(y-\bar{y}) = 9.5$, $\sum(x-\bar{x})^2 = 5$, $\sum(y-\bar{y})^2 = 18.75$. $r = \dfrac{9.5}{\sqrt{5 \times 18.75}} = \dfrac{9.5}{9.682} \approx 0.981$.

2 Means $\bar{x} = 3$, $\bar{y} = 13$. Products total 17, $\sum(x-\bar{x})^2 = 10$, $\sum(y-\bar{y})^2 = 34$. $r = \dfrac{17}{\sqrt{340}} \approx 0.922$.

3 Means $\bar{x} = 3$, $\bar{y} = 16.8$. Products total -18, $\sum(x-\bar{x})^2 = 10$, $\sum(y-\bar{y})^2 = 38.8$. $r = \dfrac{-18}{\sqrt{388}} \approx -0.914$.

4 A moderately strong negative linear association: students who watch more television tend to report fewer minutes of moderate exercise. The value does not show that television causes inactivity; diet, schedule, health and age could influence both.

5 The formula is symmetric in x and y: the product $(x-\bar{x})(y-\bar{y})$ is unchanged when the variables are swapped, and so are the two squared sums. The correlation measures the same linear pattern whichever variable is called explanatory.

6 r is not a percentage or a proportion of strength. Squaring gives $r^2 = 0.25$, so the linear model accounts for about 25% of the variation in the response; 75% is unexplained. Strength is also about scatter around a line, not a fraction of a maximum.

7 One valid choice is $(10, 2)$, far to the right of the original points and below the trend. The new mean point is $(4, 4.2)$; the point's deviations have a large negative product that outweighs the positive products, giving $r \approx -0.199$. Other points far below the trend also work; the key is a large negative contribution to $\sum(x-\bar{x})(y-\bar{y})$.

8 In the first case the points lie close to a line throughout its length; in the second, most points may be scattered while one extreme point creates the line. A scatter plot reveals the difference: the correlation coefficient alone cannot, because both data sets can give $r = 0.85$.

PRACTICE 9.3 · THE LINE OF BEST FIT

1 Gradient $\dfrac{19-7}{5-1} = 3$; $7 = 3(1) + c$, so $c = 4$ and $y = 3x + 4$.

2 $y = 4(6) + 3 = 27$; $y = 4(10) + 3 = 43$.

3 (a) 12 is inside 5 to 25: interpolation. (b) 30 is above the range: extrapolation. (c) 4 is below the range: extrapolation.

4 Predictions: $3(1) - 2 = 1$, $3(2) - 2 = 4$, $3(3) - 2 = 7$, $3(4) - 2 = 10$, $3(5) - 2 = 13$. Residuals $y_{\text{obs}} - y_{\text{pred}}$: $1, -1, 1, -1, 1$. Their total is 1, not 0, because the line does not pass exactly through the mean point $(3, 7.2)$; it gives 7 there, a residual of 0.2.

5 If the line passes through the mean point, the positive and negative vertical distances balance, so the residuals sum to zero. Moving the line away from the mean point makes the total positive or negative, which means the line is systematically too high or too low. The least-squares line also chooses the gradient that minimises the sum of squared residuals.

6 The trend in 1950–2020 may not continue: birth rates, migration, ageing and policy can change the rate of growth, and a straight line will eventually predict implausible values. A prediction for 2200 is far outside the observed range, so it is extrapolation with no data to check it.

7 The intercept 4 is the fixed charge when $d = 0$, likely a booking or minimum fare. The gradient 2.1 is the cost per kilometre. At $d = 12$: $C = 4 + 2.1(12) = 29.2$, so about \$29.20. At $d = 1$: $C = 6.10$; this may be valid only if the company charges the model's minimum, since the observed data may start at longer journeys.

8 Calculate or estimate the residuals at several well-spaced x-values and compare the sums of squared residuals; the line with the smaller total fits better overall. The lines cross where $2x + 5 = 1.8x + 8$, so $0.2x = 3$ and $x = 15$. If the data lie mostly below 15, the second line is higher; above 15, the first is higher. The better choice depends on where the data actually are, not on algebra alone.

PRACTICE 9.4 · CORRELATION AND CAUSE

1 (a) Age: older children have larger feet and read better. (b) Hot weather: more ice cream is sold and more people swim. (c) Size of the fire: it requires more firefighters and causes more damage.

2 An observational study records variables as they occur without intervening; a controlled experiment deliberately changes the explanatory variable and compares outcomes under otherwise similar conditions.

3 People who are already healthy or motivated may be more likely to download and use a fitness app, so better health could lead to app use rather than the reverse.

4 Plausible confounders include family income, parental education and support, sleep, prior attainment and school quality. One way to reduce their influence is to compare students with similar backgrounds, or to randomise access to breakfast; a controlled comparison is stronger than an unadjusted group comparison.

5 Group averages remove the variation within each group, so a correlation between averages can be stronger or weaker than the correlation between individuals. A pattern across countries, for example, need not hold for individuals within a country; this is the ecological fallacy.

6 Evidence would include a longitudinal design measuring sociability before and after gaming, a comparison group with similar baseline sociability, controls for age, sleep and family context, a clear measure of sociability, and a plausible mechanism. A single correlation cannot distinguish gaming causing isolation from isolated teenagers choosing gaming.

7 Use several plants of the same variety and age, assign half at random to plant food and half to water only, and keep light, soil, pot size, water and temperature the same. The response is tomato yield, for example total mass harvested over a fixed period. Compare mean yields and the variability within each group.

8 House prices may have risen for other reasons: wider market growth, new shops or schools, or buyers anticipating the tram and moving in before it opened. Areas near the line may also differ from other areas in ways that affect price. A stronger comparison would use similar areas without a tram and compare price changes before and after.

PRACTICE 9.5 · READING CLAIMS CRITICALLY

1 (a) 30% to 36%: $+6$ percentage points; relative increase $\frac{6}{30} = 20\%$. (b) 80% to 72%: -8 percentage points; relative decrease $\frac{8}{80} = 10\%$. (c) 2.0% to 2.5%: $+0.5$ percentage points; relative increase $\frac{0.5}{2.0} = 25\%$.

2 On an axis from 90 to 100, a change from 92 to 95 uses 3 of the 10 units, so the bars differ by about 30% of the axis height. On an axis from 0 to 100, the same change is 3% of the height. The truncated axis magnifies the visual difference.

3 "Doubles" hides the tiny absolute risk: it rises by 0.2 percentage points, from 2 in 1,000 to 4 in 1,000. The relative change is large, but the chance of injury remains small; both facts should be reported.

4 An outlier far from the other points in both variables can create a strong linear trend that the remaining points do not follow. The outlier has high leverage and can dominate $\sum(x-\bar{x})(y-\bar{y})$; removing it may leave little or no pattern, so the correlation should be reported with and without it.

5 Minor cases: Clinic A $\frac{87}{350} \approx 24.9\%$; Clinic B $\frac{270}{350} \approx 77.1\%$. Clinic B treats mostly minor cases, which have a higher recovery rate, so its overall rate is pulled up. Clinic A has the higher rate in both subgroups, yet the lower overall rate: the comparison reverses because the case mix differs.

6 The sample is self-selected from readers of a fitness magazine, who are likely to be more interested in exercise than adults in general. The 90% describes that readership, not all adults; a larger sample of the same readers would not fix the bias. A random sample of adults, with a good response rate, is needed.

7 Students were selected because their scores were low, so some of the rise is regression to the mean: extreme scores tend to move towards the average on retesting even without an intervention. A control group of similarly low-scoring students who did not receive the programme is needed to separate the programme's effect from this natural variation.

8 A confounder is overall health or fitness: healthier people can walk more and also tend to have lower blood pressure. Reverse causation is also plausible: people with lower blood pressure may feel able to walk more. The association alone does not show that walking lowers blood pressure.

9 Ask: How many staff were in the programme, and how many in a comparison group? What was the previous average, so that the 12% can be interpreted? Over what period was it measured, and were other changes introduced at the same time? Also ask whether 12% is relative or percentage points, and whether staff chose to take part.

10 Each axis can be scaled independently, so two series with different units can be made to rise and fall together visually; the apparent match is partly an artefact of the chosen scales. Plot both series as standardised values or index both to the same starting value, or use a scatter plot of the two variables with a correlation. State the scales explicitly.

11 The large trial is more trustworthy: a sample of 4,000 has much less random variation than one of 40, and small trials are more likely to produce extreme results by chance. A larger measured effect is not automatically more important: a 30% relative improvement in a tiny trial may be imprecise, and the absolute benefit and the outcome being measured also matter.

12 Example: "Ice cream sales cause more swimming accidents." A accurate rewrite is: "Ice-cream sales and swimming accidents rise together in hot weather; supervision and water safety matter." The rewrite replaces "cause" with "rise together", names the common cause (hot weather), and avoids implying that buying ice cream is dangerous.

INVESTIGATION 9 · ARM SPAN AND HEIGHT

Model results for the sample table. Individual measurements will differ; the reasoning and method are the assessable points.

2 Plot arm span on the horizontal axis and height on the vertical axis, both in centimetres, with a scale that uses most of the grid. The points rise from left to right, follow a straight line closely and show no obvious outliers, so the association is positive, linear and very strong.

3 Mean arm span $\bar{x} \approx 171.7$ cm; mean height $\bar{y} = 173.0$ cm. The mean point $(171.7, 173.0)$ lies near the centre of the pattern. For the sample, $r \approx 0.998$: an extremely strong positive linear association.

4 A line through the mean point with gradient about 0.97 has equation $y \approx 0.97x + 6.4$. The gradient means that each extra centimetre of arm span is associated with about 0.97 cm more height; the intercept is the model's prediction at arm span 0, which has no physical meaning.

5 At $x = 170$: $y \approx 0.970(170) + 6.43 \approx 171.4$ cm. This is interpolation, since 170 lies within 152 to 190. At $x = 120$: $y \approx 122.9$ cm; at $x = 210$: $y \approx 210.2$ cm. Both are extrapolation and unreliable: a 120 cm arm span is far below the sample, and a 210 cm span far above it.

6 Predictions for another pair's students should be compared with measured heights; small residuals of varying sign suggest the model generalises. A run of large residuals in one direction suggests the other sample differs systematically, perhaps in age or measurement technique.

7 If arm span equals height on average, the gradient should be about 1: a 1 cm increase in arm span corresponds to a 1 cm increase in height. The fitted value 0.97 is close to 1 and supports the conjecture approximately. Individual variation, posture, shoes and measurement error prevent an exact match, and the intercept absorbs the average offset.

8 Limitations include the narrow age range, the small sample of 12, posture and footwear, timing of measurement, and who volunteered. A larger random sample across ages, with a standardised measurement protocol, would be more reliable. The data cannot support a claim that arm span causes height, nor that one measurement determines the other.

Marking guide. Criterion B: a clear plan for measurement and plotting, correct use of r and a line of best fit, and an evaluation that distinguishes interpolation from extrapolation. Criterion C: axes labelled with units, the model stated with its equation, predictions reported with units, and limitations described precisely rather than as a generic list.

APPLICATION · THE SCREEN-TIME CLAIM

1 The gradient -0.42 means that, in this sample, each additional hour of screen time is associated with 0.42 fewer hours of sleep on average. It does not show that an extra hour causes 0.42 hours less sleep: the study is observational, and other differences between students could explain part or all of the association.

2 The intercept 9.1 is the model's predicted sleep for a student with zero screen time: 9.1 hours. No student in the sample has zero screen time; the observed range starts at 0.5 hours, so $t = 0$ is outside the data. The value is an extrapolation and should not be interpreted as a real "no-screen" sleep duration.

3 At $t = 2$: $s = 9.1 - 0.42(2) = 8.26$ hours. At $t = 5$: $s = 9.1 - 0.42(5) = 7.0$ hours. Both 2 and 5 lie within the observed range 0.5 to 5.5, so both are interpolation.

4 Predicted sleep at $t = 5.5$ is $9.1 - 0.42(5.5) = 6.79$ hours. Residual $= 5.0 - 6.79 = -1.79$ hours, so the point lies 1.79 hours below the line. The outlier may strengthen the apparent negative correlation and pull the line towards itself; its influence should be checked by refitting without it.

5 $r^2 = (-0.62)^2 = 0.3844$, so about 38% of the variation in sleep is associated with the linear model; about 62% is unexplained. This is a statement about fit, not cause: sleep, screen time and other variables such as homework, health and age may all influence one another.

6 Volunteers may be more health-conscious or have more extreme sleep habits than non-volunteers, and self-report can under- or over-estimate both screen time and sleep. A better method is a random sample from the school roll, with anonymous diaries or device logs and follow-up of non-responders.

7 Plausible confounders include caffeine or energy-drink use, stress and homework load, age and chronotype, and physical or mental health. For example, heavy homework may delay bedtime and increase screen use, producing both less sleep and more screen time without either causing the other.

8 Sample headline: "Survey of 40 students links more screen time with less sleep." Changes: "links" replaces "steals", which asserts causation; the sample size and the word "survey" are stated; the claim is limited to the students studied. An editorial note can add that the correlation is moderate and that other factors may explain part of it.

9 A randomised follow-up could assign students to a screen curfew or their usual routine, then compare sleep measured by a device over several weeks. Explanatory variable: curfew condition; response: sleep duration; comparison: curfew versus usual routine; keep constant: bedtime, caffeine rules, homework load and measurement method. Random allocation and a control group make the comparison stronger than the original survey.

CRITERION TASK · THE BREAKFAST CLAIM

Sample response. Marks depend on the quality of the reasoning, not on matching this wording.

1 Explanatory: breakfast frequency (days per week). Response: test score. The study is observational: students were asked about their usual breakfast and their scores were supplied; nothing was changed by the researchers.

2 $r = 0.31$ is a weak-to-moderate positive linear association: students who eat breakfast more often tend to score somewhat higher, but the points are widely scattered. The group difference is $68 - 56 = 12$ points, which is $\frac{12}{64} = 18.75\%$ of the overall mean of 64. This is a noticeable difference but not a precise individual effect.

3 $r = 0.31$ gives $r^2 \approx 0.096$, so the linear model accounts for only about 10% of the variation in scores; other factors explain the rest. The "12 points higher" compares the averages of two groups, not the effect of changing one student's breakfast. Group differences can arise from confounding and selection, so the headline's causal reading is not supported.

4 Three plausible confounders: family income (affects food security, resources and school support); parental education and support (affects routines and help with schoolwork); sleep and prior attainment (affect both morning appetite and test performance). School quality or attendance could also influence both. Each could raise breakfast consumption and scores without breakfast causing the score.

5 Selection affects who eats breakfast: students with crowded mornings, long journeys, food insecurity or low appetite may skip it, and these circumstances also affect learning. A breakfast club may attract students who are already well supported. One way to reduce selection bias is to randomise the offer of breakfast provision and compare all eligible students, including those who decline.

6 A stronger study would randomly allocate breakfast provision to some classes or schools and not others, then compare changes in test scores over a fixed period. Keep the test, teaching, attendance and socio-economic mix as similar as possible, monitor whether students actually eat, and use a control group with usual provision. The comparison should be between score changes, not just final averages.

7 Newsletter summary: "A survey of 600 students in 12 schools found that those who usually ate breakfast scored about 12 points higher on average than those who did not (64 overall), but the link is modest (correlation 0.31). The study cannot show that breakfast raises attainment: family income, sleep and prior support may also matter. Offering breakfast may still be worthwhile for reasons of equity and nutrition. We recommend a small trial that randomly offers breakfast and compares progress before making a whole-school claim."

Marking guide. Criterion C: correct identification of variables and design, accurate use of r, r^2 and the group difference, and clear, cautious communication. Criterion D: confounders explained rather than listed, selection and non-response

discussed, a workable randomised design, and a recommendation that matches the strength of the evidence. Band 4 requires three specific confounders with mechanisms and a design with allocation, comparison, response variable and controls.

MIXED PRACTICE 9

1 Means $\bar{x} = 6$, $\bar{y} = 10.4$. $\sum(x-\bar{x})(y-\bar{y}) = 52$, $\sum(x-\bar{x})^2 = 40$, $\sum(y-\bar{y})^2 = 69.2$. $r = \dfrac{52}{\sqrt{40 \times 69.2}} = \dfrac{52}{52.61} \approx 0.988$.

2 Gradient $\dfrac{23-11}{7-3} = 3$; $11 = 3(3) + c$, so $c = 2$ and $y = 3x + 2$. At $x = 9$, $y = 29$.

3 Predicted $y = 2.5(4) + 6 = 16$; observed 17; residual $= 17 - 16 = +1$, so the point lies 1 unit above the line.

4 $30 - 24 = 6$ percentage points; relative increase $\dfrac{6}{24} = 25\%$.

5 r is not a percentage: $r^2 = 0.7744$, so the linear model accounts for about 77% of the variation, not 88%. The remaining 23% is unexplained, and a strong correlation still does not establish causation.

6 "5%" is ambiguous: it may mean 5 percentage points (for example 40% to 45%) or a 5% relative increase (for example 40% to 42%). The two differ substantially. A pass rate rise from 40% to 45% is 5 percentage points and a 12.5% relative increase; the report should say which.

7 A randomised experiment with books is difficult because families cannot be assigned a reading culture, and giving books only to some children may be impractical or ethically fraught. Observational evidence can still be strengthened by longitudinal measurement (scores before and after books arrive), controlling for income and parental education, looking for a dose–response pattern, and identifying a plausible mechanism such as shared reading time.

8 A line fitted to a candle's height over time may predict a negative height after enough hours, which is impossible. Similarly, a linear model of a child's height against age could predict unrealistic adult heights. The model is only supported within the observed range; outside it, the mechanism may change or the line may cross a natural boundary.

9 $r = -0.84$: strong negative linear association between age and price. The gradient -2.1 means each extra year is associated with about \$2,100 lower price; the intercept 18 is the model's price for a new car, about \$18,000. At 5 years: $y = 18 - 2.1(5) = 7.5$, about \$7,500. A classic car may rise in value with age, so the linear model fails outside the ordinary used-car market; it is also extrapolation if the data do not include classics.

10 The larger study is more trustworthy because a sample of 500 has much less random variation than one of 50. The first study's 0.45 may be an unusually strong result from a small or selected sample; the second suggests the true association is weak. Neither proves causation, and the difference between the studies should be investigated (age range, measure, setting).

11 Absolute reduction: $0.8 - 0.5 = 0.3$ percentage points. Relative reduction: $\dfrac{0.3}{0.8} = 37.5\%$. Poster sentence: "The disease rate fell from 8 in 1,000 to 5 in 1,000 — a fall of 3 cases per 1,000 people." This reports the natural frequencies and avoids implying a 37.5% drop in individual risk.

12 Observational comparison: compare students who chose to use the app with those who did not, recording scores before and after. Randomised experiment: allocate app access at random, compare score gains, and keep teaching and testing the same. The randomised experiment gives stronger evidence for causation because it balances unmeasured confounders such as motivation and prior attainment.

13 The correlation is between country averages, not individuals. A country with higher average income may also have better health systems, education, nutrition and safety, which affect life expectancy. Raising one person's income by a fixed amount need not add a fixed number of years; the relationship may be nonlinear and the individual circumstances differ. This is the ecological fallacy.

14 Plot the data and look for a point far from the main cluster; calculate r with and without it and compare. If the correlation collapses, report that it depends on one influential point. Investigate whether the point is a recording error or a genuine case; do not silently remove it. Report both analyses and describe the point and its effect.

Common errors. A correlation is not a percentage; square it before interpreting explained variation. A line of best fit is a summary inside the observed range, not a law. "Correlation does not imply causation" needs a named alternative explanation. Percentage points and relative percentages are different quantities. Group averages can reverse or exaggerate individual patterns, and one influential point can create an entire correlation.

Chapter 10 · Probability, sampling and evidence

PREREQUISITE CHECK

1 $\dfrac{3}{8} = 0.375 = 37.5\%$.

2 $P(\text{red}) = \dfrac{6}{10} = \dfrac{3}{5}$.

3 $0.07 = \dfrac{7}{100} = 7\%$.

4 $0.15 \times 240 = 36$.

5 $0.6 = \dfrac{3}{5}$ and $0.65 = \dfrac{13}{20}$, so $0.6 = 0.60$, then 0.65, then $\dfrac{2}{3} \approx 0.667$. Order: $0.6 = \dfrac{3}{5} < 0.65 < \dfrac{2}{3}$.

6 Mean $= \dfrac{12+18+15+20+15}{5} = \dfrac{80}{5} = 16$.

GUIDED PRACTICE · USE THE DICE TABLE

(a) Sum 5: $(1,4), (2,3), (3,2), (4,1)$, so $\dfrac{4}{36} = \dfrac{1}{9}$.

(b) Sum 12: only $(6,6)$, so $\dfrac{1}{36}$.

(c) Sum not 7: $1 - P(\text{sum } 7) = 1 - \dfrac{6}{36} = \dfrac{5}{6}$.

(d) At least one 3: the complement is "no 3", with $5 \times 5 = 25$ outcomes, so $1 - \dfrac{25}{36} = \dfrac{11}{36}$.

PRACTICE 10.1 · SAMPLE SPACES AND PROBABILITY

1 Sample space $\{H, T\}$; $P(\text{head}) = \frac{1}{2}$.

2 Primes $2, 3, 5, 7$: $P = \frac{4}{8} = \frac{1}{2}$. Multiples of 4: $4, 8$, so $P = \frac{2}{8} = \frac{1}{4}$. Not 3: $P = \frac{7}{8}$.

3 Total $4 + 6 + 2 = 12$. $P(\text{red}) = \frac{4}{12} = \frac{1}{3}$; $P(\text{not black}) = \frac{10}{12} = \frac{5}{6}$; $P(\text{red or yellow}) = \frac{10}{12} = \frac{5}{6}$.

4 Sum 9: $(3, 6), (4, 5), (5, 4), (6, 3)$, so $\frac{4}{36} = \frac{1}{9}$. Sum at most 3: sums $2, 3$ have $1 + 2 = 3$ outcomes, so $\frac{3}{36} = \frac{1}{12}$. Both odd: $3 \times 3 = 9$ outcomes, so $\frac{9}{36} = \frac{1}{4}$.

5 A forecast probability describes the chance that rain occurs at some point in the forecast area and period, based on a model. It is not a statement that it will rain for 30% of the day; the rain could fall in a short burst or not at all.

PRACTICE 10.2 · COMPOUND EVENTS

1 Sample space $\{HH, HT, TH, TT\}$. $P(\text{two heads}) = \frac{1}{4}$; $P(\text{exactly one head}) = \frac{2}{4} = \frac{1}{2}$.

2 $P(\text{odd and tail}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$. $P(\text{odd or tail}) = \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}$.

3 Mutually exclusive, so $P(A \cup B) = 0.5 + 0.3 = 0.8$. $P(\text{neither}) = 1 - 0.8 = 0.2$.

4 $P(A \cup B) = 0.4 + 0.5 - 0.2 = 0.7$. Since $P(A) \times P(B) = 0.4 \times 0.5 = 0.2 = P(A \cap B)$, the events are independent.

5 Total 80. $P(F) = \frac{30}{80} = \frac{3}{8}$; $P(B) = \frac{25}{80} = \frac{5}{16}$; $P(F \cap B) = \frac{10}{80} = \frac{1}{8}$; $P(F \cup B) = \frac{30 + 25 - 10}{80} = \frac{45}{80} = \frac{9}{16}$; $P(\text{neither}) = \frac{35}{80} = \frac{7}{16}$.

6 $P(GG) = \frac{4}{9} \times \frac{3}{8} = \frac{12}{72} = \frac{1}{6}$. $P(\text{at least one red}) = 1 - P(GG) = 1 - \frac{1}{6} = \frac{5}{6}$.

PRACTICE 10.3 · EXPERIMENTAL PROBABILITY

1 Relative frequency $= \frac{36}{80} = 0.45$. The theoretical probability is 0.5; the result is 0.05 below, which is plausible variation for 80 tosses.

2 Relative frequency $= \frac{60}{250} = 0.24$. Expected number $= 250 \times 0.2 = 50$, so 60 is 10 above expectation, consistent with ordinary variation.

3 Modelled probability $= \frac{4}{10} = 0.4$. Successes in the list: $1, 0, 2, 3, 3, 0, 2, 1, 3$, which is 9 of 15, so the estimate is 0.6. It is above 0.4, but 15 trials are too few to conclude that the model is wrong.

4 With 10 rolls, each face is expected only about 1.67 times. Random variation is large: even a fair die can show several repeats or a missing face. Many more rolls are needed before a persistent departure from $\frac{1}{6}$ would be evidence of bias.

5 $P(\text{head}) = \frac{1}{2}$. The coin has no memory, and the tosses are independent; the previous run does not change the next toss. The law of large numbers describes the long-run proportion, not a debt that must be repaid.

PRACTICE 10.4 · SAMPLING AND BIAS

1 Population: all 800 students. Sample: the 50 students surveyed. Sample size: 50.

2 Advantage: a census avoids sampling error because every member is measured, and it can describe small subgroups exactly. Disadvantage: it is costly and slow, and non-response or measurement error can still bias the result.

3 Number the students 1 to 200. Use a random-number generator to produce numbers between 1 and 200, ignoring repeats, until 10 different students are selected. Every student then has the same chance of inclusion.

6 "Win or lose" are not necessarily equally likely. A stronger player or a biased game makes one outcome more probable; a football team that wins most matches has a win probability well above $\frac{1}{2}$. Equally likely outcomes must be justified by the setup, not by counting two descriptions.

7 $P(\text{red}) = \frac{3}{3 + B} = \frac{1}{4}$, so $12 = 3 + B$ and $B = 9$. There are 9 blue counters.

8 Red: $\frac{1}{3} \times 12 = 4$ sectors. Blue: $\frac{1}{4} \times 12 = 3$ sectors. Green: $12 - 4 - 3 = 5$ sectors. Check: $\frac{4}{12} + \frac{3}{12} + \frac{5}{12} = 1$.

9 Examples: $P(\text{even sum}) = \frac{18}{36} = \frac{1}{2}$, because each of the 6 even face values on the first die can pair with 3 even and 3 odd values to make an even sum, or by counting the table. $P(\text{sum is a multiple of } 3) = \frac{12}{36} = \frac{1}{3}$: sums $3, 6, 9, 12$ occur $2 + 5 + 4 + 1 = 12$ times.

7 Multiplication applies to independent events because the first outcome does not change the second probability: for a die and a coin, $P(6 \text{ and } H) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}$. Without replacement the bag changes: from 4 red and 3 blue, $P(\text{red then red}) = \frac{4}{7} \times \frac{3}{6} = \frac{2}{7}$, not $\left(\frac{4}{7}\right)^2$.

8 Tree: first draw red $\frac{3}{5}$, white $\frac{2}{5}$; second draw denominators 4.
$P(\text{same}) = P(RR) + P(WW) = \frac{3}{5} \times \frac{2}{4} + \frac{2}{5} \times \frac{1}{4} = \frac{6}{20} + \frac{2}{20} = \frac{8}{20} = \frac{2}{5}$.
$P(\text{different}) = 1 - \frac{2}{5} = \frac{3}{5}$.

9 Winning sums 7 and 11: $6 + 2 = 8$ outcomes, so $P = \frac{8}{36} = \frac{2}{9}$. Expected wins in 180 games: $180 \times \frac{2}{9} = 40$.

10 $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.6 + 0.5 - 0.9 = 0.2$. But $P(A) \times P(B) = 0.6 \times 0.5 = 0.3 \neq 0.2$, so the events are not independent. They overlap less than independence would predict.

6 Use digits 0–4 for a boy and 5–9 for a girl (or a coin). Read three digits per family to represent the three children; repeat for many families, for example 200. Count the families with all three the same sex (BBB or GGG) and divide by the number of simulated families. The theoretical value is $\frac{2}{8} = \frac{1}{4}$.

7 Twenty plays are too few: a 70% win rate could easily arise by chance, and the website may report only successful periods or define "win" loosely. A better test uses a large, pre-specified number of plays, records every outcome, checks that plays are independent, and reports a range of uncertainty.

8 Two samples of 20 can easily give 0.30 and 0.45 from the same true probability; the values do not prove that the truth lies between them. Combining the trials gives $\frac{6 + 9}{40} = \frac{15}{40} = 0.375$, a better estimate, though still subject to sampling variation. A range or interval, not a claim of proof, is appropriate.

4 Respondents choose themselves, so people with strong views, easy internet access or an interest in the news site are over-represented; some may vote more than once, and the readership may not resemble the population. Tens of thousands of responses reduce random variation but do not remove this systematic bias.

5 Year 7: $\frac{240}{600} \times 90 = 36$. Year 8: $\frac{200}{600} \times 90 = 30$. Year 9: $\frac{160}{600} \times 90 = 24$. Check: $36 + 30 + 24 = 90$.

6 The sample of 80 is likely to be more accurate: a larger random sample has less sampling variation, so its estimate tends to be closer to the population value. Both could still be biased if the sampling frame is incomplete, but random selection removes the main source of systematic error.

7 The item is leading and undefined: "love" sets a positive tone, and "a lot / a little" gives no neutral option or frequency. A better item asks about behaviour with balanced options: "How often do you take part in sport or physical activity in a typical week? Never / 1–2 times / 3–4 times / 5 or more times." A separate enjoyment item could use a balanced scale from "not at all" to "very much".

8 Population: all students at the school. Sampling frame: the school roll, numbered. Method: a stratified random sample by year group, proportional to size, with 100 students selected. Bias to monitor: non-response and absentees, especially if students with long journeys are more likely to be absent; follow up non-responders and compare the achieved sample with the roll.

PRACTICE 10.5 · RISK AND EVIDENCE

1 $\frac{3}{1000} = 0.003 = 0.3\%$.

2 4 in 1,000 $= 0.4\%$; 6 in 1,000 $= 0.6\%$. Absolute increase $= 0.2$ percentage points. Relative increase $= \frac{0.2}{0.4} = 50\%$.

3 $0.03 \times 5000 = 150$ expected cases.

4 The absolute increase is 1 case per 100,000, or 0.001 percentage points, which is tiny. A fair report says: "The risk doubled in relative terms, from 1 in 100,000 to 2 in 100,000 — one extra case per 100,000 people."

5 Of 20 with the condition, 90% detected: 18 true positives and 2 false negatives. Of the 980 without it, 5% false positives: 49 false positives and 931 true negatives. Positives total $18 + 49 = 67$, so $P(\text{condition} \mid \text{positive}) = \frac{18}{67} \approx 0.269$, about 26.9%.

6 99% specific means 1% of healthy people test positive: $0.01 \times 10{,}000 = 100$ false positives. When the condition is rare, these false positives can outnumber the true cases, so a positive result is not a diagnosis.

7 Benefits: early detection and treatment, reassurance for those who test negative, and information for planning support. Harms: anxiety and stigma from false positives, unnecessary treatment and cost, false reassurance from false negatives, and privacy risks. Needed data: prevalence, sensitivity and specificity, the effectiveness and harms of treatment, the cost of confirmatory testing, and how many people would be affected by each type of error.

8 Poster: "A doubled risk — from 1 in 10,000 to 2 in 10,000. In a group of 10,000 people, this means one additional case. The relative risk has doubled, but the absolute risk remains small; the change may still matter for those affected." The statement gives both the natural frequencies and the relative comparison, and says why the change matters without exaggeration.

9 The fall is $8 - 6 = 2$ injuries per 1,000, and $\frac{2}{8} = 0.25$, so "fell by 25%" is correct relative to the starting risk. In percentage points, the risk falls from 0.8% to 0.6%, a decrease of 0.2 percentage points. A fairer headline: "Injury risk fell from 8 to 6 per 1,000 players — two fewer injuries per 1,000."

10 Clinic A: $2{,}000 \times 0.02 = 40$ people have the condition, so 38 true positives and 2 false negatives; of the 1,960 healthy people, 4% is 78.4, about 78 false positives. Then $P(\text{condition} \mid \text{positive}) = \frac{38}{38 + 78} \approx 0.328$, about 32.8%. Clinic B: $500 \times 0.12 = 60$ have the condition, so 57 true positives; of 440 healthy people, 4% is 17.6, about 18 false positives; $P = \frac{57}{75} = 0.76$, or 76%. The test is identical; a higher prevalence means the people tested are more likely to have the condition, so a positive result is more informative. Clinic B tests fewer people but a higher-risk group.

11 Two acceptable explanations: (i) **lead-time bias** — screening finds the disease earlier without delaying death, so the five-year clock starts sooner and "survival" rises although lives are not extended; (ii) **overdiagnosis** — some detected cases would never have caused symptoms or death, so they inflate the survival figure. Other answers: the statistic says nothing about quality of life or about people who die within five years; "survival" is not "cure"; and without a comparison group the figure cannot show that treatment caused the outcome.

12 In 10,000 people, 1% is 100 with the condition: 90 true positives and 10 false negatives. Of the 9,900 healthy people, 5% is 495 false positives, leaving 9,405 true negatives. Positives total $90 + 495 = 585$, so $P(\text{condition} \mid \text{positive}) = \frac{90}{585} \approx 0.154$, about 15.4%. Patient wording: "Of every 100 people with a positive result, about 15 have the condition and about 85 do not. A positive result is therefore a reason for a confirmatory test, not a diagnosis." Harm of omitting the base rate: patients may assume a positive result almost certainly means disease, causing unnecessary anxiety or treatment, or may distrust a test once they discover the true figure.

INVESTIGATION 10 · HOW REPRESENTATIVE IS A SAMPLE?

Model results for the pilot sample. Students' own samples will differ; the method and interpretation are the assessable points.

1 Population: all 900 students. Variable: minutes of homework on a typical school night. A census is impractical because it would require contacting every student and would take too long; "typical" must be defined, for example "the most recent night that was not before a test or holiday", so that answers are comparable.

2 Number the 900 students on the roll. Use a random-number generator to produce numbers from 1 to 900, ignoring repeats, until 30 different students are chosen. This gives every student the same chance and avoids choosing only volunteers.

3 A neutral question: "On your most recent typical school night, how many minutes did you spend on homework? 0–30 / 31–60 / 61–90 / more than 90." Avoid "Don't you think homework takes too long?"; pilot the wording and revise any option that suggests a preferred answer.

4 Record refusals and absentees, not just responses. Compare the achieved sample with the school by year group; if Year 9 is under-represented, report it and consider following up or weighting. A high response rate matters as much as a large nominal sample.

5 Pilot mean $= \frac{600}{12} = 50$ minutes. Sorted data: 20, 30, 35, 40, 45, 45, 50, 55, 60, 60, 75, 85; median $= \frac{45 + 50}{2} = 47.5$ minutes. More than 60 minutes: 75 and 85, so $\frac{2}{12} \approx 16.7\%$. A student's own sample will give its own values.

6 Two random samples can differ because each selects different students and homework times vary. Neither is "wrong"; each is an estimate with sampling variation. A larger sample or repeated samples would give a more stable picture, and the spread between samples indicates the uncertainty.

7 Sources of bias: non-response (students who are busy or disengaged may not reply); self-report (students may round or misremember); ambiguous wording ("typical"); absentees. Remedies: follow up non-responders, use a diary or app, pilot and clarify the question, and include absent students on another day.

8 Report: "We surveyed a random sample of 30 students from the school roll about homework on a typical school night. The mean was about 50 minutes and the median 47.5 minutes in our pilot; our own sample gave a similar estimate. The estimate is not exact: non-response and self-report may bias it, and another sample would differ. We recommend using the figure as a guide, not a precise count, and repeating the survey each term." State clearly that the data cannot tell us why homework takes that long or what would happen if it changed.

Marking guide. Criteria B and C: a complete design with population, frame, random method and sample size; correct calculation of mean, median and proportion; comparison of samples with an explanation of sampling variation; three specific sources of bias with remedies; a report that states the estimate, method, limitations and a modest recommendation.

APPLICATION · THE SCREENING TEST

1 Two-way table: has condition — positive 45, negative 5, total 50; no condition — positive 95, negative 855, total 950; column totals 140 and 860; grand total 1,000.

2 $P(\text{positive}) = \dfrac{140}{1000} = 0.14 = 14\%$.

3 $P(\text{condition} \mid \text{positive}) = \dfrac{45}{140} \approx 0.321$, about 32%. A positive result means there is roughly a one-in-three chance that the person actually has the condition; most positives are false positives.

4 The test detects 90% of true cases (sensitivity) and correctly clears 90% of healthy people (specificity), but "90% accurate" does not say which. Because only 50 in 1,000 have the condition, the 95 false positives outnumber the 45 true positives. A positive result therefore needs confirmation, and the leaflet should report sensitivity, specificity and the positive predictive value.

5 With 10% prevalence, 100 of 1,000 have the condition: 90 true positives and 10 false negatives. Of the 900 healthy, 90 false positives and 810 true negatives. Positives total 180, so $P(\text{condition} \mid \text{positive}) = \dfrac{90}{180} = 0.5$, or 50%. The test is unchanged; the higher prevalence means a positive result is more likely to be genuine.

6 Clinic attendees are more likely to have symptoms or a referral, so prevalence is higher than in the general population; the positive predictive value estimated here would not apply to everyone. Attendees may also differ in age, health and risk factors, which affects sensitivity and specificity. Screening the general population would produce more false positives and a lower predictive value.

7 A false positive can cause anxiety, stigma, unnecessary treatment, cost and time off work; a false negative can delay treatment, give false reassurance and, if the condition is infectious, allow onward transmission. Screening everyone is justified only if the condition is serious, treatment is effective and the harms of false positives are manageable; otherwise test people with symptoms or high risk, and confirm every positive with a second test.

8 Patient explanation: "This test is good at finding the condition: it detects about 9 in 10 true cases. But it also gives a positive result for about 1 in 10 people who do not have it. In 1,000 people like those tested here, about 45 with the condition and 95 without it would test positive — so of every 140 positives, only about 45 are true. A positive result means you need a second test; it does not mean you have the condition. A negative result is reassuring but not a guarantee."

CRITERION TASK · THE TRAVEL SURVEY

Sample response. Marks depend on the quality of the design and communication, not on matching this wording.

1 Population: all 900 students. Sample: the 60 students surveyed at the gate. Variable: usual mode of travel to school. Claim: that 75% of students travel actively, so more bike racks than car spaces are needed.

2 The gate sample misses students who arrive by bus or car through another entrance, students who arrive later, absentees and those who use a different gate. Walkers and cyclists may be more visible or more likely to use that gate, so 75% over-represents active travel. The weather and the particular morning also affect who walks.

3 Use the school roll as the sampling frame, number all 900 students and select with a random-number generator. A stratified sample of 90 in proportion to year size: Year 7 $\dfrac{320}{900} \times 90 = 32$; Year 8 $\dfrac{300}{900} \times 90 = 30$; Year 9 $\dfrac{280}{900} \times 90 = 28$. Check: $32 + 30 + 28 = 90$.

4 Categorical item: "How do you usually travel to school? Walk / Cycle / Bus / Car / Other (please state)." Numerical item: "How long does your journey to school usually take? _____ minutes (one way)." Both avoid leading wording, offer balanced options and define "usually".

5 Non-response: students who are absent or busy may differ from responders, so the achieved sample may be biased; remedy by following up, offering several ways to respond and reporting the response rate. Self-report: students may misremember or give socially desirable answers; remedy with a travel diary, a parent check or objective counts at each entrance.

6 $\dfrac{54}{90} = 0.6$, so the estimate is $0.6 \times 900 = 540$ active travellers. The gate survey's 75% would suggest 675, a difference of 135 students. The random sample is more trustworthy because it covers all year groups and entrances and does not depend on who was at the gate. Neither is exact: sampling variation, non-response and definition of "active" remain.

7 Report: "We surveyed a stratified random sample of 90 students from the school roll. About 60% (54 of 90) usually travel actively, which suggests roughly 540 of 900 students. This is an estimate, not a count: non-response and self-report may affect it, and another sample would give a slightly different figure. The earlier gate survey's 75% probably over-represented walkers and cyclists. We recommend adding bike racks in stages and repeating the survey after installation, rather than treating 540 as exact."

Marking guide. Criteria B and C: precise identification of population, sample and claim; two specific biases tied to missed groups; a genuine random method and correct stratified numbers 32, 30, 28; neutral questionnaire items; non-response and self-report explained with workable remedies; correct estimate 540 and a justified comparison with 675; a report that states the method, limitations and a modest recommendation.

MIXED PRACTICE 10

1 Total 15. $P(\text{red}) = \frac{7}{15}$; $P(\text{not blue}) = \frac{10}{15} = \frac{2}{3}$; $P(\text{red or green}) = \frac{10}{15} = \frac{2}{3}$.

2 Sum 4: 3 outcomes, so $\frac{3}{36} = \frac{1}{12}$. Sum 8: 5 outcomes, so $\frac{5}{36}$. Even sum: 18 outcomes, so $\frac{18}{36} = \frac{1}{2}$.

3 Mutually exclusive: $P(A \cup B) = 0.35 + 0.4 = 0.75$; $P(\text{neither}) = 1 - 0.75 = 0.25$.

4 Relative frequency $= \frac{36}{240} = 0.15$. Expected successes in 800: $800 \times 0.15 = 120$.

5 Each toss is independent, so the coin has no memory and $P(\text{head}) = 0.5$ again. The law of large numbers says the long-run proportion will approach 0.5; it does not require the next toss to compensate for the past ten heads.

6 $P(A) \times P(B) = 0.4 \times 0.5 = 0.2 = P(A \cap B)$, so the events are independent. $P(A \cup B) = 0.4 + 0.5 - 0.2 = 0.7$.

7 Readers of a gardening magazine are a self-selected group with a strong interest in growing plants, so 80% describes them, not all adults. The sample is biased in a known direction; a larger sample of the same readers would not fix it. A random sample of adults is needed.

8 Absolute increase: 1 case per 20,000, or 0.005 percentage points. A fair report says: "The risk rose from 1 in 20,000 to 2 in 20,000 — one extra case per 20,000 people." This gives the natural frequencies alongside the relative doubling.

9 Each year has 280 students. A sample of 84 takes $\frac{280}{840} \times 84 = 28$ from each year. Walkers: $\frac{35}{84} = \frac{5}{12}$, so $840 \times \frac{5}{12} = 350$ estimated walkers.

10 100 have the condition: 95 true positives and 5 false negatives. Of 1,900 healthy, 10% false positive gives 190 false positives and 1,710 true negatives. Positives total 285, so $P(\text{condition} \mid \text{positive}) = \frac{95}{285} = \frac{1}{3} \approx 33.3\%$.

11 Digits 0, 1, 2 represent success in 10 equally likely digits, so the modelled probability is 0.3. Expected successes in 40 trials: $40 \times 0.3 = 12$. Estimate the probability by repeating many trials and dividing the total successes by the total trials; one run of 40 is only an estimate.

12 Class A's result is stronger evidence: its improvement is based on 20 students, while Class B's 10-mark improvement is based on only 5 and could easily be chance. A larger measured effect is not automatically more reliable; precision depends on sample size and variability, and practical importance depends on the outcome, not just the size of the number.

13 Increasing the sample does not remove bias because the same people remain more likely to respond. For example, an online poll on a fitness site may receive 50,000 responses, but its respondents are still disproportionately fitness-oriented; the estimate remains systematically different from the population.

14 Of 10,000 people, 1% = 100 have the condition: 99 true positives and 1 false negative. Of 9,900 healthy, 1% false positive gives 99 false positives and 9,801 true negatives. Positives total 198, so $P(\text{condition} \mid \text{positive}) = \frac{99}{198} = 0.5$, or 50%. Even a 99% sensitive and specific test gives a positive result that is correct only half the time when prevalence is 1%.

Common errors. Do not assume outcomes are equally likely just because there are two descriptions. "And" usually multiplies only for independent events; without replacement, update the denominator. A larger sample reduces random variation but not bias. Absolute risk and relative risk answer different questions, and a positive test result depends on prevalence as well as accuracy.

Cumulative review · Worked solutions

PAPER A · KNOWING AND UNDERSTANDING

1 $4.7 \times 10^{-3} = 0.0047$.

2 $98 = 49 \times 2$, so $\sqrt{98} = \sqrt{49}\sqrt{2} = 7\sqrt{2}$.

3 $(3 \times 2) \times 10^{4-6} = 6 \times 10^{-2}$, that is 0.06.

4 $2^{-2+5-(-1)} = 2^4 = 16$.

5 $x^2 + 7x - 4x - 28 = x^2 + 3x - 28$.

6 Difference of two squares: $x^2 - 9 = (x - 3)(x + 3)$.

7 $4x - 2x = 5 + 7$, so $2x = 12$ and $x = 6$.

8 From the second equation $y = x - 1$. Substituting: $2x + (x - 1) = 11$, so $3x = 12$, $x = 4$ and $y = 3$.

9 $y = -3x + c$ with $-1 = -3(2) + c$, so $c = 5$; hence $y = -3x + 5$.

10 Dividing by 2: $y = 3x - 4$; gradient 3, y-intercept -4.

11 $3x - 6 \le x + 8$, so $2x \le 14$ and $x \le 7$.

12 $7^2 + 24^2 = 49 + 576 = 625 = 25^2$. Since the square on the longest side equals the sum of the squares on the other two, the converse of Pythagoras gives a right angle opposite the 25 cm side.

13 Opposite $= 20 \sin 35^\circ = 11.5$ cm to three significant figures.

14 $V = \pi r^2 h = \pi(5^2)(12) = 300\pi \approx 942\ \text{cm}^3$.

15 $\frac{135}{360} = \frac{3}{8}$ of a full turn.

16 Mean $= \frac{4+6+6+7+12}{5} = \frac{35}{5} = 7$; the middle of five ordered values is the third, the median 6.

17 Strong negative correlation: as one variable increases, the other tends to decrease, and the points lie close to a downward line.

18 The primes on a die are 2, 3, 5, so $P(\text{prime}) = \frac{3}{6} = \frac{1}{2}$.

19 $P(\text{red}) = \frac{4}{10} = \frac{2}{5}$; $P(\text{not red}) = \frac{6}{10} = \frac{3}{5}$.

20 $|x - 8| \le 3$ means the distance from x to 8 is at most 3, so $5 \le x \le 11$.

Marking guide (Paper A, 37 marks). Award the method mark for a correct first step even if arithmetic slips, and the answer mark only for the correct final value with units where required. Common errors: writing $\sqrt{98} = 49\sqrt{2}$ (a factor taken out that is not a square root); dropping the negative sign in $2^{-2} \times 2^5 \div 2^{-1}$ when subtracting the index -1; using the two shorter sides in Q12 without checking which side is longest; leaving a volume without its cubic unit.

PAPER B · INVESTIGATING PATTERNS

1 Figure 1: 1 tile, perimeter 4. Figure 2: 3 tiles, perimeter 8. Figure 3: 6 tiles, perimeter 12. Figure 4: 10 tiles, perimeter 16. Figure 5: 15 tiles, perimeter 20.

2 First differences of T: 2, 3, 4, 5. Second difference: 1, 1, 1, constant. A constant second difference means T is a quadratic function of n, so the rule looks like $T = an^2 + bn + c$, not a direct proportion.

3 The tiles form the triangular numbers, $T = \frac{n(n+1)}{2}$. For $n = 6$: $T = \frac{6 \times 7}{2} = 21$. Counting: $1 + 2 + 3 + 4 + 5 + 6 = 21$, which agrees.

4 $P = 4n$. For $n = 4$: bottom edge 4, right side 4, left side 1, four top edges 4, and three vertical drops between the columns 3; these total $4 + 4 + 1 + 4 + 3 = 16 = 4(4)$, as required.

5 Need $\frac{n(n+1)}{2} > 100$, i.e. $n(n+1) > 200$. $n = 13$ gives $13 \times 14 = 182$ and $T = 91$; $n = 14$ gives $14 \times 15 = 210$ and $T = 105$. The least figure is Figure 14.

6 $P = 4n$ is four times a whole number, so it is always divisible by 4 and therefore always even. The claim is true; the rule for P explains why and generalises the observation from a few figures.

7 The model assumes unit squares that fit exactly with no gaps or trimming, and counts an ideal perimeter ignoring the wall or grout. To check a large figure, use the rule $T = \frac{n(n+1)}{2}$ in a spreadsheet and compare with a careful drawing of, say, Figure 10 on squared paper.

Marking guide (Paper B, 24 marks). Look for an ordered table, differences computed correctly, a rule that is verified rather than merely guessed, and a conclusion that names a limitation. A rule stated without verification earns the pattern mark but not the verification mark. Common errors: treating the first differences as constant; assuming $T \propto n$ rather than $T \propto n^2$; claiming a perimeter of $2n$ by counting only two sides.

PAPER C · COMMUNICATING

1 $OA = OB = OC$, because OA, OB and OC are all radii of the same circle, and the radius of a circle is constant. AB is a diameter, so O is its midpoint.

2 Triangle OAC is isosceles since $OA = OC$; triangle OBC is isosceles since $OB = OC$. In each case the equal sides are radii.

3 In triangle ABC: $\angle A = \alpha$, $\angle B = \beta$, and $\angle C = \angle OCA + \angle OCB = \alpha + \beta$, because OC splits the angle at C. The angles of a triangle sum to 180°, so $\alpha + \beta + (\alpha + \beta) = 180^\circ$, giving $2(\alpha + \beta) = 180^\circ$ and therefore $\alpha + \beta = 90^\circ$.

4 By step 3, $\angle ACB = \alpha + \beta = 90^\circ$. Since C was any point on the circumference (other than A and B), the angle in a semicircle is a right angle in every case.

5 Let the integers be n, $n + 1$ and $n + 2$. Their sum is $3n + 3 = 3(n + 1)$, which is 3 multiplied by the integer $n + 1$, so it is divisible by 3.

6 The invalid step is dividing both sides by $(a - b)$. Because $a = b$, $(a - b) = 0$, and division by zero is undefined, so the step is not permitted. For example, with $a = b = 1$, the factor $a - b = 0$; multiplying by zero has already destroyed the information that distinguished the two sides, and no legitimate operation can reconstruct it. The conclusion $2 = 1$ is therefore spurious, and the assumption $a = b$ never allows the division.

Marking guide (Paper C, 24 marks). Statements must be ordered and connected; each must be justified by a named fact (radii are equal; base angles of an isosceles triangle are equal; angles of a triangle sum to 180°). Award full credit for a proof that reaches $\alpha + \beta = 90^\circ$ with every step justified, even if the wording differs. In step 6 the crucial insight is that $(a - b) = 0$, so the division is undefined; a response that only says "because $a = b$" without naming the zero factor is incomplete.

PAPER D · APPLYING MATHEMATICS

1 $C_X = 8 + 0.90k$ and $C_Y = 20 + 0.40k$ euros per week.

2 $k = 0$: X = 8, Y = 20. $k = 10$: X = 17, Y = 24. $k = 20$: X = 26, Y = 28. $k = 30$: X = 35, Y = 32. $k = 40$: X = 44, Y = 36.

3 Setting $8 + 0.90k = 20 + 0.40k$ gives $0.50k = 12$, so $k = 24$ km. The graphs cross at $k = 24$, where each cost is €29.60.

4 For distances below 24 km, Tariff X is cheaper; for distances above 24 km, Tariff Y is cheaper. At exactly 24 km the two tariffs cost the same.

5 At 900 km: $C_X = 8 + 0.90(900) = 818$ and $C_Y = 20 + 0.40(900) = 380$. Tariff Y is cheaper, saving $818 - 380 = 438$ euros per week.

6 Now $C_Y = 20 + 0.60k$. Setting $8 + 0.90k = 20 + 0.60k$ gives $0.30k = 12$, so the break-even distance rises to $k = 40$ km. Tariff Y is now cheaper only above 40 km. At 900 km: $C_X = 818$ and $C_Y = 20 + 0.60(900) = 560$, so Y is still cheaper, saving $818 - 560 = 258$ euros. The recommendation is unchanged, but the advantage is smaller and the fixed weekly saving has shrunk.

7 Assumptions: the charge is based on exactly the kilometres driven, and the fixed and per-kilometre rates stay constant for the whole year, with no other fees. A real quotation might include tolls, fuel surcharges, waiting time, vehicle-size bands or annual price reviews, any of which would change the total.

8 Two equal sides use $2x$ m, so the third fenced side is $40 - 2x$ m. The area is $A = x(40 - 2x) = 40x - 2x^2$ square metres.

9 $x = 4 \Rightarrow 128$; $6 \Rightarrow 168$; $8 \Rightarrow 192$; $10 \Rightarrow 200$; $12 \Rightarrow 192$; $14 \Rightarrow 168$; $16 \Rightarrow 128$. The greatest area in the table is 200 m^2, at $x = 10$ m. (The table is symmetric, which is the sign of a maximum at the middle value.)

10 The table tests only every 2 m, so the true maximum may lie between two rows; the answer is therefore an estimate. Test closer values such as $x = 9,\ 9.5,\ 10,\ 10.5,\ 11$, or plot A against x and read the highest point.

Marking guide (Paper D, 33 marks). A model must define its variables and units, produce correct calculations and end with a justified recommendation. Full credit for the evaluation requires named assumptions and a reason they might fail, not a general remark. Common errors: swapping the fixed and per-kilometre parts of a tariff; reading the break-even from the intercepts instead of the intersection; forgetting to subtract to find the saving; reporting the area without units.

YEAR 3 EXIT TASK · DOES THE REVISION APP WORK?

1 With hours on the horizontal axis and score on the vertical axis, the points rise together and lie close to a straight line: a strong positive correlation. This matches $r \approx 0.985$, a value close to 1.

2 A line through the middle of the cloud gives a gradient of about $\frac{90 - 52}{12 - 1} \approx 3.5$ and an intercept near 48, so $y \approx 3.5x + 48$. Accept any line with gradient between about 3.4 and 3.6 and intercept between about 47 and 50, provided it passes through the cloud.

3 Substituting $x = 14$: $y \approx 3.5(14) + 48 = 97$. The prediction is unreliable: 14 hours lies beyond the largest value in the data (12 hours), so this is extrapolation, and 97 is close to the top of the scale, leaving little room for the trend to continue.

4 The two students differ by 5 hours, so the expected difference is $5 \times 3.5 \approx 17.5$ marks. Using the line, 12 hours gives about 90 and 7 hours gives about 72.5, a gap of about 17.5. The claim of "about 12 marks more" understates the trend; the true difference is nearer 17 to 18 marks.

5 A high correlation does not establish cause. A third factor — for example how motivated or able a student already is — could increase both app use and test score, producing the pattern without the app causing it. To test causation fairly, run a controlled experiment: assign students at random to use the app or to a comparison group, keep other conditions the same, and compare the change in scores.

6 About 30% of 200 is $0.30 \times 200 = 60$ students. This may be too high because the 30% figure came from a different school with a different intake, or because it recorded full-price purchases in another context; a free offer still costs students time, and other schools may respond differently.

7 Sample recommendation: *strength* — the data show a strong positive association ($r \approx 0.985$) and a consistent upward trend; *limitation* — the sample is small (12 students), the data are observational, and the prediction for 14 hours is an extrapolation; *request* — a larger random sample, a controlled trial, and data on prior attainment to separate the app's effect from other factors.

Marking guide (exit task, 30 marks). Credit clear axes and a sensible line of best fit; correct substitution and honest comment on extrapolation; an explanation of correlation versus causation that names a specific confounding variable and a realistic experiment; and a recommendation that balances a strength against a limitation. Common errors: reading the gradient from the first and last points without regard to the spread; claiming the data "prove" the app works; predicting beyond the data without comment; and reporting a percentage estimate as an exact count.

REFERENCE · FORMULA BANK

Formula bank

A compact reminder of the relationships used in this course. A formula is a tool, not a substitute for reasoning: name the quantities, check the units and ask whether the result is sensible in the context.

NUMBER AND STRUCTURE

Rational number	$\frac{a}{b}$, with integers a, b and $b \neq 0$
Terminating decimal	denominator has only prime factors 2 and 5
Recurring decimal	all denominators with another prime factor
Surd product	$\sqrt{a}\sqrt{b} = \sqrt{ab}\ (a, b \geq 0)$
Surd quotient	$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}\ (a \geq 0,\ b > 0)$
Rationalise a denominator	$\frac{1}{\sqrt{a}} = \frac{\sqrt{a}}{a}$
Absolute value	$\lvert x\rvert = x$ if $x \geq 0$; $\lvert x\rvert = -x$ if $x < 0$
Distance on a line	$\lvert b - a\rvert$
Density	between any two rationals there is another rational
Set operations	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

POWERS, ROOTS AND SCIENTIFIC NOTATION

Product law	$a^m \times a^n = a^{m+n}$
Quotient law	$a^m \div a^n = a^{m-n}$
Power of a power	$(a^m)^n = a^{mn}$
Product to a power	$(ab)^n = a^n b^n$
Zero and negative	$a^0 = 1, \quad a^{-n} = \frac{1}{a^n}\ (a \neq 0)$
Rational exponent	$a^{1/n} = \sqrt[n]{a}, \quad a^{m/n} = \sqrt[n]{a^m}$
Scientific notation	$a \times 10^n$, where $1 \leq a < 10$
Order of magnitude	the nearest power of ten
Relative error	$\frac{\text{error}}{\text{true value}} \times 100\%$

ALGEBRA, FUNCTIONS AND GRAPHS

Linear function	$y = mx + c$
Gradient from two points	$m = \frac{y_2 - y_1}{x_2 - x_1}$
Direct proportion	$y = kx$, so $\frac{y}{x} = k$
Function notation	$x \mapsto f(x)$, for x in the domain
Arithmetic sequence	$u_n = u_1 + (n-1)d$
Difference of squares	$a^2 - b^2 = (a+b)(a-b)$
Expand a product	$(x+p)(x+q) = x^2 + (p+q)x + pq$
Parallel lines	$m_1 = m_2$
Perpendicular lines	$m_1 m_2 = -1$

EQUATIONS, INEQUALITIES AND SYSTEMS

Solve a linear equation	$ax + b = c \Rightarrow x = \frac{c-b}{a}\ (a \neq 0)$
Inequality rule	reverse the sign when multiplying or dividing by a negative
Simultaneous equations	eliminate one variable, or substitute
Graphical solution	the intersection point satisfies both equations
No solution	parallel lines with different intercepts
Infinitely many	the equations describe the same line

TRANSFORMATIONS, CONGRUENCE AND SYMMETRY

Translation	$(x, y) \mapsto (x + a,\ y + b)$
Reflection in the x-axis	$(x, y) \mapsto (x,\ -y)$
Reflection in the y-axis	$(x, y) \mapsto (-x,\ y)$
Rotation 180° about O	$(x, y) \mapsto (-x,\ -y)$
Rotation 90° anticlockwise	$(x, y) \mapsto (-y,\ x)$
Enlargement	scale factor k, centre O: distances from O multiply by k
Congruence tests	SSS, SAS, ASA, RHS
Tessellation	angles meeting at each point sum to 360°

SIMILARITY, PROOF AND TRIGONOMETRY

Similar figures	equal angles; corresponding lengths in ratio k
Scale factor effects	lengths $\times k$, areas $\times k^2$, volumes $\times k^3$
Similarity tests	AA, SSS (ratios), SAS (included angle)
Pythagoras	$a^2 + b^2 = c^2$, c the hypotenuse
Sine	$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$
Cosine	$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$
Tangent	$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$
Exact values	$30°, 45°, 60°$ from the standard triangles

SOLIDS AND MEASUREMENT

Prism volume	$V = Bh$, B area of cross-section
Cylinder	$V = \pi r^2 h, \quad SA = 2\pi r^2 + 2\pi rh$
Pyramid and cone	$V = \frac{1}{3}Bh$
Sphere	$V = \frac{4}{3}\pi r^3, \quad SA = 4\pi r^2$
Cone surface area	$SA = \pi r^2 + \pi rl$, slant height l
Capacity	$1\ \text{cm}^3 = 1\ \text{mL}, \quad 1000\ \text{cm}^3 = 1\ \text{L}$
Density	$\rho = \frac{m}{V}$
Euler's formula	$F + V = E + 2$ for a convex polyhedron

DATA AND BIVARIATE RELATIONSHIPS

Mean	$\bar{x} = \frac{\sum x}{n}$
Range	largest value minus smallest value
Line of best fit	$y = mx + c$, chosen to keep residuals small
Residual	observed value minus predicted value
Correlation coefficient	$-1 \leq r \leq 1$; sign gives direction
Coefficient of determination	r^2, the fraction of variation explained
Interpolation	predicting inside the data range
Extrapolation	predicting outside the data range: unreliable

PROBABILITY, SAMPLING AND RISK

Probability	$P(A) = \frac{\text{number of outcomes in } A}{\text{total outcomes}}$, equally likely
Range and complement	$0 \le P(A) \le 1, \quad P(A) + P(A') = 1$
Addition rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Independent events	$P(A \cap B) = P(A) \times P(B)$
Expected frequency	$n \times p$
Relative frequency	$\frac{\text{times the outcome occurred}}{\text{number of trials}}$
Relative risk	risk after ÷ risk before
Sensitivity	$\frac{\text{true positives}}{\text{all with the condition}}$
Specificity	$\frac{\text{true negatives}}{\text{all without the condition}}$
Positive predictive value	$\frac{\text{true positives}}{\text{all positive results}}$

Notation at a glance

$=, \neq, \approx$	is equal to, is not equal to, is approximately equal to
$<, >, \le, \ge$	is less than, greater than, at most, at least
$\lvert x \rvert$	the absolute value (modulus) of x
$\sqrt{a},\ \sqrt[3]{a}$	square root, cube root of a
$a^n,\ a^{-n},\ a^{1/n}$	powers, reciprocals and roots
$a \times 10^n$	scientific notation
$f(x)$	the value of the function f at x
u_n	the nth term of a sequence
$\in,\ \notin,\ \subseteq$	is an element of, is not an element of, is a subset of
$\varnothing,\ \xi,\ n(A)$	empty set, universal set, number of elements in A
$A \cup B,\ A \cap B,\ A'$	union, intersection, complement
$\Rightarrow,\ \therefore$	implies, therefore
$\cong,\ \sim$	is congruent to, is similar to
$\parallel,\ \perp$	is parallel to, is perpendicular to
$\angle ABC$	the angle at vertex B
θ	a labelled angle, often in trigonometry
$\sin\theta,\ \cos\theta,\ \tan\theta$	trigonometric ratios of θ
π	pi, about 3.14159
$\bar{x},\ \sum x$	the mean, the sum of the values
$r,\ r^2$	correlation coefficient, coefficient of determination
$P(A)$	the probability of event A
$P(A \mid B)$	the probability of A given that B has occurred

Units matter. Write the unit with every measurement and read it as part of the answer. A number without its unit is often incomplete, and a probability without its context can mislead.

REFERENCE · VOCABULARY

Glossary

Working definitions of the terms used in this course. Where a term has a precise technical meaning, the definition states the condition the term depends on.

absolute risk — a risk stated as a number out of a group, such as 3 in 1,000.

absolute value — the distance of a number from zero; written $|x|$, always non-negative.

addition rule — $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ for two events.

angle of depression — the angle measured downwards from the horizontal to a line of sight.

angle of elevation — the angle measured upwards from the horizontal to a line of sight.

arithmetic sequence — a sequence with a constant difference between consecutive terms.

base rate — the prevalence of a condition in the group tested; it shapes what a positive result means.

bias — a systematic tendency for a sample or measurement to differ from the population in one direction.

bivariate data — data recorded as pairs of values, so that a relationship between two variables can be examined.

capacity — the amount a container can hold, measured in litres or millilitres.

causation — a relationship in which a change in one variable produces a change in another; it is not established by correlation alone.

census — a survey that measures every member of a population.

coefficient of determination — r^2, the proportion of the variation in one variable explained by the linear model.

complement (probability) — the event that A does not occur, written A'; $P(A') = 1 - P(A)$.

complement (sets) — the elements of the universal set that are not in the given set.

composite solid — a solid made by joining two or more simpler solids.

compound event — an event formed by combining two or more simple events.

cone — a solid with a circular base and a single vertex; its volume is $\frac{1}{3}\pi r^2 h$.

confounding variable — a variable related to both others in a study, which can create a misleading association.

congruent — identical in shape and size; corresponding lengths and angles are equal.

conjecture — a statement believed to be true on the evidence so far but not yet proved.

continuous data — data that can take any value in an interval, such as a length or a time.

convenience sample — a sample chosen because it is easy to reach; not random and usually biased.

converse — the statement formed by swapping the condition and the conclusion.

correlation — the strength and direction of a linear association between two variables.

correlation coefficient — the number r, between -1 and 1, measuring the fit of a linear relationship.

corresponding — matching in position or role between two figures or two sets of data.

counterexample — a single case that shows a general statement is false.

cross-section — the shape seen when a solid is cut parallel to its end.

cube root — the number that, multiplied by itself three times, gives the original number.

dependent events — events whose probabilities change when one of them occurs.

direct proportion — a relationship of the form $y = kx$, so the ratio $y : x$ is constant.

domain — the set of values that may be used as the input of a function.

ecological fallacy — drawing conclusions about individuals from grouped data.

enlargement — a transformation that multiplies every distance from a centre by the same scale factor.

equally likely — outcomes with the same probability, justified by the physical setup rather than by counting.

Euler's formula — for a convex polyhedron, faces + vertices = edges $+2$.

event — a set of outcomes from a sample space.

expected frequency — the number of times an outcome is expected in n trials: $n \times p$.

explanatory variable — the variable plotted on the horizontal axis, thought to explain the other.

extrapolation — using a model to predict outside the range of the observed data; it is unreliable.

false negative — a negative test result for someone who has the condition.

false positive — a positive test result for someone who does not have the condition.

fair — describing a die, coin or spinner whose outcomes are equally likely.

function — a rule that assigns exactly one output to each input in the domain.

gambler's fallacy — the mistaken belief that past independent trials change the probability of the next one.

gradient — the steepness of a line: the change in y divided by the change in x.

hypotenuse — the side opposite the right angle in a right-angled triangle.

independent events — events for which $P(A \cap B) = P(A) \times P(B)$.

index laws — the rules for multiplying and dividing powers.

interpolation — using a model to predict inside the range of the observed data.

intersection — $A \cap B$, the outcomes belonging to both A and B.

invariant — unchanged by a transformation, such as a length under a translation.

irrational number — a number that cannot be written as a fraction of two integers.

law of large numbers — informally, relative frequencies settle near theoretical probabilities as the number of trials grows.

line of best fit — the straight line that best describes a scatter plot of bivariate data.

line symmetry — a shape has line symmetry if a reflection in a line maps it onto itself.

linear equation — an equation in which each variable appears to the first power only.

linear relationship — a relationship whose graph is a straight line.

magnitude — the size of a quantity, often expressed as a power of ten.

mean — the sum of a set of values divided by how many there are.

median — the middle value when the data are arranged in order.

modal — describing the most frequent value or category in a data set.

mutually exclusive — describing events that cannot occur together, so $P(A \cap B) = 0$.

natural frequency — a risk or probability expressed as counts, such as "45 in 1,000".

net — a flat pattern that folds to make a solid.

non-response — failure of selected individuals to reply to a survey, a common source of bias.

order of magnitude — the nearest power of ten to a quantity.

outcome — one possible result of a probability experiment.

outlier — a value that lies far from the general pattern of the data.

percentage point — a unit for the difference between two percentages.

polyhedron — a solid whose faces are flat polygons.

population — the whole group about which information is wanted.

positive predictive value — the proportion of positive results that are true positives.

precision — how close repeated measurements are to each other.

prevalence — the proportion of a population that has a condition at a given time.

prism — a solid with a constant cross-section along its length.

probability — a number between 0 and 1 measuring how likely an event is.

proof — a chain of reasoning that establishes a statement beyond doubt from agreed starting points.

proof by contradiction — a proof that assumes the opposite of the claim and shows it leads to an impossibility.

range (function) — the set of values that a function produces as outputs.

range (statistics) — the largest value in a data set minus the smallest.

random sample — a sample in which every member of the population has a known, non-zero chance of selection.

rational number — a number that can be written as a fraction of two integers.

reflection — a transformation that flips a shape across a line without changing lengths or angles.

regression to the mean — the tendency of extreme measurements to be followed by less extreme ones.

relative frequency — the number of times an outcome occurred divided by the number of trials.

relative risk — one risk divided by another, used to compare them as a ratio.

representative — describing a sample with roughly the same composition as the population.

residual — the observed value minus the value predicted by a model.

response variable — the variable plotted on the vertical axis, measured in response to the other.

risk — the probability of an unwanted outcome.

rotation — a transformation that turns a shape about a fixed point by a given angle.

rotational symmetry — a shape has rotational symmetry if some rotation maps it onto itself.

sample — the part of a population that is actually observed.

sample space — the set of all possible outcomes of a probability experiment.

sampling frame — the list from which a sample is drawn.

sampling variation — the differences between estimates from different random samples of the same population.

scatter plot — a graph plotting paired bivariate data as points.

scientific notation — a number written as $a \times 10^n$ with $1 \le a < 10$.

self-selection — bias caused by individuals choosing whether to take part.

sensitivity — the proportion of people with a condition that a test correctly identifies.

significant figures — the digits of a number that carry meaning about its precision.

similar — having the same shape but not necessarily the same size.

simulation — a model that imitates a probability experiment, often using random digits.

simultaneous equations — two or more equations that must be satisfied at the same time.

solid — a three-dimensional shape with faces, edges and vertices.

specificity — the proportion of people without a condition that a test correctly clears.

sphere — the set of points at a fixed distance from a centre; its volume is $\frac{4}{3}\pi r^3$.

standard form — another name for scientific notation, $a \times 10^n$.

stratified sample — a sample that selects from each subgroup in proportion to its size.

subset — a set whose elements all belong to a larger set.

surd — an expression containing a root of a number that is not a perfect power, such as $\sqrt{2}$.

surface area — the total area of all the faces of a solid.

systematic sample — a sample formed by taking every kth member of a frame after a random start.

tangent — the trigonometric ratio opposite divided by adjacent; also a line touching a curve at one point.

tessellation — a tiling of the plane by shapes with no gaps or overlaps.

theorem — a mathematical statement that has been proved.

transformation — a change in the position, size or orientation of a shape.

translation — a transformation that slides every point the same distance in the same direction.

tree diagram — a diagram showing the stages of a multi-stage experiment and their probabilities.

trial — one repetition of a probability experiment.

two-way table — a table recording counts for two categorical variables at once.

union — $A \cup B$, the outcomes belonging to A or B or both.

universal set — the set ξ containing every outcome or element under discussion, shown as a rectangle.

validity — the extent to which a method measures what it claims to measure.

Venn diagram — a diagram using overlapping regions to show sets and their relationships.

vertex — a corner of a solid or polygon where edges meet.

volume — the space occupied by a solid, measured in cubic units.

voluntary-response sample — a sample made of people who choose to reply; strongly self-selected.

with replacement — describing a draw in which the item is returned before the next draw, keeping probabilities constant.

without replacement — describing a draw in which the item is not returned, changing later probabilities.

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REFERENCE · SOURCES AND CREDITS

References and acknowledgements

This coursebook is an original work. Public framework material was consulted to align the curriculum architecture; no third-party text, exercise, illustration, layout or page design has been reproduced.

FRAMEWORK AND CURRICULUM

- International Baccalaureate Organization. *Middle Years Programme: Mathematics subject brief.* IB Organization. Public information on aims, objectives and assessment criteria.
- International Baccalaureate Organization. *MYP: From principles into practice.* IB Organization. Public guidance on the inquiry cycle, key and related concepts, global contexts and approaches to learning.
- International Baccalaureate Organization. *Mathematics guide: Middle Years Programme.* IB Organization. Public framework structure consulted for concept strands and criterion language; all wording here is independently written.

SCOPE REFERENCE

The publicly listed contents of Oxford University Press's *MYP Mathematics 3* were consulted only to check the typical Year 3 scope and sequence. No proprietary wording, exercise, illustration, layout, page design or answer text was copied or adapted.

PEDAGOGY AND ASSESSMENT

- Wiggins, G., and McTighe, J. *Understanding by Design.* 2nd ed. ASCD, 2005.
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AUTHOR

SADIKOU Moukoddim

HEAD OF MATHEMATICS DEPARTMENT | MATHEMATICS TEACHER | CATEGORY 1

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This is an independent educational publication. It is not published, endorsed, sponsored or approved by the International Baccalaureate Organization, Oxford University Press or any examination board or publisher. "IB" and "MYP" describe only the curriculum framework to which the material is mapped.

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SUPPLEMENTARY SECTION

MYP Mathematics 3 — Student Coursebook

Spec-gap additions and verified corrections · 2026 verification pass

- How to use this section
- Chapters 1–10 — Try-it checks and Stretch & Apply boxes, with answers
- Answers-section entries to add
- Corrections applied in this edition

Appended to the coursebook — the original pagination is unchanged.

How to use this section

Same gap as MYP1/MYP2: the "How the labels work" page lists Definition, Key idea, Common error, ATL and Criterion-strip boxes, but the spec also requires **Try-it checks** and **Stretch & Apply (SaA)** boxes, of which the book has none. One of each per content chapter (10 chapters; the Part E cumulative review is exam practice and is excluded).

Placement convention

- **Try-it check**: after the first worked example of the chapter's central section.
- **Stretch & Apply**: at the end of the concept sequence, just before the chapter's investigation/application.

Label-page text to add (front matter, "How the labels work"): same two definitions as ../MYP1/chapter_additions.md.

> Try-it checks ask one or two rapid questions the moment a method is first shown. They test whether the new move is safe before you carry on; answers are given, so you can correct yourself in seconds. Stretch & Apply boxes move the method one step into a new context. They are deliberately short, sit at the end of a concept sequence, and reward connecting the idea to something you have already seen.

Chapter 1 — The real number system

Trial placement: after the first worked example in 1.4 (working with surds); SaA before "Chasing √2 with fractions".

Try-it check

1. Simplify $\sqrt{50}$.
2. Simplify $\sqrt{12} + \sqrt{27}$.

Answers: (1) $5\sqrt{2}$. (2) $2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$.

Stretch & Apply

Is $\sqrt{7}$ rational or irrational? Explain using the definition of a rational number.

Answer direction: irrational. If $\sqrt{7} = p/q$ in lowest terms, then $p^2 = 7q^2$, so 7 divides p, hence 7 divides q — contradicting "lowest terms" (same structure as the $\sqrt{2}$ proof).

Chapter 2 — Powers, roots and scientific notation

Trial placement: after the first worked example in 2.1 (exponent laws); SaA before "A scale model of the solar system".

Try-it check

1. Simplify $2^5 \times 2^3$.
2. Simplify $(3^2)^4$.

Answers: (1) 2^8. (2) 3^8.

Stretch & Apply

Write 0.0047 and 470 000 000 in standard form. Then decide which is larger: 4.7×10^{-3} or 47×10^{-4} — and explain why the comparison is not obvious from the forms.

Answer: 4.7×10^{-3} and 4.7×10^{8}; they are equal ($47 \times 10^{-4} = 4.7 \times 10^{-3}$) — standard form is only unique when the front digit is 1–9.

Chapter 3 — Functions, expressions and algebraic reasoning

Trial placement: after the first worked example in 3.3 (expanding and factorising); SaA before "The matchstick triangles".

Try-it check

1. Expand $3(2x - 5)$.
2. Factorise $x^2 + 7x + 12$.

Answers: (1) $6x - 15$. (2) $(x + 3)(x + 4)$.

Stretch & Apply

A rectangle has area $x^2 - 9$ and one side $x - 3$. Find the other side.

Answer: $(x^2 - 9)/(x - 3) = (x - 3)(x + 3)/(x - 3) = x + 3$.

Chapter 4 — Linear relationships and graphs

Trial placement: after the first worked example in 4.2 (gradient as a rate of change); SaA before "A library of things".

Try-it check

1. Find the gradient of the line through (1, 2) and (4, 8).
2. Write the equation of the line with gradient 2 and y-intercept -3.

Answers: (1) $(8 - 2)/(4 - 1) = 2$. (2) $y = 2x - 3$.

Stretch & Apply

A taxi charges a £3 flag-fall plus £2 per kilometre. Write the cost rule for a journey of d km, and say what the y-intercept and the gradient each mean in the situation.

Answer: $C = 3 + 2d$; intercept = the charge before any distance is travelled, gradient = the rate per km.

Chapter 5 — Equations, inequalities and systems

Trial placement: after the first worked example in 5.1 (solving linear equations); SaA before "The three weighings".

Try-it check

1. Solve $2x - 3 > 5$.

2. Solve the system $y = x + 1$ and $y = 2x - 1$.

Answers: (1) $x > 4$. (2) $x = 2$, $y = 3$.

Stretch & Apply

A café sells coffee at £2 and tea at £1.50. On one day it sold 40 drinks and took in £75. How many of each did it sell?

Answer: $c + t = 40$ and $2c + 1.5t = 75 \rightarrow c = 30$, $t = 10$ (check: $60 + 15 = 75$).

Chapter 6 — Transformations, congruence and symmetry

Trial placement: after the first worked example in 6.3 (rotating about a point); SaA before "The angle that fits".

Try-it check

1. Rotate the point (2, 1) by 90° anticlockwise about the origin.
2. Translate (0, 0) by $(3, -4)$, then reflect the image in the x-axis.

Answers: (1) $(-1, 2)$. (2) $(3, -4) \rightarrow (3, 4)$.

Stretch & Apply

A triangle has vertices (1, 1), (4, 1), (1, 4). A second triangle has vertices $(-1, -1)$, $(-4, -1)$, $(-1, -4)$. Name a transformation that maps the first onto the second, and say whether the image is congruent to the original.

Answer: rotation of 180° about the origin (equivalently, reflection in the origin); yes — rotations preserve lengths and angles.

Chapter 7 — Similarity, proof and trigonometry

Trial placement: after the first worked example in 7.3 (Pythagoras' theorem); SaA before "The shadow rule".

Try-it check

1. Find the hypotenuse of a right-angled triangle with legs 6 and 8.
2. In a right-angled triangle, the side opposite θ is 5 and the hypotenuse is 13. Find

sin θ and the angle θ to 1 decimal place.

Answers: (1) 10. (2) $\sin\theta = 5/13 \approx 0.385$; $\theta \approx 22.6°$.

Stretch & Apply

A ladder 5 m long leans against a wall, its foot 1.4 m from the wall. How high up the wall does it reach? (1 decimal place)

Answer: $\sqrt{(5^2 - 1.4^2)} = \sqrt{23.04} \approx 4.8$ m.

Chapter 8 — Three-dimensional solids and measurement

Trial placement: after the first worked example in 8.4 (volume of prisms and cylinders); SaA before "The rainwater tank".

Try-it check

1. Find the volume of a cone of radius 4 cm and height 9 cm, in terms of π.
2. Find the volume of a sphere of radius 3 cm, in terms of π.

Answers: (1) ⅓π(16)(9) = 48π cm³. (2) 4/3 π(27) = 36π cm³.

Stretch & Apply

A rainwater tank 1 m by 0.8 m by 0.6 m reaches 60% of its capacity after a storm. How many litres of water did it collect?

Answer: capacity 0.48 m³ = 480 L; 60% of 480 = 288 L.

Chapter 9 — Bivariate data, correlation and causation

Trial placement: after the first worked example in 9.2 (measuring linear association); SaA before "The screen-time claim".

Try-it check

1. A correlation of r = 0.9 between study time and test score — describe the association.
2. If r = 0 for two variables, what can and cannot be concluded?

Answers: (1) Strong positive linear association. (2) There is no linear association; a non-linear relationship may still exist, and correlation never establishes cause.

Stretch & Apply

Ice-cream sales and sunburn cases both rise over the summer. A headline claims ice-cream "causes" sunburn. Explain the flaw in the claim and name the hidden factor that explains both.

Answer: confounding — warmer weather (more outdoor time) raises both; association is not causation.

Chapter 10 — Probability, sampling and evidence

Trial placement: after the first worked example in 10.2 (compound events); SaA before "The screening test".

Try-it check

1. Two fair dice are rolled. Find P(both even).
2. Find P(the sum is 7).

Answers: (1) (1/2)(1/2) = 1/4. (2) 6 of 36 outcomes, so 1/6.

Stretch & Apply

A screening test is 95% accurate. In a population where 1% have the condition, estimate the fraction of positive results that are false positives, using a population of 10 000.

Answer: 100 have it → 95 true positives; 9 900 are healthy → about 495 false positives; of 590 positives, about 495/590 ≈ 84% are false positives (base-rate effect).

Answers-section entries to add

Append under a new heading in the answers part (page 239 onwards):

> TRY-IT & STRETCH & APPLY — CH 1: 1 (5√2; 5√3) 2 (irrational; contradiction in lowest terms). CH 2: 1 (2^8; 3^8) 2 (4.7×10^{-3}; 4.7×10^8; equal). CH 3: 1 ($6x - 15$; $(x+3)(x+4)$) 2 ($x + 3$). CH 4: 1 (2; $y = 2x - 3$) 2 ($C = 3 + 2d$; intercept = flag-fall, gradient = £/km). CH 5: 1 ($x > 4$; (2, 3)) 2 (30 coffee, 10 tea). CH 6: 1 ((−1, 2); (3, 4)) 2 (180° rotation; congruent). CH 7: 1 (10; 5/13, ≈ 22.6°) 2 (≈ 4.8 m). CH 8: 1 (48π cm^3; 36π cm^3) 2 (288 L). CH 9: 1 (strong positive; no linear association, no causation) 2 (confounding: weather). CH 10: 1 (1/4; 1/6) 2 (≈ 84%).

Corrections applied in this edition

Every answer-key entry and worked example in this book was independently recomputed during the verification pass. **No corrections were required.** The Try-it and Stretch & Apply boxes in this section were likewise recomputed before inclusion.

MIDDLE YEARS PROGRAMME
YEAR 3 · MATHEMATICS

FIRST EDITION
2026

ABOUT THIS COURSEBOOK

An inquiry-led course in number, algebra, proof, space and evidence — built around concepts, contexts and criteria, and designed as the third volume of a coherent five-year series.

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Real numbers · Powers and scientific notation · Functions and algebra
Linear relationships · Equations and systems · Transformations and congruence
Similarity and trigonometry · Solids and measurement · Bivariate data · Probability and evidence

COMPLETE WORKED SOLUTIONS · INVESTIGATIONS · CRITERION-MAPPED ASSESSMENT TASKS
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SADIKOU Moukoddim
HEAD OF MATHEMATICS DEPARTMENT · MATHEMATICS TEACHER · CATEGORY 1

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